Learning speech sounds from naturalistic input

Kasia Hitczenko (University of Delaware)

Infants learn about the speech sounds of their language despite extensive acoustic variability in natural speech. This talk explores infants' early speech sound learning in two parts. First, I discuss speech sound discrimination, where I show that traditional models of speech sound learning do not work on naturalistic speech data. I propose a new account of how infants learn about speech sounds, based on the idea that contextual regularities present in natural speech (e.g., phonotactics) can help infants distinguish speech sounds that are acoustically overlapping. Then, I turn to children's speech sound production, where I examine the development of canonical babbling in children around the world, and show how linguistic factors (e.g. the phonotactics of the target language), intrinsic factors (e.g., child vocalization practice), and extrinsic factors (e.g., caregiver contingency and speech) influence the developmental trajectory of children's early speech productions. Together, this work shows the importance of studying naturalistic speech for arriving at better models of early language learning.

Syllabification is more abstract than it appears: Initial codas and final onsets

Heather Goad (McGill University)

This presentation examines several puzzling patterns in the distribution of consonants across languages and argues that they indicate that syllables are more abstract than they appear. At the left edge of words: If branching onsets optimally rise in sonority, why are reverse sonority clusters tolerated word-initially in Selayarese? If word-initial sC clusters are branching onsets, why are the only sC clusters permitted in French flat-sonority s+stop? At the right edge of words: If syllable-final consonants are codas, why are there languages where word-final singletons and rising-sonority clusters have an onset profile, as observed in the history of French? In CVC reduplication, why does the reduplicant-final C get repaired in Ponapean but not in Mokilese? It will be proposed that puzzles such as these indicate that syllables/words in some languages can begin with codas and/or end with onsets. Additional questions include: Why do initial and final syllabification anomalies arise in the first place? Do syllabification anomalies in initial and final position arise under the same conditions? Are they stable across time?

Learning new speech sounds: Intersections of phonology and second language acquisition

Melissa Baese-Berk (University of Chicago)

Learning a second language is a complex task, requiring acquisition of syntactic structure, lexical items, and phonological structure. Perception and production of novel speech sounds are notoriously difficult for non-native speakers. Existing theories have provided hypotheses about how the relationship between first and second languages predicts perception and acquisition of contrasts in the second language. In this talk, I summarize a series of projects which examine several factors that may influence perception and production of novel speech sounds. The work I will present addresses several factors including: the relationship between perception and production during learning, the role of variability during training, and how this relationship might shift both over time and across contrasts. I will discuss the implications of the results of these studies for our understanding of speech perception and production more broadly and how these results inform theories of second language learning and phonological theory.

Computational Complexity or Typological Naturalness? Learning At-Least-One and Rightmost Vowel Harmony

Regine Lai[1] and Chunzi Li[1]

[1] *Department of Linguistics and Modern Languages, The Chinese University of Hong Kong*

Introduction. Phonological learning may be shaped by both computational complexity and typological naturalness. According to Subregular Hypothesis [1], the patterns that belong to the lower classes within the Subregular Hierarchy are less complex and thus easier to learn than in higher classes. Artificial Grammar Learning studies show that computationally simpler patterns are often learned more readily than complex ones [2–3], while learners may also favor typologically natural patterns [4–5]. Because these factors often align, it remains difficult to determine whether learning advantages reflect computational simplicity, typological naturalness, or both. At-Least-One (ALO) and Rightmost vowel harmony dissociate these factors. In ALO, suffix selection depends only on whether a triggering vowel occurs anywhere in the stem; in Rightmost, it depends on which non-neutral vowel is rightmost. Under the Subregular Hypothesis, ALO belongs to the *Locally Testable* class, which is lower in the Subregular Hierarchy than the *Non-Counting* class to which Rightmost belongs. Thus, ALO is computationally simpler: it only requires detecting whether a triggering vowel occurs in the stem, whereas Rightmost requires identifying the relative position of relevant vowels. However, ALO appears unattested in vowel harmony, although a similar ALO dependency is found in Yaka nasal consonant harmony. Rightmost, by contrast, is attested in vowel harmony systems, as in Hungarian [6]. Hwangbo [7] reported that adults learned ALO but not Rightmost vowel harmony, suggesting that computational complexity can guide learning despite typological unattestedness. However, all stems in Hwangbo's design ended in a neutral vowel, which may have made the Rightmost rule harder to detect by requiring learners to attend to the penultimate non-neutral vowel. To address this, the present study revisits the ALO–Rightmost contrast using a more varied set of stem types and asks whether this relationship changes across development by testing adults and 14- and 20-month-old toddlers.

Method. Stimuli were trisyllabic stems followed by a suffix, with syllables drawn from CV and CVN structures. Vowels included front vowels /i, e/, back vowels /u, ɔ/, and the low central vowel /a/ as the neutral vowel, because /a/ has no corresponding [+front] or [+back] counterpart in the Cantonese vowel inventory. Consonants were drawn from the set /p, t, k, kʰ, pʰ, s, ts, tsʰ, f, h, l, m, n/ to create nonwords. The suffix alternated between [-pim] and [-pum] (Table 1). In ALO, [-pim] was selected if the stem contained at least one front vowel; otherwise, [-pum] was selected. In Rightmost, the suffix agreed in backness with the rightmost non-neutral vowel. Adults (N = 59; 29 ALO, 30 Rightmost) completed a pretest, training, and test phase. The pretest assessed initial suffix-selection biases. During the exposure phase, adults heard each stem followed by its suffixed form and repeated it; the training included 12 stem types, with two words per type repeated five times. In the pretest and test, adults completed a two-alternative forced-choice task. The toddlers included 54 14-month-olds (27 ALO, 27 Rightmost) and, 33 20-month-olds (16 ALO, 17 Rightmost; data collection ongoing). Toddlers were tested in a central-fixation paradigm using Habit2. Each toddler was familiarized with four stem types, two words per type repeated four times (~138 s), followed by 12 test trials: 6 grammatical items following the trained rule and 6 ungrammatical items violating it. The exposure phase included fewer stem types for toddlers than for adults to accommodate toddlers' limited attention span.

Table 1. Stimuli design with the stem types and the suffixes according to each condition

Condition	Stem types with suffix /-pim/	Examples	Stem types with suffix /-pum/	Examples
At-Least-One	**FBB**	lenputʰɔm-pim	**BBL**	puŋtutʰaŋ-pum
	FLB	kʰinfaŋsu-pim	BLB	tsʰɔnpampʰu-pum
	BFL	tsʰɔnfeŋpʰam-pim	**BLL**	lɔnpʰampʰam-pum
	BLF	mufampem-pim	LBB	hamtsɔntsʰu-pum

	LFB	pamtempuŋ-pim	LBL	mamts^{h}ufaŋ-pum
	LBF	faŋlɔnke-pim	LLB	p^{h}amfamp^{h}u-pum
Rightmost	**BBF**	tsɔmt^{h}uken-pim	**FBB**	lenput^{h}ɔm-pum
	LBF	faŋlulem-pim	LFB	faŋk^{h}isɔn-pum
	BLF	nɔnfamheŋ-pim	FLB	kenhamlu-pum
	LFL	faŋts^{h}enmam-pim	LBL	faŋlumam-pum
	BFL	ts^{h}ɔnfeŋp^{h}am-pim	**FBL**	k^{h}innɔnham-pum
	FLL	fiŋt^{h}aŋmam-pim	BLL	sɔnt^{h}aŋfaŋ-pum

Note. F = front vowel (/i, e/), B = back vowel (/o, u/), and L = neutral vowel (/a/). Bolded stem types were used in the toddler experiment; all listed types were used in the adult experiment.

Results. Adult pretest accuracy did not differ from chance in either condition, indicating no initial suffix-selection bias. A logistic mixed-effects model with Condition (ALO versus Rightmost), Phase (Pretest versus Test), and their interaction as fixed effects and random intercepts for participants and items showed a significant Phase effect, $\chi^2(1) = 15.31$, $p < .001$, but no Condition × Phase interaction, $\chi^2(1) = 2.06$, $p = .151$. Planned comparisons showed significant improvement in Rightmost ($p < .001$), but only marginal improvement in ALO ($p = .072$) (Figure 1), suggesting stronger evidence of adult learning in Rightmost. For toddlers, linear mixed-effects models on log looking times showed no Grammaticality effect at 14 months, but a significant effect at 20 months in ALO, $F(1, 16.55) = 5.16$, $p = .037$, but not Rightmost, $F(1, 13.80) = 0.28$, $p = .608$ (Figure 2). These preliminary findings suggest emerging sensitivity to ALO at 20 months, with no evidence of discrimination for Rightmost.

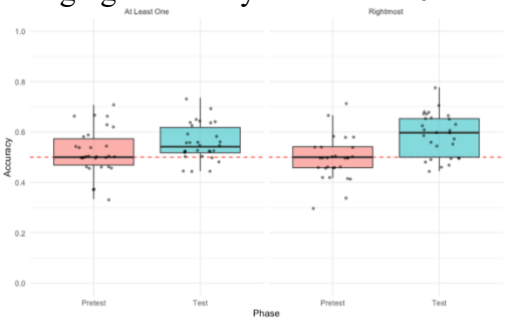


Figure 1. Adult accuracy in the pretest and test phases of the At-Least-One (ALO) and Rightmost conditions. Boxplots show the distribution of accuracy scores, with each point representing an individual participant. The dashed horizontal line indicates chance performance. (0.50).

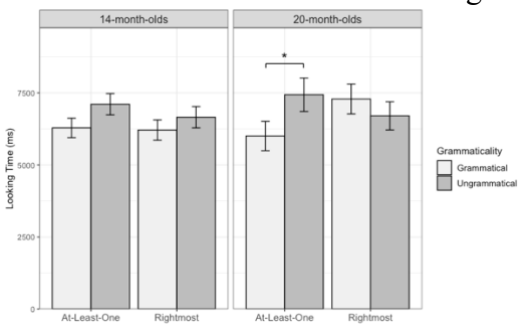


Figure 2. Toddlers' mean looking times to legal and illegal test items in the At-Least-One (ALO) and Rightmost conditions, shown separately for 14- and 20-month-olds. Bars show mean looking time in milliseconds, and error bars indicate ±1 standard error.

Discussion. Our results suggest that computational complexity and typological naturalness both shape phonological pattern learning, but their relative influence differs across development. Adults showed evidence of learning Rightmost than ALO, which contrasts with Hwangbo [7], who reported adult learning of ALO but not Rightmost. This pattern suggests that computational simplicity alone may not fully account for adult learning. Adults may also have been sensitive to typologically natural, position-dependent suffix alternations. In contrast, the current 20-month-old results suggest emerging sensitivity to ALO but not Rightmost, indicating that simpler pattern may be more accessible earlier in development. Another possibility is that learning was influenced by the number and distribution of stem types. The adult experiment used a more varied set of stem types than Hwangbo [7], allowing the rightmost non-neutral vowel to vary across positions and potentially making the Rightmost rule more accessible. By contrast, toddlers were exposed to four stem types, comparable to Hwangbo [7], which may have made the ALO pattern easier to detect under limited exposure. Further work is needed to investigate whether the observed differences reflect developmental change, differences in stimulus distribution, or both.

Selected References [1] Heinz, J. (2018). The computational nature of phonological generalizations. Phonological typology, phonetics and phonology, 23, 126-195. [2] Lai, R. (2015). Learnable vs. unlearnable harmony patterns. *Linguistic Inquiry, 46*(3), 425–451. [3] McMullin, K., & Hansson, G. Ó. (2019). Inductive learning of locality relations in segmental phonology. Laboratory Phonology, 10(1). [4] Finley, S., & Badecker, W. (2010). Linguistic and non-linguistic influences on learning biases for vowel harmony. *Proceedings of the 32nd Annual Conference of the Cognitive Science Society*, 706–711. [5] Finley, S. (2015). Learning nonadjacent dependencies in phonology: Transparent vowels in vowel harmony. *Language, 91*(1), 48–72. [6] Hayes, B., & Londe, Z. C. (2006). Stochastic phonological knowledge: The case of Hungarian vowel harmony. Phonology, 23(1), 59–104. [7] Hwangbo, H. J. (2020). Computational complexity and learnability of two vowel harmony patterns with neutral vowels. *Studies in Phonetics, Phonology and Morphology, 26*(1), 205–230.

Sara Fong (sara.fong@utah.edu); Yang Wang (yangx.wang@utah.edu) *University of Utah*

Why reduplication, not reversal? A phonological memory account

Background. Reduplication (Indonesian *buku-buku 'books'*) is typologically well-attested, but reversals (logically possible form *buku-kubu*) are rare and largely confined to language games (Bagemihl, 1989). In two artificial grammar learning experiments, Moreton et al. (2021) showed adults readily learned a reduplication pattern but largely failed to learn its syllable-reversal counterpart. They proposed phonological working memory (PWM; Baddeley 1998) as a possible explanation for such an asymmetry, but did not measure individual working memory capacity.

Our contribution. We provide a direct test of this PWM account, and we connect this behavioral asymmetry to a mechanistic account of phonological copying. Recent computational work decomposes copying into phonological form buffering and a one-step matching operation (Wang & Hunter 2023). This architecture provides no primitive for reversal, and predicts that the bottleneck in copying should shift with base length: form encoding/matching capability affects short bases, while the effects of memory limit should emerge for longer ones. We tested these hypotheses in a between-subjects 2x2 design, crossing patterns (**Red** vs **Rev**) with base length (3σ vs 5σ), as detailed in Table 1. We also separately measured two PWM components: an audio digit-span task for memory limit, and a continuous non-word recognition memory test targeting form encoding and matching capacity (cf. Hulme et al. 1991).

Our findings. Reduplication and reversal are not learned by the same mechanism. **Red**uplication is learned implicitly, with different PWM components becoming the bottleneck at different base lengths: form encoding/matching capability matters at 3σ-base, while the effects of memory limit emerge for 5σ-bases. The rare **rev**ersal, by contrast, shows no memory-driven learning rate modulation at any length: stronger encoding capability gives a small baseline boost in overall accuracy, but does not steepen the learning curve, and **Rev** learning is largely restricted to participants who explicitly verbalize the rule. The reduplication-reversal asymmetry thus reflects a deeper learnability divide between patterns that can be acquired through implicit phonological learning and patterns that appear to require explicit rule discovery (Moreton & Pertsova 2024).

Methods. 401 native speakers of English completed the two working memory tasks, and were randomly assigned to one of four conditions (N per cell in Table 1). The main task consisted of 50 nonce English forms: 25 targets and 25 foils. Within each trial, participants were asked to decide whether the auditory stimulus followed the rule (yes/no) and received explicit feedback. 3σ stimuli matched the structure of Moreton et al. (2021) with newly generated items, with stress on the second syllable of both copies; 5σ stimuli were 11 syllables, stress on the antepenult of both copies. Foils were random scramblings of the base syllables with the same stress pattern, while preserving neither the target pattern, nor any regularities along prominence positions. The study ended with a survey asking for strategies and inferred rules.

	Red	**Rev**
3σ base (7σ total)	N = 100 (0 excluded) ***Target:*** fuˈsɑ.zoʊ.bæ.fuˈsɑ.zoʊ ***Foil:*** toʊˈgu.bi.dɑ.guˈbi.toʊ	N = 100 (2 excluded → 98) ***Target:*** fu.ˈsɑ.zoʊ.bæ.zoʊ.ˈsɑ.fu ***Foil:*** toʊˈgu.bi.dɑ.guˈtoʊ.bi
5σ base (11σ total)	N = 101 (1 excluded → 100) ***Target:*** su.fʊˈgi.pɪ.keɪ.dɑ.su.fʊˈgi.pɪ.keɪ ***Foil:*** tɑ.piˈkɛ.ʃæ.gu.bɪ.kɛ.tɑˈgu.ʃæ.pi	N = 100 (1 excluded → 99) ***Target:*** su.fʊˈgi.pɪ.keɪ.dɑ.keɪ.pɪ.ˈgi.fʊ.su ***Foil:*** tɑ.piˈkɛ.ʃæ.gu.bɪ.kɛ.tɑˈgu.ʃæ.pi

Table 1. Between-subjects 2 × 2 design.

NWR and digit span were weakly correlated ($r = 0.15$ pooled; range 0.01–0.20 across cells), supporting their treatment as dissociable PWM components. After response-bias exclusions (≥

Sara Fong (sara.fong@utah.edu); Yang Wang (yangx.wang@utah.edu) *University of Utah*

90% same button), N = 397. Trial-level accuracy was modelled in R (*brms*; Bürkner 2017) with Bayesian mixed-effects logistic regression. We included Trial × PWM interactions and by-subject and by-item random effects. Below, we report posterior means with 95% credible intervals; *pd* indicates the posterior probability of direction.

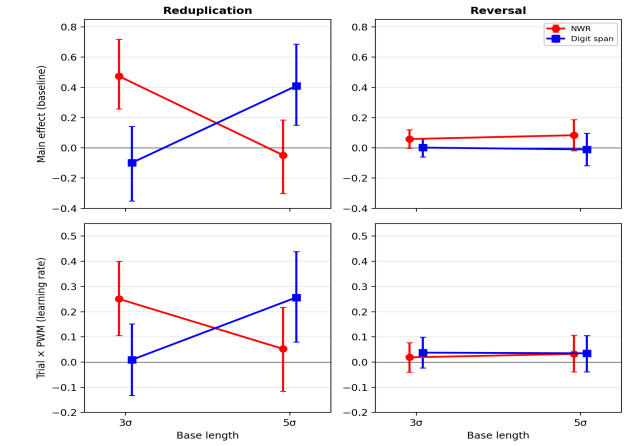


Fig 1 (top). The parameter estimates: Baseline accuracy (top) and learning rate (bottom) across base lengths. Red: NWR; Blue: max digit span. The crossover in the **Red** panels (left) shows the predicted length-dependent dissociation; the **Rev** panels show no comparable PWM modulation of learning rate.

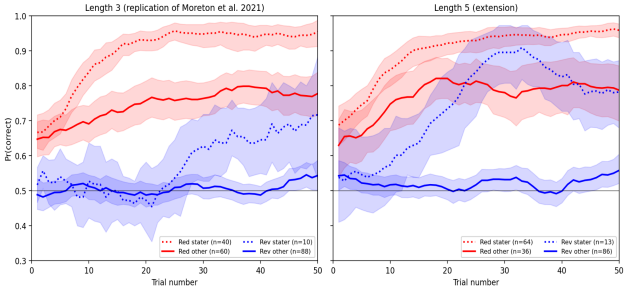


Fig 2 (bottom). Reduplication shows learning overall whereas reversal learning is only among rule-staters. Curves are moving averages (radius 5); bands are 95% bootstrap CIs.

Result 1: A length-dependent PWM dissociation within reduplication. Non-word recognition memory drove learning at 3σ-**Red** (main: β = 0.50 [0.24, 0.77]; Trial × NWR: β = 0.28 [0.12, 0.45]; both *pd* = 1.00), with no credible digit-span effects. Digit span drove learning at 5σ-**Red** (main: β = 0.46 [0.15, 0.77]; Trial × DigitSpan: β = 0.29 [0.08, 0.50]; *pd* ≥ 0.99), with no credible NWR effects.

Result 2: Reversal learning is largely restricted to participants who can later articulate the rule. Across all four conditions, learning rates were credibly positive (*pd* ≥ 0.96), but learning was much faster in **Red** than **Rev** (β ≈ 0.8 vs 0.07), and only **Red** ended credibly above chance (see Fig 2). In **Rev**, only NWR showed a small positive main effect on baseline accuracy (length 3: β = 0.06 [-0.00, 0.13], *pd* = 0.97; length 5: β = 0.09 [-0.01, 0.21], *pd* = 0.95), with no Trial × PWM interactions**.** Following Moreton et al. (2021), we coded participants as correct rule staters if they described the target pattern in the final survey. In **Red,** both rule staters and non-staters showed credible learning effects (non-staters: learning rate β = 0.32-0.34, *pd* ≥ 0.998). Staters learned much faster (Trial × Stater: β = 0.64-0.76, *pd* = 1.00). In **Rev**, learning was only found within rule staters (Trial × Stater: β = 0.24-0.57, *pd* ≥ 0.99); non-staters' trial slopes were near zero for both experiments (β = 0.01-0.04, *pd* ≤ 0.92), consistent with the possibility that **Rev** relies more on explicit rule discovery while **Red** is learnable through implicit phonological learning.

Conclusion and implication. These results support and refine a working-memory account of the reduplication-reversal asymmetry. The within-Red finding provides behavioral support for a buffered architecture of copying (Wang & Hunter 2023). In line with Moreton et al. (2021), the typological rarity of reversal may reflect, in part, a learnability asymmetry: reduplication aligns with memory operations that support implicit learning, whereas reversal appears more closely tied to explicit rule discovery. More broadly, the findings bear on a central question in phonological theory, i.e. why some formally possible patterns are natural while others are not, by identifying a memory-based mechanism that favors copying over reversal.

Selected references. Moreton, E., Prickett, B., Pertsova, K., Fennell, J., Pater, J., & Sanders, L. (2021). Learning repetition, but not syllable reversal. In R. Bennett et al. (eds.), *Supplemental Proceedings of AMP 2020*. Moreton, E., & Pertsova, K. (2024). Implicit and explicit processes in phonological concept learning. *Phonology* 40 (2):1--53. Wang, Y., & Hunter, T. (2023). On regular copying languages. *Journal of Language Modelling*, 11(1), 1–66.

Decomposing bounded iterativity in Kazakh rounding harmony

In vowel harmony, the harmonic feature tends to spread as far as possible within some domain. In a handful of cases, though, harmony has been argued to be non-iterative, assimilating a single target vowel (Jurgec 2011; McCollum & Kavitskaya 2022; McCollum 2024; cf. Kaplan 2008). More problematic is the logically possible scenario where harmony targets some specific number of vowels within a larger domain, what McCollum (2024) calls *bounded iterativity*. Three- or four-syllable domains for harmony have been reported in a handful of languages across Eurasia, Estonian dialects (Uralic; Wiik 1988), Veps (Uralic; Zaiceva 1981; Wiik 1988), and Kazakh (Turkic; Kenesbayev 1962; Aralbayev 1988). In this paper, we report on an experiment with speakers of Kazakh, finding that, although rounding may spread up to the third syllable only, harmony is decomposable into two distinct patterns, without necessary reference to the third syllable or bounded iterativity.

Kazakh has nine contrastive vowels, /ɪ ʏ e: ø: æ: ɑ: o: ɯ ʊ/, where the historical contrast between mid and high vowels has become a length contrast (McCollum 2018). Although backness harmony is quite robust, rounding harmony is restricted. Concerning targets, reports indicate that rounding harmony may target high or non-high front vowels, but only high back vowels; /ɑ:/ does not alternate for rounding. Concerning triggers, McCollum reports that harmony depends on the trigger vowel, as well, with /ʏ/ being the best trigger and /o:/ being the poorest. Extant descriptions further suggest that harmony may reach the third syllable if both V2 and V3 are high/short (Balakayev 1962; McCollum 2018).

Given these descriptions, we conducted a production experiment with 12 university students. We elicited 2- and 3-syllable words, where V2 was high/short, /I/, or non-high/long, /A:/. V3 was always high/short. V2 was preceded by the lateral and followed by a dorsal obstruent, /K/. V3 was always the 3rd person possessive suffix, /I/, followed by /-n/ 'ACC', as in (1). Each speaker produced 336 words embedded within the same carrier phrase.

(1)

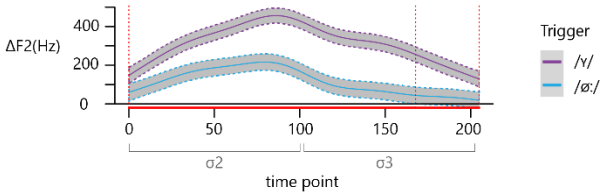
a.	/ɪlIK/	'kinship'	g.	/tɪlA:K/	'wish'
b.	/ɪlIK-I-n/		h.	/tɪlA:K-I-n/	
c.	/tʏlIK/	'cattle'	i.	/tʏlA:K/	'chick'
d.	/tʏlIK-I-n/		j.	/tʏlA:K-I-n/	
e.	/ø:lIK/	'corpse'	k.	/bø:lA:K/	'separate'
f.	/ø:lIK-I-n/		l.	/bɪlA:K/	

To assess the extent of lip rounding, we extracted formant measurements from 21 points during each vowel. We concatenated non-initial F2 as well as F3 measurements to create a single contour for each formant for each word, using a GAMM with fixed effects for trigger vowel and V2 height, along with mixed effects for speaker and word. We generated difference plots to evaluate significant differences in contours after round and unround triggers, as well as to distinguish between potential phonological harmony and phonetic coarticulation (e.g., ΔF2 on V3 when harmony may apply, /qulIK-I-n/ 'demeanor' versus when harmony is blocked by V2, /qulA:K-I-n/ 'ear').

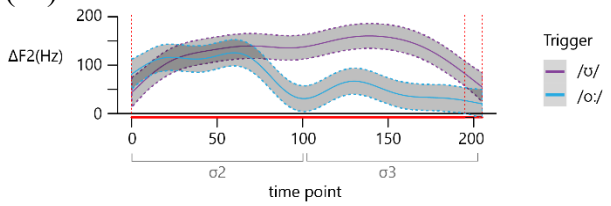
Results indicate that, when the targets are high, V2 and V3 are rounded after short vowels, while only V2 is rounded after long trigger vowels, which is evident in (2a,b), Significant differences in F2 are retained throughout second- and third-syllable vowels after short triggers (purple). In

contrast, differences in F2 after long vowels decrease substantially after the second syllable. When V2 is non-high, V2 is rounded only after /ʏ/, with harmony also applying to V3. F3 was not a reliable indicator of lip rounding. Diminishing ΔF2 at the end of V3 reveals the edge of the harmonic domain. This is not a word-edge effect, as no comparable decline in ΔF2 accompanies V2 in two-syllable words. Results from 3-syllable words are summarized in (3).

(2a)



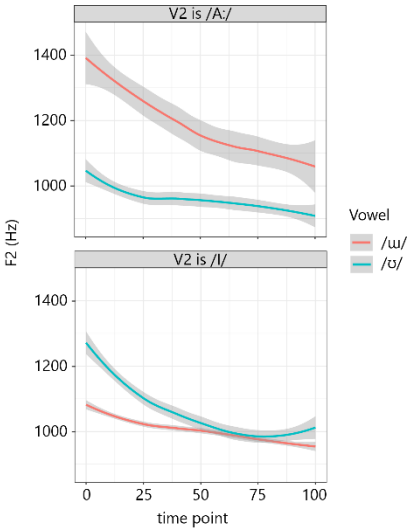
(2b)



(3)

Targets	Short triggers		Long triggers	
	/ʏ/	/ʊ/	/øː/	/oː/
/I…I/	tɪlʏgʏn	qɯlʊʁʊn	øːlʏgɪn	toːlʊʁɯn
/A…I/	tɪløːgʏn	qʊlɑːʁɯn	øːleːgɪn	ʃoːlɑːʁɯn

(4)



Inspection of initial-syllable vowels supports a length contrast between formerly high and non-high vowels. F1 does not distinguish between historically mid and high vowels. Moreover, high/short vowels were approximately 35 ms., while mid/long vowels were 75 ms. in duration. Significant reduction of short vowels is readily attested in the language (Vajda 1994; Muhamedowa 2016). However, our results show that rounding distinctions on initial-syllable short vowels are typically lost. The rounding of short vowels surfaces on V2, except when V2 is [ɑː], (3). In such cases, rounding is retained on V1. This neutralization is seen in (4), where F2 of /ʊ/ is substantially lower when followed by /A:/, but is raised, approximating F2 of /ɯ/, when followed by /I/. Long initial-syllable vowels never lose their rounding, and harmony targets only second-syllable /I/ after long triggers.

These data thus suggest that rounding harmony may affect the third syllable as the byproduct of two distinct processes, (i) [round] shifting from initial-syllable short vowels to V2, and (ii) non-iterative harmony from V2 to V3. We model this in Optimal Domains Theory (ODT; Cole & Kisseberth 1994; Cassimjee & Kisseberth 1998), deriving shifting from a high-ranked markedness constraint against [rd] in prosodically weak initial syllables. Non-iterative spreading obtains due to a constraint on non-binary domains outranking general featural alignment.

Despite some claims that suggest rounding harmony in Kazakh may require bounded iterativity, our findings indicate that spreading to third syllables falls out from typologically well-attested patterns of local shifting and spreading typically found in tone, especially Bantu tone (e.g., Cassimjee & Kisseberth 1998; Odden 2001; Hyman 2011). Furthermore, the apparent ternarity of spreading here is the outcome of two separate, formally distinct phenomena, in a manner similar to previous analyses of bounded shifting and spreading in tone (e.g., Goldsmith 1985; Odden 2001; Bickmore & Kula 2013).

Search and Copy/Change with Boolean Monadic Recursive Schemes

Overview Search and Copy/Change (SC) theories of phonology are based around the idea that the target of a phonological process initiates a "search" for elements that will trigger some type of change within the target (Mailhot & Reiss 2007, Nevins 2010, Dabbous *et al.* 2021). Mailhot & Reiss (2007) describe their underlying philosophy in the following way: "The basic idea of a procedural approach is that the grammar should specify, not what is well-formed or ill-formed, but how to map an input to an output, as an explicitly characterized function." Relatedly, Nevins (2010) writes, "The no-lookahead property of the Search procedure seems to require a target-initiated, procedural search, which cannot obviously be replicated in declarative models." We reject these characterizations in their strong forms and show that Boolean Monadic Recursive Schemes (BMRS; Bhaskar *et al.* 2020) are declarative and define well-formedness conditions on transformations, but also are target-oriented, and exhibit search-like behavior.

Search & Copy/Change Nevins (2010) introduces a search procedure such that morphemes which undergo harmony are lexically underspecified for the harmonizing feature and need to search for a donor to provide a value. A similar approach has been championed by the "Concordia School" which originated as Search & *Copy*, but later updated to Search & *Change* (Mailhot & Reiss 2007, Dabbous *et al.* 2021). For this approach, computation also originates at a target position and looks outward for a trigger. Unlike Nevins, The Concordia School proposal is more general and does not require copying a value from the trigger nor underspecification. Here, we show that BMRS can directly implement both of these search-based theories.

BMRS BMRS define string-to-string functions as subregular logical transductions and corresponds with different classes of subregular functions. A string is a structure $\mathbf{s}$ over a model signature $\mathbf{S} = \langle f \mid f \in \mathcal{F}, p(x), s(x) \rangle$, where f are unary labeling relations drawn from a set $\mathcal{F}$, and $s(x)$ and $p(x)$ are successor and predecessor ordering functions. A BMRS program ρ is a collection of recursive functions $\mathtt{f}_1, \ldots, \mathtt{f}_n$ where each function has output type boolean and a single argument (monadic). A subset of these functions are labeled as $\phi_f(x)$ and indicate that domain element x carries property f in the output. Following a suggestion by Chandlee & Jardine (2021), we loosen the boolean restriction and extend these functions to have type $\mathtt{index} \rightarrow \{0, -, +\}$. BMRS functions are built using $\mathtt{IF} \ldots \mathtt{THEN} \ldots \mathtt{ELSE} \ldots$ syntax and logical connectives are freely insertable since they are BMRS definable. $\phi_f(x) := \mathtt{IF}\ g(x) \approx + \wedge g(s(x)) \approx +\ \mathtt{THEN}\ f(s(x))\ \mathtt{ELSE}\ f(x)$ computes assimilation of feature f between local $+g$ segments, while $\phi_f(x) := \mathtt{IF}\ g(x) \approx -\ \mathtt{THEN}\ \phi_f^{-g}(x)\ \mathtt{ELSE}\ f(x)$ and $\phi_f^{-g}(x) := \mathtt{IF}\ \mathtt{edge}(x)\ \mathtt{THEN}\ f(x)\ \mathtt{ELSE}\ \mathtt{IF}\ g(s(x)) \approx -\ \mathtt{THEN}\ f(s(x))\ \mathtt{ELSE}\ \phi_f^{-g}(s(x))$ compute non-local harmony for feature f between $-g$ segments. Figure 1 partially illustrates these functions in action.

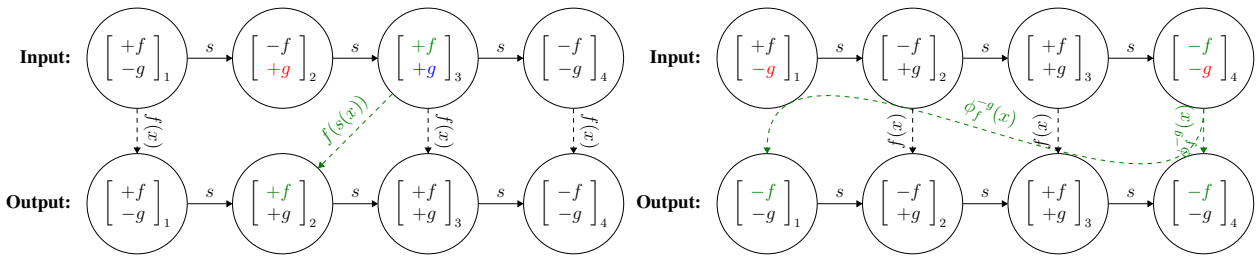


Figure 1: Local (left) and long-distance (right) string-to-string mappings.

Embedding SC in a BMRS Program The Nevins approach describes a process in which an underspecified element searches in a particular direction for a different element that meets specific termination requirements, then analyzes that character for particular copy requirements and if those are met proceeds with copying that feature value to the underspecified element. If no terminating element is found, or the required copy features are not met on one that is, the initiator is given a default value. Nevins models this with a procedural algorithm, but it is also captured with a declarative BMRS program as shown in Figure 2 (left).

This program is built from three predicates. The primary predicate is $\phi_{f_i}(x)$ which checks if an initiator is seen, runs the search procedure ($\phi^{\iota}_{f_i}$) if so, and makes no change if not. An element being an initiator is determined by the second predicate ($\iota_i(x)$) which states that an element initiates search for the feature f_i if it is underspecified for that feature. If an element is on an edge, it has nowhere to search and therefore automatically returns a default value. Otherwise, the functions looks to the adjacent element in the string. If the adjacent element satisfies the termination requirement established by ψ, the predicate moves forward with checking if that character has the required features for copying, R_i. If so, f_i is copied over to the initiating element, otherwise it takes the default value for f_i. If the adjacent element is not a terminator, the predicate recurses until a terminator is found or the procedure reaches the end of the string. In these functions, δ stands for the successor or predecessor functions which determine the direction of recursion/"search".

The Concordia School version of Search and Change can be captured with BMRS in a similar manner as shown in Figure 2 (right). Importantly, the initiator elements for this procedure do not need to be underspecified. Instead, the initiators are defined with a set of features on a domain element ($P(x)$), which trigger a search in direction (δ) for a terminating element (`term`) which has further conditions (`cond`) similarly to Nevins. If these conditions are met, the initiating element takes a feature value for f according to a change procedure. Unlike the Nevins version, there is no requirement that the change be related directly to the terminator. This can be seen on line 5 in blue where we represent change as v which stands for a feature $v \in \{+, -, 0\}$. The other major difference between the two approaches is what happens when a search fails (in this case, nothing).

$$
\begin{aligned}
\phi_{f_i}(x) &:= \texttt{IF } \iota_i(x) \texttt{ THEN } \phi^{\iota}_{f_i}(x) \texttt{ ELSE } f_i(x) \\
\iota_i(x) &:= \texttt{IF } f_i(x) \approx 0 \texttt{ THEN } \top \texttt{ ELSE } \bot \\
\phi^{\iota}_{f_i}(x) &:= \texttt{IF edge}(x) \texttt{ THEN def ELSE} \\
&\qquad \texttt{IF } \psi(\delta(x)) \texttt{ THEN} \\
&\qquad\quad \texttt{IF } R_i(\delta(x)) \texttt{ THEN } f_i(\delta(x)) \texttt{ ELSE def} \\
&\qquad \texttt{ELSE } \phi^{\iota}_{f_i}(\delta(x))
\end{aligned}
\qquad\qquad
\begin{aligned}
\phi_{f_i}(x) &:= \texttt{IF } \iota_i(x) \texttt{ THEN } \phi^{\iota}_{f_i}(x) \texttt{ ELSE } f_i(x) \\
\iota_i(x) &:= \texttt{IF } P(x) \texttt{ THEN } \top \texttt{ ELSE } \bot \\
\phi^{\iota}_{f_i}(x) &:= \texttt{IF edge}(x) \texttt{ THEN } f_i(x) \texttt{ ELSE} \\
&\qquad \texttt{IF term}(\delta(x)) \texttt{ THEN} \\
&\qquad\quad \texttt{IF cond}(\delta(x)) \texttt{ THEN } v \texttt{ ELSE } f_i(x) \\
&\qquad \texttt{ELSE } \phi^{\iota}_{f_i}(\delta(x))
\end{aligned}
$$

Figure 2: Functions for Nevins (left) or Concordia School (right) style "search" using BMRS.

Conclusion Despite small semantic differences, the general structure of the BMRS functions are identical across both approaches. This highlights their computational similarity and the recursive structure of the functions captures the inherent procedural-like "search" property (no-lookahead and all) despite ultimately being a declarative model that describes well-formedness on change.

Bhaskar, Siddharth, Jane Chandlee, Adam Jardine & Christopher Oakden. 2020. Boolean monadic recursive schemes as a logical characterization of the subsequential functions. In *Language and automata theory and applications: 14th international conference, lata 2020, milan, italy, march 4–6, 2020, proceedings 14*, 157–169. Springer. Chandlee, Jane & Adam Jardine. 2021. Computational universals in linguistic theory: Using recursive programs for phonological analysis. *Language* 97(3). 485–519. Dabbous, Rim, Leduc Marjori, Fatemeh Mousavi, Charles Reiss & David Ta-Chun Shen. 2021. Satisfying long-distance relationships (without tiers): A strictly anti-local approach to phonology. Https://ling.auf.net/lingbuzz/006329. Mailhot, Frédéric & Charles Reiss. 2007. Computing long-distance dependencies in vowel harmony. *Biolinguistics* 1. 028–048. Nevins, Andrew. 2010. *Locality in vowel harmony*. MIT Press.

Learning Biases in Tone-Segment Complexity Asymmetry: An Artificial Language Learning Experiment

Yue Yin, Micha Elsner, Rebecca Morley

Background. There is a well-documented tone–segment asymmetry in phonological complexity. As Hyman (2011: 199) states, "Tone can do everything that segmental or metrical phonology can do, but the reverse is not true." One manifestation concerns unbounded circumambient (UC) processes (Jardine, 2016): while UC processes are robustly attested in tonal phonology, for example, in tone plateauing (Kisseberth and Odden, 2003; Hyman, 2011; Hyman and Katamba, 2010) and sour-grapes-like tonal spreading (Bickmore and Kula, 2013), they remain typologically rare in segmental phonology.[1] However, whether this typological asymmetry stems from representational or learning biases in human cognition remains an open question. Artificial language learning experiments have shown that some typologically common patterns tend to be more learnable (see Moreton and Pater, 2012a,b, for an overview), and Cui et al. (2026) provided direct evidence that tonal counting rules are acquired more readily than segmental ones. Whether a similar bias extends to UC processes is the focus of the present study.

Methods. *Design.* The experiment adopted a 2×2 between-subjects design crossing two variables: Feature type (consonant vs. tone) and Complexity (local circumambient vs. unbounded circumambient), resulting in four artificial languages: LCC (local circumambient consonantal), LCT (local circumambient tonal), UCC (unbounded circumambient consonantal), and UCT (unbounded circumambient tonal). For each language, there are two target patterns and four filler patterns. In Language UCT, the two target patterns are: (a) Low tone becomes high when there are adjacent high tones on both sides (HLH → HHH). (b) When multiple low tone units occur between the two high tones, all the low tones between the two high tones become high (HL...LH → HH...HH). The four filler patterns were created by removing the high tone trigger on the left or on the right side from the two target patterns. In Language UCC, the pattern is parallel to Language UCT, but the trigger and undergoer were sibilant /ʃ/ and /s/, respectively. In Language LCT and Language LCC, only pattern (a) would lead to the alternation, while in pattern (b), the undergoers remain unchanged.

Procedure. The experiment began with a pre-test (four minimal pair learning-and-test sessions) to assess participants' ability to distinguish the two target contrasts, followed by two rounds of training and testing on the phonological alternation (Round 1: 16 training trials then 16 test trials; Round 2: 32 training trials then 32 test trials). In each training trial, participants were presented with a noun (image with audio), a color term (color patch with audio), and their combination (colored image with audio). Testing used a 2AFC paradigm: participants heard the noun and color term in isolation with images, and selected between the alternated and non-alternated combined form. Test items included both trained and novel nouns to assess generalization of the learned rule. Figure 1 illustrates the full procedure.

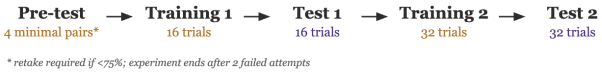


Figure 1: Experiment procedure.

Stimuli. All stimuli words used CVCV structure with the inventory of vowels /i, u, a/, consonants /s, ʃ, k/, and tones /H, L/. To ensure comparability across tonal and consonantal conditions, all four languages share the same tonal and segmental contrasts; syllables with onset /ʃ/ are assigned high tone, while syllables with onset /s/ and /k/ are assigned low tone. Nouns and color terms each fall into three classes, creating both alternation-triggering and non-triggering combinations (Table 1).

[1] McCollum et al. (2020) reported a sour-grapes-like ATR harmony case in Tutrugbu, but segmental UC processes are still relatively rare.

Participants. 24 native American English speakers (LCC: 5; LCT: 7; UCC: 7; UCT: 5) participated in the pilot study; they did not have experience with any tonal languages. To further examine the effect of native language background, the following experiments will additionally recruit native Mandarin speakers alongside English speakers.

Table 1: Training item patterns by color and noun class, with the number of trials in each round (R1, R2), and alternation applicability in UC (unbounded circumambient) and LC (local circumambient) languages.

Color	Noun	R1	R2	UC	LC
/k–ʃ/ (LH)	/s–ʃ/ (LH)	4	8	✓	✓
/k–ʃ/ (LH)	/s–k/ (LL)	2	4	×	×
/k–s/ (LL)	/ʃ–k/ (HL)	2	4	×	×
/ʃ–s/ (HL)	/s–ʃ/ (LH)	4	8	✓	×
/k–s/ (LL)	/s–ʃ/ (LH)	2	4	×	×
/ʃ–s/ (HL)	/s–k/ (LL)	2	4	×	×

Preliminary Results. Data from 2 participants were excluded for pressing the same key throughout all Round 2 test trials, leaving 22 participants for analysis (LCC: 5; LCT: 7; UCC: 5; UCT: 5). All four groups performed above chance on overall accuracy (LCC: .63, LCT: .55, UCC: .59, UCT: .69).

However, overall accuracy is not the optimal measure, since the correct-response distribution is unbalanced for the two LC grammars (non-alternation: alternation = 3:1), potentially introducing a non-alternation-response bias. We therefore examined alternation and non-alternation accuracy separately (Figure 2). We find clear evidence for such a response bias in the LC grammars: participants predict few alternations correctly (LCC: .30, LCT: .43), but many non-alternations (LCC: .73, LCT: .59). The UC grammars showed no such dissociation (UCC: .55 vs. .63; UCT: .65 vs. .73). Therefore, to obtain a bias-free measure of sensitivity, we computed d' (Macmillan and Creelman, 2005) for each participant, and the results show a clear ordering: UCT (1.27) > UCC (0.63) > LCC (0.27) > LCT (0.06). These trends, especially UCT > UCC, align with our hypothesized ordering, though pairwise comparisons did not reach significance, likely due to the small pilot sample. We are continuing to collect data to achieve our full target sample size.

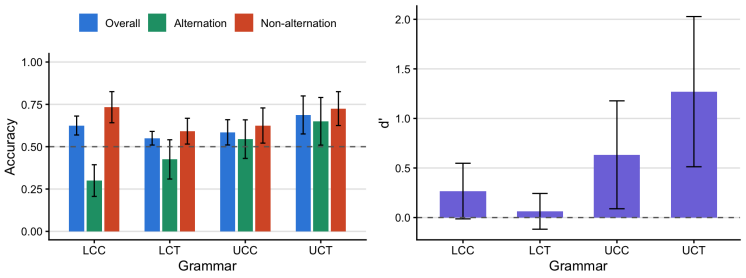


Figure 2: Mean accuracy by condition. Dashed line: chance (50%).

Figure 3: Mean d' by condition. Dashed line: $d' = 0$.

Discussion. The preliminary results showed that the tonal UC process is easier to learn than the consonantal one, as suggested by both accuracy and sensitivity measures. This aligns with previous findings that typological asymmetries stem from synchronic learning biases (Finley, 2012; Huang and Do, 2021; Cui et al., 2026, among many others). As Cui et al. (2026) suggested, the learning advantage of tonal rules may reflect substantive bias, such as greater perceptual salience of tonal changes over consonantal ones (Liu et al., 2024), and/or structural bias, given that tone is simpler regarding phonetic implementation and featural inventory. Our results suggest a more nuanced picture: the tonal advantage emerged specifically for UC processes, but not for local circumambient ones. This interaction between feature type and computational complexity points to an additional source of learning bias, which is tied to the formal representational demands of UC processes themselves, rather than to tonal properties alone.

Selected References. Cui, J., H. Shine, Y. Do, and J. Snedeker. 2026. Investigating the tone-segment asymmetry in phonological counting: A learnability experiment. In *Proc. AMP 2025*. Hyman, L. M. 2011. Tone: Is it different? In *Blackwell Companion to Phonology*, 197–239. Jardine, A. 2016. Computationally, tone is different. *Phonology* 33(2):247–283. Moreton, E., and J. Pater. 2012a/b. Structure and substance in artificial-phonology learning. *Language and Linguistics Compass* 6(11):686–718.

Silent prosody and the Bolinger effect

Overview. The received view in generative phonology (Chomsky and Halle 1968 = SPE, Liberman and Prince 1977) holds that sentential prominence in English is a matter of stress rules that operate mechanically on syntactic representations. This was challenged by Bolinger (1972) who proposed that prominence depends on information: unpredictable words are accented; predictable words are unaccented. His best examples come from verbs: in *I have a póint to make* the verb is deaccented because it is predictable, whereas in *I have a pòint to émphasize* it is accented because it is unpredictable and therefore informative. Bolinger (1957) further suggested that stress and information are aligned even in writing: good writers are aware of an implicit stress contour and maneuver semantically heavy words into syntactically stressed positions (Bolinger 1957, 235). This predicts that stress peaks and information peaks should be aligned in good prose. We explored this hypothesis in two written corpora and found it to be supported under a version of the SPE theory of phrasal stress and multiple (both context-sensitive and context-invariant) formulations of information content, but crucially only in verbs. This parallels the finding that perceived stress is sensitive to information in verbs and function words, but not in nominals (Anttila et al. 2020; Clapp and Anttila 2021). More generally, our results support the reality of silent prosody (Breen and Clifton Jr 2011; Breen 2014) and stress-information alignment in the domain of verbs.

Theory. We operationalized silent prosody using the publicly available software MetricalTree (`metricaltree.stanford.edu`) that applies a version of the Nuclear and Compound Stress Rules (NSR/CSR, Chomsky and Halle 1968, Liberman and Prince 1977) to written texts. We evaluated two distinct measures of information content in the tradition of Shannon (1948): bigram informativity (e.g., Piantadosi et al. 2011, Cohen Priva 2015) as a context-invariant measure of information, as well as experimental next-token probabilities computed by Large Language Models, which give an on-the-fly, context-aware estimate of informativity (Biderman et al., 2023).

Data and models. We used two corpora: (1) inaugural addresses of six recent U.S. presidents (Woolley and Peters 1999) curated by the Presidents Project (Anttila et al. 2020, $N = 15,531$ words) that represent oratorical prose delivered to an audience, and (2) the California DMV Handbook (California DMV 2024, $N = 12,756$ words), that represents well-written but obviously non-oratorical prose. Using (mixed effects) linear regression (Bates et al. 2015), we modeled the informativity of the word as a function of several predictors: mechanical stress predicted by syntax (= NSR/CSR), the relative position of the word in the sentence, the number of segments and syllables in the word, part of speech, and the interaction of mechanical stress and part of speech. A positive correlation between mechanical stress and informativity would constitute evidence for stress-information alignment.

Results. We found that syntax-driven stress and informativity are indeed strongly aligned, but crucially only in verbs. In the Presidents Corpus, the verb-specific slope of mechanical stress was strongly positive ($b = 1.08, SE = 0.086, t = 12.52, p < 2 \cdot 10^{-16}$; all inputs z-scored) and significantly steeper than the slopes for nouns ($b = 0.10, SE = 0.081, t = 1.19, p = 0.24$; not statistically significant) and adjectives ($b = 0.15, SE = 0.069, t = 2.17, p = 0.03$). An F-test for nested model comparison shows that adding mechanical stress to the model accounts for 9.21% of the residual variance in bigram informativity in verbs ($F(1, 2319) = 235.2, p < 2 \cdot 10^{-16}$) in the expected direction, but only 0.16% in nouns ($F(1, 3265) = 5.22, p = 0.022$) and no significance for adjectives ($p = 0.56$). The same pattern holds in the DMV corpus, with mechanical stress accounting for 16.72% of residual variance in verbs (cf. 0.93% in adjectives, 0.88% in nouns, and 0.24% in function words). These results also hold when information content is operationalized using LLM contextual probabilities.

We interpret this result as a ceiling/floor effect. In nominals (nouns, adjectives) and function words, mechanical stress drowns informativity: nominals have a high degree of mechanical stress, while function words have a low degree; both leave little room for information. This leaves the intermediate category of verbs, where there is significant variability in both mechanical (syntax-driven) stress and information content, in which the Bolinger effect emerges.

Finally, our study bears on the Prague School hypothesis (e.g., Firbas 1992) that old information precedes new information, implying that information should increase towards the end of the sentence and therefore be positively correlated with the word's linear position. This turns out to be false when the stress variable is included in the model ($b = -0.067, SE = 0.018, p = 10^{-4}$), suggesting that information is distributed in the sentence not by linear position but rather by the stress contour.

Conclusion. We have shown that Bolinger (1957)'s proposal of stress-information alignment in written prose holds robustly in both oratorical and non-oratorical prose, but only for verbs. This relationship exists independently of the influence informativity has on audibly produced/perceived stress, and evidences the reality of stress-information alignment in silent prosody. Further, it is syntax, and not linear position, that drives the informativity distribution in English sentences.

```
informativity  ~  mechanical stress (mech) +
                  word position (wpos) +
                  number of segments (nseg) +
                  number of syllables (nsyll) +
                  part of speech (pos) +
                  mechanical stress (mech):part of speech (pos) +
                  (1|president) (only in the Presidents Corpus)
```

Figure 1: Model formula: Predicting the informativity of a word

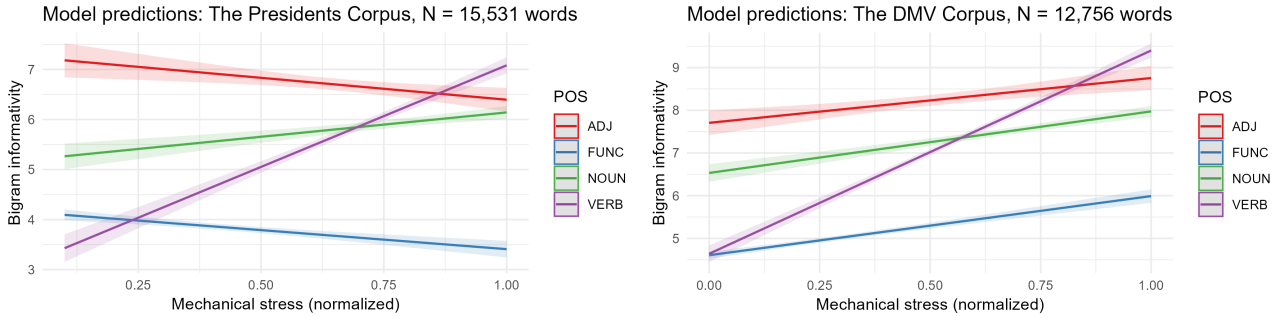


Figure 2: The interaction of mechanical stress and information (with sjPlot, Lüdecke 2025). Verbs show the strongest alignment in both datasets.

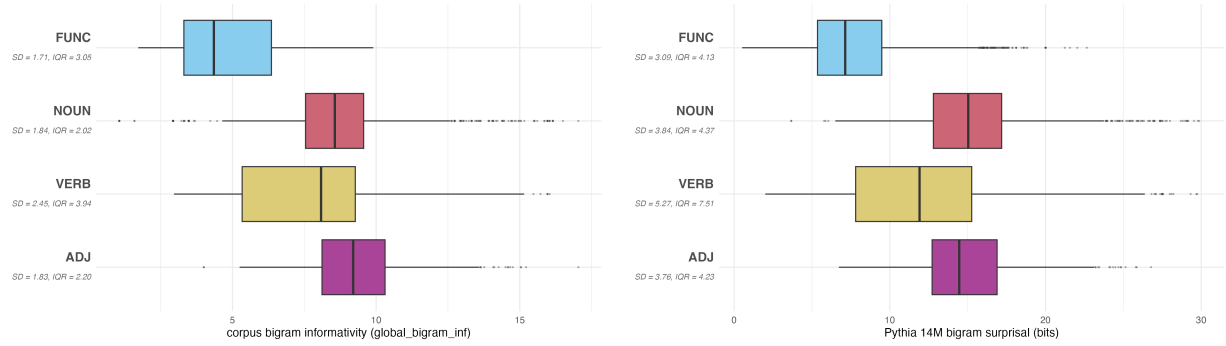


Figure 3: Informativity distributions in the Presidents Project data. Verbs exhibit significantly more variation than other parts of speech.

Noisy Harmonic Grammar and MaxEnt bound differently: An argument for MaxEnt

Aljoša Milenković, University of Southern California milenkov@usc.edu

Constraint-based frameworks differ in their handling of simple harmonic bounding (SHB), where one candidate's violations are a proper subset of its rival's. Whereas some frameworks enforce ABSOLUTE SHB (categorically excluding the bounded), two stochastic variants of Harmonic Grammar (HG) — MaxEnt HG (Goldwater & Johnson 2003) and censored Noisy HG with random tie resolution (cNHG; Boersma & Pater 2016) — observe only RELATIVE SHB: The bounded is not invariably assigned zero probability, but it cannot have a higher frequency than the bounder (Jesney 2007, Hayes 2017, Hayes & Kaplan 2025). Stochastic HGs make divergent predictions, which are difficult to evaluate empirically (Flemming 2021) because ideal test cases are not easily found in naturalistic data. This paper (i) uses SHB to identify new testable predictions of stochastic HGs and (ii) evaluates them against South Slavic corpus data.

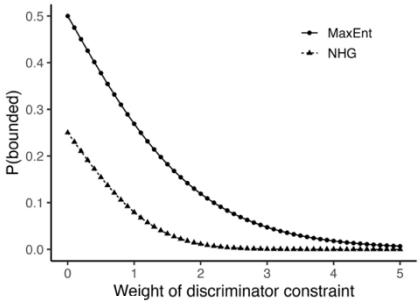


Figure 1

cNHG produces stronger SHB effects: For all weight differences, cNHG assigns a lower probability to the bounded, P(bounded), than MaxEnt (Figure 1). For example, cNHG imposes a lower ceiling value on P(bounded): 25% vs. MaxEnt's 50%. Though MaxEnt's permissiveness seems problematic, potentially leading to overgeneration (Anttila & Magri 2018), our data suggest that this flexibility is a virtue of MaxEnt (Hayes & Moore-Cantwell 2011; Hayes & Schuh 2017; Hayes 2022).

Serbian folk meter (Jakobson 1966) displays phonologically conditioned syntactic variation akin to Shih & Zuraw (2017): Adjectival modifiers are prenominal by default ([bijɛlɔ:ga gra:da] 'white city'), but phonological factors can trigger inversion: [gra:da bijɛlɔ:ga] 'city white.' Figure 2 shows the effects of two key phonological constraints: END-WEIGHT and *LIGHT/FINALSTRONG. END-WEIGHT requires syllabically longer words to be line-final. Adjective-finality rate is therefore positively correlated with adjective length (y-axis) and negatively correlated with noun length (x-axis). *LIGHT/FINALSTRONG, whose effect is evident from different inversion rates across the panels, dictates that the line-final foot head, i.e. the penultimate position of the line, needs to be assigned a heavy syllable (Zec 2008).

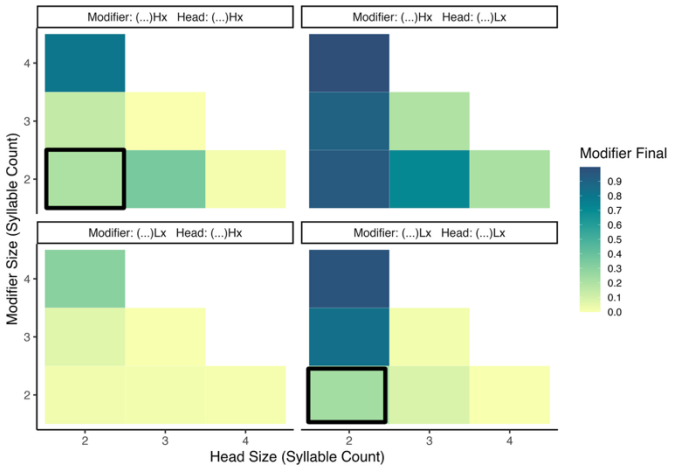


Figure 2 Inversion propensity as a function of word size and weight profile (N lines = 10,894)

The key environments are boxed. The syntactic constraint ADJECTIVEFIRST penalizes the non-canonical, adjective-final order, and no phonological constraint favors the non-canonical over canonical order, rendering the inversion candidate harmonically bounded. The bounded is attested in over 20% of lines in both conditions. Discussion of SHB is inherently confounded by the possibility that the apparently bounded candidate is favored by a constraint not included in the current set. This, I argue, is not the case here. The adjective and noun are identical along all relevant phonological parameters, and rhyme and other stylistic factors are controlled for.

I conducted MaxEnt and cNHG simulations, comparing the frameworks' ability to predict poetic word order (prenominal [default] vs. postnominal Adj). (The full constraint set, models, and

replication data are available in an [anonymous repository].) For MaxEnt, the maximum-likelihood (ML) weights are found algorithmically in R. For cNHG, weights are learned using the Gradual Learning Algorithm (Boersma 1998), which is not guaranteed to find the ML weights. I therefore report the weights from the best-fitting cNHG model (lowest residual deviance) across 100 runs in OT Soft (each 999 training and 1,000,000 test trials). cNHG achieves a better quantitative fit than MaxEnt in the SHB condition (1), but at the expense of poor fit elsewhere (2-3). Due to its aggressive bounding (Fig. 1), cNHG must effectively nullify the contribution of the discriminator constraint ADJFIRST to produce a high P(bounded) in (1). This is inconsistent with ADJFIRST's independent effect in the data. ADJFIRST has a tangible effect on word-order probability by counteracting phonological constraints (2), and through ganging-up interactions (3).

(1) SHB: [ʃuːta guːja] 'angry snake'

		ADJ1ST	END-W	*L/FINS			
	w(MaxEnt)	0.96	2.52	2.02			
	w(cNHG)	0.26	2.88	2.14	p(MaxEnt)	p(cNHG)	observed
Adj-N	ʃuːta guːja				0.72	0.80	0.80
N-Adj	guːja ʃuːta	−1			0.28	0.20	0.20

(2) [bijɛlɔːga graːda] 'white city(GEN.SG)'

		ADJ1ST	END-W	*L/FINS			
	w(MaxEnt)	0.96	2.52	2.02			
	w(cNHG)	0.26	2.88	2.14	p(MaxEnt)	p(cNHG)	observed
Adj-N	bijɛlɔːga graːda		−1		0.17	0.03	0.20
N-Adj	graːda bijɛlɔːga	−1			0.83	0.97	0.80

(3) [bijɛlijɛm graːdɔm] 'white city(INS.SG)'

		ADJ1ST	END-W	*L/FINS			
	w(MaxEnt)	0.96	2.52	2.02			
	w(cNHG)	0.26	2.88	2.14	p(MaxEnt)	p(cNHG)	observed
Adj-N	bijɛlijɛm graːdɔm		−1		0.61	0.44	0.69
N-Adj	graːdɔm bijɛlijɛm	−1		−1	0.39	0.56	0.31

The findings support MaxEnt's flexibility vis-à-vis SHB. Under no weighting conditions can cNHG simultaneously accommodate (1) and (2-3). Capturing SHB entails grossly suppressing ADJFIRST's contribution. In MaxEnt, no such trade-off is observed: ADJFIRST can be weighted commensurate with its contribution to meter (2-3), but this syntactic constraint still produces a sufficiently high P(bounded) in (1). The argument also applies to cNHG with trial cancelation (Kaplan 2025; Hayes & Kaplan 2025), which produces absolute SHB.

This paper contributes to an underexplored topic in phonological theory. While ample work has shown that stochastic HGs are to be preferred over stochastic OT (Zuraw & Hayes 2017, Smith & Pater 2020), the work probing the differences between stochastic HGs is scarce (Flemming 2021).

Finally, I concede that MaxEnt's permissiveness is a double-edged sword (Anttila & Magri 2018). The solution, I surmise, lies outside the grammar: MaxEnt learners are biased against assigning probability to the bounded (O'Hara 2022), which may explain the apparent rarity (but not complete absence) of simply harmonically bounded forms in gradient phonological patterns.

Selected references. Flemming E. (2021). Comparing MaxEnt and noisy HG. *Glossa*. • Hayes, B. & A. Kaplan (2025). Zero-weighted constraints in noisy HG. *LI*. • Shih, S. & K. Zuraw (2017). Phonological conditions on variable adjective and noun word order in Tagalog. *Lg*. • Zuraw, K. & B. Hayes (2017). Intersecting constraint families. *Lg*.

The phonology-phonetics interface in code-switched tone sandhi

Yueke Zeng (yueke@umich.edu) and Huteng Dai (huteng@umich.edu)
University of Michigan

Mandarin Tone 3 Sandhi (T3S) is a productive phonological process that applies even when its trigger context is non-Mandarin. In Mandarin–English code-switching, English words are also able to, though not always, trigger T3S in its preceding T3 (Cheng 1968). What license T3S when the trigger is non-Mandarin, and how much of the conditioning is phonological? To address this, we jointly estimate three cues—two phonological (lexical stress, onset phonotactics) and one phonetic (F0 similarity to expected environment)—in a multivariate Bayesian multilevel model of T3S.

Background. In Mandarin T3S, Tone 3 (T3, low tone) surfaces as a rising tone (T2) before another T3 ($T3^{21} \rightarrow T2^{35}$ / ___ $T3^{21}$; Xu 1994) or before some English words such as *professor*, as shown in Table 1. However, T3S rarely applies to English words such as *profile* (#3). It remains unclear whether this divergence reflects (1) the categorical lexical-stress status of the English initial syllable; (2) tonal mapping from Mandarin to English words; or (3) surface pitch similarity between English prosody and the tonal target in Mandarin T3.

#	Underlying form	Surface form	Context
1	/ni^{21} hao^{21}/	[ni^{35} hao^{21}]	Mandarin T3 + T3
2	/hao^{21} professor/	[hao^{35} professor]	Mandarin T3 + English
3	/hao^{21} profile/	[hao^{21} profile]	Mandarin T3 + English

Table 1: Example of Mandarin T3S triggered by Mandarin T3 or English words.

Method. The analysis uses a speech corpus of Mandarin T3 + English words of 23 speakers (Lovenia et al. 2022). 289 data points were force-aligned with the Montreal Forced Aligner (Mandarin dictionary; McAuliffe and Sonderegger 2024), then manually adjusted by analysts. Pitch contours of T3 and of the English initial syllable were extracted using Praat, and tokens with more than one-third missing f0 values were excluded, leaving 250 tokens for analysis, representing 196 English words. T3S application was labeled by two Mandarin speakers.

Two predictor classes were included. *Phonological* predictors—comprise (i) dictionary-coded stress status of the English initial syllable from the CMU Pronouncing Dictionary (baseline: unstressed) and (ii) onset consonant cluster in the English word's initial syllable (baseline: no cluster). And *phonetic* predictor—the similarity between the F0 contour of the English word's initial syllable and a T3 reference contour. F0 was first $\log_2$-transformed and then z-normalized within speaker over the Mandarin tonal and English vowel contour. The T3 reference was computed as the mean contour across all T3 (low) syllables in the analysis data. Similarity is defined as $S = r \cdot e^{-\mathrm{RMSE}}$, where r is the Pearson correlation (capturing contour shape) and $e^{-\mathrm{RMSE}}$ is an exponential transform of root-mean-square error (capturing pitch-height proximity); high S thus requires both shape correlation and small height difference. We additionally include the z-standardized temporal interval between the end of T3 and the beginning of the English syllable as a control and allowed it to interact with each primary predictor.

A multivariate Bayesian multilevel model jointly evaluates the phonetic and phonological predictors of T3S probability. The model comprises two sub-models: a Gaussian sub-model regressing pitch similarity on stress (testing whether the categorical effect of stress is reducible to its predicted phonetic shadow), and a Bernoulli sub-model (logit link) regressing T3S occurrence

on pitch similarity, stress, onset complexity, and the temporal interval. Both sub-models include random intercepts for speakers, coupled via a shared correlation matrix to capture between-speaker variability and the joint correlation between the two outcomes. Weakly informative priors were used throughout ($\texttt{normal}(0,3)$). Lexical items were not included as a random-intercept grouping factor due to data sparsity (196 items across 250 observations).

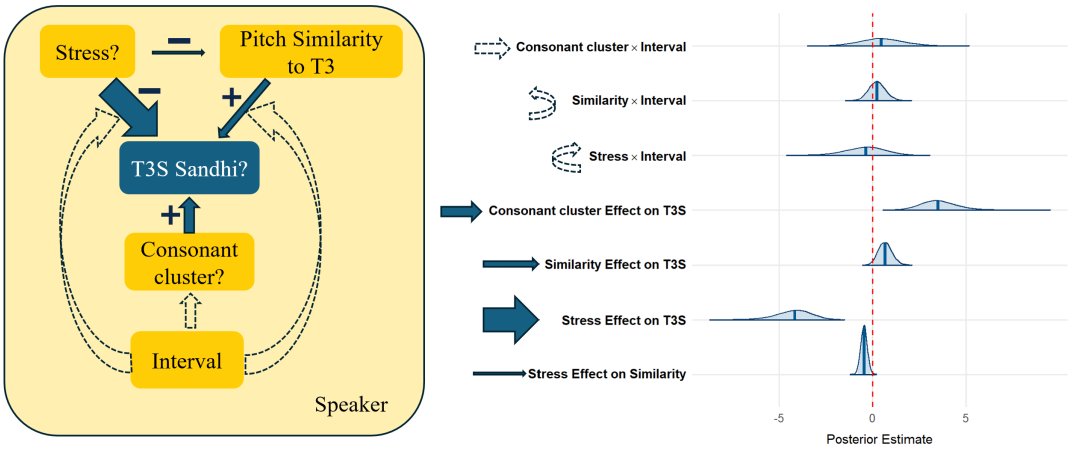


Figure 1: Representation of the model results (left) and posterior distributions from the model (right).

Results. The Bayesian model showed excellent convergence ($\hat{R} = 1.002$, all ESS > 3000) and predictive robustness (all Pareto $\hat{k} < 0.7$). The fixed-effect posteriors are shown in Figure 1. At the mean standardized temporal interval, stress exerts a **strong inhibitory effect** on T3S ($\beta = -4.15$; odds ratio ≈ 0.017): syllables bearing English lexical stress are substantially less likely to trigger sandhi, consistent with prior proposals that stress conditions tonal mapping (Cheng 1968). Onset consonant clusters show a substantial positive effect ($\beta = 3.50$), indicating that segmental structure contributes information beyond stress alone. F0 contour similarity also shows graded evidence for a positive effect ($\beta = 0.68$; posterior probability $P(\beta > 0) \approx 0.976$), with the direction consistent with a gradient phonetic contribution whereby F0 contours more closely resembling Mandarin T3 raise sandhi probability. The Gaussian sub-model further showed that stressed syllables had lower T3 similarity than unstressed syllables ($\beta = -0.45$, 95% CI $[-0.79, -0.10]$). Thus, stress explains part of the variation in F0 similarity. However, similarity retained a positive association with T3S after stress and onset were controlled, while stress retained a negative association with T3S after similarity was controlled. The interactions between temporal interval and the main effects show no credible evidence of an effect, with all corresponding posterior distributions spanning zero. Substantial between-speaker variability (sd $= 0.84$ on the log-odds scale, corresponding to roughly four-fold variation in T3S odds across speakers) suggests that sandhi triggering is non-uniform across individuals.

Conclusion. The results converge on two core generalizations. First, phonological and phonetic factors both contribute to T3S application, supporting a model in which sandhi application is governed by the weighted interaction of concurrent constraints rather than a single categorical trigger. Second, stress predicts F0 similarity, indicating that the phonological and phonetic predictors are related rather than fully independent. Nevertheless, the stress association with T3S is not statistically exhausted by the measured similarity effect. Together, these findings point to a phonetics-phonology interface in which gradient phonetic and categorical phonological knowledge jointly shape sandhi application.

Reference.

Cheng (1968). English stresses and Chinese tones in Chinese sentences. *Phonetica*.

Lovenia et al. (2022). ASCEND: A spontaneous Chinese-English dataset for code-switching in multi-turn conversation. *Proc. LREC*.

McAuliffe & Sonderegger (2024). *Mandarin (China) MFA dictionary v3.0.0*. Tech. rep.

Xu (1994). Production and perception of coarticulated tones. *Journal of the Acoustical Society of America*.

A synchronic scalar vowel shift in Lobi (Gur/Mabia) plural formation

Hannah Sande and Dylan Worden

Introduction: We present novel data on vowel alternations in plural nouns in Lobi (lob; Gur/Mabia; Burkina Faso, Ghana, and Côte d'Ivoire). Lobi shows a rarely attested synchronic vowel shift, which we analyze in a constraint-based account without the need for controversial scale-specific or anti-faithfulness constraints.

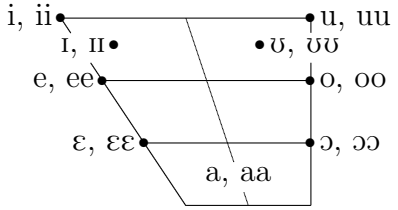
Cases of synchronic scalar shifts among vowel heights are rare, and the majority of attested cases come from a single language family: Bantoid (Guthrie, 1968; Fointein, 1986; Hyman, 1988; Clements, 1991; Schmidt, 1996; Walker, 1997); see Moreton (2004) and Neasom (2016) for comprehensive lists of attested synchronic shifts. We present another case of an apparent synchronic vowel shift, from a different language and family.

The data come from a Lobi documentation project that began in 2022. For this presentation, we elicited over 350 nouns in the singular and plural, and consulted previous documentation, especially Becuwe (1982); Maïmouna Le Men (2007); Miehe and Tham (2007).

The vowel system of Lobi: The contrastive vowels of Lobi are given in (1). We set aside long vowels and diphthongs for the abstract, but the analysis extends straightforwardly.

(1) **Vowel inventory with distinctive features**

	i	ɪ	e	ɛ	u	ʊ	o	ɔ	a
High	+	+	-	-	+	+	-	-	-
ATR	+	-	+	-	+	-	+	-	-
Back	-	-	-	-	+	+	+	+	-
Low	-	-	-	-	-	-	-	-	+



Plural formation via vowel shift: Count nouns in Lobi are interpreted as singular in their bare form. For an interpretation of two or more, there are several morphological pluralization strategies depending on the phonological, morphological, and semantic properties of the noun base, likely historically related to a Proto-Gur noun class system whose remnants in Lobi survive only in the multiple pluralization strategies (Miehe and Tham, 2007). A common plural allomorph is an /-a/ suffix that has a nasal allomorph after nasal-final roots. Another pluralization strategy involves a change in root-final vowel quality, which can be the sole marker of plurality or can co-occur with the /-a/ suffix. There is alternatively a separate word generalized plural morpheme /sʊnɔ/; for some nouns, this is the only licit pluralization option, while for others there is optionality between the morphological and generalized plural.

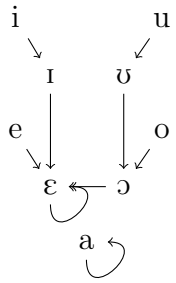
We focus on the vowel shift plural allomorphy, exemplified for nouns with distinct final vowel qualities in (2). Generally, vowels in final syllables of the root (regardless of whether they are followed by a consonant), lower and/or become [-ATR] in the plural (2a-f). /ɔ/ fronts to [ɛ] (2g). /ɛ/ and /a/ do not shift (2h-i); the only possible plural formation strategy for /ɛ/- and /a/-final nouns is the generalized plural word /sʊnɔ/ (2h). Nouns that end with a [+high, +ATR] vowel and a nasal coda exceptionally undergo both a change in height and ATR (2j,k), whereas other [+high, +ATR] vowels only undergo an ATR change (2a,d).[1] These alternations can be represented as a scalar shift, as in (3).

[1]The only example we have found so far of an /ɪ/ to [ɛ] shift is before a nasal.

(2) **Plural vowel shift examples**

	Singular	Plural	Meaning
a.	tib(i)l	tíbl-a	'person'
b.	manɟín	manɟén-án	'papaya'
c.	ɓundé	ɓundɛ́	'rat'
d.	$k^huɓ$	$k^hʊɓ$-a	'bone'
e.	bʊ	bɔ	'goat'
f.	kór	kór-á	'hole'
g.	nɔ́ɔ́	nɛ́ɛ́	'leg'
h.	$t^hɛɛ$	$t^hɛɛ$ sʊnɔ	'in-law'
i.	j'àá	j'àá sʊnɔ	'market'
j.	cin	cɛn-an	'guinea fowl'
k.	kun	kɔn-an	'man'

(3) **Vowel shift diagram**



Analysis: The shift in (2, 3) could be modeled by the (plural-specific) ordered rules in (4).

(4) **An account using ordered rules**

1. *ɔ-Fronting:* /ɔ/ → [-Back]
2. *Pre-nasal laxing:* [+ high] → [-ATR] | __ n
3. *Lowering:* [-ATR] → [-high]
4. *Laxing:* V → [-ATR]

However, the shift does not involve the same feature change at each step. The first step involves a change in ATR, the second and fourth a change in height, and the third a change in backness. The rule-based analysis involves multiple, seemingly unrelated, rules that together result in an epiphenomenal, conspiratorial vowel shift in a single morphological context.

While these rules account for the facts, they miss a generalization: Each underlying vowel quality becomes one feature more similar to /a/ (which is also the form of the overt plural suffix), without ever becoming [a]. /ɛ/ cannot become more similar to /a/ without becoming [a], so it does not change, and can only be pluralized with the periphrastic generalized plural word. We propose a constraint-based analysis that does not require scalar-specific constraints (cf. Kirchner 1996; Mortensen 2006), or controversial RealizeMorph (Kurisu, 2001) or Anti-faithfulness (Alderete, 2001), but rather involves a constraint motivating assimilation to (floating) /a/, which interacts with feature faithfulness, including undominated Ident-Low preventing new [a] vowels. There is a preference to change ATR over height over backness, and two violations of feature-faithfulness constraints are worse than failing to become more [a]-like, which can be analyzed via constraint conjunction or weighted constraints. Lowering high vowels before a nasal onset is found to be a language-wide process.

Implications: We describe the vowel shift in Lobi plural contexts, adding a new language (and language family) to the limited number of attested cases of synchronic vowel shifts. We argue that the shift is best analyzed via the interaction of basic markedness and faithfulness constraints rather than rules to avoid a conspiracy. No scalar-specific machinery is necessary to account for this shift, consistent with recent findings by Neasom (2016) that scalar shifts are not a unified phenomenon and are best modeled in a variety of other ways, in this case as assimilation to [a], one feature at a time.

References

Alderete, John D. 2001. Dominance effects as transderivational anti-faithfulness. Phonology 18:201–253.

Becuwe, Jacques. 1982. Eléments de phonologie et de grammaire du lobiri (parler de Bouna, Cote-d'ivoire). Doctoral Dissertation, Université de la Sorbonne Nouvelle (Paris 3).

Clements, George N. 1991. Vowel height assimilation in Bantu languages. In Annual meeting of the Berkeley linguistics society, 25–64.

Fointein, Jacenta Nsang. 1986. Phonology of Esimbi. Phd thesis, Université de Yaoundé.

Guthrie, Malcolm. 1968. Notes on Nzebi (Gabon). Journal of African Languages 7:101–129.

Hyman, Larry M. 1988. Underspecification and vowel height transfer in Esimbi. Phonology 5:255–273.

Kirchner, Robert. 1996. Synchronic chain shifts in Optimality Theory. Linguistic Inquiry 27:341–350.

Kurisu, Kazutaka. 2001. The phonology of morpheme realization. Doctoral Dissertation, University of California Santa Cruz.

Maïmouna Le Men, Fané. 2007. Parlons Lobiri: Langue et culture des Lobis du Burkina Faso. L'Harmattan.

Miehe, Gudrun, and Florian Tham. 2007. Lobiri. In Noun class systems in Gur languages, volume 1. Rüdiger K
opper Verlag.

Moreton, Elliott. 2004. Synchronic chain shift (counterfeeding) examples.

Mortensen, David Roland. 2006. Logical and substantive scales in phonology. Doctoral Dissertation, University of California, Berkeley.

Neasom, Nicholas Charles. 2016. Against synchronic chain shifting. Doctoral Dissertation, UCL (University College London).

Schmidt, Deborah Schlindwein. 1996. Vowel raising in Basaa: A synchronic chain shift. Phonology 13:239–267.

Walker, Rachel. 1997. Faith and markedness in Esimbi feature transfer. Phonology at Santa Cruz 5:103–115.

Beyond Wug Tests: Sublexical Phonotactics Affect Real and Nonce Word Processing

Across languages, speakers use phonological generalizations over stems to determine which morpho-phonological processes to apply to novel words (e.g. Albright & Hayes, 2003; Becker & Gouskova, 2016). For example, the velar-final nonce verb [splɪŋ] is more likely to be [splʌŋ] in the past tense compared to the fricative-final [plɪf] is to be [plʌf]. Evidence for these "islands of reliability" or "sublexical phonotactic generalizations" primarily come from Wug tasks (Berko 1958), where participants are asked to produce a nonce word in a context where the morpho-phonological process could apply ("*What's the past tense of [splɪŋ]?*") or asked to choose between alternatives ("*Is it [splɪŋd] or [splʌŋ]?*"). While such tasks are useful in testing productivity, they inherently introduce two issues. First, the task promotes explicit metalinguistic reasoning, as participants are explicitly aware of the process being studied ("*I must make the verb past tense*") and can thus generate task-specific strategies (Schütze 2005). Second, the task obscures the cognitive source of these generalizations: grammatical inference or early lexical processing (see below). As an alternative to wug tasks, I demonstrate that a lexical decision task (LDT) can be used to assess whether sublexical phonotactic generalizations are part of a speaker's grammar.

As a test case, I focus on the three Spanish verbal inflectional/conjugation classes (IC), whose membership cannot be categorically predicted by the stem's phonology (table 1). However, per Montgomery (1979), there may be generalizations over the final syllable vowel height for these three ICs. I first **simulate** whether this vowel height difference is statistically significant. In **Exp. 1**, I test whether this generalization affects speed/accuracy of responses in a Spanish visual lexical decision task megastudy (Aguasvivas et al. 2018). In this task, participants are asked whether a given real/nonce word exists in Spanish. As the task contained words from various parts-of-speech and required no active application of morpho-phonological processes, effects are unlikely to be due to task-specific strategies or metalinguistic reasoning. **Exp. 2** tests whether Exp. 1 results extend beyond vowel height to a full set of phonotactic constraints.

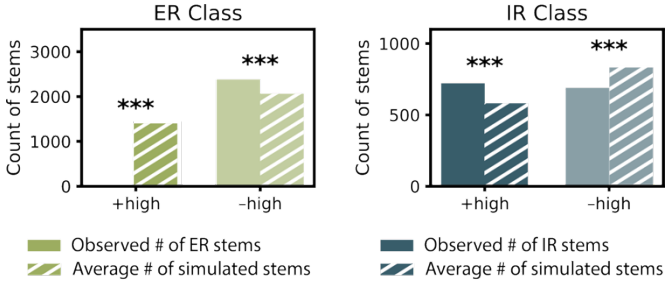
AR	ER	IR
sed-ar	ced-er	ped-ir
[seð-ar]	[seð-er]	[peð-ir]
'to sedate'	'to cede'	'to ask'

Table 1: Spanish verbal IC Examples

Cognitive Source. Since the lexical decision task must have both real and nonce words, it serves as a natural task to test the cognitive source of sublexical phonotactic generalizations. While sublexical phonotactic well-formedness has been argued to be a product of **grammatical inference** (e.g. Becker & Gouskova 2016), i.e. determines how people extend morphophonological rules to nonce words, this theory makes no predictions for real words. Alternatively, neurolinguistic studies have shown that the initial stages of form-based lexical processing (Fruchter & Marantz 2015) don't distinguish between real and nonce words (Dehaene et al. 2002). Thus, if sublexical phonotactic generalizations arise from this **early stage of processing** rather than being a product of grammatical inference, they should equally apply to real and nonce words.

Simulation: *Do the Spanish ICs significantly differ in vowel height of their stems' final syllable?*

Methods. Montgomery (1979) argues that the ER class only contains -high vowels [a, e, o] in stem final syllable while the IR class has a gradient tendency towards +high vowels [i, u]. To test this generalization, we ran a 10,000 trial Monte Carlo simulation (Gouskova 2025)—drawing verbs at random to recreate each IC and calculating the likelihood of recreating the observed counts.



Results. Observed values and simulated averages are given above. The IR class has a significant skew towards +high (*viv-ir*, 'to

live') and away from -high (*abr-ir*, 'to open') stems ($p < 0.001$). The ER class has the opposite skew ($p < 0.001$), with many -high stems (*ced-er,* 'to cede') and no +high stems (**bim-er*). This confirms Montgomery (1979)'s claims that ER/IR have significant vowel height generalizations.

Exp. 1: *Do IC vowel height generalizations modulate real/nonce word processing in an LDT?*

Methods. The visual lexical decision task megastudy had 676,910 nonce and 1,440,165 real verb observations over 169,390 participants. As real and nonce words differ in their "correct" responses, their data was analyzed separately. Logistic regressions were fit for accuracy and linear regressions for response time, with main effects for avg. bi-letter freq., stimulus length, lemma log freq., interaction term between IC and vowel height, and random intercept for subject.

Results. Results of interest are the pairwise contrasts of model-adjusted marginal means. Effects are shown for ref. level (-high). As expected, since ER and IR classes have the opposite vowel height generalizations, they show the opposite direction of effects for the -high vowel. Participants are less accurate/slower at rejecting -high ER nonce words (*bem-er)* than +high (*bim-er*) but are more accurate/faster at rejecting -high IR nonce words (*bem-ir*) than +high (*bim-ir*). Given the significance of these effects, we have evidence that participants treat these nonce words as morphologically complex verbs. For real IR verbs, participants were less accurate at accepting -high (*abr-ir*, 'to open') than +high (*viv-ir,* 'to live') verbs. The results show a class typicality effect for all items: *a nonce word that looks like it belongs to its class is hard/slow to be rejected* while *a real word that looks like it belongs to its class is easy to be accepted.*

	Nonce				**Real**			
	Accuracy		**Response Time**		**Accuracy**		**Response Time**	
	Estimate	**P-Val**	**Estimate**	**P-Val**	**Estimate**	**P-Val**	**Estimate**	**P-Val**
ER	-0.333	<0.001	0.069	<0.001	-	-	-	-
IR	0.145	<0.001	-0.009	<0.05	-0.245	<0.001	0.002	0.376

Exp. 2: *Do Exp. 1 results extend to a full set of IC-level phonotactic constraints?*

Methods. Exp. 1 was replicated with IC level sublexical phonotactic violation scores computed using the UCLA Phonotactic Learner (Hayes & Wilson 2008). This feature-based inductive algorithm was trained on real verbal stems (excluding stems w/ derivational suffixes) subset by their IC, resulting in a 500 markedness constraint phonotactic grammar for each IC. The weighted sum of constraint violations for the corresponding IC grammar was computed for each real/nonce word, resulting in an IC-violation score. Models were identical to Exp. 2, but included a Part-of-Speech phonotactic violation score, neighborhood density, and avg. neighborhood freq.

Results. The AR/IR class results confirm the results from Exp. 1: participants were more accurate/faster at rejecting ill-formed nonce words and less accurate/slower at accepting ill-formed real words. The ER class, however, showed the opposite pattern.

	Nonce				**Real**			
	Accuracy		**Response Time**		**Accuracy**		**Response Time**	
	Estimate	**P-Val**	**Estimate**	**P-Val**	**Estimate**	**P-Val**	**Estimate**	**P-Val**
AR	0.0112	<0.001	-0.0008	<0.001	-0.245	<0.001	0.0039	<0.05
ER	0.0029	<0.001	0.0003	<0.05	0.189	<0.001	-0.0259	<0.001
IR	0.0069	<0.001	-0.0001	0.20	-0.355	<0.001	0.0144	<0.001

Conclusion. IC-level phonotactic generalizations affect processing of both real and nonce words in a visual LDT. As early stages of lexical processing do not distinguish between real and nonce words, these results suggest that sublexical phonotactic effects arise during these stages rather than later grammatical inference. The directionality is largely consistent with typicality—the more typical an item is of an IC, the more likely the system is to treat it as a member of the class.

Laryngeal phasing in voiceless and breathy sonorants

Madeline Gilbert, *University of Georgia*

Introduction: This study investigates laryngeal timing in non-modal sonorants. Laryngeal timing has been studied extensively for stops, where the presence and timing of a laryngeal gesture distinguish phonological categories (Browman & Goldstein 1992). Less research focuses on sonorants with laryngeal contrasts. What does exist focuses on *voiceless sonorants*, which are typically reported to have two phases, usually a voiceless frication and voiced sonorant phase (e.g,. Burmese, Bhaskararao & Ladefoged 1991). Their salient bi-phasal structure seems to differentiate them from *breathy voiced sonorants*, where subsegmental organization is rarely discussed. In this paper, I synthesize laryngeal timing patterns reported in the literature for non-modal sonorants and analyze acoustic data for a new type of breathy sonorant: those derived through synchronic gestural shifts in Sevillian Spanish. The typological findings combined with the Sevillian data suggest that laryngeal timing is variable in both breathy and voiceless sonorants, but apparently more so in breathy ones. Subsegmental organization of non-modal sonorants differs in constrained ways and requires gradient phasing representation to account for this variability.

Typology: 19 languages with non-modal sonorants were investigated in published papers containing production data. Out of 11 phonologically *voiceless* sonorants, 8 have two acoustically clear parts: a voiceless sonorant and voiced sonorant phase (e.g., Burmese; Bhaskararao & Ladefoged 1991). More rarely they are entirely voiceless but have distinct phases within that (e.g., Angami, *ibid*). Phonologically *breathy* sonorants can consist of two phases (breathy voiced sonorant + more modally voiced sonorant (e.g., Marathi, Berkson 2019), one phase (breathy sonorant), or variable phasing (e.g., Tsonga; UCLA Phon. Lab Archive). However, voiceless sonorants could be more like breathy sonorants than immediately apparent. Recent studies show variability in the timing of non-modality in voiceless sonorants too (e.g., Guillemot & Lee 2024).

Voiceless sonorants appear to be a more uniform category, possibly because they require a voiced portion for sonorant POA to be perceptible (Ohala & Ohala 1993). If that is true, timing of breathiness in breathy voiced sonorants *should* be less constrained because continuous voicing satisfies that need. Indeed, much of the laryngeal variability reported for phonologically 'voiceless' sonorants occurs in acoustically voiced tokens (Guillemot & Lee 2024). The Sevillian data on breathy sonorants also support this idea.

Seville /SN/: Breathy sonorants in Sevillian arise synchronically from a shift in gestural alignment. In Sevillian /sC/ clusters, oral and glottal gestures realign after coda /s/ debuccalizes to [h]. /s//ptk/ clusters result in metathesis (/ka**sk**o/→[ka**kh**o]; Ruch & Peters 2016). /s/+/mnl/ clusters have not received experimental treatment but are claimed to be partially voiceless (Martinez-Gil 2012).

***Methods**:* Data come from 12 female Sevillian speakers reading sentences with words and phrases containing intervocalic sonorants /m, n, l/ (/N/) and /sm, sn, sl/ sequences (/SN/) (e.g., se**m**ana, e**sm**alte). The duration of /N/ and /SN/ was measured as the interval between vowels. In those same intervals, cepstral peak prominence (CPP; Hillenbrand & Houde 1996) was extracted with PraatSauce (Kirby & Puggaard-Rode 2026) at 20 equidistant points, to quantify breathiness corresponding to glottal spreading. The study analyzes 356 /N/ intervals and 431 /SN/ intervals and GAMMs will be run to compare CPP trajectories for /N/ vs. /SN/.

Results: Fig. 1 shows two timing patterns found in Sevillian breathy voiced sonorants: two-phase [ɦN] and one-phase overlapped [ɦ]/[N]. Others include one-phase modal [N] (laryngeal deletion), three-phase [ɦNɦ], and two-phase inversed order [Nɦ]. Fig. 2 shows CPP trajectories for /N/ and /SN/; low CPP indicates weaker modality and stronger breathiness. Speakers' patterns were divided into three categories (4 per category), each illustrated here by one speaker. *Two-phase*

[ɦN] sequences are reflected by initial low CPP that rises sharply from [h] to a modal sonorant (Fig. 2, left). *One-phase overlapped* productions [ɦ]/[N] are reflected by a /SN/ CPP trajectory that is flat and lower than /N/ (Fig. 2, middle). CPP rises slightly for some speakers in this category but is lower for /SN/ than /N/ throughout. *One-phase productions with /s/ deletion* are reflected by overlapping CPP curves for /N/ and /SN/, indicating no difference in breathiness (Fig. 2, right). Duration also distinguishes the categories for all speakers (/N/: mean = 63ms; /SN/: mean = 110ms).

Discussion: Both voiceless and breathy sonorants show variability in the timing of non-modality within the sonorant and variability seems to be tied to voicing. Voiceless sonorants most frequently have two phases, consistent with the claim that a voiced portion provides important perceptual information. When they contain a single phase, this phase is acoustically voiced and breathy. Voiced breathy sonorants (allophonic and phonemic) with voicing throughout permit freer phasing across and within languages.

The Sevillian data demonstrate that coarticulatory breathiness can fall anywhere in relation to the sonorant or overlap it completely, analogous to Hakha Lai (Ziegler et al. 2023) and Drenjongke (Guillemot & Lee 2024). In terms of gestural representations, variable laryngeal-sonorant alignment requires a flexible phasing relationship, such as in a phase-window account (Byrd 1996).

Diachronically, non-modal sonorants are thought to develop from /SN/ sequences through an intermediate stage of laryngeal-sonorant coarticulation (Blevins 2018; Berkson 2019). As seen in the Seville data, synchronic coarticulation between a sonorant and breathy laryngeal gesture allows many alignment possibilities. Given this pool of variation from which phonologization can occur, breathy sonorants *should* differ across and within languages. Voiceless sonorants may be constrained by perception in the coarticulatory, pre-phonologization phase, resulting in more consistent outcomes.

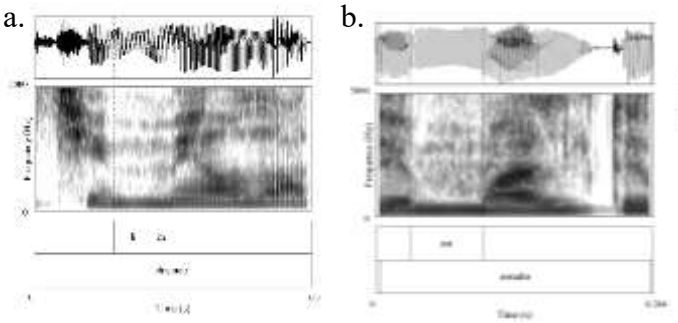


Fig. 1 (a) 2-phase [ɦN]; (b) 1-phase overlap [ɦ]/[N]

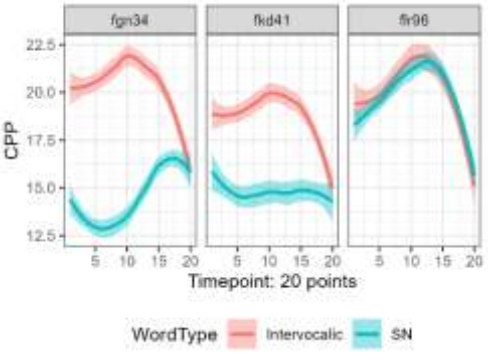


Fig. 2 CPP for Sevillian /N/ vs. /SN/. L: 2-phase [ɦN]; Middle: 1-phase [ɦ]/[N]; R: [N] (laryngeal deletion)

Selected references

Berkson, Kelly Harper. 2019. Acoustic correlates of breathy sonorants in Marathi. *Journal of Phonetics* 73. 70–90.

Blevins, Juliette. 2018. Evolutionary Phonology and the life cycle of voiceless sonorants. In S. Cristofaro & F. Zúñiga (eds.), *Typological Studies in Language*, vol. 121, 31–58. Amsterdam: John Benjamins.

Chirkova, Katia, Patricia Basset & Angélique Amelot. 2019. Voiceless nasal sounds in three Tibeto-Burman languages. *JIPA* 49(1). 1–32.

Ford, Catherine, Benjamin V. Tucker & Tsuyoshi Ono. 2023. Voiceless nasals in the Ikema dialect of Miyako Ryukyuan. *JIPA* 53(3). 694–711.

Guillemot, Céleste & Seunghun J. Lee. 2024. Phonetic Variation of Voiceless Nasals in Drenjongke (Bhutia). *Journal of Asian and African Studies, Supplement* 4. 107–127.

Ohala, John & Manjari Ohala. 1993. The phonetics of nasal phonology: Theorems and data. In Marie Huffman & Rena Krakow (eds.), *Nasals, nasalization, and the velum*, 225–249. San Diego: Academic Press.

Ziegler, Grayson, Stefon Flego & Kelly Harper Berkson. 2023. Variable production of voiceless sonorants in Hakha Lai. Poster presentation presented at The 184th Meeting of the ASA, Chicago, IL.

Gestural transparency in apparent opacity:
Lenis stop lenition and vowel devoicing in Seoul Korean

Suyeon Yun (Chungnam National University; suyeon.yun@cnu.ac.kr)

Variation has come to occupy a central place in phonological theory (Anttila 2002, Coetzee & Pater 2011), and Anttila (2006) shows that variation and opacity can co-occur within the same grammar. This paper extends that line of inquiry by asking whether opaque interactions may be probabilistic, phonetically grounded, and variable across speakers and tokens. I address this question through a corpus study of two phonological processes in Seoul Korean that are independently variable in their application: intervocalic lenis stop lenition (e.g., /ika/ → [iɰa]; Lee 1995) and post-obstruent vowel devoicing (e.g., /si/ → [s̥i̥]; Jun & Beckman 1994, a.o.). I show that the interaction between these processes yields apparently opaque patterns that are gesturally transparent and probabilistically variable, providing evidence that opacity-like effects can emerge from gradient phonetic coordination rather than categorical rule ordering.

The data come from the Speech Corpus of Reading-Style Standard Korean (National Institute of Korean Language 2005). A subset of 51 sentences from text #1 was analyzed for 20 speakers (5M in 20s, 5M in 30s, 5F in 20s, 5F in 40s). The data were hand-labeled using the LEXI framework (Huilgol et al. 2019), which allows segmental modifications to be coded in terms of acoustic cues. Lenition of a lenis stop to an approximant was coded as loss of stop closure and release cues together with the addition of a glide cue (<Sc>-x, <Sr>-x, and <G>-+), and full vowel devoicing was coded as deletion of the vowel landmark (<V>-x).

Out of 4,840 underlying lenis stops in intervocalic position, 707 cases of lenition were identified (14.6%). Lenition was more frequent among male than female speakers, and among females it was more frequent in the 40s group than in the 20s group. Dorsal /k/ showed the highest lenition rate (23.5%), followed by labial /p/ (13.5%) and coronal /t/ (1.6%). Low vowels in both the preceding and following positions were associated with higher lenition rates.

For vowel devoicing, 573 instances of total devoicing were identified among 17,853 vowels following a voiceless obstruent (3.25%), without significant differences across gender or age group. High vowels devoiced at a substantially higher rate (7.4%) than low vowels (0.3%) or mid vowels (0.6%). For the preceding consonant, fricatives triggered vowel devoicing more frequently than affricates, and affricates more frequently than stops. Within stops, aspirated stops favored devoicing more than lenis and fortis stops.

The main theoretical interest lies in contexts where the two processes interact. In a sequence /obstruent-V-lenis stop-V/ (e.g., /siko/), the first vowel is both the target of vowel devoicing and also part of the intervocalic environment conditioning lenition of the following stop. Depending on which rule applies first, the output may be transparent with vowel devoicing only (1a) or opaque with lenition following the surface voiceless vowel (1b).

(1) a.	/siko/	b.	/siko/
Devoicing	i̥	Lenition	ɰ
Lenition	NA	Devoicing	i̥
	[si̥ko] (bleeding)		[si̥ɰo] (counterbleeding)

The corpus showed that both types of output were attested. Out of 1,271 eligible target sequences, vowel devoicing occurred in 24 tokens (1.9%), of which 20 tokens (83.3%) showed the transparent pattern (1a), while four tokens (16.7%) showed the opaque pattern (1b). Because the relevant interaction was relatively rare in the main corpus, I conducted an additional targeted analysis of 200 tokens from two frequent morpheme sequences: /su+ka/ 'possibility marker + subject marker' and /si+ko/ 'honorific marker + conjunctive'. In this subset, vowel devoicing occurred in 36 tokens (18%), of which 25 tokens (69.4%) were transparent (1a), and 11 tokens (30.6%) showed apparent opacity: seven were approximantized as in (1b), and four surfaced as [si̥go] with intervocalic voicing, not fully lenited.

Both transparent and apparently opaque surface patterns are observed across speakers.

Eight speakers consistently showed the transparent pattern; five consistently showed the apparently opaque pattern; and five alternated between transparent and opaque outputs. The existence of within-speaker alternation is crucial; it suggests that the variation cannot be fully reduced to categorical dialectal or idiolectal differences in rule ordering.

A parallel, surface-segmental OT analysis does not straightforwardly capture opaque interactions like (1b) because a markedness constraint such as *VTV cannot refer to the surface-devoiced vowel. Adopting Articulatory Phonology (Browman & Goldstein 1989, 1992), this paper argues that the apparent opacity dissolves once constraints are evaluated against the gestural score rather than surface segmental strings. In both [s̥iko] (Figure 1) and [s̥iɥo] (Figure 2), the first vowel retains its oral vocalic gesture; what differs is whether its laryngeal voicing gesture is suppressed by overlap with the glottal opening gesture of the preceding obstruent. Thus, the vowel may be voiceless acoustically while still participating in the oral V-C-V gestural configuration that conditions lenition. A markedness constraint penalizing robust lenis stop closure between oral vowel gestures applies equally to [siko] and [s̥iko]. The apparent opaque form [s̥iɥo] results when the lenition pressure and the devoicing pressure are both satisfied in the same gestural score.

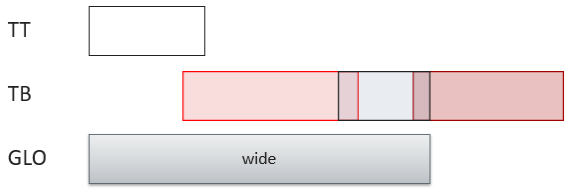


Figure 1. Gestural score of transparent [s̥iko]

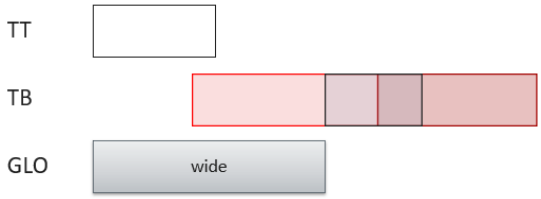


Figure 2. Gestural score of opaque [s̥iɥo]

Variation among [s̥iko] and [s̥iɥo]/[s̥igo] is modeled probabilistically using a MaxEnt (Hayes & Wilson 2008) framework with constraints grounded in Articulatory Phonology, assigning probabilities to competing gestural configurations. In this model, surface variation arises because the grammar weighs multiple, potentially conflicting pressures. MAX(Constriction) penalizes loss or reduction of the stop constriction gesture, REDUCE(Constriction) penalizes robust stop constriction between oral vowel gestures, ALIGN(Glottal-Spreading, Peak, vowel, L) favors overlap between a preceding obstruent's glottal opening gesture and the following vowel, and MAX(Glottal-Adduction) penalizes suppression of vocalic voicing. Speakers who produce [s̥iko] assign relatively high weight to MAX(Constriction), whereas speakers who produce [s̥iɥo] assign relatively higher weight to REDUCE(Constriction), allowing the lenis stop to weaken even when the preceding vowel is devoiced. Speakers producing [s̥igo] reflect an intermediate grammar, with voicing but no full approximantization. Within-speaker alternation reflects comparable probabilities for competing gestural configurations.

These findings argue that opacity, at least in this case, is not a categorical property of grammar but emerges from probabilistic, phonetically grounded gestural coordination. The within-speaker alternation is particularly diagnostic; a single grammar produces both transparent and apparently opaque outputs in the same morphological context, which is straightforward in MaxEnt over articulatory constraints but requires stipulation in any categorical account. More broadly, the results support a model in which apparent opacity, gradient phonetic variation, and inter-speaker differences arise from a single mechanism—probabilistic weighting of articulatory pressures—rather than from distinct components of grammar.

Selected References. Anttila, A. 2006. Variation and opacity. *NLLT* 24(4), 893-944. Browman, C. P. & Goldstein, L. 1989. Articulatory gestures as phonological units. *Phonology* 6, 201-251. / Browman, C. P. & Goldstein, L. 1992. Articulatory Phonology: an overview. *Phonetica* 49, 155-180. / Coetzee, A. & Pater, J. 2011. The place of variation in phonological theory. In The Handbook of Phonological Theory. / Hayes, B. & Wilson, C. 2008. A Maximum Entropy model of phonotactics and phonotactic learning. *LI*, 39(3), 379-440. / Huilgol, S., Baik, J. & Shattuck-Hufnagel, S. 2019. A framework for labeling speech with acoustic cues to linguistic distinctive features.

Gradient bilingual phonotactic judgments reflect dominance-weighted grammar mixture

Katerina A. Tetzloff & Han Li – Department of Linguistics, Stony Brook University

Introduction: Monolingual speakers typically make gradient well-formedness judgments that reflect the phonotactics of their native language [1]. Recent work has shown that English-Spanish bilinguals give intermediate acceptability ratings to sequences in Spanish that have conflicting phonotactic pressures from each of their two languages: sC-clusters are licit in English but illicit in Spanish and yield acceptability ratings that fall between English and Spanish monolingual patterns [2]. This suggests that bilinguals' two phonotactic systems interact, particularly when the languages provide contradictory evidence. While gradient phonotactic judgments in monolinguals have been modeled extensively [1, 3-4], previous work has not attempted to computationally model such bilingual gradient judgments. Here, we used corpus-learned phonotactic grammars, Maximum Entropy Phonotactic Learner (MaxEnt) [3] and the Bottom-Up Factor Inference Algorithm (BUFIA) [5], to derive monolingual well-formedness predictions in English and Spanish and combined them via an α-weighted mixture model, testing a constant α model and a speaker-specific model where α reflects each listener's language dominance. The two constraint-learning approaches were compared because they differ in whether phonotactic knowledge is represented as gradient weighted constraints (MaxEnt) or categorical binary constraints (BUFIA), allowing us to assess which type of representation better captures bilingual gradient phonotactic judgments.

Behavioral methods: Behavioral data came from previously published acceptability ratings from 33 English-Spanish bilinguals and 8 Spanish monolinguals, who rated from 1 - 4 the well-formedness of Spanish nonce words belonging to one of four categories: esC-initial (licit in both languages), CC-initial (licit in both languages), sC-initial (licit in English but not Spanish), and *CC-initial (illicit in both languages) [2].

Modeling grammars: Separate English and Spanish grammars were learned from attested onset clusters in each language using two approaches. In the first approach, constraints and their weights were learned using the MaxEnt learner; test forms were evaluated by computing a weighted sum of constraint violations; higher violation scores indicate lower predicted well-formedness. In the second approach, constraints were learned deterministically from the same attested data using BUFIA; test forms were evaluated by determining if test forms violate any induced constraints, yielding binary accept/reject predictions ($P \in \{0, 1\}$) for each test form. Crucially, both methods incorporated a syllable boundary in esC (parsed /es.C/, licit in Spanish) from bare sC (illicit in Spanish), allowing the grammars to correctly assign grammaticality across cluster types. For bilinguals, predicted well-formedness (pred) was computed as $\text{pred}_{\text{bilingual}} = \alpha \cdot \text{pred}_{\text{Spanish}} + (1 - \alpha) \cdot \text{pred}_{\text{English}}$. The constant α model used a fixed value ($\alpha = 0.6$, weighted toward Spanish, the task language), while the speaker-specific α model derived listener-specific weights from each listener's dominance score (English proficiency minus Spanish proficiency [2] scaled to $[0, 1]$ via the logistic function). Only Spanish grammar predictions were used for Spanish monolingual judgment data.

Model evaluation: We fit ordinal regressions for each of the four models (constant α and speaker-specific α for MaxEnt and BUFIA) with model-predicted well-formedness and its interaction with listener group (bilingual vs. monolingual) as fixed effects. Models were compared using AIC.

Results: All four models significantly predicted human acceptability ratings ($p < .0001$). The BUFIA speaker-specific α model provided the best fit (AIC $= 17568$; Figure 1), outperforming the BUFIA constant α model (ΔAIC $= 40$) and substantially outperforming both MaxEnt models (ΔAIC > 872). Onset-level Pearson correlations for the winning BUFIA model were $r = 0.96$ for

bilinguals and $r = 0.93$ for monolinguals, compared to $r \leq 0.62$ for MaxEnt models. Additionally, all models showed a significant interaction between predicted well-formedness and listener group ($p < .0001$), with phonotactic predictions more strongly modulating acceptability ratings for monolinguals ($\beta = 3.17$) than for bilinguals ($\beta = 1.98$). Thus, while the mixture models capture the central tendency of bilingual acceptability ratings, individual variability in bilinguals, beyond what is predicted by a dominance score alone, remains a source of unexplained variance.

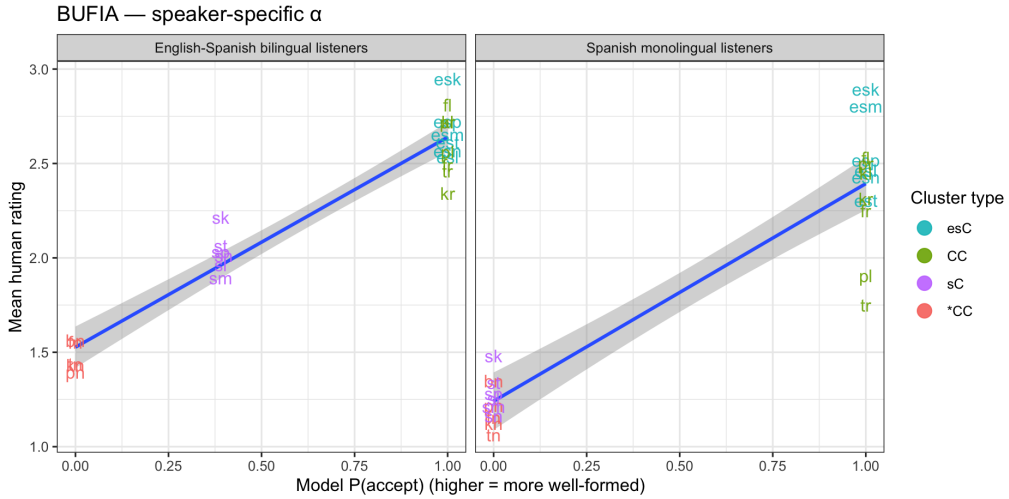


Figure 1: *Relationship between model-predicted well-formedness from the BUFIA speaker-specific α mixture model and mean human acceptability ratings for bilinguals and monolinguals.*

Discussion: The modeling results demonstrate that non-monolingual-like gradient well-formedness judgments of bilinguals can be captured by weighted combinations of their two independent grammars. The superiority of the speaker-specific α model suggests that bilingual phonotactic interactions may vary systematically with an individual's language dominance, rather than reflecting a uniform cross-speaker weighting. The substantial advantage of BUFIA over MaxEnt suggests that categorical constraint induction captures the relevant phonotactic distinctions more effectively than gradient harmony scores for this task [6]. These results provide the first computational evidence that bilingual phonotactic judgments can be modeled as mixture grammars with dominance-weighted contributions from each language system, a representation that is neither categorically language-specific nor globally merged, but reflects an experience-sensitive integration of the two phonotactic systems. Future work will extend this mixture grammar approach to other bilingual populations to assess the generalizability of dominance-weighted phonotactic integration across distinct bilingual language pairings.

References

[1] Daland, R., Hayes, B., White, J., Garellek, M., Davis, A., & Norrmann, I. (2011). Explaining sonority projection effects. *Phonology, 28*(2), 197–234.

[2] Tetzloff, K. A. (2024). Variability in English–Spanish bilingual phonotactics. *Journal of Monolingual and Bilingual Speech, 6*(1), 1–20.

[3] Hayes, B. & Wilson, C. (2008). A maximum entropy model of phonotactics and phonotactic learning. *Linguistic Inquiry*, *39*(3), 379–440.

[4] Albright, A. (2009). Feature-based generalisation as a source of gradient acceptability. *Phonology*, *26*(1), 9–41.

[5] Chandlee, J., Eyraud, R., Heinz, J., Jardine, A., & Rawski, J. (2019). Learning with partially ordered representations. *Proceedings of the 16th Meeting on Mathematics of Language*, 91-101.

[6] Swanson, L., Heinz, J., & Rawski, J. (2026). Phonotactic learning and constraint selection without statistics. *Linguistic Inquiry*, 1-24.

Formalizing stability, convergence and reanalysis in heritage language phonology: MaxEnt modeling of word-final obstruent devoicing in homeland and heritage Polish

Background. Heritage languages (HLs) offer a unique window into phonological malleability under multilingualism and language contact. Three main outcomes have been proposed for the acquisition of HL phonology: (i) stability: the representations resemble the monolingual L1 variety [9] ; (ii) convergence: majority-language features are transferred into the HL [7]; and (iii) reanalysis: the HL grammar is restructured, often in ways that reflect the path of cross-linguistic typological universals [11]. This paper argues that these outcomes can be distinguished formally in MaxEnt (Maximum Entropy) constraint grammars [2]: stability is reflected in homeland-like constraint rankings, convergence in the promotion of constraints active in the majority language, and reanalysis in the loss of language-specific constraints or the promotion of phonetically natural constraints (as also documented in L2 phonology [1]). This study uses MaxEnt grammar [2] to test this proposal with word-final obstruent devoicing in Polish ([-sonorant] > [-voice]/__#) among homeland speakers in Poland and two generations of Toronto heritage speakers [5] (Table 1). HerGen1 speakers received regular input and schooling in Polish before age 18 and are therefore expected to show stability, according to the critical period hypothesis [10]. HerGen2 speakers, who had less regular input before adulthood, may show both convergence and reanalysis. Since English does not neutralize word-final voicing contrasts (IDENTVOICE: penalizing segments with different voicing >> *VOICEDCODA: penalizing voiced codas), convergence toward English is expected to appear as the promotion of IDENTVOICE.

Methods. We analyzed 3,201 tokens with underlyingly voiced word-final obstruents [5]. Separate MaxEnt grammars were trained for the three groups, with parameters optimized through cross-validation to avoid overfitting [6]. Following [3], we started with a baseline model with AGREEVOICE (penalizing disagreement of voicing between coda and the following sound) and IDENTVOICE [4], and then added markedness constraints targeting final voicing by place, manner, and prosodic alignment, then removing constraints that did not improve log-likelihood.

Results. Table 2 summarizes the constraint rankings and weights. Compared to Hom speakers, (i) HerGen1 shows stability with limited convergence toward English, while (ii) HerGen2 shows reanalysis without clear evidence of further convergence. HerGen1 speakers preserve homeland-like high-weight markedness and assimilation constraints: *VOICEDCODAPWORD (penalizing voiced coda in the prosodic word), *VOICEDCODAVELAR (penalizing voiced velar coda), and AGREEVOICE. Meanwhile, IDENTVOICE enters the final grammar with a positive weight, consistent with English pressure against neutralization and continued plasticity in adult phonological representations [12]. HerGen2, however, does not move further toward English: IDENTVOICE is not retained in the final model. Instead, HerGen2 reweights the grammar around markedness based on phonetic naturalness. The highest-weight constraint is *VOICEDCODAFRICATIVE, a manner markedness constraint penalizing voiced fricative codas. By contrast, the prosodic-word-specific *VOICEDCODAPWORD that characterizes Hom and HerGen1 is absent. This pattern suggests reanalysis from a prosodically specified final-devoicing grammar toward a grammar organized by the aerodynamic difficulty of maintaining voicing, reflecting the

directionality of phonological malleability under contact and reduced L1 input: grammars restructure along paths constrained by phonetic substance—mirroring the directionality observed in cross-linguistic typological change that fricative final devoicing may phonologize before final devoicing of stops [1]. Taken together, the results bear on the architecture of phonological malleability in multilingualism: stability, convergence, and reanalysis reflect not different degrees of L1 attrition, but the dynamic promotion and demotion of faithfulness constraints, language-specific markedness constraints, and phonetically grounded constraints in MaxEnt.

Group	Definitions [8]	# speakers	# tokens	Devoicing rate %
Homeland (Hom)	Polish L1 speakers with primary residence in Poland	10	1210	66 %
Heritage generation 1 (HerGen1)	Polish L1 speakers who immigrated to Toronto after the age of 18 and lived in Toronto for more than 20 years.	8	981	67%
Heritage generation 2 (HerGen2)	Children of the HerGen1 speakers; either born in Poland and immigrated to Toronto before six or born in Toronto	9	1010	74%

Table 1. Description of speaker groups, number of speakers, number of tokens, and averaged devoicing rates.

Constraint (Hom)	Weights	Constraint (HerGen1)	Weights	Constraint (HerGen2)	Weights
*VoicedCodaPword	0.529	AgreeVoice	1.243	*VoicedCodaFricative	0.985
*VoicedCodaVelar	0.438	*VoicedCodaVelar	1.133	AgreeVoice	0.833
AgreeVoice	0.404	*VoicedCodaPword	0.665	*VoicedCodaStop	0.697
*VoicedCodaStop	0.277	*VoicedCodaFricative	0.458	*VoicedCodaLabial	0.400
*VoicedCodaFricative	0.259	IdentVoice	0.241		
*VoicedCodaClitics	0.007				

Table 2. Constraint rankings, listed in order of decreasing weights in the final MaxEnt models for the three groups.

[1] Broselow, E. (2018). Laryngeal contrasts in second language phonology. In L. M. Hyman & F. Plank (Eds.), *Phonological Typology* (pp. 312–340). De Gruyter. https://doi.org/10.1515/9783110451931-009

[2] Hayes, B., & Wilson, C. (2008). A maximum entropy model of phonotactics and phonotactic learning. *Linguistic Inquiry*, *39*(3), 379–440.

[3] Katsuda, H. (2025). A probabilistic model of loanword accentuation in Japanese. *Phonology*, *42*, e9.

[4] Lombardi, L. (1999). Positional faithfulness and voicing assimilation in Optimality Theory. *Natural Language and Linguistic Theory*, *17*, 267–302.

[5] Łyskawa, P., Maddeaux, R., Melara, E., & Nagy, N. (2016). Heritage Speakers Follow All the Rules: Language Contact and Convergence in Polish Devoicing. *Heritage Language Journal*, *13*(2), 219–244. https://doi.org/10.46538/hlj.13.2.7

[6] Mayer, C., Tan, A., & Zuraw, K. R. (2024). Introducing maxent.ot: An R package for Maxi-mum Entropy constraint grammars: an R package for Maximum Entropy constraint grammars. *Phonological Data and Analysis*, *6*(4), 1–44. https://doi.org/10.3765/pda.v6art4.88

[7] Nagy, N., & Kochetov, A. (2013). VOT across the generations: A cross-linguistic study of contact-induced change. In P. Siemund, I. Gogolin, M. E. Schulz, & J. Davydova (Eds.), *Multilingualism and language contact in urban areas* (pp. 19–38). John Benjamins Publishing.

[8] Nagy, N. (2024). Heritage languages: Extending variationist approaches. Cambridge University Press.

[9] Oh, J. S., Jun, S.-A., Knightly, L. M., & Au, T. K. (2003). Holding on to childhood language memory. *Cognition*, *86*(3), B53–B64.

[10] Penfield, W., & Roberts, L. (1959). *Speech and brain mechanisms*. Princeton University Press.

[11] Polinsky, M. (2016). Structure vs. Use in heritage language. *Linguistics Vanguard*, *2*(1). https://doi.org/10.1515/lingvan-2015-0036

[12] Sankoff, G. (2019). Language change across the lifespan: Three trajectory types. *Language*, *95*(2), 197–229.

How language experience reshapes the P-Map: Final nasalization in Serudung Murut

Jian-Leat Siah, UCLA (jlsiah@g.ucla.edu)

Background: Across languages, voiced stops are dispreferred in word-final position for both articulatory and perceptual reasons (Myers 2012). When they arise in paradigms or loanwords, they are typically repaired, as in German final devoicing. Although several repairs are possible (e.g., final nasalization, vowel epenthesis), final devoicing is by far the most common cross-linguistically. Steriade (2001) accounts for this typological asymmetry with the P-Map hypothesis, proposing that learners favor repairs that minimize perceptual differences between input and output forms, and that final devoicing is perceptually the minimal repair. Yet there are languages that behave contrary to the P-Map. One such case is Serudung Murut, discussed below. We suggest that the Murut pattern is less anomalous than it might seem: Murut speakers seem to have acquired a different P-Map from their native language. To test this hypothesis, we conducted perceptual experiments with two populations: American English listeners, whose phonology lacks a final devoicing alternation, and Serudung Murut listeners, speakers of an endangered Austronesian language reported to exhibit a counter-P-Map repair, namely final nasalization, e.g., /alid-in/ → [ali**d**-in] 'to buy (Location Focus)', but /aN-alid/ → [aŋ-ali**n**] 'to buy (Actor Focus)' (Townsend 2017; Lobel 2016).

Methods: 136 American English and 25 Serudung Murut listeners participated in a perceptual AXB task remotely and in person respectively, identifying whether the middle stimulus (X) matched the first (A) or the last (B), with an interstimulus interval of 1500 ms. The stimuli (Table I), recorded by three phonetically trained speakers (2F, 1M) in a sound-proof booth, consisted of disyllabic forms with an unreleased final voiced stop (alveolar vs. labial) and five repair strategies discussed in Steriade (2001). All forms carried initial stress and contained either the [a] or [u] vowel. Each trial (e.g., [alap̚]$_A$-[alab̚]$_X$-[alab̚]$_B$) compared an unrepaired and a repaired form across four counterbalanced orders (AAB, ABB, BAA, BBA), with tokens produced by different speakers. Participants completed 80 trials in total (5 repairs × 2 places of articulation × 2 vowels × 4 orders).

Results: The left and right panels of Figure 1 show the results for American English and Serudung Murut listeners, respectively. The y-axis plots the proportion of incorrect responses, and the x-axis represents repair strategies. Higher proportion of incorrect responses indicates greater perceptual similarity to the unrepaired form. As can be seen in the figure, American English listeners find final devoicing to be the most perceptually confusable repair, in line with P-Map predictions, yielding the hierarchy: Metathesis < Insertion, Nasalization < Deletion < Devoicing, where "<" indicates perceptually less similar than. In contrast, Serudung Murut listeners find both devoicing and nasalization equally confusable. When results are further broken down by language proficiency (Figure 2), less proficient Murut listeners pattern more like the English control group, whereas more proficient listeners exhibit a different hierarchy: Metathesis, Insertion < Deletion, Devoicing < Nasalization, in which nasalization surpasses devoicing in confusability. All results are supported by Bayesian mixed-effects logistic regression models using the *brms* package in R (Bürkner 2017).

Implications: Overall, American English listeners show patterns consistent with the P-Map hypothesis. In contrast, results from Serudung Murut listeners suggest that the perceptual biases posited by the P-Map are malleable and can be reshaped through alternations that contradict the P-Map rankings. This effect is most evident among highly proficient speakers, for whom final nasalization emerges as the most confusable repair, indicating a synchronic restructuring of the P-Map and underscoring the role of experience in phonological learning. Taken together, these findings suggest that the P-Map functions as a soft bias in guiding phonological learning and typological tendencies (White 2017). They also support the view that sounds in an allophonic relationship are perceived as less distinct than those in contrastive opposition (e.g., [d]-[n] are contrastive in English in final position but allophonic in Serudung Murut; Boomershine *et al.* 2008).

Table I. Audio stimuli used in the perceptual AXB task (Left: Alveolar; Right: Bilabial).

ˈalad˥	ˈalat˥	Final Devoicing	ˈalap˥	ˈalab˥
	ˈala	Consonant Deletion	ˈala	
	ˈalan	Final Nasalization	ˈalam	
	ˈaladə	Vowel Insertion	ˈalabə	
	ˈadal	Metathesis	ˈabal	

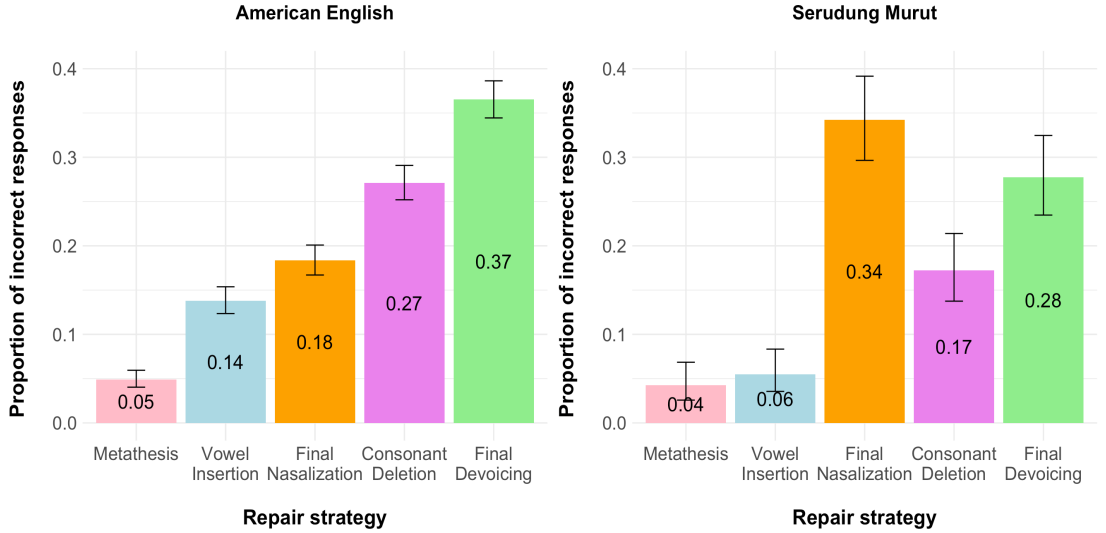


Figure 1. Proportion of incorrect responses (Left: American English; Right: Serudung Murut).

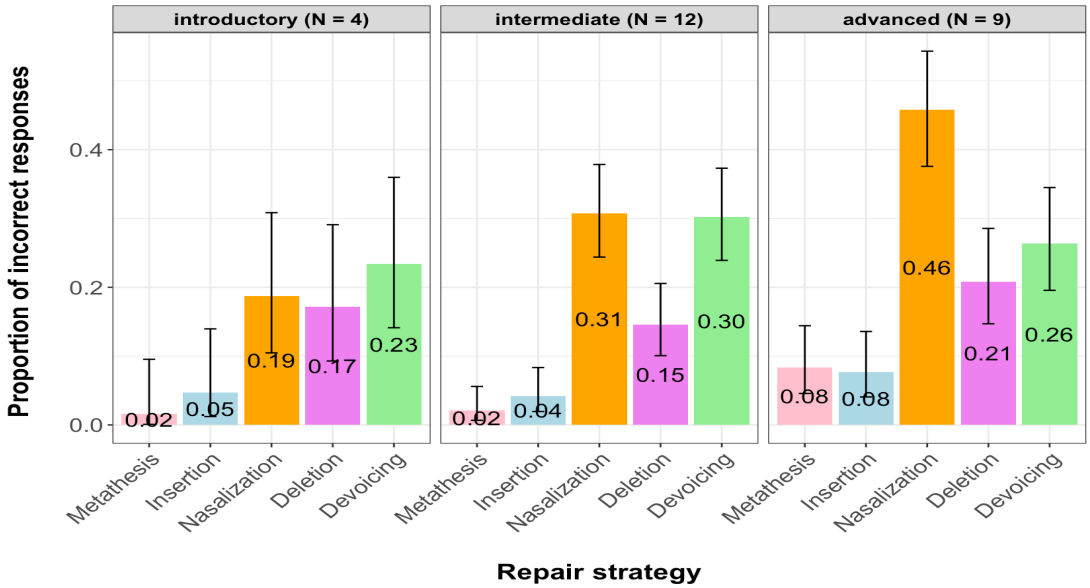


Figure 2. Results of Serudung Murut listeners by language proficiency.

Boomershine, A., Hall, K. C., Hume, E., Johnson, K. (2008). The impact of allophony versus contrast on speech perception. In *Contrast in phonology: Theory, perception, acquisition*, *13*, 145-172. **Lobel, J. W. (2016)**. *North Borneo sourcebook: Vocabularies and functors*. UH Press. **Myers, S. (2012)**. Final devoicing: Production and perception studies. In *Prosody matters: Essays in honor of Elisabeth Selkirk*, 148-180. **Steriade, D. (2001).** The phonology of perceptibility effects: The P-map and its consequences for constraint organization. *Ms., UCLA*. **Townsend, J. (2017)**. *Reduplication in Serudung Murut* [MA Thesis]. Leiden University. **White, J. (2017).** Accounting for the Learnability of Saltation in Phonological Theory: A Maximum Entropy Model with A P-Map Bias. *Language*, *93*(1), 1-36.

Arriving at Hidden Structure: Mutually Reinforcing Syllabification and Stress Learning

Kaustubh Ghoshal, New York University

The Problem: Hidden Structure and Learning. Weight-sensitive stress presents a classic "hidden structure" problem: a learner must determine which syllables are heavy to arrive at the right stress grammar, but weight is often opaque and requires reference to the distribution of stress (Tesar & Smolensky 1998). I show how this circularity can be resolved by a learner that allows syllabification and stress to operate in a feedback loop, using stress position as a diagnostic for structural parsing. This is the *first implemented proposal* for learning syllabification and syllable weight along with stress.

Weight-sensitive stress. Languages can refer to syllable shape to determine the position of stress. In Kabardian, stress is on the ultima if it has a long vowel or a coda (CVX), and on the penult otherwise. But in Huallaga Quechua, stress is on the ultima only when it has a long vowel (CVV), not when it is CVC. What counts as 'heavy' varies across languages.

(1) CVX: Kabardian (Gordon 2017)

tɐpʃə́g	'plate'
saːbíː	'baby'
máʃɐ	'bear'
χɐrzánɐ	'good'

(2) CVV: Huallaga Quechua (Weber 1983)

ajwaʃáː	'bear'
ʧajʌaráː	'just a moment ago'
ajwájkaŋ	'they are going'
manáraχ	'not yet'

The learning problem is compounded by syllabification ambiguity outside final position: /VCCV/ may be syllabified as [VC.CV] or [V.CCV], affecting the shape of medial syllables. A learner must identify both the *correct syllabic parse* as well as *which syllables are heavy.*

The Model: Feedback-Driven Learning. The model combines a MaxEnt constraint induction engine (Hayes & Wilson 2008) with a parameter-based syllabification module.

▷ *Parametric Parsing*: The learner is equipped with syllabic parameters (e.g., max_onset) and structural projections ([NUC2], [RIME2]), representing branching at the rime and nucleus levels. Structural branching is then the key discriminator for weight (Dresher 2001).

▷ *The Feedback Loop*: The learner evaluates competing syllabic parameter settings by inducing a stress grammar for each parse. Based on the learned grammar, scores are assigned to a set of test forms and the results are evaluated using two metrics:

Selection Metric: The % of legal stress forms among the n highest-scored items, where n is the actual number of legal stress forms (for the target pattern) in the testing set.

Prediction Metric: Mean relative probability assigned to legal stress form for a given word, out of all its possible stress forms. $P(\text{legal}|\text{word}) \simeq \frac{P(\text{legal})}{P(\text{stress } \sigma_1)+P(\text{stress } \sigma_2)...+P(\text{stress } \sigma_n)}$

Result 1: Resolving ambiguous syllabification. I test the model on Petrarki (42K words), an artificial language where intervocalic clusters are either TR (stop-rhotic) or RT. The data was syllabified in two ways and assigned stress (3). By design, there were no word-initial clusters and no final consonants, so syllabification of intervocalic clusters is indeterminate and depends on stress (Table 1).

Word	RDTS-NoCC	RDTS-OR
tatrita	tát.ri.ta	ta.tri.tá
tartepri	tar.tép.ri	tár.te.pri
riːrotrere	riː.rót.re.re	rí́ː.ro.tre.re
toːpeːre	toː.péː.re	toː.péː.re
topetu	to.pe.tú	to.pe.tú

Table 1: Stress patterns in Petrarki

(3) Right-aligned default-to-same (RDTS): Stress the rightmost CVX, else the final syllable.
   a. RDTS-NoCC: All clusters are heterosyllabic (VT.RV, VR.TV).
   b. RDTS-OR: TR clusters are homorganic (V.TRV), RT heterosyllabic (VR.TV).

The training data contained surface forms with stress marking but no syllabic information.

Target pattern	Learner's syllabification	With long vowels		Without long vowels	
		Selection	Prediction	Selection	Prediction
RDTS-NoCC	None	90.92	96.7	62.83	82.0
RDTS-NoCC	NoCC	96.26	98.1	88.56	100
RDTS-NoCC	OR	69.00	93.1	62.44	81.5
RDTS-OR	None	85.72	96.0	79.49	93.6
RDTS-OR	NoCC	73.33	96.0	69.31	90.2
RDTS-OR	OR	96.40	100	98.17	100

Table 2: Learning syllabification and stress in a feedback loop

The learner successfully uses surface stress to bootstrap syllabification (Table 2). By comparing the grammars learned under different parameter settings (max_onset=0 vs. 1), the learner identifies the parse that maximizes the evaluation metrics for stress learning. For example, finding initial stress in tátrita allows the learner to reject the onset parse (tá.tri.ta) in favor of the heterosyllabic parse (tát.ri.ta), leading to a stress grammar with better fit.

The role of the feedback loop is even more crucial in the absence of long vowels. Stress in the original data often fell on a long vowel, making it easier to capture (*[-Stress,+Long]) without syllabification. Without this, the learner must rely entirely on the correct coda parse. Accordingly, when long vowels were removed from Petrarki, the gap between correct and incorrect syllabification results widened, demonstrating the need for correct parsing.

Result 2: Representational failure of linear models. Even in final position, where syllabification is invariable, the current model is an advance over linear induction models like the UCLA Phonotactic Learner (Hayes & Wilson 2008). To show this, I used artificial data based on Latin (86K words), with stress on the ultima if heavy (*perspectúm*), else the penult (*detésto*). Weight was coded in turn as CVV-only and CVX.

System	Linear	Syllabic
CVX	68.4	93.5
CVV	96.2	90.0
highV	89.9	53.8

Table 3: Selection accuracy

Table 3 shows that while CVV is learned relatively easily, the default UCLAPL struggles with CVX as it prefers shorter constraints, biasing towards unigram-based systems (*[-Stress,+Long] for CVV). On the other hand, with no structural encoding, analyzing CVX requires disjunction and longer constraints. Adding syllabic features like [Rime2] allows reference to CVV and CVC together, greatly improving learning for CVX. Cross-linguistically, CVX is the most common type of system, occurring more frequently than CVV (Gordon 2006). The syllabic learner captures this typological asymmetry. Furthermore, the default UCLAPL *overgenerates*: unattested systems such as highV-heavy are predicted to be more common than CVX, since any feature can trigger stress (*[-Stress,+High]). For the syllabic model, such systems are essentially unlearnable (53.8=chance accuracy).

Conclusion. This proposal integrates a syllabification module in a feedback loop with a constraint induction learner, thus overcoming the hidden structure impasse for syllable weight. The learning results demonstrate that structural representations are not just an analytical convenience but a computational prerequisite for learning weight-sensitive stress.

References. Dresher, B.E. 1999. Charting the Learning Path: Cues to Parameter Setting. *Linguistic Inquiry* 30. Gordon, M. 2006. *Syllable weight: phonetics, phonology, typology.* Hayes, B. & C. Wilson 2008. A Maximum Entropy model of phonotactics and phonotactic learning. *Linguistic Inquiry* 39. Tesar, B. & P. Smolensky 1998. Learnability in Optimality Theory. *Linguistic Inquiry* 29.

The Representation of Complex Onsets in Bangbing Bulang: Evidence from a Fieldwork-Based Phonological Sketch

Rui Liu*, Yijin Gui*

**These authors contributed equally to this work.*

1. Introduction. Complex consonant sequences at the left edge of the syllable raise a familiar representational issue for Austroasiatic phonology: which sequences are true onset clusters, which are prenasalized segments, and which belong to sesquisyllabic word structure? This paper examines this issue in Bangbing Bulang, an underdescribed Palaungic Austroasiatic variety spoken in Shuangjiang County, Yunnan (Li, Nie & Qiu 1986; Sidwell 2015). The central claim is that Bangbing licenses complex onset structure: stop-liquid sequences provide independent evidence for branching, while nasal material can co-occur with this independently motivated complexity.

2. Fieldwork basis. The data come from original elicitation with five native speakers using a classified wordlist of approximately 1,000 items, repeated and slow productions, phrase contexts, and targeted phonological questions. The diagnostic used here is whether an apparent cluster remains vowelless during careful repetition. Such forms pattern differently from overt presyllabic forms such as /a31mɛʔ35/ ‘who’ and /a31tʰak55/ ‘tongue’.

3. Initial consonant sequences. The corpus contains nasal-stop, stop-liquid, and nasal-stop-liquid initial sequences, summarized in Table 1 with tone numerals as transcribed in elicitation.

Table 1. Initial consonant sequences in Bangbing Bulang.

Type	Representative sequences	Examples	Analytical issue
Nasal-stop	mp, mpʰ, mb, nt, ntʰ, nd, ŋk, ŋkʰ, ŋq, ŋqʰ	/mbɤʔ35/ ‘porridge’; /ntʰu31/ ‘wash’	Cluster analysis or prenasalized stop analysis
Stop-liquid	pl, pʰl, bl, br, gr, kl, kʰl, gl, ql, qʰl	/briʔ51/ ‘mountain’; /grɔŋ31/ ‘road’	Independent evidence for branching onsets
Nasal-stop-liquid	mbl, mpl, mbr, ŋgr, ŋgl	/mbli31/ ‘louse’; /ŋglɔŋ31/ ‘lake’	Cooccurrence of nasal material and liquid clustering

The three types differ in how directly they motivate onset complexity. Nasal-stop sequences by themselves are analytically ambiguous because they can receive a prenasalized-segment analysis in many languages. Stop-liquid sequences provide

independent evidence for complex onset structure: such sequences recur across multiple places of articulation, motivating more than one prevocalic consonantal unit. Nasal-stop-liquid sequences show that nasal material can co-occur with this independently motivated complexity. They do not by themselves require an analysis with three independent segmental positions, and the internal representation of individual nasal-stop sequences remains open.

4. Diagnostics and analysis. Three representational possibilities are relevant. First, a synchronic hidden-vowel analysis predicts an intervening vowel, but /mbli31/, /grɔŋ31/, /pʰli31/, and /ŋqʰɔʔ35/ remained vowelless during careful repetition. Second, some NC sequences, including the NC portion of NCL forms, may be prenasalized consonants; this does not eliminate the branching independently motivated by non-nasal stop-liquid sequences. Third, some initial sequences may involve minor-syllable structure. However, Bangbing also has overt presyllabic forms such as /a31mɛʔ35/ and /a31tʰak55/, which contrast with the vowelless sequences in careful repetition. This diagnostic does not exclude every minor-syllable analysis.

The surface patterns are NCV(C), CLV(C), and NCLV(C). NC sequences may be bisegmental or prenasalized, but this uncertainty does not affect the complex-onset conclusion because non-nasal stop-liquid sequences independently motivate branching. Careful repetition distinguishes vowelless initial sequences from overt presyllabic forms without excluding all minor-syllable analyses.

5. Implications. This analysis contributes new fieldwork evidence from a local Bulang variety whose phonology remains underdescribed. The attested initial sequences cannot all be reduced to either uniform prenasalization or a uniform minor-syllable account. The grammar must therefore license complex onset structures. This result contributes to broader Palaungic discussions of consonant complexity and syllable structure (Watkins 2002; Sidwell 2015). Bangbing shows that complex initial sequences and overt presyllabic forms can coexist in one synchronic system while displaying distinct surface patterns in careful repetition.

References. Li, D., Nie, X. & Qiu, E. 1986. Bulangyu jianzhi [A sketch of the Bulang language]. Minzu Chubanshe. Sidwell, P. 2015. The Palaungic languages: Classification, reconstruction, and comparative lexicon. Lincom Europa. Watkins, J. 2002. The phonetics of Wa. Pacific Linguistics.

Affix classes are gradient: the effect of weight on demonym selection in English

Yanran Mou[1] & Heather Goad[2]

[1]University of Massachusetts, Amherst, [2]McGill University

Introduction: English derivational affixes are traditionally described as falling into two classes [2, 3, 5]: class 1 (c1) affixes trigger phonological changes such as a 'shift' in stress location, and stress in class 2 (c2) affixed forms stays faithful to the base [4]. Affix class is assumed to be stored in the lexicon. However, several proposals contra the traditional c1/c2 distinction have been made, as certain c2 suffixes display gradient behaviour (e.g. *-able*: *írrigate/írrigable, cómpensate/compénsable*; cf. [2, 8]). It remains unclear if lexical encoding of the c1/c2 division is necessary, and if it isn't, what criteria speakers believe to be relevant for determining the behaviour of specific suffixes. We experimentally probe this question using novel town-demonym pairs and propose that the differences in phonological behaviour of c1 and c2 affixed forms may largely be derivable from the segmental and prosodic shapes of the suffixes themselves.

Background: English word stress is assigned within a three-syllable window from the right edge, and is affected by factors like syllable weight and suffix type. Suffixation with c1 yields words that respect the same phonotactic and rhythmic constraints as hold of monomorphemic words. If a new heavy penult is created via c1 suffixation, stress shifts onto it (*párent-paréntal*), and if c1 suffixation causes faithful stress to be too far from the right word edge, stress shifts rightward such that it falls within the three-syllable window (e.g., *phótograph*, c1: *photógraphy*). c2 suffixes are invisible to stress calculation, and can yield stress outside of the three-syllable window (*phótograph*, c2: *phótographless*) as well as clusters that are phonotactically unusual in monomorphemic words (e.g., adjacent fricatives, c2: *wi<u>sh</u>ful*).

However, direct comparison between classes 1 and 2 can be difficult: A) c1 suffixes are mostly vowel-initial, while c2 are mostly consonant-initial; resyllabification in the former case can change the weight of the base-final syllable. B) c1 and c2 are generally not compatible with the same bases: c2 affixes can sometimes attach to a c1 base but not vice-versa (e.g., c1: *stupid, stupidity, *?stupidness* vs. c2: *clever, cleverness, *cleverity*).

Objectives: Heterogenous behaviour in the stress of c2 suffixed forms is lexically specific and modulated by the stress patterns of morphologically related words [2]. Testing generalizability separate from these confounds requires the use of novel bases. Most demonyms in English are formed from town names using one of three suffixes: *-ian*, *-er*, *-ite* (e.g., *Cálgary*, *Calgárian*; *Wínnipeg*, *Wínnipeger*; *Vancóuver*, *Vancóuverite*). We contend that these suffixes are an excellent test for generalizability because: A) the shape of a demonym affix is not predictable from the base (i.e., town name); B) the affixes fall into different classes (c1 *-ian*; c2 *-er, -ite*); C) all three suffixes are vowel-initial, so we avoid the possible creation of phonotactically unusual clusters at the morpheme boundary; and D) the suffixes have different shapes (*-ian* is LL, *-er* is L, *-ite* is H).

Predictions: We predict that differences in suffix shape (i.e., weight and length) better capture the stress patterns of derived words than traditional affix classes. We expect participants to prefer shift to the antepenult in demonyms formed with bisyllabic *-ian*, and predict that the stress patterns of demonyms formed with *-er* and *-ite* will be modulated by the weight of the final syllable, contra the traditional account of c2 as categorically faithful.

Methods: The stimuli comprised 16 town names ending in a branching coda (heavy), yielding a heavy penult in the suffixed form, and 16 town names ending in a simplex coda (light); see table overleaf. Of light town names, 8 were fillers, where one variant shifted stress onto a different syllable than the penult/antepenult. We used a Latin-square design; each town name occurred with all three suffixes, and each participant heard one town-suffix pair.

20 native speakers of North American English were recruited via Prolific. In each trial, participants were simultaneously exposed to the auditory and orthographic representation of a

novel town name. They then heard two alternative demonyms. They signalled their preferences on a continuous scale between two end points labelled *Option 1* and *Option 2,* corresponding to the order in which they heard the demonyms. The demonyms derived from a given town name

Town	Final weight	Base stress	Shape	Suffix	Faithful	Shifted
Múrkendile	Heavy	antepen	HHH	-er	Múrkendiler	Murkendíler
Ránodresp	Heavy	antepen	LLH	-ite	Ránodrespite	Ranodréspite
Siróndirk	Heavy	penult	LHH	-ian	Sirondírkian	Siróndirkian
Pánarin	Light	antepen	LLL	-er	Pánariner	Panaríner
Cáltrenas	Light	antepen	HLL	-ite	Cáltrenasite	Caltrenásite
Andóltan	Light	penult	HHL	-ian	Andóltanian	Andoltánian

differed from each other only in stress; one was stressed on the same syllable as their town-name base, and the other had stress shifted to the root-final syllable (to the antepenult for base + bisyllabic *-ian*, to the penult for base + monosyllabic *-er/-ite*). We randomized the order of presentation of items and demonym variants (i.e., which demonym they heard first).

Results: We analyzed participants' preference for stress shift using a linear mixed-effects model (R, lme4 package) with fixed factors Suffix (-ian, -er, -ite), Weight (light vs. heavy), and Base Stress (initial (1) and peninitial (2)), and random intercepts for Town name and Participant. The model revealed lower shift preference for *-er* than *-ian*, $\beta = -30.10$, SE = 11.55, $t(433.65) = -2.61$, $p = .009$, and for *-ite* than *-ian*, $\beta = -46.92$, SE = 11.55, $t(433.62) = -4.06$, $p < .001$.

The effect of Weight differed by suffix (Fig. 1): Shift preference was not impacted by weight for *-ian* and *-er*; heavy forms increased shift preference for *-ite*, $\beta = 34.89$, SE = 13.08, $t(431.87) = 2.67$, $p = .008$. This weight effect for *-ite* was further qualified by Base Stress, $\beta = -48.28$, SE = 23.95, $t(443.73) = -2.02$, $p = .044$, indicating that heavy *-ite* forms showed greater shift preference mainly when base stress was on the initial syllable in the unsuffixed form.

Discussion: Participants' preference for stress shift with demonyms affixed with c1 *-ian* is weight-insensitive. With the c2 forms, though, the shape of the suffix itself (L vs. H), and of the preceding syllable, seem to matter; while *-er* affixed forms show no weight sensitivity, for *-ite,* suffix weight interacts with the weight of the final syllable. While there were no effects of weight for *-er* suffixed forms, the preference for shift was higher than the categorical faithful behaviour expected. This could potentially be due to the presence of various homophonous forms of *-er* in English; e.g., c1 agentive *-er* (*photograph-photographer*), and c2 comparative *-er* (*long, longer*).

In conclusion, preferences for the stress patterns of novel derived words are not determined by suffix identity alone. The perhaps once categorical affix class divide may be fading in favour of greater sensitivity to prosodic factors. Our findings suggest that while c1/c2 distinctions do still exist in the synchronic grammar, speakers have gradient representations of these categories sensitive to base and suffix weight.

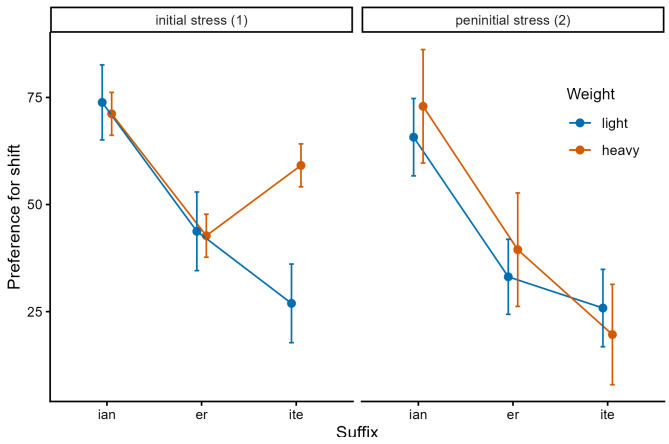


Fig 1. Preference for stress shift by suffix, base weight, and base stress. Points represent estimated marginal means from the mixed-effects model, with error bars showing ±1 SE.

Select references: **[1]** Bermúdez-Otero (2012) The architecture of grammar and the division of labour in exponence. In *The morphology and phonology of exponence: The state of the art*. OUP. **[2]** Breiss (2024) When bases compete: A voting model of lexical conservatism. *Lg* 100: 308-358. **[3]** Chomsky & Halle (1968) *The sound pattern of English.* Harper & Row. **[4]** Halle & Vergnaud (1987) Stress and the cycle. *LI* 18. **[5]** Kiparsky (1982) Lexical Morphology and Phonology. In

Linguistics in the morning calm. Hanshin. **[6]** R Core Team (2023) R: A Language and Environment for Statistical Computing. Version 4.3.0. R Foundation for Statistical Computing. **[7]** Siegel (1974) *Topics in English morphology.* PhD diss, MIT. **[8]** Steriade (1997) Lexical conservatism. *Linguistics in the morning calm.*

Modeling segment strength in Q Theory: A case study of Ticuna

Authors

We define *phonological strength* as a cluster of properties that manifest across two domains: (i) metricality (e.g., stress, minimality), and (ii) segmenthood (e.g., asymmetric assimilation, resistance to neutralization). While mora count captures the former (Hayes 1995), it does not readily extend to the latter. We propose a two-tiered model of phonological strength integrating subsegments into the Prosodic Hierarchy (Nespor & Vogel 1983, Selkirk 1981), grounded in Q Theory's (Shih & Inkelas 2019) decomposition of segments (Q) into quantized units (q). A case study of /ɨ/ assimilation in Ticuna shows that a q-theoretic analysis outperforms a moraic account, supporting a division of labor in which (sub)segments account for fine-grained segmental patterns and are subordinate to moras, which capture metrical weight: μ >> Q >> q.

The Ticuna puzzle: Ticuna (Skilton 2023), an isolate spoken in Peru, has six vowels /i, e, ɨ, a, o, u/, each contrasting for nasality. All vowels except /ɨ/ also exhibit a glottalization contrast, and /ɨ/ alone assimilates to the quality and nasality of a preceding stressed non-low vowel (1). These properties indicate that /ɨ/ is phonologically weaker than the other vowels in the inventory.

(1) a. /tʃa³=na³=w**i**⁵¹=ʔ**ɨ**¹ra¹/ → [tʃa³na³wi⁵¹ʔi¹ra¹] 'I sort of cut it with a knife'
b. /na⁴=b**u**³¹=ʔ**ɨ**¹ra¹/ → [na⁴bu³ʔu¹ra¹] 's/he's sort of a child'
c. /ni⁴¹=**ẽ**¹=ʔ**ɨ**¹ra¹/ → [ni⁴¹ʔẽ¹ʔẽ¹ra¹] 'It [a wound] sort of heals'
d. /na⁴=ɲa³=**õ**²=ʔ**ɨ**¹ra¹/ → [na⁴ɲa³ʔõ²ʔõ¹ra¹] 's/he sort of believes it'

When /ɨ/ follows a diphthong (/ai, au/), the assimilation pattern exhibits three outcomes: most commonly, the diphthong splits into two syllables and /ɨ/ takes on the quality of the diphthong's non-low portion (e.g., /aV.ɨ/ → [a.V]); alternatively, /ɨ/ fails to assimilate entirely ([aV.ɨ]); or the diphthong remains while /ɨ/ assimilates to its non-low portion ([aV.V]). Based on text data and grammaticality judgments from three speakers, these outcomes are estimated to occur at 50%, 37.5%, and 12.5%, respectively—frequencies that form the empirical basis for our MaxEnt HG model (Goldwater & Johnson 2003, Hayes & Wilson 2008, George et al. 2006).

Shortcomings of a Moraic Analysis: First, we demonstrate that a moraic analysis of /ɨ/ assimilation in Ticuna is insufficient to capture the attested data. We propose moraic representations that assign different mora counts to the three vowel classes: /ɨ/ is monomoraic, other monophthongs /i, e, a, o, u/ are bimoraic, and diphthongs /ai, au/ are trimoraic. The moraic model performs well for assimilation of /ɨ/ to monophthongs and its non-assimilation to /a/, but falls short for assimilation to diphthongs: It incorrectly predicts that [aʔi] and [aiʔɨ] should surface with equal frequency, when in fact [aʔi] is more commonly observed.

	/aiʔɨ/	MAX-μ	*WEAK-[a]	CORR-VV	DEP-μ	IDENT-IO	ℋ	Obs.	Pred.
	weights	*16.40*	*15.34*	*8.38*	*1.25*	*0*			
a.	aʔi			-1		-1	*-8.38*	*0.5*	*0.437*
b.	aiʔɨ			-1			*-8.38*	*0.375*	*0.437*
c.	aiʔi			-1	-1	-1	*-9.63*	*0.125*	*0.122*
d.	aiʔa		-1	-1	-1	-1	*-24.97*	*0*	*0*
e.	aiʔai		-1		-2	-1	*-17.84*	*0*	*0*
f.	ɨʔɨ	-2				-1	*-32.80*	*0*	*0*

Tableau 1: Tableau for input /aiʔɨ/ sequence in Ticuna

Beyond its empirical shortcomings, the moraic analysis is also theoretically unmotivated. Moras are designed to model metrical structure, not patterns of segmental alternation. Accordingly,

mora count should be determined by weight-sensitive phenomena, rather than by the extent of featural specification or susceptibility to assimilation. In this case, there is no independent evidence that Ticuna diphthongs are trimoraic, nor that /ɨ/ differs in moraic weight from other monophthongs. Alternative approaches either require multiple disparate mechanisms or effectively reintroduce subsegmental structure—precisely what Q Theory's architecture provides.

Analysis: We propose a q-theoretic account of vowel strength directly encoding subsegmental structure. Vowels are represented on a monotonic four-point scale, where stronger vowels consist of more subsegments and weaker vowels of fewer: (v), (v v), (v v v), (v v v v). Phonological strength emerges from these representations: the more subsegments a vowel contains, the more faithfulness violations modifications incur, making it harder to alter. For example, assimilation of (vv) incurs two MAX-q violations, whereas assimilation of (vvv) incurs three. We represent Ticuna diphthongs with four subsegments (a a a i), monophthongs with three (e e e), and /ɨ/ with two (ɨ ɨ)—making /ɨ/ the preferred assimilation target. We employ three subsegmental constraint families in a MaxEnt model: (i) IDENT-IO-q; (ii) CORR-vv, CORR-v^2v^2, and CORR-(vv), which enforce agreement at different levels (subscript numbers index subsegmental corresponding pairs); and (iii) *WEAK-[a], which blocks assimilation to /a/. In this analysis, /ɨ/ straightforwardly assimilates to a preceding monophthong to satisfy CORR constraints, while assimilation to /a/ is blocked by the high-weighted *WEAK-[a] constraint penalizing weak (a a) vowels in the output. Diphthongs surface with three competing surface forms, and the model correctly predicts their observed frequencies with near-perfect accuracy by capturing how the subsegmental structure of diphthongs interacts with the correspondence and faithfulness constraints.

	/(a a a i) ʔ (ɨ ɨ)/	CORR-vv	IDENT-q	*WEAK-[a]	CORR-v^2v^2	CORR-(vv)	$\mathcal{H}$	Obs.	Pred.
	weights	*25.00*	*13.05*	*14.52*	*7.48*	*1.39*			
a.	$(a_1\ a_{126}\ a_{23}\ i_{34})ʔ(i_{45}\ i_{56})$	-1	-2		-1	-1	*-59.97*	*0.125*	*0.125*
b.	$(a_1\ a_{126}\ a_{23}\ i_{34})ʔ(ɨ_{45}\ ɨ_{56})$	-2			-1	-1	*-58.87*	*0.375*	*0.375*
c.	$(a_1\ a_{126}\ a_{23}\ i_{34})ʔ(a_{45}\ a_{56})$	-2	-2	-1		-1	*-92.01*	*0*	*0*
d.	$(a_1\ a_{126}\ a_{23})ʔ(i_{34}\ i_{456}\ i_5)$	-1	-2		-1		*-58.58*	*0.5*	*0.5*

Tableau 2: Tableau for input /(a a a i) ʔ (ɨ ɨ)/ sequence in Ticuna

Implications: Our proposed model presents a transparent hierarchy of strength through an interaction between subsegmental representations and constraint violations. The case study shows that Q Theory is better suited than Moraic Theory to modeling patterns of segmental strength. More broadly, we argue for a principled and complementary division of labour: Moraic Theory excels at weight phenomena like stress and tone, while Q Theory is better suited for handling segmental phenomena, including asymmetric assimilation, featural specification, and (sub)segmental structure. Augmenting the Prosodic Hierarchy to incorporate subsegmental structure below the mora, therefore, enables a more fine-grained and unified—yet still discrete and constrained—framework for characterizing phonological strength.

References: George, B., Hayes, B. & Wilson, C. 2006. Maxent Grammar Tool. ◆ **Goldwater, S. & Johnson, M.** 2003. Learning OT constraint rankings using a Maximum Entropy model. *Proc. of Variation within OT*. ◆ **Hayes, B.** 1995. Metrical stress theory: Principles and case studies. UCP. ◆ **Hayes, B. & Wilson, C.** 2008. A maximum entropy model of phonotactics and phonotactic learning, *LI 39*. ◆ **Nespor, M. & Vogel, I.** 1983. *Prosodic structure above the word*. ◆ **Selkirk, E.** 1981. On the nature of phonological representation ◆ **Shih, S. S. & Inkelas, S.** 2019. Autosegmental Aims in Surface-Optimizing Phonology. *LI 50:1*. ◆ **Skilton, A.** 2023. Tone, stress, and their interactions in Cushillococha Ticuna. *PD&A 5:5*.

Bora Nominalisation Across Strata

Andrés Napurí
New York University

This paper develops a Stratal Optimality Theory analysis (Bermúdez-Otero, 2017) of nominalisation in Bora (ISO 639-3: boa, Boran), arguing that floating low tones associated with category-changing morphology are cyclically evaluated at distinct grammatical strata. Bora exhibits a floating low tone associated with category change at the word level: verbal bases become nouns through tonal modification (e.g. (1) *májchò* [máxt͡ʃò] 'to eat' → *màjchó* [màxt͡ʃó] 'food') (Napurí & Panduro, 2018). The same low tone appears at the left edge of nominal subordinate clauses, lowering and lengthening the clause-initial pronoun (e.g. (2) /ú/ '2SG' → [ˈùː] '2SG\NOM'). I propose that both phenomena instantiate a single nominalising ◌̀ 'NOM' morpheme inserted at different structural heights—within the word at the lexical stratum and within a nominalising clausal head at the phrasal stratum.

(1) Word-level: RM, IDENT-μ, MAX-H >> IDENT-Tone

(01)	◌̀+/máxt͡ʃò/	RM	IDENT-μ	MAX-H	IDENT-Tone
☞	a) [màxt͡ʃó]				**
	b) [màxt͡ʃò]			*!	**
	c) [măːxt͡ʃò]		*!		*
	d) [mâːxt͡ʃó]		*!		**
	e) [máxt͡ʃò]	*!			

(2) *Nàníyò méé ùù méîyónè*
nà.ní.jò mɛ́ː ˈùː méj.-ì.-jó.-nɛ̀
uncle collect 2SG\NOM collect-FUT-FRU-CLF:thing
'The uncle collected what you should have collected'

Phrase-level: *CONTOUR, RM, PRWD-[μμ]$_σ$ >> IDENT-μ, MAX-H

(02)	/ú/+◌̀	*CONTOUR	RM	PRWD-[μμ]$_σ$	IDENT-μ	MAX-H
☞	a) [ùːː]				*	*
	b) [ù]			*!		*
	c) [ú]		*!	*		
	d) [ûː]	*!			*	
	e) [ŭː]	*!			*	
	f) [úː]	*!			*	

At the word level, nominalisation is evaluated in a grammar where the realization of the nominalising morpheme is prioritized while the prosodic structure of the lexical word remains fixed. The relevant ranking is RealiseMorpheme, IDENT-μ, MAX-H >> IDENT-Tone. Under this ranking, the floating low tone associated with nominalisation must surface. At the same time, the moraic structure of the lexical base is preserved and underlying tonal specifications must be retained. Because IDENT-μ dominates IDENT-Tone, the grammar does not permit vowel lengthening as a strategy for accommodating additional tonal material. Instead, the nominalising low tone is integrated through tonal reassociation within the existing prosodic structure. The result is systematic tonal modification in derived nouns while maintaining the moraic shape of the lexical base.

At the phrase level, nominal subordination is evaluated under a different grammar in which prosodic requirements associated with clausal structure become active. The ranking at this level is *CONTOUR, RM, PRWD-$[\mu\mu]_{\sigma}$ >> IDENT-μ, MAX-H. As at the lexical level, the floating low tone must surface. However, the phrasal grammar also requires the head of the subordinate clause to form a bimoraic prosodic word. Because PRWD-$[\mu\mu]_{\sigma}$ dominates IDENT-μ, lengthening becomes necessary to satisfy the prosodic requirements of the construction. At the same time, the constraint *CONTOUR prevents the realization of multiple tones on a single syllable, favouring a uniform low tonal melody across the lengthened vowel. Consequently, a short high-toned pronoun surfaces as long and low in nominal subordinate clauses.

The contrast between these two grammars reflects genuine stratal reranking. At the lexical level, preservation of moraic structure is prioritized, preventing lengthening and forcing the nominalising tone to be realized through tonal reassociation within the word. At the phrasal level, however, a prosodic requirement for bimoraicity outranks moraic faithfulness, allowing the grammar to satisfy the same tonal morpheme through lengthening and tonal overwriting. The same floating nominalising low tone therefore produces distinct phonological outcomes depending on the domain of evaluation. Bora nominalisation thus provides evidence that a single tonal exponent can underlie both derivational and clausal nominalisation while being interpreted by different constraint rankings across strata. This analysis supports a cyclic organization of phonological grammar in which prosodic and tonal constraints are reorganized between lexical and phrasal phonology, contributing new empirical evidence from Amazonian tonology to the study of Stratal Optimality Theory.

References

Bermúdez-Otero, R. (2017). Stratal Phonology. In S. J. Hannahs & A. Bosch (Eds.), *The Routledge Handbook of Phonological Theory*. Routledge. https://doi.org/10.4324/9781315675428

Napurí, A., & Panduro, W. (2018). Sobre partes del cuerpo bora: polisemia y derivación nominal con cambio tonal. *Línguas Indígenas Americanas*, *18*(2), 302-315. https://doi.org/10.20396/liames.v18i2.8651025

Why do marked words persist? Corpus and experimental evidence for competition between phonological and semantic well-formedness in word-formation and the lexicon

Po-Hsuan Huang (pohsuanh@sas.upenn.edu) & Canaan Breiss (cbreiss@uchicago.edu)

Introduction. From a probabilistic grammar perspective, grammatically marked forms can be underrepresented but nevertheless tolerated (Kempen & Harbusch, 2008; Martin, 2011; Breiss & Hayes, 2020, *i. a.*). This study examines why marked forms persist in the lexicon rather than disappearing over time (Martin, 2011). We focus on phonologically marked compounds in English, Finnish, and Hungarian and propose that *word form flexibility* (henceforth, flexibility), measured as the number of alternative forms compatible with the target semantics, competes with phonological well-formedness to determine what form enters the lexicon. We propose that marked forms survive because there are fewer viable semantic alternatives available for speakers to use. While intuitive, no study has systematically examined this claim. We present corpus studies and a nonce compound formation experiment. Both yield results in support of our hypothesis.

English, Finnish, and Hungarian corpus studies. We first examined whether attested phonologically marked compounds are more likely to have lower flexibility in English, Finnish, and Hungarian. The markedness constraints investigated were *GEMINATE in English (e.g., *bookkeeper*; Martin, 2011) and *vowel backness harmony* in Finnish (e.g., *kesäloma* "summer vacation"; Duncan, 2015) and Hungarian (e.g., *kőpor* "rock flour"; Siptár & Törkenczy, 2000). We collected two-member compounds in these languages from Wiktionary and published datasets (Gagné *et al.*, 2019; Günther & Marelli, 2019). First, we found compounds containing marked structures (geminates, disharmonic members) were underrepresented by a chi-squared test, suggesting they are indeed lexically disfavored (Figure 1). Second, each compound's flexibility was calculated as the product of the summed cosine similarities between each member and all the other words in a language-specific word2vec model. This metric served to indicate the semantic similarity of the compound across the lexicon in the language. In each language, a linear mixed effects model found that the probability of a compound being marked increased as its flexibility decreased (Figure 2), suggesting that flexibility likely modulates the extent of phonological optimization in the lexicon.

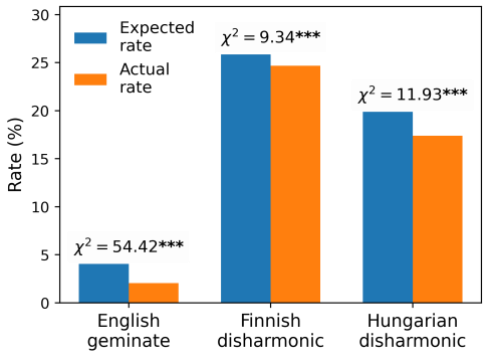
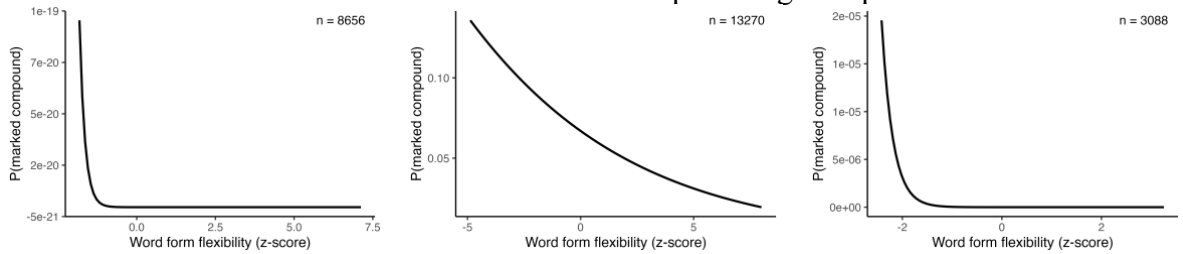


Figure 1. The underrepresentation of geminate and disharmonic compounds in the three languages.



(a) English (b) Finnish (c) Hungarian

Figure 2. Corpus study results: the model-derived interaction between a compound's word form flexibility and the markedness rate.

English nonce compound formation experiment. To gather complementary evidence for the real-time competition of phonological markedness and semantic flexibility in word-formation, we conducted two experiments examining English nonce compound formation, where participants were instructed to form nonce compounds given combinations of two noun concepts (cf. Figure 3). In order to estimate the relationship between the flexibility of a noun concept, vocabulary size, and education level, we first conducted a norming study. 80 American

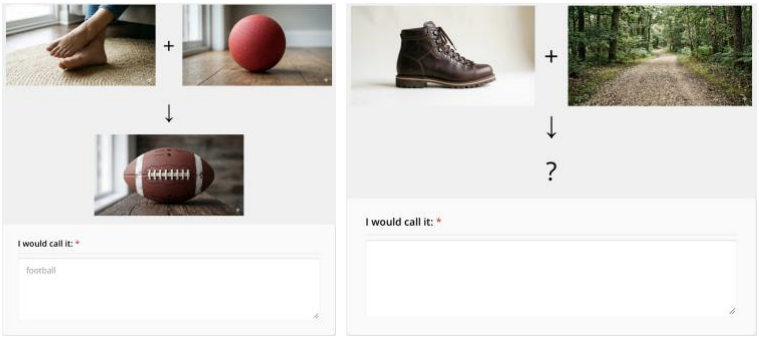


Figure 3. Example trials in the compound formation experiment. The trials prompt the participant to create a real compound (left) and a nonce compound (right).

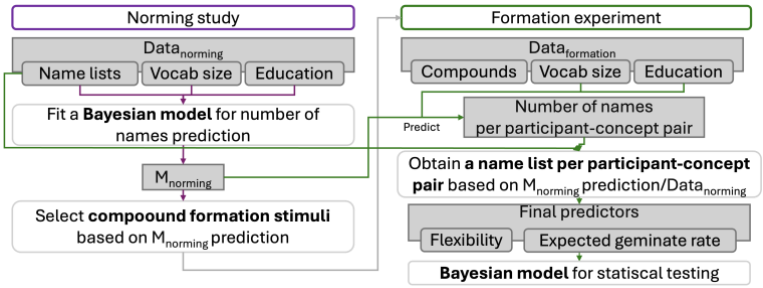


Figure 4. Experimental flow and data processing.

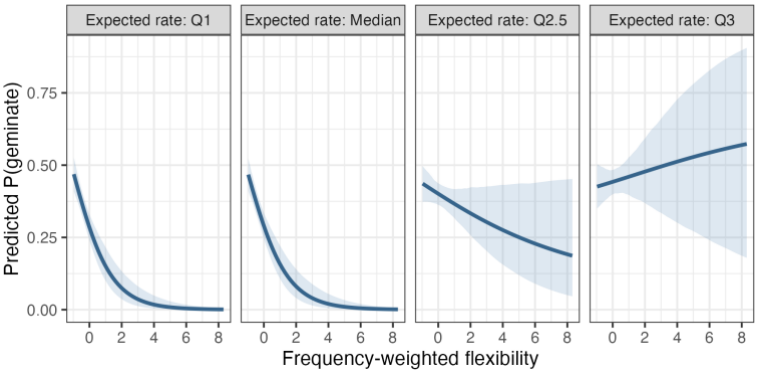


Figure 5. Compound formation experiment results: the model-derived interaction between a compound's flexibility and the markedness rate of the name offered by the participant.

English speakers offered as many names as possible for pictures of 80 noun concepts selected to vary in their number of word forms based on WordNet (Miller, 1995). Participants also provided demographic information and completed a *MINT sprint* vocabulary size assessment (Garcia & Gollan, 2022). A Bayesian model was fit to predict the number of names from concept identity, education level, and vocabulary size. Concepts were then classified as high- or low-flexibility and paired to maximize expected geminate rates in novel compounds. We obtained 20 high- and 20 low-flexibility concept pairs. In the formation experiment, a separate group of 86 American English speakers named these concept pairs, after being familiarized with the process with real compounds, such as *football*. Demographic and vocabulary size were also collected. To best estimate the specific names that were available for each participant when seeing a concept, for each participant-concept pair, we predicted a list of names based on the norming model and the concepts' name distributions in the norming study. Flexibility was then calculated as the number of names, weighted by their frequencies in the norming data. To account for the baseline geminate rate, the expected geminate rate for each participant-concept pair was also estimated. The results of a Bayesian linear mixed effects model (plotted in Figure 5) showed a significant negative correlation between flexibility and geminate rate. This suggests that when a novel concept pair has a lower flexibility, the participant is more likely to violate the phonological constraint in the name they offer. A competing effect from expected geminate rate also surfaced, where the flexibility effect weakened at extremely high expected geminate rates, suggesting a "ceiling."

Discussion Both the corpus and experimental results supported the hypothesis that flexibility competes with phonological well-formedness in marked forms. A compound is more likely to violate the markedness constraint when the language user has fewer alternatives to express the target meaning. This suggests that semantic flexibility is a moderating constraint on phonological avoidance of marked structures that shapes both static lexical distribution and online language use.

Biases do not help with the problem of maximum entropy's unrestrictiveness

Giorgio Magri (SFL, CNRS, and University of Paris 8)

OT or HG satisfy the Greenbergian **implicational universal** $(x, y) \rightarrow (\widehat{x}, \widehat{y})$ provided every OT or HG grammar that realizes the antecedent underlying form x as the surface form y, also realizes the consequent underlying form $\widehat{x}$ as the surface form $\widehat{y}$. Anttila & Magri (2018, 2025; AM) say that ME or NHG satisfy this universal $(x, y) \rightarrow (\widehat{x}, \widehat{y})$ provided the probability of realizing $\widehat{x}$ as $\widehat{y}$ in the consequent is always at least as large as the probability of realizing x as y in the antecedent. To illustrate, the universal (/cost us/, [cos'us]) → (/cost me/, [cos'me]) says that the probability of t-deletion is always at least as large before consonants as before vowels. AM show that probabilistic NHG and categorical HG share the same universals while ME misses scores of them. AM conclude that ME is a less restrictive probabilistic extension of categorical HG than NHG is.

Yet, AM's conclusion only holds when all non-negative weights contribute to the ME typology. Could we rescue ME by pruning the weights that yield ME counterexample grammars? Weight pruning can be hard or soft. Hard pruning restricts the ME typology by replacing the sheer assumption that weights are non-negative with some stronger assumption (say, weights are within certain values). Soft pruning sticks with the standard ME typology but nudges the learner away from certain regions of weight space through learning biases (Wilson 2006; White 2013, 2014; Culbertson, Smolensky & Wilson 2013). This abstract shows through three examples that weight pruning does not help, no matter whether it is hard or soft. AM's conclusion thus stands strong.

[1] Following Lombardi (1999), we consider the underlying forms /at/, /ad/, /asta/, /azta/, /asda/, and /azda/ with the candidates obtained by changing underlying obstruent voicing in all possible ways. The constraints are *[+voice], AGREEVOICE (that penalizes clusters of obstruents that disagree in voicing), $\text{IDENT}_{[\text{voice}]}$, and its restriction $\text{IDENT}^{\text{onset}}_{[\text{voice}]}$ to onset consonants. The first (second) consonant of each surface intervocalic cluster is syllabified as a coda (an onset).

OT and HG satisfy 132 implicational universals. We sort them into three classes. Class I consists of 96 universals that have an impossible antecedent mapping (no OT/HG grammar contains it) and thus hold in OT/HG simply because the material implication that defines Greenbergian implicational universals has a false antecedent. To illustrate, the universal (/asta/, [azda]) → (/azda/, [asta]) says that devoicing is better than voicing and holds in OT/HG simply because the antecedent mapping is impossible (no OT/HG grammar voices the voiceless cluster of /asta/). Class II consists of the 22 universals that compare each possible antecedent mapping with the consequent mappings (/atta/, [atta]) or (/at/, [at]), capturing the generalization that the latter two mappings are necessary (every OT/HG grammar contains them because they violate no constraint). Class III consists of the remaining 14 universals plotted in figure 1 that compare antecedent and consequent mappings that are both contingent (some but not all OT/HG grammars contain them). These 14 universals are all sensible: the five blue arrows govern devoicing when voicing agreement is not at issue; the four red arrows govern voicing preservation; the three orange arrows govern devoicing in the face of disagreement; finally, the two violet arrows govern regressive assimilation.

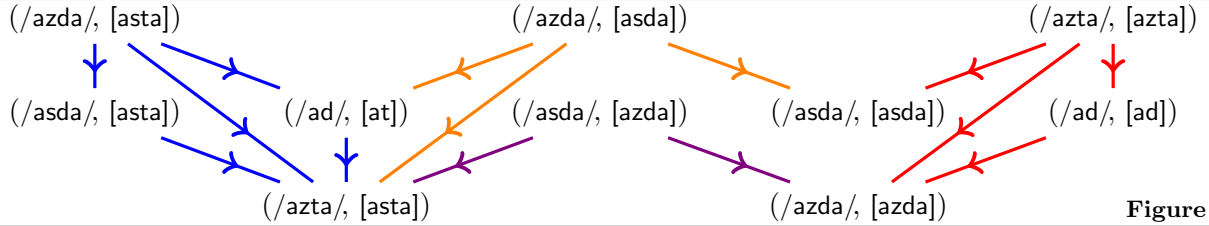


Figure 1

NHG satisfies all 132 of these OT/HG universals, because AM show that HG and NHG always share the same universals. ME instead flouts 45 of these universals. We thus ask: which non-negative weights need to be pruned to arrive at a subset of ME grammars that satisfy all 132 OT/HG/NHG universals? To answer, we used a Python script to check all weights for the four constraints between 0 and 10 at increments of 0.5. The script reveals that the ME grammars corresponding to the resulting 160,000 weight vectors *all* contradict at least one OT/HG/NHG universal. We have no reasons to expect a different result if the increment 0.5 were decreased or the upper bound 10 increased. We conclude that weight pruning does not help ME to satisfy all OT/HG/NHG universals because we would need to prune all weights leaving us with no ME grammars! As a result, the biases added to ME in the recent literature cannot rescue ME from AM's objection that ME is un unrestrictive probabilistic extension of categorical HG.

[2] Following Prince & Smolensky (2004), we consider the nine forms V, VC, VCC, CV, CVC, CVCC, CCV, CCVC, CCVCC, each a candidate of the other. The constraints are NoCoda, Onset, NoCompOnset, NoCompCoda, Dep, and Max. Onset and coda consonants are never in correspondence (to illustrate, (/VC/, [CV]) violates both Dep and Max). OT satisfies 1,635 universals. Since these universals are too many to plot, we content ourselves with plotting in figure 2 the universals that compare faithful mappings, that are interpreted by AM as a markedness hierarchy. Of these 1,635 OT universals, 1,258 fail in ME. In particular, all the markedness universals listed in figure 2 fail in ME. Again, we ask: which non-negative weights need to be pruned to arrive at a subset of ME grammars that satisfy all 1,635 OT universals? To answer this question, we used again our python script (with the increment set equal to the larger value 2, to keep the search feasible despite the larger number of constraints). The script reveals that among the resulting 1,953,125 weight vectors, only the one with all weights equal to zero yields a ME grammar that satisfies all OT universals (because this ME grammar assigns to all mappings the same probability 1/9). We conclude again that biases cannot rescue ME from AM's objection of unrestrictiveness.

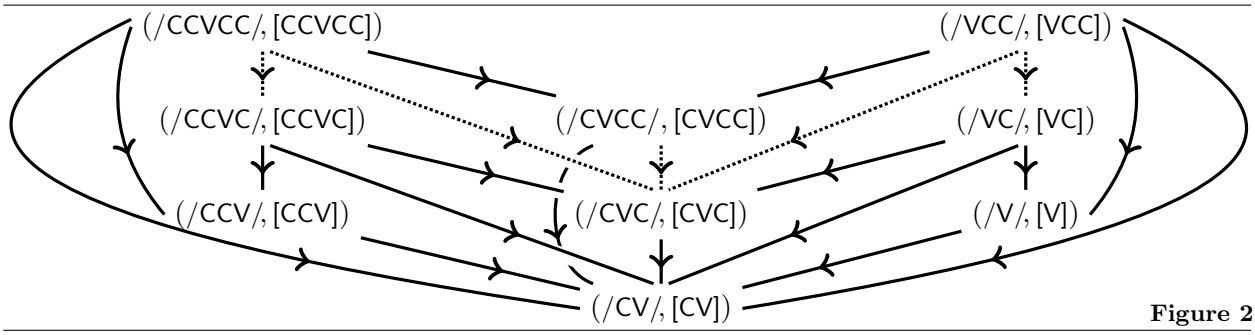


Figure 2

Of the 1,635 OT universals, 18 fail in HG and therefore also in NHG (because HG and NHG always share the same universals). These 18 OT universals that fail in HG all involve complex codas and fail because of a gang effect between NoCoda and NoCompCoda. To illustrate, among the OT universals among faithful mappings in figure 2, the five universals that involve complex codas and are represented by dotted arrows fail in HG. To see why, we focus on the OT universal (/CVCC/, [CVCC]) → (/CVC/, [CVC]). It fails in HG because the weight conditions in figure 3 are feasible and any HG grammar corresponding to such weights contains the antecedent mapping (the weight conditions (a) and (b) ensure that [CVCC] beats [CVC] and [CV], respectively) but not the consequent mapping (the weight condition (c) ensures that [CV] beats [CVC] as the surface realization of /CVC/).

(a)	$w_{\text{NoCompCoda}} < w_{\text{Max}}$
(b)	$w_{\text{NoCompCoda}} + w_{\text{NoCoda}} < 2w_{\text{Max}}$
(c)	$w_{\text{Max}} < w_{\text{NoCoda}}$

Figure 3

Yet, the HG grammars that contradict these 18 OT universals can all be pruned by requiring NoCompCoda to have a larger weight than NoCoda. This is a simple weight pruning condition that can be easily implemented as a learning bias. This weight condition can be made sense of as follows. In OT, the generalization that complex codas are worse than simple codas can be formalized by having the two corresponding constraints in a stringency relation *á la* Prince (1997): NoCoda is violated whenever NoCompCoda is. In HG instead, stringency does not suffice and we need the constraint NoCompCoda that penalizes the worse violation to have a larger weight.

[3] Following Hayes (2004), we consider the underlying forms /ta/, /da/, /tʰa/, /dʰa/; /at/, /ad/, /atʰ/,/adʰ/; and /ata/, /ada/, /atʰa/, /adʰa/ with the candidates obtained by changing underlying voicing and aspiration in all possible ways. The constraints are *[+voiced], *[+spread glottis], *[+voiced, +spread], *VTV (against intervocalic voiceless stops) together with $\text{Ident}_{[\text{voice}]}$, $\text{Ident}_{[\text{spread}]}$ and their restrictions $\text{Ident}^{\text{onset}}_{[\text{voice}]}$, $\text{Ident}^{\text{onset}}_{[\text{spread}]}$ to onsets. OT and HG satisfy 661 implicational universals. NHG satisfies them as well (because HG and NHG always share the same universals). ME instead flouts 406 of these universals. Which non-negative weights need to be pruned to arrive at a subset of ME grammars that satisfy all 661 OT/HG/NHG universals? To answer this question, we used again our python script (with the larger increment 2). The script reveals that, out of the 390,625 weight vectors considered, only a mere 449 weight vectors yield ME grammars that comply with *all* OT/HG/NHG universals. We conclude once again that counterexample weights are pervasively spread, so that biases are unlikely to rescue ME from AM's objection of unrestrictiveness.

Touch is enhanced and encoded in the phonology of signing minds

Shengyun Gu, Occidental College

The visual-manual modality is commonly regarded as the primary medium through which signers perceive and produce linguistic expressions. In this study, I investigate another modality, i.e., the tactile modality, with respect to its relevance to sign language phonology. Going beyond the existing findings in the cognitive and socio-emotional domains, such as enhanced tactile sensitivity in congenitally deaf individuals and culturally valued tactile communications like tapping in the deaf communities [1-2], I ask whether the tactile feature, namely contact, is enhanced and encoded in the phonology of American Sign Language (ASL). The contact feature refers to either the two hands of the signer ever touching (i.e., hand contact) or the hand(s) touching any location on the signer's body (i.e., body contact) during sign production. Contact is considered an important property of the location parameter in ASL phonology [3-4]. A search over the 2723 ASL signs in the lexical database ASL-LEX 2.0 [5] revealed that 62.6% of ASL signs involve contact (also noted in [4]). This suggests that the contact feature is rich in the ASL lexicon. In addition to this kind of language-internal evidence, I present evidence from an ASL pseudosign repetition task with both ASL-signing children (and ASL-signing adults) and sign-naïve adults. I show that compared with non-signing minds, touch is enhanced and encoded in the phonology of signing minds.

Stimuli. The ASL pseudosign repetition task assesses phonological skills and teases away the effect of vocabulary knowledge [6]. The stimuli are 39 nonsense words that are phonotactically permissible in ASL. Among these 39 pseudosigns, 23 involve body contact (face, ear, chest, arm), 8 involve hand contact, 4 involve both body and hand contact, and 4 are produced in the air without any contact. **Participants**. The study involved two groups. The sign-naïve adult group consisted of 31 hearing college students who had never been exposed to any sign language. The ASL-signing children group comprised 22 children (ages: 4;0–8;10, mean age: 6;04, 9 deaf and 13 hearing) who were exposed to ASL from birth by their deaf signing parents and acquired ASL as their first language (L1). By the time of writing, I had also recruited two deaf adult ASL signers as controls. **Procedure**. Each participant watched a video of a deaf signer producing the 39 pseudosigns and was asked to copy each pseudosign as similarly as possible. All test items were shown only once. **Results**. I coded the accuracy in reproducing contact in each participant. To analyze the association between contact accuracy (overall accuracy and accuracy for each contact type) and group, I performed generalized linear mixed-effects models with participant and item levels as random effects. In **Figure 1**, ASL-signing children (mean = 97.43%, SD = 3.12%) outperformed sign-naïve adults (mean = 81.23%, SD = 15.7%) in overall contact accuracy ($p<0.001$). Crucially, signing children consistently reproduced contact, whereas sign-naïve adults showed considerable variability. A further analysis of the subtypes of contact (i.e., body contact and hand contact) yielded the results shown in **Figure 2**. Sign-naïve adults were substantially less accurate than ASL-signing children in both body contact ($p<0.001$) and hand contact ($p = 0.001$). In addition, body contact was significantly less accurate than hand contact in sign-naïve adults ($p<0.001$), but no evidence of a significant difference in the accuracy of body contact and hand contact was found in ASL-signing children ($p = 0.660$). I also coded the unexpected body contact and hand contact added by the participants themselves, rather than copied from the test items. As indicated in **Table 1**, sign-naïve adults produced more unexpected hand contact than unexpected body contact. The ASL-acquiring children showed an opposite pattern: Unexpected body contact occurred more than twice that of unexpected hand contact in ASL-acquiring children, and overall, unexpected body contact occurred more dominantly among ASL-acquiring children than among sign-naïve adults. The deaf adult signers achieved 100% accuracy in contact and did not add any unwanted contact.

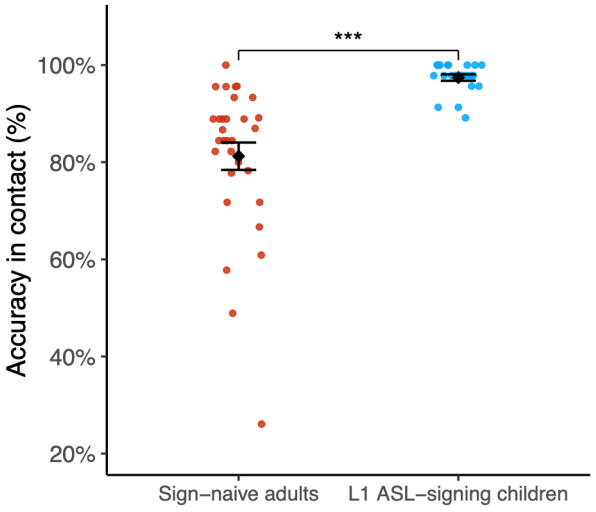


Figure 1 Overall accuracies of contact by group

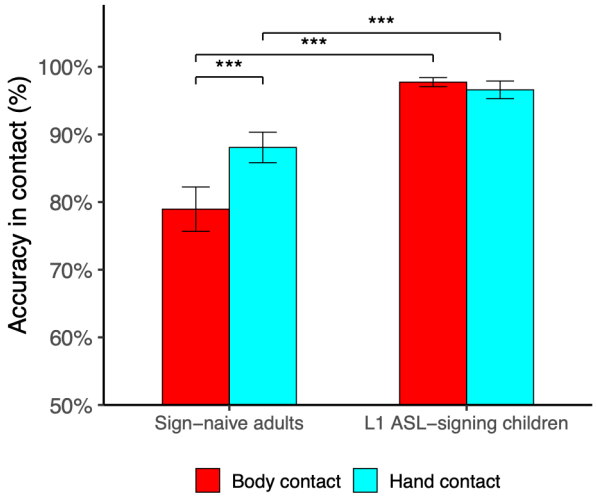


Figure 2 Body and hand contact accuracies by group

Table 1 Total number and ratio of unexpected body/hand contact productions by group

Group	Unexpected body contact	Unexpected hand contact
Sign-naïve adults (N = 31)	6 (0.19)	9 (0.29)
ASL-signing children (N = 22)	12 (0.55)	5 (0.23)

Discussion. The ceiling performance of the ASL-signing children on the contact features in the pseudosign repetition task shows that contact was controlled by L1 ASL-signing children by about age 6;04. Another ASL study revealed that contact can be mastered around age 2;05 [7]. In contrast, sign-naïve people performed worse and showed substantial individual variation in reproducing contact. A closer inspection of the contact errors showed a further division. Children largely missed contact production because they deleted the entire motion to/from the target location. In contrast, sign-naïve adults mostly successfully reproduced other location features such as side and height, but they failed to make physical contact with the target location. The performance in body contact may also be influenced by extralinguistic factors, such as touch being a cultural taboo for some adults, which can lead to reluctance to self-touch. However, the study instructions were to copy as similarly as possible. The group difference in hand contact accuracies further suggests that contact is systematically encoded as a phonological feature for ASL-exposed children, but not so for non-signing people. In addition, the unexpectedly added body contact seen in sign-acquiring children but not in deaf adult signers indicates that, on the one hand, even though contact is acquired early, the adult-like way of controlling contact takes longer to fully develop; on the other hand, tactile feedback plays an important role in the development of child sign phonology.

Conclusion. Based on these findings, I argue that touch is encoded and enhanced in the phonology of signing minds, echoing the proposal that contact matters in sign language phonology. This study also calls for considering touch as contributing to the multimodality of human language capacity.

References. [1] Villwock, A., & Grin, K. (2022). Somatosensory processing in deaf and deafblind individuals: How does the brain adapt as a function of sensory and linguistic experience? A critical review. [2] Napoli, D. J. (2014). A magic touch: Deaf Gain and the benefits of tactile sensation. [3] Wilbur, R. B. (2010). The role of contact in the phonology of ASL. [4] Corina, D. P. (2021). Articulatory postures and forward models in American Sign Language: Linguistic and neuroscience evidence. [5] Sehyr, Z. S. et al. (2021). The ASL-LEX 2.0 Project: A database of lexical and phonological properties for 2,723 signs in American Sign Language. [6] Quadros, R. M. et al. (2015). Methods in bimodal bilingualism research: Experimental studies. [7] Gu et al. (2025). Early development in ASL phonology: A longitudinal study of deaf children with hearing parents.

Phonetic consequences of gradient phonological activation

Manasvi Chaturvedi, Michael C. Stern, & Jason A. Shaw

Recent research in phonology has posited gradient activation, such that a phonological representation, instead of being fully active or inactive, can have a continuous range of activation values. Gradient activation has shed new light on phonological patterning, including vowel harmony, morphophonology, prosodification, and stress (e.g., Lionnet, 2017; McCollum, 2019; Smolensky & Goldrick, 2016; Smolensky et al., 2019; Rosen, 2022; Hsu, 2019). To date, there is no general theory of the relationship between gradient phonological activation and the phonetic signal. This gap has led to inconsistencies in how phonetic detail has been interpreted in terms of phonological representation; e.g., some authors have assumed that reduced phonological activation causes shorter duration (Goldrick & Chu, 2014) or phonetic reduction (McCollum, 2019). Using phonetic measurements to infer phonological structure, including gradient activation, requires an explicit theory of the relation between these domains. Here, we present such a theory and enumerate its predictions about the phonetic consequences of gradient phonological activation.

We implement the theory as a neural process model in the framework of Dynamic Field Theory (DFT: Schöner et al., 2016). In the model, discrete phonological categories are represented by neural nodes, continuous phonetic dimensions are represented by neural fields, and mappings between them are represented by synaptic coupling. For the sake of demonstration, we consider two arbitrary phonological features ([+α] and [–α]) which map to an arbitrary phonetic dimension (see Figures 1-3). When a phonological node becomes active (corresponding to the speaker's intention to produce that feature), the node sends input to the phonetic field, increasing activation in a specific region of the field. Node-to-field input has a Gaussian shape. Here, [+α] is coupled to the high end of the phonetic field, and [–α] is coupled to the low end of the field. The amplitude of input (shown with a blue-green color gradient) corresponds to the activation of the node. When field activation crosses the threshold for interaction (shown with a grey dashed horizontal line on the lefthand plots), a stabilized peak of activation forms. The position of the peak in the field determines the phonetic target for articulation (see Stern (2025) for a review of DFT models of speech and language). Below, we simulate gradient phonological activation under three conditions in order to demonstrate the predicted phonetic consequences.

Figure 1 demonstrates the result of varying the activation of [+α] (left) on the phonetic target (right) when there is no co-active phonological representation (i.e., [–α] has an activation of 0). Under these conditions, gradient activation has no influence on the phonetic target. The reason is that the location of phonological input (from [+α]) to the phonetic dimension, dictated by synaptic coupling, is fixed and independent of the activation of [+α]. Thus, in the general case, when there is only one phonological input to a phonetic dimension, gradient activation is independent from the phonetic target.

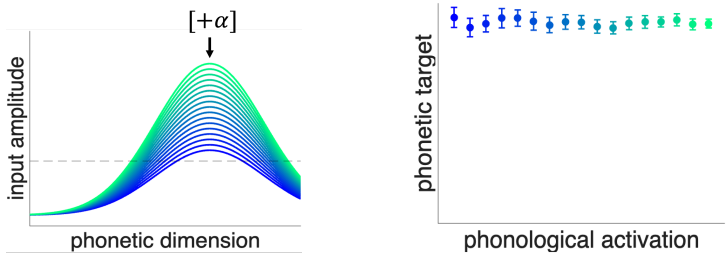


Figure 1. With a single active phonological representation (left), continuous variation in phonological activation does not cause systematic variation in the phonetic target (right).

Figure 2 demonstrates the phonetic consequences of [+α]'s activation in the presence of a co-active representation, [–α]. Crucially, these representations overlap in their mapping to the

phonetic field. With increasing activation of [+α], the field stabilizes at values that are closer to the center of the [+α] node's coupling to the field. This is because the overlapping representation [–α] exerts an influence on where the field stabilizes, and this influence is greater when [+α] has lower activation. Thus, gradient activation of [+α] results in gradient variation of the phonetic target in the presence of a co-active representation [–α] which overlaps in its mapping to the phonetic field.

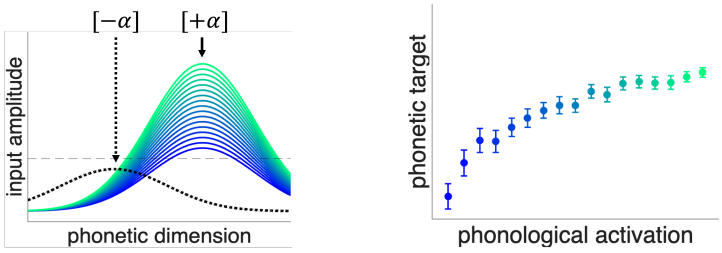


Figure 2. When multiple overlapping phonological representations are simultaneously active (left), decreased phonological activation leads to hypoarticulated phonetic targets (right).

Results of the third simulation are in **Figure 3**. In this case, there are two non-overlapping inputs to the field. Under these conditions, gradient activation of [+α] has a non-linear influence on the phonetic target. When the activation of [+α] is weaker than the activation of [–α], the peak forms at the center of the [–α] input; when [+α] is stronger than [–α], the peak forms at the center of the [+α] input. The field dynamics of DFT impose a categorical selection, rather than a linear sum of inputs, allowing gradient phonological activation to yield a categorical shift in phonetic target.

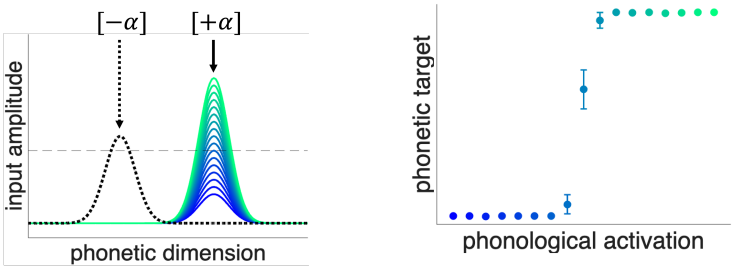


Figure 3. When co-active phonological representations do not overlap in phonetic space (left), continuous variation in phonological activation can lead to a categorical change in the phonetic target (right).

The model simulations demonstrate that, in general, gradience in phonological activation is only predicted to cause systematic malleability of the phonetic signal when there are multiple co-active representations coupled to the same phonetic field. Moreover, the details of phonetic malleability are predicted to depend on the mappings from each phonological representation to the relevant phonetic dimensions. When the mappings overlap, a gradient shift in phonetic target is predicted; when the mappings do not overlap, only a categorical change in phonetic target is possible. Thus, on this theory, phonetic gradience, as observed by, e.g., McCollum (2019), cannot be explained by gradient phonology alone—it requires blending of co-active phonological representations.

References: Goldrick, M., & Chu, K. (2014). Gradient co-activation and speech error articulation: Comment on Pouplier and Goldstein (2010). *Language, Cognition and Neuroscience*, *29*(4), 452-458. | Hsu, B. (2019). Exceptional prosodification effects revisited in Gradient Harmonic Grammar. *Phonology*, *36*(2), 225-263. | Lionnet, F. (2017). A theory of subfeatural representations: The case of rounding harmony in Laal. *Phonology*, *34*(3), 523-564. | McCollum, A. (2018). Gradient morphophonology: Evidence from Uyghur vowel harmony. In *Proceedings of the Annual Meetings on Phonology*. | Rosen, E. R. (2022). Gradient Symbolic Computation (Smolensky & Goldrick, 2016) does derive A'ingae stress patterns. In *Proceedings of the Annual Meetings on Phonology*. | Schöner, G. (2016). *Dynamic thinking: A primer on dynamic field theory*. Oxford University Press. | Smolensky, P., & Goldrick, M. (2016). Gradient symbolic representations in grammar: The case of French liaison. *Rutgers Optimality Archive*, *1552*, 1-37. | Smolensky, P., Rosen, E., & Goldrick, M. (2019). Learning a gradient grammar of French liaison. In *Proceedings of*

the Annual Meetings on Phonology. | Stern, M. C. (2025). Dynamic Field Theory unifies discrete and continuous aspects of linguistic representations. *Proceedings of the Linguistic Society of America*, *10*(1).

Vowel ablaut alternations are floating features: a case against readjustment

Nonconcatenative morphology like root vowel alternations has been analyzed using readjustment rules (Chomsky & Halle 1968, Halle & Marantz 1993), floating features (Wolf 2005, Zoll 2001 a.o.), and special devices like antifaithfulness (Alderete 2001). I present an argument against against readjustment and in favor of the insertion of floating feature allomorphs that are docked on the root in the phonological component.

Lithuanian verbal roots show vowel alternations (henceforth ablaut) in several grammatical contexts. Causativization is one, as in (1), where /ɪ/ of the inchoative 'rise' has a lower alternant /e/ in the causative 'raise'. Another ablaut context is the present/past tense. In (2), [gɪntɪ] 'to defend' has a long vowel /iː/ in the past tense (I treat the vowel length constrast in Lithuanian as [±long] (Kushnir 2019), see Kenstowicz 1971 for an alternative).

(1) a. kɪl-tɪ
rise-INF
'to rise'

b. kel-tɪ
rise.CAUS-INF
'to raise'

(2) a. gɪn-tɪ
defend-INF
'to defend'

b. giːn-eː
defend-3.PAST
'(they) defended'

Two concomitant vowel alternations are thus possible: causative-related, and tense-related. The causative [keltɪ] 'raise', (1b), shows ablaut in the past tense; the inchoative [kɪltɪ] 'rise' shows ablaut in the present tense (blue cells in the table below). Maintaining that underlyingly both 'rise' and 'raise' involve the same root (inchoatives and causatives are derivationally related), a single root $\sqrt{kl}$ is then associated with four different forms: /kɪl/, /kiːl/, /kel/ and /keːl/.

	'rise'	'raise'
Pres	kiːl	kel
Past	kɪl	keːl
Fut	kɪl	kel

Moreover, present and past tense ablaut both involve the same device: vowel length. Assuming /kel/ is derived from /kɪl/ (causative from inchoative, but directionality is not crucial here), and long vowels from short vowels, descriptively there are at least two phonological ablaut devices available: *lengthening* and *lowering* of the vowel.

Generalization: causativization is associated with lowering, and tense features are associated with lengthening (for some set of roots). The syntax-phonology mapping must mediate this indexation between CAUS/PRES/PAST and lowering/lengthening.

I argue that the only way to account for the pattern elegantly is by floating features that dock on the root. Therefore, in the syntax-phonology mapping, ablaut is not due to (readjustment) rules that target the root (below in (7)), but due to (Vocabulary Insertion) rules like in (3), that target functional items like CAUS, which can be spelled out as floating features.

(3) *Vocabulary Insertion of floating features.*

a. CAUS ⟷ [–high] / __ $\{\sqrt{kl}, \ldots\}$

b. PRES ⟷ [+long] / __ $\{\sqrt{kl}, \ldots\}$

c. PAST ⟷ [+long] / __ $\{\{\sqrt{kl}, \text{CAUS}\}, \sqrt{\ }, \ldots\}$

VI insertion rules can be conditioned by a span – a sequence of contiguous heads (Merchant 2015). (3c) states that the feature PAST is spelled out with the phonological feature [+long] in a set of contexts, including the context of a span of the root $\sqrt{kl}$ and the causativizer. Crucially, (3c) allows us to capture the generalization that the past tense feature is associated with [+long], which is impossible in a readjustment analysis (below in (7)-(8)).

(4) kaːr-tɪ - koːr-eː
hang-INF hang-3.PAST

(5) PAST ⟷ { [+long], [–low] } / __ $\{\{\sqrt{kl}, \text{CAUS}\}, \sqrt{gn}, \sqrt{kr} \ldots\}$

	kaːr + [+long]$_2$ / [+low, +long]$_1$ [–low]$_3$	NoVacDoc	MaxFlt
▶ a.	koːr / [+long$_1$, –low$_3$]		*
b.	kaːr / [+low$_1$, +long$_2$]	*!	*

Past tense ablaut has several features, including [+long] and [–low], depending on the root (cf. (4) 'hang'). I unify these types with a single VI rule (5) that hosts all floating features, the relevant one docking (see tableau) based on phonological

generalizations (machinery inspired by Wolf 2005, where vacuous docking is penalized). Such unification is impossible using readjustment.

Vowel alternations in the past tense occur not only in roots, but also in suffixes, cf. (6), where the diphthong /uo/ alternates with /av/, regardless if it is in a root or in a suffix. Morphologically simplex and complex verbs gain a unified account assuming floating features: a floating feature docks on an adjacent syllable; whereas readjustment (root-specific) cannot capture that the same process applies to both roots and suffixes.

(6) a. duo-tɪ - dav-e:
give-INF give-3.PAST
'to give' '(they) gave'

b. dain-uo-tɪ - dain-av-o
sing-VBLZ-INF sing-VBLZ-3.PAST
'to sing' '(they) sang'

While VI rules, in (3), treat PAST as a target for insertion, readjustment rules treat PAST as a context, and this runs into several problems. Readjustment statements in (7) target the rime of a set of roots, and they capture the generalization: for some roots Y, the rime becomes long in the past tense; *mutatis mutandis* for other ablaut. This overgenerates, since 'rise' and 'raise' involve the same root: $\sqrt{kl}$ is in all three sets of roots Y, which generates the distribution in Table 1.

(7) *Readjustment rules.*

a. $\text{Rime}(Y_1) \rightarrow$ [+long] / ___ PAST, where $Y_1 = \{\sqrt{}\dots\}$

b. $\text{Rime}(Y_2) \rightarrow$ [+long] / ___ PRES, where $Y_2 = \{\sqrt{}\dots\}$

c. $\text{Rime}(Y_3) \rightarrow$ [–high] / ___ CAUS, where $Y_3 = \{\sqrt{}\dots\}$

Table 1: Overgenerated paradigms.

Pres	ki:l	ke:l
Past	ki:l	ke:l
Fut	kɪl	kel

We can enrich the context, as in (8). There are three problems with this adjustment: (i) it still overgenerates for 'raise': causative needs to block PRES-ablaut, but the context of (8b) is satisfied by the causative verb. An adjacency fix is controversial for readjustment (Embick 2010), but not for VI in (3b). (ii) This obscures the generalization that lengthening is associated with tense, and has nothing to do with the causative; in (8a), PAST and CAUS have equal trigger status for [+long]. (iii) We still need the past-only rule (7a) for many other roots that are not causative but still undergo past-tense ablaut (cf. 'defend' in (2)). The statement in (8a) obscures the generalization that these roots are in the same class. NB: a spanning 'fix' will not work for the readjustment analysis because it is impossible to state a disjoint environment in the same way as for VI: Y by hypothesis only includes roots. Hence, duplicate rules (7a) and (8a) are needed.

(8) *Readjustment: second pass.*

a. $\text{Rime}(Y_1) \rightarrow$ [+long] / ___ PAST, CAUS, where $Y_1 = \{\sqrt{},\dots\}$

b. $\text{Rime}(Y_2) \rightarrow$ [+long] / ___ PRES, where $Y_2 = \{\sqrt{},\dots\}$

c. $\text{Rime}(Y_3) \rightarrow$ [–high] / ___ CAUS, where $Y_3 = \{\sqrt{},\dots\}$

Conclusion: Lithuanian provides a ripe testing ground for theories of so-called process morphology: large classes of roots undergo various types of ablaut. I argue that a readjustment mechanism cannot handle cases where a single root has four alternants, like $\sqrt{kl}$ 'rise' /kɪl/, /ki:l/, /kel/, /ke:l/. I show that the syntax-phonology mapping fares better with span-conditioned Vocabulary Insertion, which involves insertion of floating phonological features; this output is shipped off to phonology where the features dock, producing ablaut. This reduces the amount of morphological mechanisms needed by eliminating readjustment from the inventory. I also show that a floating feature account elegantly handles ablaut that shows up on suffixes, and can unify several phonological types of ablaut under the same account.

Sel. refs.: Halle, M. & Marantz, A. 1993. Distributed morphology and the pieces of inflection. Kushnir, Y. 2019. Prosodic patterns in Lithuanian morphology. Merchant, J. 2015. How much context is enough? Two cases of span-conditioned stem allomorphy. Wolf, M. 2005. For an autosegmental theory of mutation.

Loss of tone contour preserves a register distinction in Changde Mandarin reduplication

Zhenrui Zhang[1], Jason Shaw[2]

[1]University of Chinese Academy of Social Sciences; [2]Yale University

Overview: Several theoretical proposals for tone representation draw distinctions between tone and register features (e.g., Bao 1999; Duanmu 1994; Hyman, 1993; Lionnet 2022; Snider 1998; Yip 1989). We present novel evidence for this distinction from reduplication in Changde Mandarin, showing that reduplicated syllables lose tonal contour while retaining a register contrast.

Language background: Changde Mandarin (CM) is spoken by about 2,120,000 people in Changde, Hunan Province, China. CM has four contrastive lexical tones; like Standard Mandarin (SM), Tone1 (T1) is high and Tone3 (T3) is low. The other two tones differ from SM, both Tone2 (T2) and Tone4 (T4) are rising, with T2 a low to mid rise and T4 a low to high rise (cf. SM, where T4 is falling); see also Figure 1 (left). CM also differs from SM in the so-called 'neutral tone'. In SM, the 'neutral tone' refers to toneless syllables whose pitch value is dictated by tonal environment (e.g., Chao 1968), possibly due to interpolation (e.g., Chen & Xu 2006). In CM, neutral tones appear to have some lexical pitch specifications (Zhang & Hu 2020). One domain for 'neutral tones' in both varieties is the second syllable of lexical reduplicants; for example: 哥哥 [kuo¹ kuo⁰] '*brother*', 爸爸 [pa² pa⁰] '*dad*', 姐姐 [tɕiɛ³ tɕiɛ⁰] '*sister*', and 舅舅 [tɕiu⁴ tɕiu⁰] '*uncle*'. In these words, the first syllable has a full tone while the second (reduplicant) syllable shares segments with the first but differs in tone. We examined the phonetic details of the reduplicant to assess the nature of its tone specification.

Experimental methods: The materials were drawn from a production experiment with 9 native speakers of Changde Mandarin (2 male, 7 female). We report two conditions. In the full-tone baseline condition, citation tones were analyzed in word-initial position (e.g., 咖 [kʰa¹] in 新咖啡杯 'new coffee cup', 篮 [lan²] in 篮球鞋 'basketball shoes', 演 [ian³] in 演讲 'speech', 塑 [su⁴] in 塑料 'plastic'). In the reduplication condition, the second syllable of reduplicated kinship terms was analyzed (see items in Table 1). All targets were embedded in one of two carrier sentences: (1) tse⁴ ɕie¹ tsʰai² liau⁴ xo² __ iu³ kuan¹ 'These materials are related to __'; (2) uaŋ² tan¹ tsuo² tʰiɛn¹ kaŋ³ ti⁰ sɿ⁴ __ 'Wang Dan talked about __ yesterday'. For word-initial syllables (full tone), 12-24 tokens per item were included in the analysis; for reduplication, there were 36 tokens per item. Acoustic analysis was performed in Praat. The rhyme of each syllable was taken as the measurement interval where 10 equidistant points were extracted to obtain f_0 values, which were converted to semitones and then z-scored within each speaker. Generalized Additive Mixed Models (GAMMs) were then fitted to the pitch contours (z-score), with tone category as the main predictor, including smooths for measurement point, syllable duration, and their interaction, as well as random smooths for speaker and trajectory.

Full Tone (word-initial)	T1 - 咖 [kʰa] (24), 冰 [piŋ] (12), 新 [ɕin] (12), 喝 [xuo] (12)	T2 - 篮 [lan] (23), 球 [tɕʰiu] (12), 白 [pe] (24), 尝 [tʂʰaŋ] (12)	T3- **演** [ian] (24), **好** [xau] (24), **写** [ɕie] (12)	T4 – 塑 [su] (24), 烂 [lan] (24), 办 [pan] (12)
Reduplicants	T1- 哥 [kuo] (36)	T2- 爸 [pa] (36)	T3- **姐** [tɕiɛ] (36)	T4- **舅** [tɕiu] (36)

Table 1. Token counts for experimental item (FT word-initial vs. NT kinship reduplicant)

Results: The left panel of Figure 1 shows the full tones. As expected, there are distinctive contours for the four lexical tones, which replicates previous literature on CM: T1 is high; T3 is low; T2/T4 are rising. The same pattern is found for the first syllable of lexical reduplicants (not shown here for space reasons). All pairwise difference smooths were significant, confirming that every tone pair is distinct. In contrast, the right panel shows that, in reduplication, some of the tone contrasts are lost. T1 and T4 merge into one high, falling contour while T2 and T3 merge into a low, rising contour, with difference smooths confirming no significant differences within these pairs.

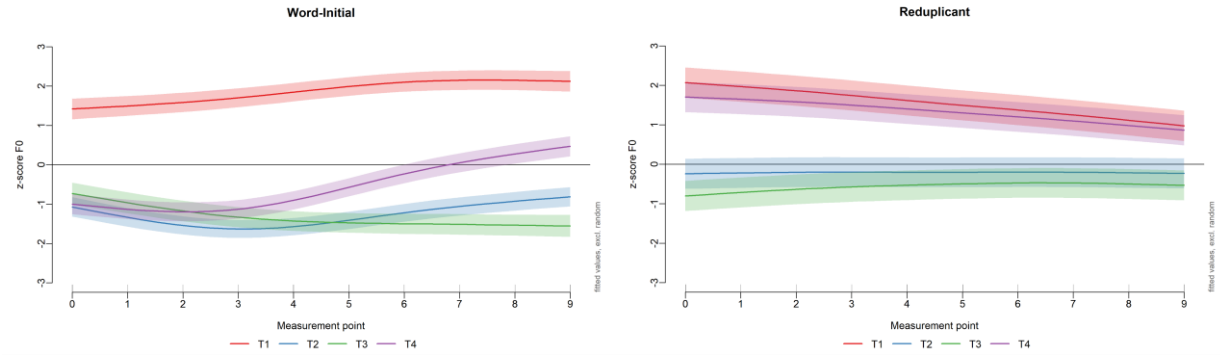


Figure 1. f_0 contours of word-initial full tones (left panel) and second-syllable neutral tones in reduplicated kinship terms (right panel) with normalized time.

Discussion: In lexical reduplicants, the four-way lexical tone distinction of CM is reduced to a two-way distinction. Moreover, the contour of the rising tones is lost. The pitch contours of the reduplicant involve high tones gradually lowering (T1, T4) and low tones gradually rising (T2, T3). We interpret the pattern in reduplicants as a distinction in high vs. low register, with the change in pitch over time reflecting a gradual progression towards a neutral position. The pattern is consistent with a representation of full tones in CM as a composite of register features and contour tones which can be independently targeted by phonological operations. We propose an autosegmental representation with a register tier distinct from a contour tier. Reduplication preserves register but loses tonal contour, through delinking. This pattern differs from SM, where the reduplicant takes on the neutral tone, i.e., no tone contrast is preserved. Thus, on this analysis, the two Mandarin varieties differ in whether de-linking in the second syllable of lexical reduplicants impacts the entire tonal complex (SM) or just the contour node (CM).

References:

Bao, Z. (1999). *The structure of tone*. Oxford University Press. **Chao, Y. R.** (1968). *A grammar of spoken Chinese*. University of California Press. **Chen, Y. & Xu, Y. (2006).** Production of Weak Elements in Speech - Evidence from F0 Patterns of Neutral Tone in Standard Chinese. *Phonetica, 63*, 47-75. **Duanmu, S.** (1994). Against contour tone units. *Linguistic Inquiry, 25*(4), 555-608. **Hyman, L. M. (1993)**. Register tones and tonal geometry. In K. Snider & H. van der Hulst (Eds.), *The phonology of tone: The representation of tonal register*, 75-108. Berlin, Boston: De Gruyter Mouton. **Lionnet, F. (2022).** Tone and downstep in Paicî (Oceanic, New Caledonia). *Phonological Data and Analysis*, *4*(1), 1-47. **Snider, K. L. (1998).** Phonetic realisation of downstep in Bimoba. *Phonology*, *15*(1), 77-101. **Yip, M.** (1989). Contour tones. *Phonology, 6*(1), 149-174. **Zhang, Z. & Hu, F.** (2020). Neutral Tone in Changde Mandarin. *Proc. Interspeech 2020,* 1923-1927.

Relative frequency cues facilitate phonological alternation learning

Darby Grachek | University of Southern California

Introduction: Many phonological alternations involve exceptions. This is well-exemplified by English stress shift, which predicts primary stress shifts rightward in a stem when a suffix is added (e.g. párent → parént-al, nárrate → narrát-ion); however, many suffixes do not trigger the expected shift (e.g. párent → párent-hood, nárrate → nárrat-ive). Prominent analyses of these patterns have relied on *a priori* categorization of suffixes, with speakers associating each category with either rule-application or non-application (Chomsky and Halle, 1968; Kiparsky 1982; 2000). However, alternative accounts opposing pre-categorization of suffixes have suggested looking to *lexical decomposition,* a parsing process occurring during lexical access, to explain the association of different phonological outcomes with different suffixes. This study uses artificial language learning (ALL) to investigate whether lexical decomposition (as indicated by *relative frequency,* a comparison of the frequency of stems and suffixed forms), aids in the learning of phonological exceptions. Preliminary results show that lexical decomposition facilitates learning of exceptional suffixed forms. This suggests a correspondence between lexical parsing and phonological derivations, which supports reference to psycholinguistic factors when considering theoretical approaches to phonological exceptionality.

Background: Though abstract, lexical decomposition can be reliably predicted by distributional factors like relative frequency, (Hay 2001, Hay and Baayen 2002, 2003). Specifically, a *decomposed parse*, where a suffix is recognized separately from its stem (e.g. /move/ + /-ment/), is associated with forms that have higher stem frequency relative to suffixed-form frequency. Conversely, a *whole-word parsed* form, where stems and suffixes are recognized together (e.g. /judgement/), is associated with forms having higher frequency suffixed-forms compared to stem-forms. Recent work on correspondence between lexical decomposition and phonological exceptionality (Dabouis 2019, Zuraw et al. 2021, Grachek 2025) finds a correlation between non-application affixes and highly decomposable words. In other words, stems and affixes which are parsed separately in the lexicon also pattern separately in phonological alternations (i.e. are exceptions), suggesting speakers utilize perceptual cues from decomposition when learning exception-filled alternations. This may explain how speakers come to associate different phonological patterns with particular suffixes. However, more behavioral studies are necessary to rule out the possibility that this correlation is due to diachronic coincidence of the English lexicon, rather than correspondence between decomposition and phonological exceptions.

(1) Rule-application:		**(stem-final voicing)**
a. porap	→	porabifa, *porapifa
b. talik	→	taligifa, *talikifa
(2) Non-application:		**(no stem-final voicing)**
a. porap	→	porapiso, *porabiso
b. talik	→	talikiso, *taligiso

Experiment: An ALL task was used to test the hypothesis that lexical decomposition (indicated by token relative frequency) is *actively* used to enhance the learning of a phonological alternation involving exceptional suffixes. In the initial learning phase, nonce stems (e.g. [porap]) and suffixes ([-ifa/-iso]) were involved in a stem-final voicing alternation which applied inconsistently across stem-suffix boundaries. Participants saw a pattern in which one nonce suffix triggered the alternation (1), and the other nonce suffix did not (2). Additionally, the task contained an artificial relative frequency distribution which encouraged decomposition for one suffix over another by manipulating stem and suffixed form frequencies. Suffix type and decomposition were then crossed to generate 3 different conditions: 1.) **a congruent condition** containing the same correlation as previous studies: The non-application suffix is associated with a decomposed parse while the rule-

application suffix is associated with a whole-word parse, 2.) **an incongruent condition** containing the opposite correlation: The non-application suffix is associated with a whole-word parse while the rule-application suffix is associated with a decomposed parse, and 3.) **a baseline condition** where the effect of decomposition is neutralized. After the learning phase, a subsequent test phase evaluated how accurately participants generalized the alternation to new words. If participants are reliant on a correspondence between cues to 'separateness' in lexical decomposition and phonological patterning, accuracy is predicted to be highest in the congruent condition, and lowest in the incongruent condition.

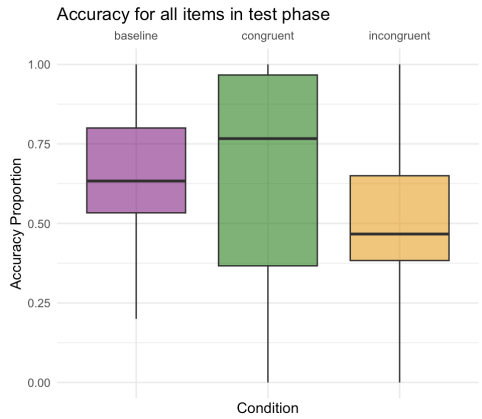


Figure 1: Participant accuracy.

Results: Preliminary analyses were performed on 26 pilot participants (full task is estimated to have 60 participants). Figure 1 shows the proportion of accurate responses in the test phase, and a mixed effects logistic regression which predicted participant accuracy based on condition (with random intercepts of participant and word) showed that the incongruent condition has numerically lower accuracy compared to the congruent condition but did not reach significance ($\beta = -0.73$, $p = 0.07$). This shows some limited support for the hypothesis: The congruent condition has numerically higher accuracy rates compared to the incongruent condition. Figure 2 breaks down participant accuracy between **rule-application suffixes** and **non-application suffixes**. In the baseline and incongruent conditions, rule-application suffixes have higher accuracy than non-application suffixes. The congruent condition, however, shows the opposite pattern: Non-application suffixes have a higher accuracy rate while rule-application suffixes are similar to other conditions, indicating that accuracy for the non-application suffix pattern improves when the suffix is highly decomposable.

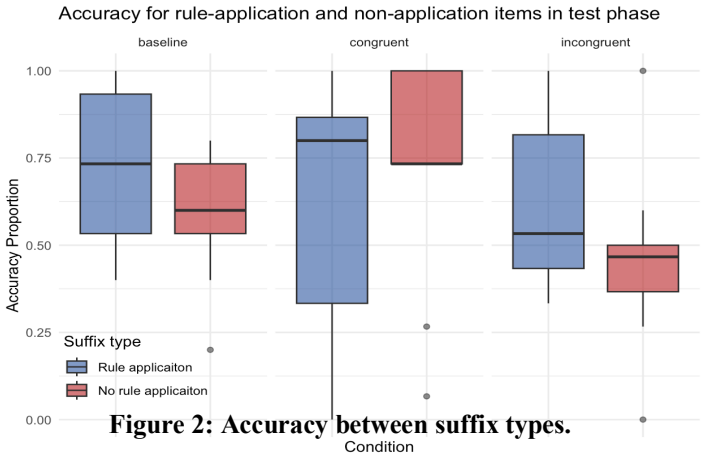


Figure 2: Accuracy between suffix types.

Discussion: The results of this study show limited evidence that a correlation in non-application suffixes and highly decomposable suffixes facilitates the learning of non-application suffixes without being detrimental to the learning of rule-application suffixes. This provides a crucial piece of evidence that supports previous work showing distributional facts about the lexicon may aid in the learning phonological alternations involving exceptions. It also provides an explanation for how speakers associate different phonological behavior with particular suffixes, as opposed to pre-categorization of suffixes. This preliminary result suggests that perceptual cues traded between phonological and lexical domains are utilized by speakers when forming grammatical generalizations in the face of phonological exceptionality.

Select References: Dabouis, Q. (2019). When accent preservation leads to clash. *English Language and Linguistics*, 23 (2):363–404; Grachek, D. 2025. Connection Between Lexical Processing and Phonological Regularity: A Comparison Between English and Turkish. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 47; Hay, J. and Baayen, H. (2003). Phonotactics, parsing and productivity. *Rivista di Linguistica*, 15.1:99–130.

A Case for Rendaku with Boolean Monadic Recursive Schemes

Daniel Yeung
University of Illinois Urbana-Champaign

Background There is a known opaque interaction in Japanese between the three processes of Rendaku, Lyman's Law, and Velar Nasalization (Kawahara, 2016). Rendaku is a process in which initial obstruents of second members of compounds become voiced. This process is blocked when there is a voiced obstruent present at any other position within that member (Lyman's Law). However, Rendaku is also blocked by a surface velar nasal [ŋ] which is not a voiced obstruent. It is such opaque interactions as this that Optimality Theory struggles to represent in its classic implementation (OT: Prince & Smolensky, 1993/2004). As a result of this opacity, Rendaku has been a common subject of focus in theoretical phonology. For example, Ito and Mester (2003) provide an account using Stratal OT, which requires the addition of serialism into Classic OT. We posit the use of Boolean Monadic Recursive Schemes to represent this interaction in a parallel manner (BMRS: Bhaskar et al., 2020; Chandlee & Jardine, 2021; Bhaskar et al., 2023). We show that BMRS properly represents these opaque processes, while being computationally more restricted than OT in general as well as maintaining parallelism, contra Stratal OT. Furthermore, we show that these long distance blocking patterns in BMRS require covert spreading functions to implement.

Examples we provide general examples below, illustrating Rendaku, Lyman's Law, and their interaction with the dialectal velar nasalization process. In this latter case it is usually said that underlying /g/ becomes [ŋ] in certain lexical items. If this process occurs after Rendaku in a serial analysis, the blocking behavior of [ŋ] is straightforward.

ooi	tako	oo-dako	Rendaku
big	*octopus*	*big octopus*	
akai	kabu	aka-kabu	Blocked by [b]
red	*radish*	*red radish*	(Lyman's Law)
saka	toge	saka-toŋe	Blocked by [ŋ]
reverse	*thorn*	*reverse thorn*	(Opaque Lyman's Law)

BMRS Boolean Monadic Recursive Schemes are a computational framework that defines phonological input-output mappings. BMRS programs are made up of boolean functions with IF THEN ELSE syntax and, when evaluated to True for a function *f(x)*, indicate that *x* has property *f* on the output. These are computationally restricted, and maximally compute rational functions (Bhaskar et al., 2023) which are (sub)regular, therefore falling within the hypothesized bounds of phonological knowledge (Heinz, 2018). Classic OT, on the other hand, has beyond regular expressive power (Gerdemann & Hulden 2012).

Analysis We posit two functions to represent the processes of Rendaku, Lyman's Law, and Velar Nasalization. *VdObR(x)* looks to the right of an input position to determine whether or not there is a voiced obstruent to its right. Mechanically, this is done by using the successor function (*s(x)*) to see if the following position contains a voiced obstruent. If not, the function is recursively called with the current successor as its input. This continues until either finding a voiced obstruent and returning True ($\top$) or defaulting to False ($\bot$) when reaching final position. *ϕvoice(x)* is the main function that determines output [voice] valuation. This first checks if the domain element being considered is the initial element of the second member of a compound (*E2I(x)*). If it is not, then the voicing valuation is faithful to the input value. If it is, then voicing is dependent on whether or not there is a voiced obstruent to the right, which is determined using

VdObR(x): = If VdOb(s(x)) then ⊤ *Else If final(x) then* ⊥ *Else VdObR(s(x))*	*ϕvoice(x):= If E2I(x) then if VdObR(x)* *Then* ⊥ *else* ⊤ *else voice(x)*

Input	s	a	k	a	t	o	g	e
Position	1	2	3	4	5	6	7	8
E2I	⊥	⊥	⊥	⊥	⊤	⊥	⊥	⊥
VdObR	⊤	⊤	⊤	⊤	⊤	⊤	⊤	⊥
ϕŋ(x)	⊥	⊥	⊥	⊥	⊥	⊥	⊤	⊥
ϕvoice(x)	⊥	⊤	⊥	⊤	⊥	⊤	⊤	⊤
Output	s	a	k	a	t	o	ŋ	e

Input	o	o	i	t	a	k	o
Position	1	2	3	4	5	6	7
E2I	⊥	⊥	⊥	⊤	⊥	⊥	⊥
VdObR	⊥	⊥	⊥	⊥	⊥	⊥	⊥
ϕŋ(x)	⊥	⊥	⊥	⊥	⊥	⊥	⊥
ϕvoice(x)	⊤	⊤	⊤	⊥	⊤	⊤	⊤
Output	o	o		d	a	k	o

the VdObR function. If there is one, Lyman's Law applies, and voicing is blocked (the function returns False). If there is not, then Rendaku applies and voicing occurs. The opacity due to [ŋ] is also captured by this function, because BMRS computation is ultimately *input* oriented.

Left: two evaluation tables. Leftmost represents the full opaque process with input form /saka-toge/, where we assume a function *ϕŋ(x)*. This function represents velar nasalization, which must apply after the check for a voiced obstruent to the right, but before we write the output, [saka-toŋe]. The rightmost table represents a standard rendaku process, with /ooi-tako /→ [oo-dako]. Although it has not been explicitly noted in prior literature, BMRS requires covert transmission of information in order to perform such “checking” actions for long distance blocking processes. Our function *VdObR(x)* demonstrates this by only returning a result upon either reaching the final position or a voiced obstruent, yet it still checks each other adjacent environment without writing a result. This process is comparable to the operations that occur in Search and Change phonology (Dabbous et al, 2024). E2I acts like a S&C initiator, but the search process performed by *VdObR(x)* is a composition of recursive adjacent checks, therefore BMRS is able to capture unbounded search processes without treating adjacency nor long distance as unique cases. They are both inherently captured within the recursive process.
Conclusion Previous constraint based approaches to opaque data require Stratal OT, introducing serialism into OT (Bakovic, 2013). This is known to loosen the desired restrictedness of Classic OT (Ito and Mester, 2003). Our BMRS account avoids this, as it does not require the separate application of their various functions on different levels such as the lexical, postlexical, and such. We have demonstrated that BMRS are able to present a working account for Japanese Rendaku and the opaque interaction it and Lyman’s Law has with Velar Nasalization. In summary, BMRS has lower generative power than Classic OT and remains parallel in contrast to Stratal OT. Because of this, we argue that BMRS are a compelling alternative framework through which we can explore long distance opaque interactions more broadly.

Baković, E. (2013). Blocking and complementarity in phonological theory. University of Toronto Press. • Bhaskar, S., Chandlee, J., & Jardine, A. (2023). Rational functions via recursive schemes. arXiv preprint arXiv:2302.03074. • Bhaskar, S., Chandlee, J., Jardine, A., & Oakden, C. (2020). Boolean monadic recursive schemes as a logical characterization of the subsequential functions. In International conference on language and automata theory and applications (pp. 157-169). Cham: Springer International Publishing. • Chandlee, J., & Jardine, A. (2021). Computational universals in linguistic theory: Using recursive programs for phonological analysis. Language 97(3), 485-519. • Dabbous, R., Leduc, M., Reiss, C., & Shen, D. T. C. (2024). Locality is epiphenomenal: Adjacency is opaqueness. In Proceedings of the 2024 annual conference of the Canadian Linguistic Association. • Heinz, J. (2018). The computational nature of phonological generalizations. Phonological typology, phonetics and phonology, 23, 126-195. • Ito, J., & Mester, A. (2003). Lexical and postlexical phonology in Optimality Theory: evidence from Japanese. Linguistische Berichte. Sonderheft, (11), 183-207. • Itô, J., & Mester, A. (2017). The phonological

lexicon. The handbook of Japanese linguistics, 62-100. • Kawahara, S. (2016). Phonological opacity in Japanese. Ms. Keio University, 1-21. • McCarthy, J. J. (1998). Sympathy and phonological opacity. Phonology, 16(3), 331–399. • Prince, A., & Smolensky, P. (1993/2004). Optimality Theory: Constraint Interaction in Generative Grammar (Technical Report no. RuCCS-TR-2). New Brunswick, NJ: Rutgers University Center for Cognitive Science.

Prosodically-conditioned contour/register alternations in Northern Toussian

Anthony Struthers-Young and Nicholas Rolle
Princeton University

This paper analyzes the variable realization of verbal grammatical tone in Northern Toussian within an indirect-reference approach to syntax-prosody interactions (Selkirk 2011), based on data collected through *in situ* fieldwork (Struthers-Young *forthcoming*).

Puzzle. In Northern Toussian – a Niger-Congo language of Burkina Faso with a H/M/L tone contrast – when a transitive verb appears with an object, it surfaces with its lexical tones only (1a). In contrast, both intransitive verbs (1b) and transitive verbs without an overt object (1c) bear an additional grammatical L tone. This grammatical tone is absent in nominalizations, where only lexical tone surfaces (e.g. ɟá-nɔ̄ watch-NOM 'watching').

(1) a. ádámá sú ɟá
Adama father watch
'Adama watched father'

b. ádámá **kɛ̌y** (</kéy/)
Adama go.**L**
'Adama went'

c. ádámá **ɟǎ** (</ɟá/)
Adama watch.**L**
'Adama watched'

Furthermore, Northern Toussian has SAuxOVX word order, where Aux contains tense, aspect, mood, and polarity (TAMP) items, some adverbs, and auxiliary verbs such as /tó/ 'again' and /pwó/ 'come'. The first auxiliary in a sequence bears a grammatical L (2) realized as a rising contour (as in 1). However, on all subsequent verbs the L is realized as downstep (cf. with objects, where there is no realization – 1a).

(2) a. ádámá **tǒ** ˈkéy
Adama again.**L** go.**L**
'Adama went again'

b. ádámá **tǒ** ˈpwó ˈɟá
Adama again.**L** come.**L** watch.**L**
'Adama came to watch again'

The realization of the L depends on the lexical tone of the verb, shown in the table below. While the grammatical L affects /H/, /HL/, and /HM/ in the 'contouring' context, in the 'downstepping' context only /H/ is affected. Note that /M/ and /L/ are never affected.

	/H/ (e.g. ɟá 'watch')	/HL/ (ɟâ 'search')	/HM/ (kó 'walk')	/M/ (ɟɛ̄ 'sweep')	/L/) (fàn 'mix')
(V :	[**LH**] (e.g. **ɟǎ**)	[**L**] (**ɟà**)	[**LHM**] (**kǒ¯**)	[M] (ɟɛ̄)	[L] (fàn)
Aux (V :	[ˈ**H**] (e.g. ˈ**ɟá**)	[HL] (ɟâ)	[HM] (kó)	[M] (ɟɛ̄)	[L] (fàn)

In addition to auxiliary verbs, inflectional particles can appear in the Aux position, in (3). Most of these particles condition the downstepping form of the verb, e.g. modals (/à/ COND), polarity (/kə́/ NEG), and some adverbials (/yē/ 'truly'). However, a small number of particles condition the contouring form, e.g. adverbials (/fáná/ 'also', /kwàn/ 'anyways') and past (/á/ PST). This latter group (particle type 1) are also positioned external to the former (particle type 2), but semantically neither type forms a natural class. Note that unlike auxiliaries, particles never bear the grammatical L (cf. 2).

(3) a. dā kə́ ˈkéy
Daou NEG go.**L**
'Daou didn't go'

b. dā á **kɛ̌y**
Daou PST go.**L**
'Daou had gone'

In sum, the puzzle of this grammatical L is accounting for its form (contouring vs. downstepping, or nothing with /L/ and /M/), its distribution (in isolation versus with

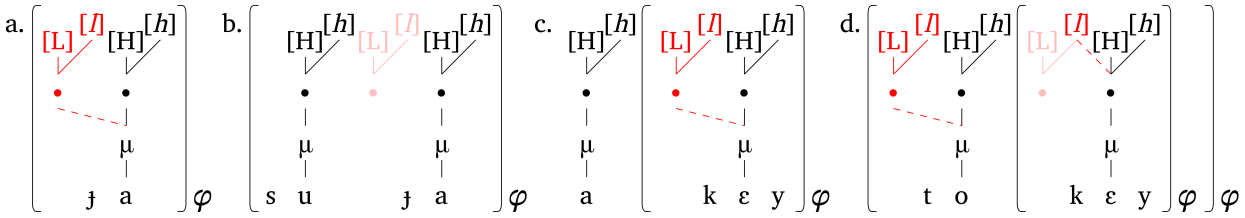
objects; with auxiliaries or particles), and ultimately its function and meaning.

Analysis. We adopt Register Tier Theory (Snider 2020) deconstructing tones into tonal and register features: /H/ is a feature bundle [H,*h*] in a high register, while /M/ and /L/ are bundles [H,*l*] and [L,*l*] respectively, in a low register (contours constitute multiple bundles). Given that the grammatical L only docks to verbs and auxiliaries, we analyze it as a floating bundle [L,*l*] which expones a verbalizing head, i.e. [*v*] ↔ [L,*l*]. This exponent, however, only surfaces at the left edge of a phonological phrase (φ).

In (4), we schematize the contexts of (1–3) in terms of phrasehood. At the syntax-prosody interface, verb phrases are uniformly mapped to a φ, prosodifying the object and verb together (this grouping is independently supported by a separate downstepping process conditioned by M tone at the right edge of a φ – Struthers-Young 2025). Furthermore, type 1 particles (e.g. á PST from 3b) do not form a phonological phrase with the verb. As stated, these particles appear external within the 'middle field' and have greater flexibility in word order than other particles, suggesting less tight prosodic coupling. Lastly, auxiliary verbs (e.g. tó 'again' from 2) and type 2 particles (e.g. kə́ NEG from 3a) form a recursive phonological phrase with the verb phrase (Bennett 2018, Ito & Mester 2021).

(4) a. (V)φ b. (O V)φ c. Part$_1$ (V)φ d. (Aux/P$_2$ (V)φ)φ

We diagram these constituents in the figure below, where we include the grammatical L, i.e. the floating [L,*l*] bundle before the verb in red (note that d. below only models the auxiliary tó – a type 2 particle would have identical prosodic structure but without the initial L). In this figure, only verbs at the left edge of a maximal φ (the outermost one) can host a rising contour, i.e. examples a., c., and the outer φ of d. (a non-recursive φ as in a. and c. is incidentally a maximal φ). Rising contours often correlate with positions of prominence cross-linguistically (Gordon 2001, Zhang 2013), which we interpret maximal φ edges to be (rises are marked in Northern Toussian, and rarely underlying).



While rises are restricted to maximal φ edges, downstep can appear at any φ edge. Building on Struthers-Young (2025), we interpret downstep as the spreading of a low register feature [*l*]. Structures where both a [*l*] and [*h*] register feature associate to a single tonal root node (•) are restricted to φ edges, as in d. This structure (manifesting as downstep) is categorically banned within a phrase in the language, reflected by the absence of a grammatical L effect in b. (i.e. (O V)φ). Moreover, at this stage in the derivation, a [*l*] register can only spread to structures which do not already contain [*l*]. The only potential host of downstep is thus a H verb bearing [H,*h*] – verbs with other tones (HL, HM, M, and L) all contains a [*l*] feature.

This case study, which investigates and analyzes the prosodic system of a minority African language, is relevant to several important issues and debates at the syntax-phonology interface. Our model supports indirect-reference approaches to this interface where cross-module interactions are mediated through a separate structure, the prosodic

hierarchy (Bennett & Elfner 2019). No direct reference to syntax is warranted. Furthermore, our model also supports the need for nested prosodic domains (i.e. prosodic recursion), as well as reference to (at least) maximal and non-maximal prosodic domains.

Eliminating a length-nasality perception bias as an explanation of contrast typology

Explaining the typological frequency of phonetically-motivated phonological patterns is an active research area. Cross-linguistically, nasal contrasts occur more commonly on low vowels than high, e.g. Amuzgo contrasts [a] and [ã], but has no nasal counterpart to [i] (Kingston, 2007; Longacre 1966). Since corresponding differences exist in both the production and perception of vowel nasality, there are multiple candidate causes of the typological asymmetry (e.g. Henderson, 1984; Kunay et al. 2022; Whalen & Beddor, 1989; Hajek 1997). We provide new evidence that disfavours one of these candidate causes, namely elevated perceptibility of phonetic nasality on long vowels. We identified a novel prediction of this hypothesized cause and tested it in a vowel discrimination experiment, where we found that greater vowel duration does not make an ambiguous degree of nasality perceptually closer to a clear degree of nasality. This finding does not support enhanced nasality perception in long vowels as a cause of the typological asymmetry.

Background. Low vowels tend to be longer than high vowels (Lindblom and Rapp 1973; Peterson & Lehiste 1960). Previous experiments suggest nasality is more perceptible on longer vowels (Whalen & Beddor 1989; Hajek & Watson 1998). Hajek (1997) proposes that this length-dependent difference in nasality perceptibility causes the elevated frequency of nasal distinctions on low vowels (henceforth the "length perception bias hypothesis"). However, little is known about the sound change mechanisms that must be involved for this to be correct. Contrastive vowel nasalization has been assumed to develop first from nasal vowel allophony (*a → a and *an → ãn) and then consonant loss (*a → a and *ãn → ã) (Hajek, 1997; cf. Beddor, 2009). For the creation of nasal contrasts to be more common in low (longer) than high (shorter) vowels, those changes would need to occur with greater probability in low vowels. However, it remains unclear (1) how this hypothesized phonetic cause would result in the required difference in probability at the level of individual learners, and (2) what the necessary input distribution properties are for learners to innovate the new phonological categories.

Approach & results. We implemented a Dirichlet process Mixture of Gaussians (MOG) model of category learning and sound change (cf. Kirby, 2013; Gubian et al., 2023). A MOG learner takes as input a set of uncategorized vowel tokens (Fig. 1) and searches for the set of categories that maximizes the likelihood of that data. The number of categories is inferred from the data. In a single generation of language transmission, the child MOG's input consists of the parent MOG's productions: noisy samples from the parent's categories. Nasal allophony arises in this model when vowels that neighbor different consonants split into two categories, a nasal and an oral allophone (Fig. 1). Because of noise in its input, the child can learn categories that are different from its parent's. The less the overlap between vowel distributions, the more likely the next generation is to learn them as separate categories. If it is true that a real-world phonetic effect causes more long nasal contrasts, then a perceptual bias shifting the distributions in the model's learning data should make MOG more likely to split long vowels into oral and nasal categories. Thus, the "length-perception bias hypothesis" predicts not only that long vowels

should overall sound more nasal, but — crucially — that there should also be a bigger difference in the distributions of perceived nasality between nasal and oral consonant contexts, a criterion that has not been directly tested by previous phonetic experiments. We conducted a discrimination experiment manipulating the duration of vowel nasality, consonant context (CVC or CVN), and vowel length with 62 adult English speakers and 288 four-interval forced choice trials per participant (cf. Zellou 2017). To compare the degree of overlap in perceptual distributions with only a limited set of stimuli, we included a vowel stimulus with an ambiguous (very small) duration of vowel nasality that would likely fall in the region of overlap and used discrimination ability to estimate its perceptual distance from unambiguously oral and nasal vowels (Fig. 2). Participants were excluded if they did not choose correctly on 60% of the control trials, resulting in 37 usable participants. Linear regression model estimation reveals that vowel length did **not** result in ambiguous tokens being perceived as significantly more nasal preceding a nasal consonant, contra the "length-perception bias hypothesis." In fact, the effect was numerically in the opposite direction (Table 1).

Discussion. This result provides evidence against one of the candidate phonetic causes of the cross-linguistic nasal asymmetry, assuming a MOG sound change model. Future work should focus on model components (for example, the exact role of distributional overlap in phonological category learning is an empirical question, cf. Feldman 2013), and on testing the other candidate phonetic causes within concrete sound change models (Henderson 1984; Kunay et al. 2022).

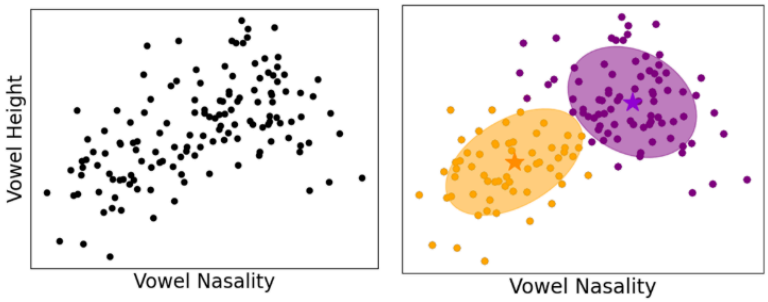


Fig 1. Example MOG uncategorized input (left) and categorized output (right). "Nasality" is abstracted here, but could be quantified by a perceptual scale (Whalen & Beddor 1989) or velar port measurements as in Henderson (1984).

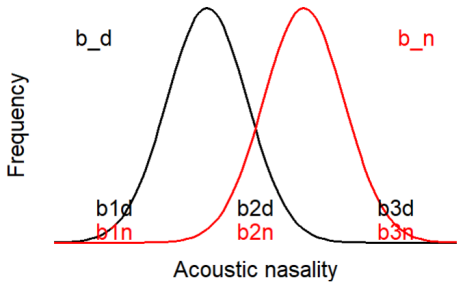


Fig 2. Location of stimuli (1: unambiguously oral vowel, 2: ambiguous vowel, 3: unambiguously nasal vowel) in schematic of assumed acoustic vowel distribution.

	Estimate	SE	t	p
Intercept	0.068	0.19	0.34	0.728
Length	-0.17	0.27	-0.60	0.548

Table 1. Linear model regression results for estimating vowel duration's effect on the relative perceptual distance of the ambiguous vowel to the oral and nasal vowels in a nasal consonant context ('b' in Fig. 3)

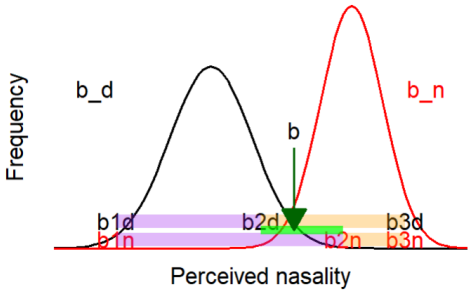


Fig 3. Perceptual distances used to calculate a change in overlap in the perceived distribution. The effect of the nasal consonant context on perceived nasality is operationalized as 'b': the relative distances of b2n and b2d to the unambiguous oral and nasal vowels in the same contexts.

The neural realization of functional load in Italian: An ERP study

Zachary M. N. McCormick Margaret E. L. Renwick
Department of Cognitive Science, Johns Hopkins University

Speech perception requires the auditory system to rapidly transform continuous acoustic input into discrete linguistic categories while simultaneously generating predictions about upcoming sounds. Although extensive work has illustrated that early auditory cortical responses reflect both sensory adaptation and experience-dependent learning (e.g., Näätänen et al., 1997), less is known about whether these responses are sensitive to the informational structure of phonological contrasts. In particular, it remains unclear the extent to which phonemic contrasts differing in functional load (FL)—i.e., the degree to which a phonemic contrast contributes to distinguishing lexical meaning (as quantified by Hockett, 1966)—are processed equivalently during early auditory processing. More precisely, the present study aims to determine whether FL influences mismatch negativity (MMN) responses during passive speech perception.

Mismatch negativity (MMN) is an event-related potential (ERP) reflecting the automatic detection of deviations in an otherwise regular sequence of stimuli (Näätänen et al., 1978) and can be observed under passive listening conditions as a fronto-central negativity around 100-250 ms post-stimulus (Sams et al., 1985). MMN has been widely used to study phonological representations, with larger amplitudes reported for native-language contrasts than for non-native contrasts, and for well-established categories relative to poorly learned ones (Ylinen et al., 2006). Furthermore, Miglietta et al. (2013) indicates that phonological structure modulates MMN responses beyond purely acoustic differences, having shown that phonemic contrasts elicit lower-latency (i.e., faster) MMN responses than allophonic contrasts. Together, these studies support the idea that auditory-phonological mapping is influenced by the listener's phonological system and that pre-attentive auditory processing is modulated by linguistic structure. However, it remains unknown whether phonemic contrasts that differ in their lexical informativeness are encoded with systematically different neural dynamics, as reflected in MMN responses.

The present study asks whether phonemic contrasts with higher FL elicit MMN responses greater in amplitude and/or lower in latency than contrasts with lower FL, consistent with the hypothesis that more lexically informative distinctions are prioritized in early auditory processing. An optimal test case for this question is provided by the vowel system of "Standard" Italian (as operationalized by Rogers & d'Arcangeli, 2004), which contains a low-FL /e/-/ɛ/ distinction yielding few lexical minimal pairs and showing greater cross-dialectal variability, as well as a high-FL /e/-/i/ contrast distinguishing many lexical items. Each of these three target vowels (i.e., /e/, /ɛ/, and /i/) are embedded in a fixed CVCV pseudoword frame (i.e., /gVba/), yielding minimal pair stimuli differing only in the tonic vowel.

Participants are healthy adult native speakers of Italian (n = 9), aged 18-40, with normal or corrected-to-normal hearing and vision. Electroencephalography (EEG) data are collected from a 32-channel scalp montage using a passive auditory oddball paradigm, in which participants are instructed to watch a series of clips from nature documentaries while ignoring the auditory stream of /gVba/ pseudowords presented via speakers. The design consists of four oddball blocks targeting two vowel contrasts (i.e., /e/-/ɛ/ and /e/-/i/), with standard and deviant roles reversed across blocks to yield identity MMN (iMMN) measures in order to maximally ensure that observed ERP morphology is not influenced by mere acoustic differences (Pulvermüller & Shtyrov, 2006). Block order is counterbalanced, with a total of 700 trials per block (2,800 total trials; 80% standards, 20% deviants).

In addition to the EEG task, participants also engage in a speech production task, during which they read aloud a list of target words (with varying tonic vowels) embedded in sentence frames (from Renwick & Ladd, 2016), allowing for the estimation of individual vowel spaces through formant analysis and quantification of within- and between-category dispersion. Participants also complete a brief AX discrimination task for the low-FL contrast, included to provide an independent behavioral measure of perceptual discriminability given known variability in this distinction. These two additional measures are used as covariates to assess whether neural sensitivity to phonological contrasts relates to individual differences in phonetic realization and perceptual access.

Current observations suggest emerging differences in MMN amplitude as a function of FL, with MMN amplitude being larger for high-FL contrasts as compared to low-FL contrasts. MMN latency trends are also broadly consistent with the expected patterns—i.e., with low-FL contrasts tending to elicit later MMN responses than high-FL contrasts—though show some individual variability. Thus, more granular analyses are required to determine whether this variability can be accounted for by individual phonetic implementation (as indexed by production measures) or by differences in perceptual discrimination (as indexed by the AX task).

To evaluate generalizability beyond Italian, data collection and analysis is also underway for native Spanish and Brazilian Portuguese speakers. Brazilian Portuguese contains a phonemic contrast between high-mid /e/ and low-mid /ɛ/, whereas Spanish lacks such a phonemic contrast and instead only maintains it allophonically (if at all). These typological differences provide a principled cross-linguistic comparison for assessing whether MMN sensitivity tracks FL in a continuous way across languages with differing phonological organization.

More broadly, the present study aims to contribute to theoretical debates concerning whether phonological representations are strictly categorical or whether they are structured by graded informational properties such as functional load. By linking FL to early neural indices of speech perception, this study seeks to provide evidence that phonological structure is reflected in the dynamics of pre-attentional auditory processing.

Hockett, C. F. (1966). *The Quantification of Functional Load—A Linguistic Problem*. https://eric.ed.gov/?id=ED011649

Miglietta, S., Grimaldi, M., & Calabrese, A. (2013). Conditioned allophony in speech perception: An ERP study. *Brain and Language*, *126*(3), 285–290. https://doi.org/10.1016/j.bandl.2013.06.001

Näätänen, R., Gaillard, A. W. K., & Mäntysalo, S. (1978). Early selective-attention effect on evoked potential reinterpreted. *Acta Psychologica*, *42*(4), 313–329. https://doi.org/10.1016/0001-6918(78)90006-9

Näätänen, R., Lehtokoski, A., Lennes, M., et al. (1997). Language-specific phoneme representations revealed by electric and magnetic brain responses. *Nature, 385*, 432–434. https://doi.org/10.1038/385432a0

Pulvermüller, F., & Shtyrov, Y. (2006). Language outside the focus of attention: The mismatch negativity as a tool for studying higher cognitive processes. *Progress in Neurobiology*, 79(1), 49–71. https://doi.org/10.1016/j.pneurobio.2006.04.004

Renwick, M. E. L., & Ladd, D. R. (2016). Phonetic Distinctiveness vs. Lexical Contrastiveness in Non-Robust Phonemic Contrasts. *Laboratory Phonology*, *7*(1). https://doi.org/10.5334/labphon.17

Rogers, D., & d'Arcangeli, L. (2004). Italian. *Journal of the International Phonetic Association*, 34(1), 117–121. https://doi.org/10.1017/S0025100304001628

Sams, M., Paavilainen, P., Alho, K., & Näätänen, R. (1985). Auditory frequency discrimination and event-related potentials. *Electroencephalography and Clinical Neurophysiology/Evoked Potentials Section*, 62(6), 437–448. https://doi.org/10.1016/0168-5597(85)90054-1

Ylinen, S., Shestakova, A., Huotilainen, M., Alku, P., & Näätänen, R. (2006). Mismatch negativity (MMN) elicited by changes in phoneme length: A cross-linguistic study. *Brain Research*, *1072*(1), 175–185. https://doi.org/10.1016/j.brainres.2005.12.004

The nature of emphasis spread in Jordanian Arabic

Hani Al Naeem | Massachusetts Institute of Technology

Arabic varieties retain a contrast between *plain* coronals /t, d, s, ð/ and *emphatics* /ṣ, ṭ, ḍ, ðˤ/, obstruents with a secondary constriction near the upper pharyngeal wall. Emphatics trigger a type of tongue root harmony which has come to be known as *emphasis spread* (ES). ES causes substantial F2 lowering in adjacent segments, the most salient effect being the backing of low vowels, which are always front otherwise. In (1), the emphatic has a dot under it, and the underline indicates that ES has affected all segments to the left.

(1)

plain	*emphatic*
/xællæːt/ → [xællæːt]	/xællæːṭ/ → [xa̲lla̲ːṭ]
'pickles'	'blender'

Previous descriptions of ES across Arabic dialects differ on some of the details, but a general agreement is that the leftward effect of ES is unbounded while the rightward effect has a narrower domain, gradiently fades out, or is blocked by high segments (Abudalbuh et al. 2022; Davis 1995; McCarthy 1997; Younes 1993; Zawaydeh 1999). The uncertainty surrounding rightward effects has led to analyses that collapse the distinction between leftward and rightward ES and/or fail to fully explain the latter. Using production data from Jordanian Arabic (JA), I show that the difference between leftward and rightward ES is robust and consistent; I then argue that ES reflects two distinct mechanisms: feature harmony and coarticulation.

Analyzed data come from 11 speakers of JA, who produced 18 minimal pairs with emphatics as initial or final consonants (e.g. /**s**ælæbæ ~ **ṣ**ælæbæ/ and /sælæ**t**æ ~ sælæ**ṭ**æ/). The general structure of the stimuli is $C_1V_1C_2V_2C_3(V_3)$, with emphatics either in C_1 or C_3; no putative blockers were present in this set of stimuli. Formant measurements were taken at 10-50-90% of vowel periods.

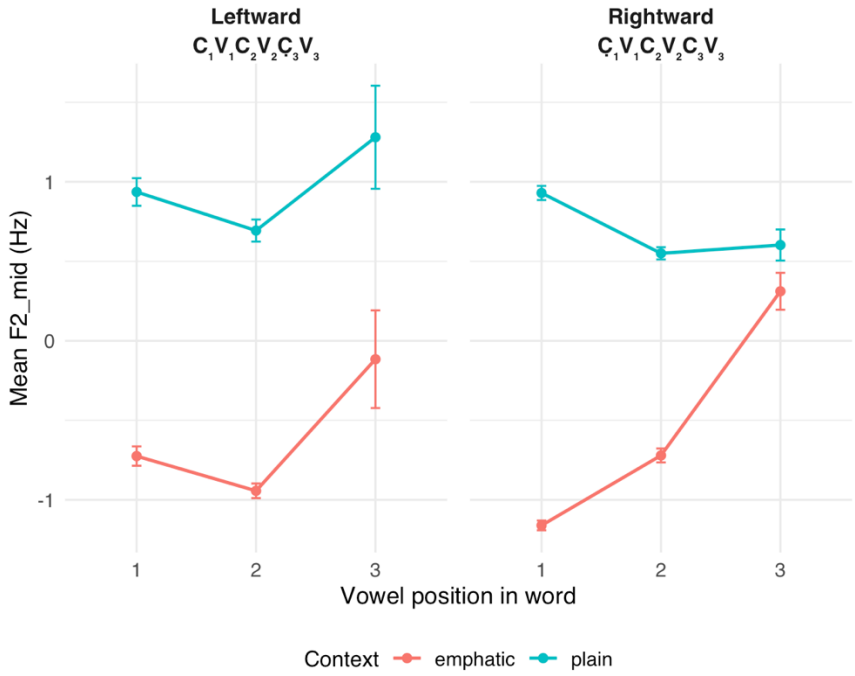


Figure 1. Mean midpoint F2 values for V1–V3 in plain and emphatic contexts under leftward and rightward ES, showing uniform effects under leftward spread, but not under rightward spread. All vowels are phonemically /æ/.

Figure (1) shows the difference in effect size between leftward and rightward ES. In the leftward direction (i.e. emphatic in C_3), the difference in F2 across plain vs. emphatic context is large (~390Hz) and consistent regardless of distance from emphatic trigger. No significant differences were found in the effect size across the three vowel positions in this leftward context. On the other hand, in the rightward direction (i.e. emphatic is in C_1), difference in F2 across plain vs. emphatic context gets smaller as distance from trigger increases, being significantly smaller in V_2 (~290Hz) and significantly smaller still in V_3 (~70Hz). In the absence of putative blockers from both contexts, I argue that this asymmetry derives from two different mechanisms.

I analyze the large effect sizes observed in the leftward direction and on V_1 in the rightward direction as feature harmony. I adapt the DISTAL and LOCAL constraints from Hayes & Londe (2006) to derive these effects. The harmonizing feature, [RTR], is realized on leftward segments, yielding robust long-distance effects, and it is realized locally on a following vowel. Departing from previous descriptions, I argue that non-low vowels do not participate in the harmony process (even in the unbounded leftward context); non-low vowels do not show a change in quality, and their F2 trajectories show transitional changes only. This is achieved by a standard feature co-occurrence constraint *[RTR, -LOW].

With the harmony effects accounted for, the remaining gradient effects in rightward ES are argued to be coarticulatory and are modeled as vowel undershoot. Following the model sketched in Flemming (2001), vowels balance pressure to reach an F2 target against effort minimization, producing target undershoot that is modulated by the segmental context. Vowel undershoot is estimated using the formula: $F2_V = \alpha F2_C + (1 - \alpha)F2_T$, where α is a slope term estimated from fitting a linear model of F2V as a function of F2 of preceding consonant (F2V~F2C), and $F2_T$ is a theoretical target for the vowel, $F2_T = intercept/(1 - \alpha)$. Current stimuli suffer from a limitation of having only labials or /l/ in C2. However, applying the coarticulation model to V_2 in the rightward context yields good predictions when compared against observed F2 values, Pearson's r = 0.73.

The full analysis provides an explanation of the directional asymmetry in ES and clarifies the nature of the long-distance rightward effects by attributing them to a phonetic mechanism.

References

Abudalbuh, M., Jaber, A., & Omari, O. (2022). *An Optimality Theoretic account for pharyngealization spread in Rural Jordanian Arabic*. Italian Journal of Linguistics, *34*(1), 3–30. https://doi.org/10.26346/1120-2726-182

Davis, Stuart (1995) Emphasis Spread in Arabic and Grounded Phonology. *Linguistic Inquiry* 26:465-498.

Flemming, E. (2001). Scalar and categorical phenomena in a unified model of phonetics and phonology. *Phonology, 18(1),* 7-44. DOI: https://doi.org/10.1017/S0952675701004006

Hayes, B., & Cziráky Londe, Z. (2006). Stochastic phonological knowledge: The case of Hungarian vowel harmony. *Phonology, 23*(1), 59-104. DOI: https://doi.org/10.1017/S0952675706000765

McCarthy, J. J. (1997). Process-Specific Constraints in Optimality Theory. *Linguistic Inquiry*, *28*(2), 231–251. http://www.jstor.org/stable/4178976

Younes, M. A. (1993). Emphasis spread in two Arabic dialects. In M. Eid & C. Holes (Eds.), *Perspectives on Arabic linguistics V: Papers from the annual symposium on Arabic linguistics* (pp. 119–145). John Benjamins.

Zawaydeh, B. A. (1999). The phonetics and phonology of gutturals in Arabic [Ph.D., Indiana University]. In *ProQuest Dissertations and Theses* (304503584). ProQuest Dissertations & Theses Global.

Modeling Sociolinguistic Variation of Loanword Stress Adaptation in Mirpur Pahari

Morgan Chubb | University of North Carolina at Chapel Hill

Adaptation of source stress patterns in Mirpur Pahari (MP) loanwords is subject to variation and, therefore, inadequately represented by non-stochastic OT models. Existing analysis identified a correlation between this variation and speakers' age and bilingual status, proposing different grammars between groups and variable adaptation patterns within groups, considering a stochastic alternative, but not formalizing it (Shafi & Hellmuth [S&H], 2024). This paper implements a formally explicit model based on Weighted Scalar Constraints (WSC) (Hsu & Jesney [H&J], 2018), where constraints scale by English exposure, operationalized using Vocabulary Size Test (VST) scores from S&H (2024). This model derives the variation of stress adaptation in MP loanwords without the need for multiple grammars or metaconditions.

Variable stress: Native MP words stress final superheavy (CVVC/CVCC) or heavy penult syllables (CVC/CV:) (Shafi [S], 2019). This is not always the case for loanwords (S&H 2024). This paper focuses on 2 adaptation patterns exhibiting variation due to 2 conflicting constraints:

Pattern 1: Stress placement reassigned to match MP preferences at the expense of faithfulness.

- **ALLFTRIGHT dominates**: Apply a violation when stress does not fall on the rightmost possible position according to native weight-to-stress preferences (S, 2019).
- [ˈsteɪ.dɪəm]Eng→**[sə.ˈte:.dɪəm]**MP: **1 violation,** stress is not on superheavy final syllable.

Pattern 2: Stress placement matches the source word while violating MP stress preferences.

- **MATCHSTRESS dominates**: Apply a violation when the stress position of the MP output does not correspond to the stress position of the English source (S, 2019).
- [ˈsteɪ.dɪəm]Eng→**[sə.te:.ˈdɪəm]**MP: **1 violation** stress shifted to the final syllable

They found that speakers' VST scores correlate with this variation. The higher a speaker's VST score, the less they exhibited Pattern 1 and the more they used the English-faithful Pattern 2.

Theoretical framework: Their alternative stochastic OT model represents these constraints as probability distributions ranked on a continuous scale by VST score, varying when distributions overlap (S&H, 2024). This model treats English exposure as an external correlation rather than a formal part of the grammar and implies pattern invariance at polar ends of the scale.

WSC can integrate these external correlations into a unified grammar. WSC is typically used to model variation in loanword adaptation patterns between words in the nativized core versus non-nativized periphery through Harmony cost sensitive to the parameterized distance (d) of a word from the lexical core. H&J (2018) model this scaling penalty as: penalty = $w + s(d)$.

- Base weight (w): Represents the intrinsic pressure of a constraint across the entire lexicon.
- Scaling factor (s): The degree to which a constraint's weight changes as d increases.

Sociolinguistic-adapted implementation: Given that the prevalence of Patterns 1 and 2 varies between speakers rather than lexical items, this logic must be adapted to represent sociolinguistic variation. In this paper, d will be used to mean the following:

- Distance (d): Represents the sociolinguistic distance resulting from greater English proficiency, operationalized as a speaker's VST score.

The 0-100 VST score was adapted to a 0-1 d value. A VST of 0 represents a speaker with zero English exposure, while 1 represents the maximum possible exposure. Conceptually, a high VST score is analogous to the "periphery," representing maximum deviation from native MP stress.

Greater English exposure moves speakers from the monolingual "core" towards a bilingual "periphery," where faithful stress patterns are preferred due to a high scaled Harmony cost.

I calculated *w* and *s* using the MaxEnt Excel "Solver" approach laid out by Hayes & Moore-Cantwell (2025). The "Solver" logic calculates s and w for each constraint by optimizing a log likelihood value that adjusts these values until the model's predicted probabilities (p) are as close to the observed values as possible.

Results: The shift from Pattern 1 to 2 can be represented by a scaling weight for the faithfulness constraint, while markedness retains a fixed weight. The Solver gave ALLFTRIGHT a *w* of 1.30 and an *s* of 0, while MATCHSTRESS has an intrinsic *w* of 0 and an *s* of 3.16. This means the Harmony cost (H) for violating source faithfulness (MATCHSTRESS) is purely a function of exposure. For a speaker at a given VST score (*d*), the penalty is $3.16 \times VST$.

d		ALLFTRIGHT	MATCHSTRESS	*H*	*e*	*p*
	w:	1,30	0			
	s:	0,00	3,16			
0,94	Pattern 1	0	1	3	0,05	0,15
	Pattern 2	1	0	1,3	0,27	0,85
0,49	Pattern 1	0	1	1,6	0,21	0,43
	Pattern 2	1	0	1,3	0,27	0,57

From this, we can predict at what VST score speakers go from preferring Pattern 1 to 2. By setting ALLFTRIGHT equal to MATCHSTRESS: $1.30 = 3.16 \times d$, we can solve for d to determine the point when the model's preferences flip: $d \approx 0.41$. Additionally, given that we are using a stochastic model, we can predict that unless a speaker were to have zero English exposure ($d = 0$), there would still be some amount of variation in pattern preferences.

Implications: Using stochastic WSC naturally predicts observed sociolinguistic variation. Even when one pattern is favored, the competing candidate retains a proportional share of the probability (e.g., p = 57% for Pattern 2 at VST), matching the gradient data observed by S&H (2024). Additionally, this WSC model makes testable predictions that go beyond mere description. The model derives a threshold VST value where the grammar's pattern preference flips ($d \approx 0.41$), and predicts variation even at low VST values, both of which can be tested empirically through collecting further native speaker data. Ultimately, this implementation provides a more parsimonious and predictive model of MP loanword adaptation, and demonstrates the modeling of sociolinguistic variation as part of a single, unified grammar.

References:

Hayes, Bruce & Claire Moore-Cantwell. 2025. *A guide to analysis in MaxEnt optimality theory*.

Hsu, Brian & Karen Jesney. 2018. Weighted scalar constraints capture the typology of loanword adaptation.

Shafi, Sehrish. 2019. A phonological analysis of English loanwords in Mirpur Pahari.

Shafi, Sehrish & Sam Hellmuth. 2024. Stress placement in English loanwords by speakers of Mirpur Pahari in the UK.

Source-language influence and cross-domain faithfulness in Nuosu Yi loanword phonology

Yao Zhang, Jennifer Kuo

Department of Linguistics, Cornell University

In loanword adaptation, input from a foreign source language is typically adapted to some degree to fit phonological constraints of the native language. An important open question is how exposure to a source language can change a speaker's loanword grammar, and even reshape their native phonology. Additionally, there is limited understanding of how segmental and suprasegmental phonology interact in loanword adaptation; suprasegmental adaptation often behaves differently from segmental adaptation [1], but it's unclear whether a speaker's strategies for the two will correlate. This study addresses these questions by examining Standard Mandarin (StM) loanwords in Nuosu Yi (Yi); we compare how loanword adaptation in a corpus, reflecting the past state of the language, differs from adaptation in a production experiment, reflecting the present state of the language. We find that speakers in the production experiment retain more structures that are phonologically marked in Yi; this is interpreted as an effect of language background, as current speakers of Yi have greater fluency and exposure to Mandarin. Additionally, rates of segmental and tonal adaptation systematically co-vary, but at the same time, speakers are more likely to preserve marked segmental structures than the marked suprasegmental structures.

Background and corpus patterns Yi, a Tibeto-Burman language mainly spoken in Sichuan, China, has a restricted (C)V syllable structure that disallows codas and permits only two diphthongs /ie, uo/ (Tab. 1). In terms of the tone inventory, Yi has four tones 55, 44, 33, 21, of which high tones (55, 44) have been shown to be marked and avoided in loanwords [2]. In contrast, StM allows for nasal codas and has a larger diphthong inventory; the tonal system has multiple high tones, of which 55 is very frequent. Patterns of loanword adaptation were first examined in a corpus of 462 established Mandarin-to-Yi loanwords; the corpus was compiled in the 1980s and represents an older generation of Yi speakers, who likely had lower Mandarin proficiency and spoke a local variety of Mandarin rather than StM. Overall, corpus items strongly conform to native constraints, where the 55 tone, diphthongs, and nasal codas are repaired instead of faithfully adapted: only 0.3% of source 55 surface as 55, all source diphthongs are either adapted as licit diphthongs (63.4%) or monophthongized (36.6%), and nasal codas are entirely absent (Fig.1) [1,3].

Table1 Phonological restrictions in Yi vs. StM

Language	Yi	StM
Syllable	(C)V	(C)V(N)
Diphthong	/ie, uo/ (monophthongal allophones [i, o])	/ ie, uo; ai, au, ia, iau, ie, iou, ou, ua, uai, uei, ye/
Tone	55(infrequent), 44 (infrequent), 33, 21	55 (frequent), 35, 51, 214

Table 2 Examples of the stimuli

Frequency	Example	Tones
frequent	/tʂoŋ^{55} kuo^{35}/ 'China'	55-35
	/kuo^{35} tɕa^{55}/ 'country'	35-55
common	/hua^{35} ɕa^{51}/ 'Cathay'	51-35
	/fu^{51} kuan^{55}/ 'adjutant'	35-51
rare	/hai^{214} ʂən^{55}/ 'trepang'	214-55
	/uei^{55} mi^{214}/ 'micrometer'	55-214

Experiment To compare current adaptation patterns to the corpus data, a production experiment was conducted. Fifteen adult native speakers of Nuosu Yi (M=3; mean age = 22.6) participated the online experiment via Labvanced. All participants are bilingual or semi-bilingual in Yi and StM, and likely have higher StM proficiency than the speakers represented in the corpus. Participants listened to 96 disyllabic real words pronounced in StM and produced their corresponding Yi borrowed forms. The stimuli (Tab. 2) were selected from existing loanwords in the corpus and balanced across tone (55/35/51/214), coda type (nasal/none) in both syllables, and frequency (frequent/common/rare) based on the SUBTLEX-CH corpus.

Results Comparison of corpus and experimental results (Fig.1) shows that in general, present-day speakers are more faithful to the source sound than suggested by the corpus in all three domains that were investigated. Faithful adaptation of tone 55 rose to 64%, nasal coda preservation to 63%, and non-native diphthong preservation to 81.2%. Overall, it appears that previously restricted forms (both segmental and suprasegmental) are now faithfully adapted, indicating a weakening of native markedness constraints on loanwords.

Figure 1 Adaptation patterns in corpus vs. experiment

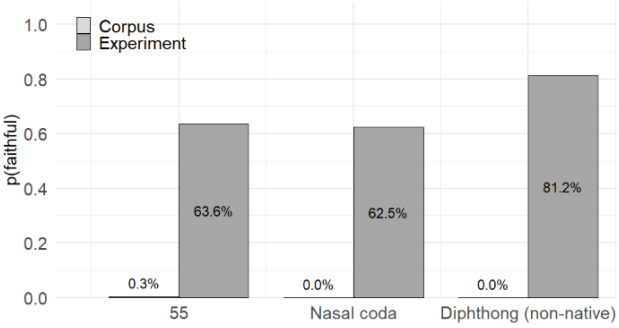


Figure 2 Gradient variation across subjects (ordered by p(55))

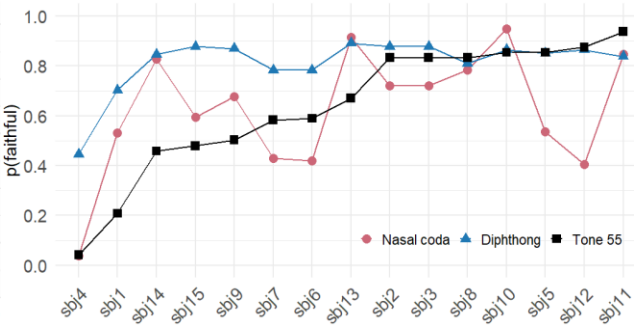
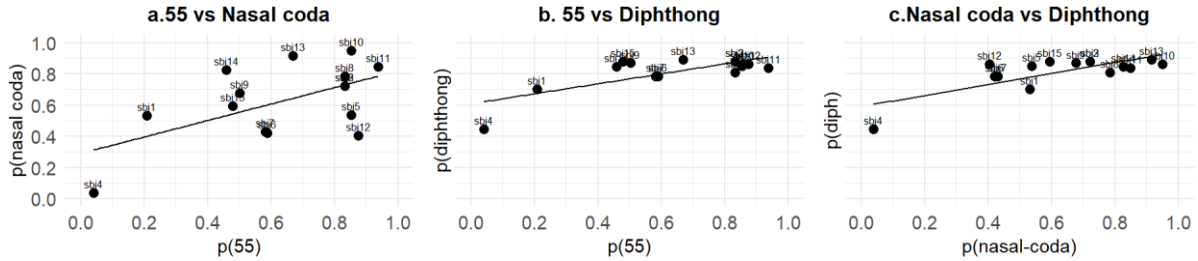


Fig. 2 shows the proportion of faithful adaptations by individual participant. These results suggest that most participants showed variable levels of adaptation. Adaptation patterns co-vary across domains, but the strength of co-variation differs by domain. Rates of 55 retention correlate weakly with nasal coda retention ($r = 0.69$, $p = .02$; Fig. 3a), but more strongly with diphthong retention ($r = 0.75$, $p < .01$; Fig. 3b); diphthong retention and nasal coda retention are also strongly correlated ($r = 0.75$, $p < .01$; Fig. 3c). These correlations indicate that speakers who are more faithful in tonal adaptation also tend to preserve segmental structure to a greater extent, supporting a cross-domain effect of source-oriented faithfulness.

Figure 3 Correlations in rate of faithful adaptations



Despite this shift towards more faithful adaptations, an asymmetry remains between segmental and suprasegmental adaptation. ⅔ of of subjects consistently show higher retention of diphthongs than 55, and about ½ show higher retention of nasal codas than 55. These patterns indicate that, although faithfulness to the source language increases across domains, it is stronger in the segmental domain than in the suprasegmental domain, and that not all segmental structures are equally privileged.

Conclusion Overall, current speakers of Yi (as reflected by experimental data) adapted marked Mandarin structures more faithfully than the speakers in an earlier state of language (as reflected by corpus data). I propose that this shift is due to speakers' increasing exposure to and proficiency in Mandarin. Native restrictions are being weakened, and faithfulness to the source is reshaping both segmental and suprasegmental patterns of loanword adaptation. Notably, this influence is coordinated across domains but not uniform, as 55 realization aligns more closely with diphthong preservation than with nasal coda retention.

Selected References [1] Kang, Y. & Katsuda, H. *In press*. Loanword phonology. In *Companion to Phonology, eds*. Wiley Blackwell. [2] *redacted*. Tone Adaptation from Mandarin to Nuosu Yi. [3] Azimo, X. 2019. *[A study on Chinese Loanwords in Liangshan Yi Language]*.

The role of indexation on Portuguese exceptional verbal patterns

Lucas Pereira Eberle
University of Campinas (Brazil)

This work is set on a Maximum Entropy framework (Goldwater & Johnson, 2003) with index constraints (Itô & Mester, 2001; Itô et al., 1999; Pater, 2000, 2009). I present data on Brazilian Portuguese vowel alternations in verbal paradigms that lead to an inconsistency problem, as opposite patterns conflict. This work argues in favor of two combined tools to solve this inconsistency: indexation and faithfulness to remote bases (Breiss, 2024; Steriade, 1997).

Portuguese verb morphology: Brazilian Portuguese exhibits a regular vowel alternation in verbs, where root mid vowels are augmented when stressed, via lowering or by glide epenthesis in a hiatus context, while remaining faithful or raising (in hiatus) when unstressed. On the other hand, high vowels are faithful throughout the entire system. There are, however, a few verbs that present an exceptional pattern of alternations, as shown below:

	Regular		Exceptional	Regular		Exceptional
	p/e/gar	map/e/ar	fr/e/ar	p/i/car	cop/i/ar	od/i/ar
1-SG	p['ɛ]go	map['ej]o	fr['ej]o	p['i]co	cop['i]o	od['ej]o
2-SG	p['ɛ]gas	map['ej]as	fr['ej]as	p['i]cas	cop['i]as	od['ej]as
3-SG	p['ɛ]ga	map['ej]a	fr['ej]a	p['i]ca	cop['i]a	od['ej]a
1-PL	p[e]'gamos	map[e]'amos map[i]'amos	fr[e]'amos fr[ej]'amos	p[i]'camos	cop[i]'amos	od[i]'amos
2-PL	p[e]'gais	map[e]'ais map[i]'ais	fr[e]'ais fr[ej]'ais	p[i]'cais	cop[i]'ais	od[i]'ais
3-PL	p['ɛ]gam	map['ej]am	fr['ej]am	p['i]cam	cop['i]am	od['ej]am

This work focuses on the hiatus context. Mid vowels tend to undergo markedness pressures, where duration is increased in stressed, i.e., longer, syllables, and duration is reduced in unstressed ones (short). The exceptional pattern in mid vowels allows the duration to increase in unstressed positions (resulting in marked forms). I argue that it happens due to faithfulness outside the paradigm, as all the exceptional verbs have a stressed [ej] in their nominal base. Regarding high vowels, they are mostly faithful even in stressed positions, which results in a marked form: short vowel [i] in a long position; but when they alternate, they change toward the less marked form. In this case, I argue that the alternating verbs do not present a stressed [i] in their nominal base; therefore, not enough faithfulness is assigned. Nonetheless, the faithfulness to nominal bases is not categorical.

Research questions: Why do these verbs behave differently? How does the grammar deal with the exceptional patterns?

Hypothesis: (1) Faithfulness to bases favors the exceptional alternations for mid vowels and blocks alternation for high vowels; (2) Because faithfulness to bases is not categorical, index constraints will be needed, and (3) a model with combined tools (indexation and faithfulness to bases) will perform better than the ones without the combinations.

Training data: 356 verbs of the [ar]-conjugation whose stem's final vowel is front were collected from the dictionary *Dicionário br.ispell* — 166 verbs ended with /e/, from which only 7 presented the exceptional pattern /e/ → [ej] (unstressed) and 190 with /i/, from which only 8 presented alternations /i/ → [ej] (stressed). The rate of alternation for verbs was established via random sampling of tokens per class of verbs (regular or exceptional) from YouTube, using the YouGlish platform, which were then categorized as [i], [e], and [ej] by a phonetically trained linguist.

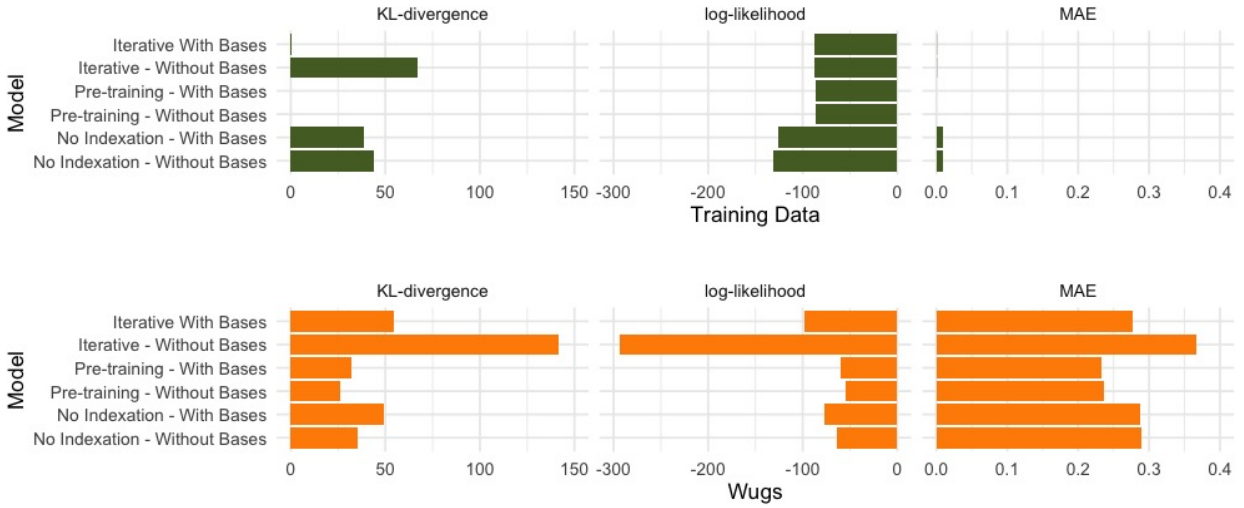
Testing data: Existing nouns that have no related verbs were used to create novel verbs. For mid vowel alternation, verbs were created using nominal bases with stressed [ej], unstressed [ej], and without [ej] to test the effect of faithfulness to bases. For high vowel alternation, verbs were created

using bases with stressed [i], unstressed [i], and without [i]. An experiment was conducted in which participants judged these novel verbs with the forms [i e ej] as 'natural' or 'unnatural'.

Learners: Six models of grammar were tested using two learners: MaxEnt grammar tool (Hayes & Wilson, 2008) and the indexation learner by Nazarov e Smith, 2022. The models tested are presented in (1). Regarding indexation, 'pre-training' assumes that each constraint has an indexed version from the start (for every class of verbs). In contrast, 'iterative indexation' adds indexed constraints one at a time until the model stops improving (as measured by KL-divergence).

(1) a. No indexation (without faithfulness to bases)
b. No indexation (with faithfulness to bases)
c. Pre-training indexation (without faithfulness to bases)
d. Pre-training indexation (with faithfulness to bases)
e. Iterative indexation (without faithfulness to bases)
f. Iterative indexation (with faithfulness to bases)

Results: The models were tested according to three measures: KL-divergence, log-likelihood, and mean absolute error. The results show that models without indexation perform worse than models with indexation. Between indexations, the 'pre-training indexation' performs better in general in learning the Lexicon (fitting the data), and both versions with and without faithfulness to bases perform equally. Regarding generalization to novel verbs, no model performs perfectly, but 'pre-training' also shows a better performance.



Concluding Remarks: Regarding the generalization, I argue that the overfitting in the training data diminishes the effect of bases, which is not categorical and affects only 4% of the verbs. However, in the experiments, speakers appear to be sensitive to this kind of faithfulness, as alternation from /e/ to [ej] was more accepted as 'natural' if the bases presented a stressed [ej], and the faithful [i] in stressed position was more accepted if the base presented a stressed [i]. Nonetheless, because alternations are exceptional, models do not learn them correctly, and generalization fails. It seems that speakers, when generalizing to novel verbs, are more sensitive to markedness constraints, and the effect of faithfulness is weaker.

References: Breiss, C. (2024). When bases compete: A voting model of lexical conservatism. Language, 100 (2), 308–358. Feofiloff, P. (2025, julho). Goldwater, S., & Johnson, M. (2003). Learning OT constraint rankings using a maximum entropy model. Proceedings of the Stockholm workshop on variation within Optimality Theory, 111, 120. Hayes, B., & Wilson, C. (2008). A maximum entropy model of phonotactics and phonotactic learning. Linguistic inquiry, 39 (3), 379–440. Itô, J., & Mester, A. (2001). Covert generalizations in Optimality Theory: the role of stratal faithfulness constraints. Studies in Phonetics, Phonology, and Morphology, 7 (2), 273–299. Itô, J., Mester, A., & Padgett, J. (1999). Lexical classes in Japanese: a reply to Rice. Nazarov, A., & Smith, B. (2022). Generalizing French schwa deletion: the role of indexed constraints. In Proceedings of the annual Meetings on phonology. Pater, J. (2000). Non-uniformity in English secondary stress: the role of ranked and lexically specific constraints. Phonology, 17 (2), 237–274. Pater, J. (2009). Morpheme-specific phonology: Constraint indexation and inconsistency resolution. Steriade, D. (1997). Lexical conservatism. Linguistics in the morning calm, 157-179.

Constraint Ranking Paradoxes in Taiwanese Circular Tone Sandhi: Evidence Against Parallel Optimality Theory

Yichen Wang
Michigan State University
wangy176@msu.edu

1. Introduction. Taiwanese Southern Min exhibits a well-known case of circular tone sandhi in which citation tones alternate in non-phrase-final position according to a closed chain (53 → 44 → 22 → 21 → 53, with an additional mapping 24 → 22). This pattern poses a fundamental challenge for Optimality Theory (OT) because, as a subtype of counterfeeding opacity, it requires reference to intermediate representations that classic parallel OT cannot access. In this abstract, I argue that even enriched parallel frameworks such as Comparative Markedness [1] fail to resolve this issue: circular tone sandhi gives rise to an inherent ranking contradiction. Approaches that appeal to serial evaluation, such as Stratal OT [2], are also ill-suited, since all mappings in the chain occur in the same morphosyntactic environment, providing no empirical basis for stratification. Similarly, OT-CC [3] predicts the possibility of longer harmonically improving chains, leading to overgeneration. The only explicit OT analysis, Mortensen [4], derives the pattern by imposing a linear scale over tones via constraints such as HIGHER and ENDMOST, but this ordering is arbitrary and lacks independent phonological motivation. I therefore ask whether circular tone sandhi can be derived using only simple, independently motivated constraints within a parallel OT framework, and argue that it cannot: even enriched OT systems inevitably produce irresolvable ranking contradictions. I adopted the framework of Comparative Markedness McCarthy [1] and pursued two analyses: an IDENT-based approach, which treats tones as featural representations, and a MAX/DEP account, which models tone sandhi as the deletion and insertion of tonal elements. Both analyses converge on the same result: regardless of representational assumptions, circular tone sandhi cannot be derived without yielding irresolvable ranking contradictions.

2. Feature Faithfulness: IDENT-based Analysis. The first analysis adopts Comparative Markedness (McCarthy 2003) with a standard IDENT-based faithfulness system. The constraint set includes positional markedness enforcing sandhi in non-final position, general tonal markedness constraints penalizing contour types and register values (each split into OLD and NEW versions), and faithfulness constraints preserving contour and register. Under appropriate local rankings, each step in the sandhi chain can be derived. A visualization of the constraint interactions that derive each tonal shift in the circular system is provided in Figure 1. In the diagram, blue arrows represent derivations in which register is preserved while tonal shape is altered. In contrast, orange arrows represent derivations in which tonal shape is preserved while register alternations are induced.

A closer examination of the constraint interactions between adjacent tones in the circular system reveals systematic ranking conflicts. For instance, the mapping /44/ → /22/ requires preservation of tonal contour, which is only possible if IDENT-CONTOUR dominates *FALL_N and *HIGH_O dominates IDENT-REGISTER, thereby blocking contour formation and maintaining a level tone. However, the adjacent mapping /53/ → /44/ requires the opposite outcome: a falling contour must be eliminated in favor of a level tone, which can only be derived if *FALL_N dominates IDENT-CONTOUR. Thus, the same pair of constraints must simultaneously support two contradictory dominance relations across different parts of the circle. Because these requirements apply to the same constraints within a single system, no consistent global ranking can satisfy them simultaneously. The result is a ranking paradox in which the grammar must enforce mutually contradictory

dominance relations.

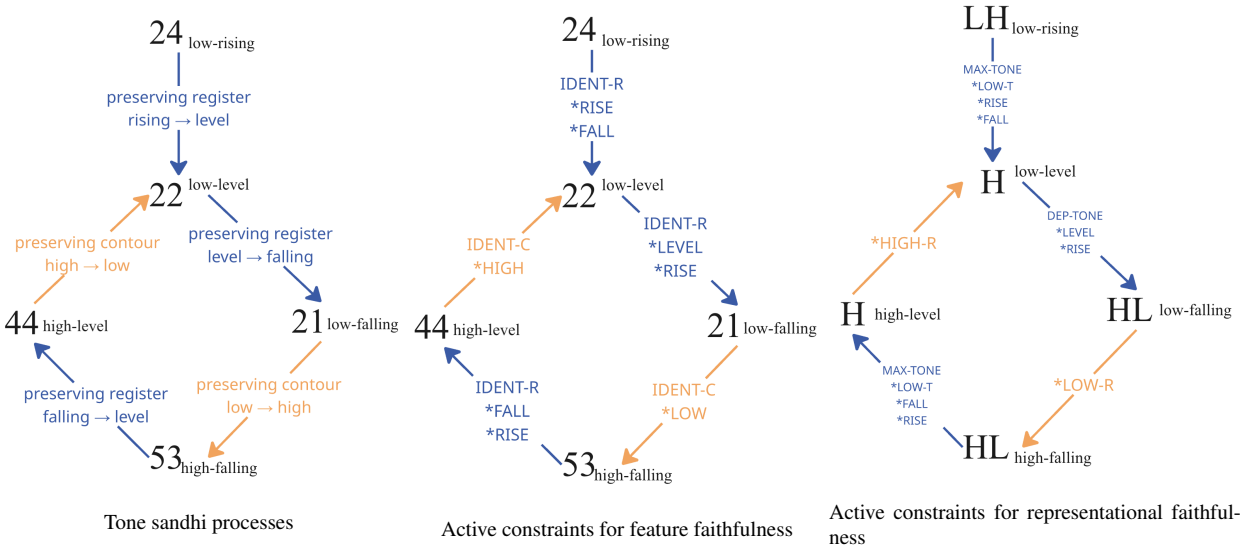


Figure 1. A visual representation of the sandhi process and their corresponding dominant constraints.

3. Representational Faithfulness: MAX/DEP Analysis. To determine whether this paradox is an artifact of feature-based faithfulness, the second analysis reformulates tonal representation in autosegmental terms, decomposing tones into sequences of H and L elements and replacing IDENT constraints with MAX and DEP constraints governing deletion and insertion. Markedness constraints are correspondingly reinterpreted as penalties on tonal sequences and register specifications, again split into OLD and NEW versions. This approach allows contour changes to be modeled as structural operations rather than feature substitutions, potentially increasing analytical flexibility. Nevertheless, the same contradiction reappears when considering the final step H (high register) → H (low register) (/44/ → /22/), which requires a register shift without insertion or deletion of tonal elements. This mapping must be driven by *HIGH-REGISTER$_N$ dominating register faithfulness, while crucially preserving the tonal element H. However, earlier in the chain, the grammar must have *LOW-REGISTER$_N$ dominating register faithfulness and permit the insertion and preservation of L elements (e.g., in /22/ → /21/) under the dominance of DEP-L by markedness constraints such as *LEVEL$_N$. The contradiction therefore persists along two independent dimensionsregister assignment and tonal element manipulationdemonstrating that the failure is not tied to a particular representation of tone or formulation of faithfulness, but is inherent to the structure of the circular system.

4. Conclusion. This paper has shown that circular tone sandhi in Taiwanese Southern Min cannot be derived within any parallel OT framework, even under enriched systems such as Comparative Markedness and across distinct representational assumptions. Both IDENT-based and MAX/DEP-based analyses converge on the same result: the circular chain necessarily induces mutually incompatible constraint rankings that no single hierarchy can satisfy. Crucially, this failure is not an artifact of a particular constraint set or tonal representation, but follows from the structural properties of circular mappings themselves.

References. [1] J. J. McCarthy. "Comparative markedness". In: *Theoretical linguistics* 29.1-2 (2003). [2] P. Kiparsky. "Stratal OT: A synopsis and FAQs". In: *Capturing phonological shades within and across languages* 2 (2015). [3] J. J. McCarthy. *Hidden generalizations: phonological opacity in Optimality Theory*. University of Toronto Press, 2007. [4] D. Mortensen. "Logical and substantive scales in phonology". In: (2006).

A tool to derive P-map constraints from infants' discriminability of phonetic contrasts

Muhammad Rehan, Megha Sundara
University of California, Los Angeles

Introduction. The influential P-map hypothesis [18, 24], that speakers possess implicit knowledge of the perceptual distance between two surface forms, has been used to motivate prior expectations on constraint weights in MaxENT models [22, 23]. These priors are derived from confusion matrices or error rates in perception experiments on adults (e.g., [9]). However, we know that adults' perception is affected by native language experience (e.g., [3–6, 8, 11, 17]). Instead, we develop a tool that uses effect sizes from young infants' discrimination, before they are re-tuned by language experience [1, 15, 19, 21], to better approximate human perceptual confusability for speech segments. **Input.** Data were obtained from 141 discrimination experiments (61 studies; $n = 4238$ infants) where the influence of language-experience could be ruled out, either because infants with and without experience behaved identically or because only older infants demonstrated effects of language experience indexed by an improvement or decrement in discrimination. Infant discriminability was indexed by $|g|$ (absolute Hedges' g). Because infant experiments are expensive and time consuming, discrimination data are available for only a small subset of contrasts of the world's languages. So we imputed the expected effect size for other pairs of segments using 3 independent routes. **Imputing effect sizes for untested contrasts.** In the "Metric Learning Route", each consonant or vowel contrast was represented with features [13], and effect sizes from the infant experiments were used as the dependent variable to derive weights for individual features:

$$d_{\mathrm{w}}(a,b) = \left(\sum_k w_k (x_{ak} - x_{bk})^2 \right)^{1/2} \qquad w_k\text{: feature weight; } x_{ak}\text{: value of feature } k \text{ for segment } a \tag{1}$$

$$\bar{g}_c = \alpha + \gamma\, d_{\mathrm{w}}(c) + \varepsilon_c \qquad \bar{g}_c\text{: mean effect size for contrast } c\text{; } \varepsilon_c\text{: residual} \tag{2}$$

Larger weights indicate more perceptually salient dimensions. For contrasts where infant discrimination data was unavailable, the feature weights were recombined to obtain effect sizes. In the "Observed Pair Graph route", the effect sizes for tested pairs were bootstrapped to estimate the effect sizes for untested pairs by calculating a perturbed shortest path along a weighted graph using the Floyd-Warshall algorithm [7, 20]:

$$\hat{g}_{ab}^{(C)} = \delta_G(a,b) \quad \delta_G\text{: shortest-path distance in graph } G \text{ between segments } a, b \tag{3}$$

Finally, for the "Bayesian Inversion Route", we first evaluated whether infant discriminability of phonetic contrasts predicts the typological frequency of those contrasts across the world's languages. This prediction emerges directly from perceptual salience based explanations for consonant and vowel typology, e.g., Dispersion theory [6, 10, 16] and diachronic-filtering accounts [2, 14]. Typological frequency was computed from PHOIBLE [12] using a *stratified* procedure that corrects for genetic and areal autocorrelation. Stratified counts were then modeled using a Bayesian GLMM. Observations were weighted by sample size to define a penalized log-likelihood function. There was a credible effect for Hedges' g (95% HDI [0.39, 0.51]), which confirms a positive relationship between infant discriminability and typological frequency (Figure 1): contrasts that infants discriminate more easily are more frequent across the world's languages. This Bayesian mapping between effect sizes in infant discrimination experiments and PHOIBLE co-occurrence frequencies was then inverted to estimate effect sizes for untested contrasts. **Precision-weighted fusion.** The estimates from each route were combined using inverse-variance weights (Equation 4) to generate a pairwise perceptual dissimilarity matrix $D_{xy} = f(\hat{s}_{(x,y)})$ where larger values indicate more perceptually distinct segment pairs.

$$\hat{g}_{ab}^{(\mathrm{comb})} = \frac{\sum_r \pi_{ab}^{(r)} \hat{g}_{ab}^{(r)}}{\sum_r \pi_{ab}^{(r)}} \qquad \pi_{ab}^{(r)}\text{: precision (inverse variance) of route } r\text{'s estimate} \tag{4}$$

This distance matrix was then represented geometrically using multidimensional scaling (MDS), which embeds each segment x as a point z_x in a low-dimensional perceptual space such that the Euclidean distance $\|z_x - z_y\| \approx D_{xy}$. The perceptual distance between x and y in the MDS-derived space can then be used

as priors on *MAP(x, y) [24] and similar constraints in MaxEnt and related learning models, with $\mu_{xy} \approx \|z_x - z_y\|$ parametrizing *MAP$(x, y) \sim \mathcal{N}(\mu_{xy}, \sigma)$. **Conclusion.** Under this formulation, mappings between highly distinct sounds receive higher expected faithfulness weights, while mappings between perceptually similar sounds receive lower expected weights based on an independently-estimated perceptual space that is grounded in infant discrimination and not confounded by language experience. A tool to generate such weights is available through an anonymized Shiny app here: https://pmap-priors.shinyapps.io/pmap-priors/. The app can be used to visualize the (2D/3D) MDS space along with exporting constraint weights. Through these inductive biases, our model links infant speech perception, typological frequency, and MaxEnt phonological learning in a single quantitative framework. We will discuss considerations for selecting the MDS space for consonants versus vowels using this new tool as well as demonstrate applications.

$$y_i \sim \text{Binomial}\Big(n_i, \text{logit}^{-1}(\beta_0 + \beta_1 \,\text{Hedges}'\text{g}_i + u_i[\text{study}_i])\Big)$$

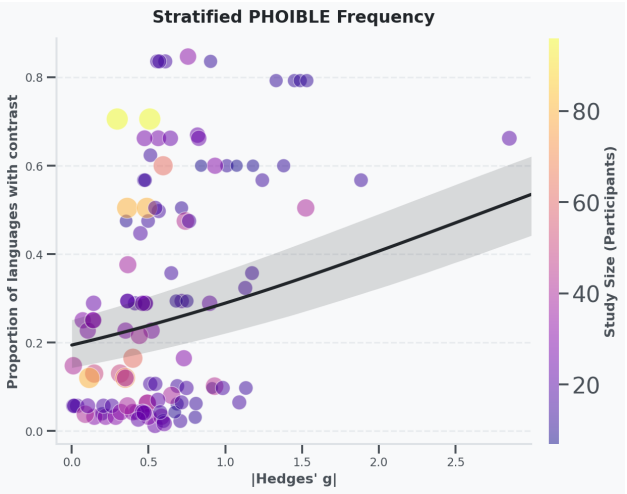


Figure 1: Relationship between infant discriminability and typological frequency.

References

[1] C. T. B. A direct realist view of cross-language speech perception. In W. S., *Speech Perception and Linguistic Experience: Issues in Cross-Language Research*, 171–204. York Press, Timonium, MD, 1995. [2] J. B. *Evolutionary Phonology: The Emergence of Sound Patterns*. Cambridge University Press, Cambridge, 2004. [3] R. D., et al. On the relation between speech perception and loanword adaptation: Cross-linguistic perception of korean-illicit word-medial clusters. *Natural Language Linguistic Theory*, 37, 08 2018. [4] L. D. and J. A. S. Sources of illusion in consonant cluster perception. *Journal of Phonetics*, 40 (2):234–248, 2012. [5] N. H. F., et al. The influence of categories on perception: Explaining the perceptual magnet effect as optimal statistical inference. *Psychological Review*, 116(4):752–782, 2009. [6] E. F. Contrast and perceptual distinctiveness. In B. H., et al., *Phonetically Based Phonology*, 232–276. Cambridge University Press, Cambridge, 2004. [7] R. W. F. Algorithm 97: Shortest path. *Communications of the ACM*, 5(6):345, 1962. [8] W. F. G. Phonetic categorization in auditory word perception. *Journal of Experimental Psychology: Human Perception and Performance*, 6(1):110–125, 1980. [9] B. H. and J. W. Saltation and the P-map. *Phonology*, 32(2):267–302, 2015. [10] B. L. Phonetic universals in vowel systems. In J. O. and J. J., *Experimental Phonology*, 13–44. Academic Press, Orlando, 1986. [11] D. W. M. and M. M. C. Phonological context in speech perception. *Perception & Psychophysics*, 34:338–348, 1983. [12] S. M. and D. M. *PHOIBLE 2.0*. Max Planck Institute for the Science of Human History, Jena, 2019. [13] D. R. M., et al. Panphon: A resource for mapping IPA segments to articulatory feature vectors. In *Proceedings of COLING 2016*, 3475–3484, 2016. [14] J. J. O. The listener as a source of sound change. In *Papers from the Parasession on Language and Behavior*, volume 17 of *CLS*, 178–203, 1981. [15] L. P. and O.-S. B. Natural Referent Vowel (NRV) framework: An emerging view of early phonetic development. *Journal of Phonetics*, 39(4):467–478, 2011. [16] J.-L. S., et al. Major trends in vowel system inventories. *Journal of Phonetics*, 25(3):233–253, 1997. [17] J. S., et al. Phonotactic constraints shape speech perception: Implications for sublexical and lexical processing. In E. D., *Language, Brain, and Cognitive Development: Essays in Honor of Jacques Mehler*. The MIT Press, 2002. [18] D. S. The phonology of perceptibility effects: The P-Map and its consequences for constraint organization. In K. H. and S. I., *The Nature of the Word: Studies in Honor of Paul Kiparsky*, 151–179. MIT Press, Cambridge, MA, 2008. [19] M. S., et al. Young infants' discrimination of subtle phonetic contrasts. *Cognition*, 178:57–66, 2018. [20] S. W. A theorem on boolean matrices. *Journal of the ACM*, 9(1):11–12, 1962. [21] J. F. W. and R. C. T. Cross-language speech perception: Evidence for perceptual reorganization during the first year of life. *Infant Behavior and Development*, 7(1): 49–63, 1984. [22] J. W. Accounting for the learnability of saltation in phonological theory: A maximum entropy model with a P-Map bias. *Language*, 93(1):1–36, 2017. [23] C. W. Learning phonology with substantive bias: An experimental and computational study of velar palatalization. *Cognitive Science*, 30(5):945–982, 2006. [24] K. Z. *map constraints. Unpublished manuscript, University of California, Los Angeles, 2013.

Accent cues language-specific phonotactics in Spanish-English bilingual perceptual repair

Background. When confronted with phonotactically-disallowed clusters, listeners sometimes perceive epenthetic segments that are not acoustically present [1]. This is called **perceptual repair**. Though exposure to an L2 that contains the marked cluster can attenuate perceptual repairs of structures that are phonotactically marked in the L1 [2], much is still unknown about multilinguals' perception when their languages' phonotactics differ. Spanish disallows #sC clusters and repairs them with /e/-insertion (e.g. [snid] → /esnid/) [3,4], while in English #sC clusters are common; Spanish-English bilinguals must reconcile conflicting phonotactic expectations when processing these clusters. This study investigates two contextual cues that may influence how L1 Spanish-L2 English bilinguals perceive #(e)sC: **accent** and **carrier language**. We hypothesize both **L2 contextual cues** and **higher L2 proficiency** will attenuate perceptual repair, on the grounds that both facilitate access to L2 phonotactic knowledge.

Method. Self-reported native Spanish speakers with English fluency were recruited via Prolific (N = 55); English proficiency of each participant was approximated using the LexTALE-ENG vocabulary test [5]. Participants completed 48 ABX trials, where they heard #esC pseudowords that varied in initial [e] duration, and judged whether these items were more like a reference word with [e] at max duration (fully "#esC') or excised (fully "#sC"). Stimuli varied by **target word accent** (Spanish or English) and **language of the carrier phrase** (Spanish or English). Half of the participants heard only Spanish-accented pseudowords while the other half heard only English; all participants heard both carrier phrases. Pseudowords were in the shape *esCVnu* (C = {p, m, t, n}; V = {i, u}), which is phonotactically permissible but highly infrequent in both languages. To generate the stimuli, a native bilingual talker first produced #esC-initial versions of each word. Glottal pulses were successively removed from the [e] of a single token to create a continuum of six durations ("gates"), from 0-5 glottal pulses (see fig. 1 for an illustration).

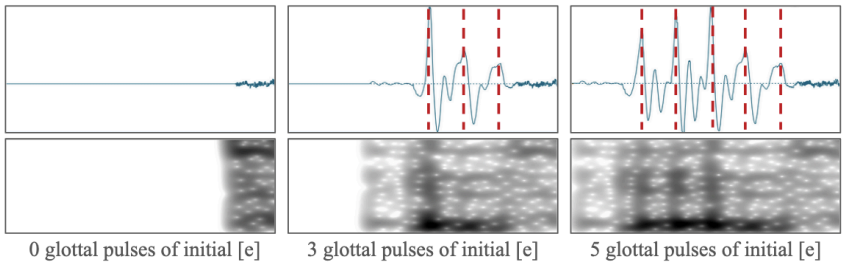


Figure 1. Examples of [esC] tokens at different durational gates

Hypotheses/predictions. We hypothesize that both **contextual cues** (H1) and **L2 proficiency** (H2) influence when listeners draw on L2 phonotactic knowledge in perceptual repair. We predict that in the **English accent/carrier conditions** participants will: i) perceive #esC less often when [e] is absent, as #sC is allowed in English (P1a); ii) perceive #sC even when a short duration of [e] is present, as #sC is more frequent in English than #esC (P1b). We predict that **high English proficiency** will correlate with higher #sC response rate across all conditions (P2).

Results. Response data were analyzed using a logistic generalized linear mixed-effects model with a dependent variable *response* (sC vs. esC) with by-subject random intercepts and by-subject random slopes for the effect of gate. Results of this analysis are summarized in figure 2. We find a significant effect of **LexTALE score** ($\beta = 0.90$, SE = 0.78, $p = .016$) and significant interactions of **gate × LexTALE** ($\beta = -0.62$, SE = 0.31, $p = .049$) and **gate × accent** ($\beta = -0.24$, SE = 0.09, $p = .013$). Specifically, responses were more sensitive to gate in the Spanish accent

condition (*Fig. 2b*) and for participants with higher LexTALE scores (*Fig. 2a*). Notably, we find no significant main effect of **carrier** and no significant interaction of **carrier x gate** (*Fig. 2c*).

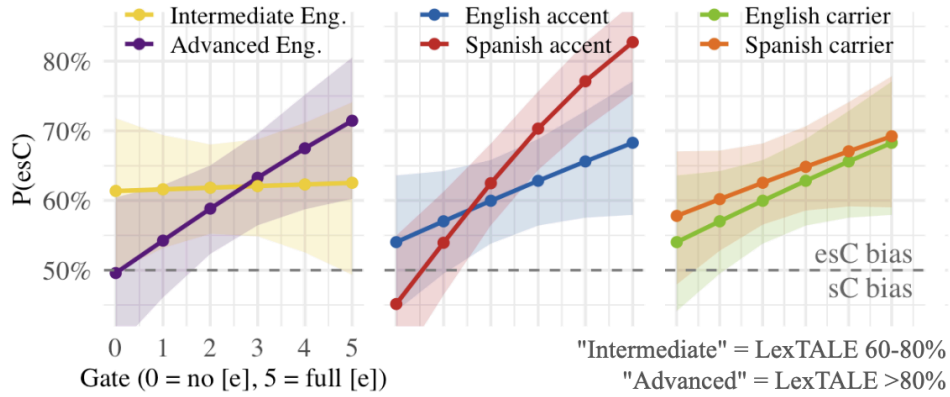


Figure 2. Predicted probability of *esC* response by gate across proficiency groups and conditions

Discussion. Results partially replicate the finding that illusory [e] perception is attenuated in Spanish-English bilinguals with higher English proficiency; in particular, there was a significantly lower proportion of esC responses for the "advanced English" (LexTALE > 80%) group at gate 0 (*Fig. 2a*). Additionally, rather than simply affecting overall bias, listeners showed significant differences in *sensitivity* to gate across the two proficiency groups (*Fig. 1a*) and accent conditions (*Fig. 2b*). Despite both #esC and #sC being phonotactically available in English, the Spanish accent condition showed higher sensitivity than the English condition. One potential explanation is that the pervasiveness of schwa in English leads listeners to treat extremely brief vocalic material as a plausible full vowel in English-accented speech, while an initial vowel of such short duration is not plausible in Spanish. Alternatively, listeners with English experience may have learned that Spanish-accented #sC productions can contain incidental vocalic material, raising their durational threshold for /e/ in Spanish-accented speech. Since English-accented speech lacks this articulatory property, any vocalic material in the English accent condition may be interpreted as more likely to be phonological /e/, especially in an experimental context where Spanish- and English-accented speech is juxtaposed. This difference in sensitivity is not seen across the carrier conditions (*Fig. 2c*), suggesting listeners may attend more closely to word-level cues than sentence-level contextual cues for perceptual repair. Alternatively, since the carrier always remained constant between the A and B stimuli in a given trial, listeners may have learned to ignore the carrier and attend only to the target.

Conclusion. Perceptual repair in Spanish-English bilinguals is influenced by multiple factors, including the talker's accent and the listener's overall English proficiency. Surprisingly, listeners were more discriminating between #sC and #esC in the Spanish-accented condition than the English-accented condition, despite only English allowing #sC. We suggest that this sensitivity may be driven by knowledge of English vowel characteristics (i.e., availability of /ə/) or by the social salience of prosthetic [e]- in Spanish-accented speech.

Selected references. **[1]** E.g., Dupoux et al. (1999): *Epenthetic vowels in Japanese: A perceptual illusion?* **[2]** E.g., Carlson et al. (2016): *Navigating conflicting phonotactic constraints in bilingual speech perception.* **[3]** Gibson (2012): *Perception-based vowel insertion by native Spanish-speaking learners of English.* **[4]** Tetzloff (2022): *Examining variability in Spanish monolingual and bilingual phonotactics: A look at sC-clusters.* **[5]** Lemhöfer & Broersma (2011): *Introducing LexTALE: A quick and valid Lexical Test for Advanced Learners of English*.

Yang Wang (yangx.wang@utah.edu) & Bruce Hayes (bhayes@humnet.ucla.edu)
University of Utah; UCLA

A learning algorithm for phonemicization in Stratal MaxEnt OT

Overview. In this study we apply the frameworks of MaxEnt OT (Goldwater and Johnson 2003) and Stratal OT (Bermúdez-Otero 1999, Kiparsky 2000) to address how classical phonemicization can be algorithmically learned. An example of classical phonemicization is the reduction of the set of similar English sounds {[tʰ], [t], [ˀt]} to a single underlying phoneme /t/, with the phonological grammar deriving the correct allophones on the surface.

The need for phonemicization. Since phonemicization has mostly been ignored in constraint-based frameworks, we mention three areas in which it appears to be applicable.

(1) Wilson and Gallagher (2018) demonstrate a problem with the phonotactic theory of Hayes and Wilson (2008): since it posits only nonconflicting Markedness constraints, it cannot account for "elsewhere" allophones. In Quechua, [e o] occur solely as uvular-adjacent allophones of /i u/. Classical OT describes this by letting context-specific markedness outrank general markedness (*IQ >> *E >> IDENT), but Hayes/Wilson's system has no access to this mechanism. While in principle one could replace *E with a long list of detailed constraints that expresses the complement set of *IQ (WG 612), this is unlikely to be an adequate answer, since the grammar ought to generalize to novel cases (say, introduced in loan words), and only fully-general constraints can do this. To express the problem in traditional terms: we want to ban [e o] completely at the phonemic level, allowing them back in Q-adjacent contexts, as allophones.

(2) Gouskova (2024) presents the same problem in a different guise. In Russian, [ɣ] exists only as an allophone of /x/, with the ranking AGREE(voice), IDENT(voice)/ __V, >> *ɣ. Thus schematic /noʃbi/ is realized as [noʤbi], with faithful realization of prevocalic [b] and the correct [ʤ] as an allophone. But since IDENT(voice) is ranked high, the grammar ends up treating [ʤ] as phonemic, wrongly allowing words like *[ʤa]. As Gouskova notes, we want to ban /ʤ/ at the phonemic level, but allow it back in as an allophone.

(3) Experimental evidence indicates that phonemes are cognitive categories. For instance, Gudschinsky et al. (1970) found that a Maxakali native speaker employed the appropriate phonemic letter (*c*) to spell an allophone that he had (by design) never encountered before, the [i̯] allophone of /k/. Jaeger (1980) trained participants to give a conditioned response whenever they heard the [kʰ] allophone of /k/; they gave the same response to the untrained allophone [ˀk].

The common basis of these problems is that they can be dealt with straightforwardly using the familiar apparatus of phonemes and allophones. This approach remains viable, but it is *not* implicit in classical OT, which by using Rich Base grammars can derive surface [noʤbi] just as easily from /noʤbi/ (with allophones in the UR) as from phonemic /noʃbi/.

Our approach. We designed a learning system which discovers allophonic patterning *after* the learning of general phonotactics, but *before* the discovery of paradigms and alternation (for evidence that this is true of real children, see Sundara et al. 2021). In our approach, allophonic variation is derived in a non-initial stratum of the type posited in Stratal OT; for similar work using strata for phonological learning see Nazarov and Pater (2017). The output of Stratum 1 consists solely of phonemes, with contextual allophones derived at Stratum 2. Thus, for Quechua the allophone [e] is banned at Stratum 1 by high-weighted *E. *IQ (high vowel before uvular) is enforced at Stratum 2, and surface [eq] is derived from /iq/ at Stratum 2 by making *iQ much stronger than either *E or IDENT(high).

Yang Wang (yangx.wang@utah.edu) & Bruce Hayes (bhayes@humnet.ucla.edu)
University of Utah; UCLA

Here are the learning steps. (1) Learning begins with discovery of a draft system of phonotactics, placed in Stratum 1; for this task we follow Hayes and Wilson (2008). For Quechua, general constraints like *VV would be installed at this level. (2) Stratum 2 starts out vacuous (no constraints, GEN limited to a single candidate identical to the Stratum 1 output), but is augmented during the learning of the phonemic system. (3) Phonemicization examines a series of hypotheses of the form "All B are derived from A," where A and B are a phonetically-similar pair. (4) To test each such hypothesis, an inviolable constraint *B is provisionally installed at Stratum 1, in effect removing B from the phoneme inventory. (5) Next, at Stratum 2, an environment-hunting algorithm seeks constraints of the form *A / P___Q; i.e. environments where only Bs and no As occur. These constraints, along with the Faithfulness constraints violated by A → B mappings, are provisionally installed in Stratum 2. (6) All constraints, in both strata, are provisionally reweighted to maximize the log likelihood of the data. (7) The last step is model comparison: the provisional grammar is adopted only if it outperforms the best earlier grammar on a metric incorporating the Bayesian Information Criterion (BIC). The latter is a model-selection metric that balances data fit and model simplicity. (8) The process of (3)-(7) is iterated, with further phoneme discovery, until no further improvements are possible. Our presentation will illustrate the implemented system with the phonemicization of Quechua [e o], using Wilson and Gallagher's lexical corpus. More briefly, we give modeling results for the Russian, Maxakali, and English cases noted above.

Simplicity in phonology. A lesson we learn from our simulations is the essential role of simplicity in phonemicization. Simplicity plays no role at all in classical OT, where every grammar is a ranking of the same universal constraint set. It is essential for *SPE* (Chomsky and Halle 1968), and for MDL approaches (Rasin et al. 2021). For us, simplicity is enforced by the BIC metric, and assures the correct outcome in two areas: (1) Choice of the underlying segment: Quechua must derive [e o] from /i u/ and not vice versa; (2) Avoidance of overphonemicization, e.g. highly complex grammars that derive English [ʒ] from /ð/ in dialects that lack minimal pairs for these sounds.

References

Bermúdez-Otero, R. 2000. *Constraint interaction in language change: quantity in English and Germanic*. Ph.D. U. Manchester.

Chomsky, N., & Halle, M. (1968). *The sound pattern of English*. New York: Harper and Row.

Hayes, B. & C. Wilson. 2008. A maximum entropy model of phonotactics and phonotactic learning. *LI* 39.379–440.

Goldwater, S. & M. Johnson. 2003. Learning OT Constraint Rankings Using a Maximum Entropy Model. *Proc. Stockholm Workshop on Variation within OT*, 111–120.

Gouskova, M. 2024. MSCs in positional neutralization: The problem of gapped inventories. *Phonology* 40229-265.

Gudschinsky, S. C., Popovich, H., & Popovich, F. (1970). Native reaction and phonetic similarity in Maxakalí phonology. *Lg*. 46:77-88.

Jaeger, J. 1980. Testing the psychological reality of phonemes." *Lg. & Speech* 23:233-253.

Kiparsky, Paul (2000) Opacity and cyclicity. *The Linguistic Review* 17:351-365.

Rasin, E., I. Berger, N. Lan, I. Shefi, & R. Katzir. 2021. Approaching explanatory adequacy in phonology using Minimum Description Length. *J. Language Modelling* 9:17-66.

Wilson, C. & Gallagher, G. 2018. Accidental gaps and surface-based phonotactic learning: A case study of South Bolivian Quechua. *LI* 49:610-623.

Anti-neutralization in Estonian

Scott Borgeson – Michigan State University – borges11@msu.edu

Estonian displays three degrees of segment length: short, long, and overlong. Within most noun classes, long vs. overlong is not contrastive; instead, overlength is derived when long segments are lengthened via compensatory lengthening (CL) (Borgeson 2022). For example, when the final vowels in (1) are deleted, their moras are preserved and resettled onto the underlined segments below, which become overlong.

(1)	a. /vi͟ːna-a/	→	vi͟ːːna-∅	"vodka-PART"	*viːnaː, *viːnna
	b. /li͟nna-a/	→	li͟nːna-∅	"city-PART"	*linnaː, *liːnna

While CL in Estonian can turn long segments into overlong ones in this fashion, it is not permitted to turn short segments into long ones. This has two effects. First, it prohibits the alternative patterns of lengthening in (1) (right) and thus forces CL to be long-distance in these words. Second, it blocks CL entirely in (2), forcing the dislodged mora to delete altogether. In short, CL in Estonian is *anti-neutralizing*: it cannot collapse an existing contrast (short vs. long), but can create a new one (long vs. overlong).

(2)	a. /ema-a/	→	ema-∅	"mother-PART"	*emaː, *emma, *eːma
	b. /jahu-a/	→	jahu-∅	"flour-PART"	*jahuː, *jahhu, *jaːhu

In this talk, I show that Comparative Markedness (CM; McCarthy 2003) can explain these facts quite easily. In this framework, every markedness constraint is split into two versions: old (M_{old}) and new (M_{new}). Old markedness constraints effectively punish marked structures that are original to the input and are preserved in the output; new ones punish those that were not originally present. This can explain the facts above as follows. We begin with *VV and *GEM, which demand that all vowels/consonants be short, but do not distinguish between long and overlong segments. Ranking $*VV_{new}$/$*GEM_{new}$ above MAX-μ suffices to ban the creation of new long segments in Estonian, and instead force moras to delete (as in 2). At the same time, ranking $*VV_{old}$/$*GEM_{old}$ fairly low will not force existing long segments to shorten, and will even permit them to lengthen so as to avoid mora deletion (as in 1).

I consider two alternative approaches and show that these encounter insurmountable problems. The first is the Positional Faithfulness account of Borgeson (2022). In this approach, long vowels and consonants are prohibited in general, but are protected within primary-stressed syllables specifically. This is enforced via the following constraints.

(3)

a.	MAX-$\mu(\acute{\sigma})$: Do not delete moras from $\acute{\sigma}$.
b.	*VV: Vowels must be monomoraic.
c.	*GEM: Consonants must belong to a single syllable.
d.	MAX-μ: Do not delete moras.

Ranking (3b-c) $\gg$ (3d) will force moras to delete rather than create new long segments; (3a) $\gg$ (3b-c) will protect existing ones within primary-stressed syllables. This system works for (1-2), but gets (4) wrong: if preserving all moras in the primary-stressed syllable is paramount, then the expected output of (4a) is *[ˈkiːri] or *[ˈkirri]. Instead, the final glide vocalizes, and the preceding heavy syllable becomes light: [kiri]. Thus, this account fails. (The CM approach succeeds, since nothing in it prohibits syllable shortening.)

(4)

a.	/kirja/	→ kiri	"book.NOM"
b.	/varju/	→ vari	"shadow.NOM"

Alternatively, we might invoke an anti-neutralization constraint such as NONEUTR(ALIZATION) in Kiparsky (2008), which prohibits any input:output mapping for which a better input could be found. In this view, a derivation like /ema-a/ → *[emma] is forbidden because the output [emma] is better derived from an input like /emma/, which possesses a long consonant underlyingly, than it is from /ema/, which does not. With /ema-a/ → *[emma] thus ruled out, /ema-a/ must surface as [ema]. The problem is that this *also* neutralizes a lexical contrast, namely between /ema-a/ and /ema-∅/ "mother (PART/NOM)," which both surface as [ema]. Crucially, then, NONEUTR cannot differentiate between the illicit lengthening in *[emma] and the attested deletion-*sans*-lengthening in [ema] because both outputs have better inputs. As a result, the decision falls to MAX-μ, which favors *[emma]. The anti-neutralization approach, then, also fails, and in the absence of other alternatives, we are forced to adopt a solution couched in Comparative Markedness.

Borgeson, S. (2022). *Long-distance compensatory lengthening* [Doctoral dissertation]. Stanford University.

Kiparsky, P. (2008). Fenno-swedish quantity: Contrast in Stratal OT. *Rules, constraints, and phonological phenomena*, *185219*.

McCarthy, J. J. (2003). Comparative markedness. *Theoretical linguistics*, *29*(1-2), 1–51.

Discriminating Allophones: Impacts of Acoustic, Contextual, and Semantic Factors

Shuhao Zhang and Youngah Do
The University of Hong Kong

Previous research indicates that listeners exhibit reduced sensitivity to the acoustic distinction between allophones compared to phonemes (Boomershine et al., 2008). This reduction in sensitivity is likely attributable to contextual and semantic cues. For allophones in complementary distribution (e.g., [tʰ] and [t] in English), high predictability enables listeners to depend on contextual information rather than solely on acoustic details (Noguchi & Kam, 2018; Peperkamp et al., 2003). Allophones in free variation (e.g., [w] and [v] in Beijing Mandarin) are also more difficult to distinguish than contrastive phonemes (e.g., /w/ and /v/ in English) because they map onto the same semantic referent (Barrios et al., 2023). In addition, acoustic distance serves as a fundamental constraint, as sounds with low acoustic proximity are less likely to be perceived as allophones (Peperkamp et al., 2006; Seidl & Cristia, 2012). The present study employs a supervised contrastive learning framework to simulate the perception of phonemes, allophones in complementary distribution, and allophones in free variation across varying levels of acoustic distance, to evaluate how contextual and semantic cues interact with acoustic distance in influencing perceptual discrimination.

The stimuli consist of CV syllables represented as 2 × 16 acoustic vectors. Ten pairs of voiceless fricatives differ in the center-of-gravity (CoG) distance and are combined with eight vowels as contexts. Three sound types are included: (1) phonemes, characterized by overlapping distributions and distinct referents for sounds within the same vowel context; (2) allophones in complementary distribution, which do not co-occur with the same vowel context; and (3) allophones in free variation, which maintain overlapping distributions but map to the same semantic category within identical vowel contexts. To keep complexity comparable across conditions, we introduce an additional sound pair consisting of a voiced counterpart and a voiced affricate counterpart of one of the target sounds, which is unrelated to the CoG dimension. This additional pair is contrastive when the target sounds are in the allophonic conditions but is non-contrastive (i.e., in complementary distribution or in free variation) when the target sounds are contrastive. We also include related pairs in the allophonic conditions where the additional contrastive sounds are the voiced counterpart of the target sounds with the same CoG distance, allowing us to examine the influence from other sound pairs in the language. This design yields 24 categories for each training condition, with 1,000 tokens per vowel per pair of sounds, totaling 16,000 tokens per condition.

The core encoder uses a fully connected neural network architecture, in which a flattening layer is followed by two linear layers with ReLU activations, projecting the input vectors into a four-dimensional feature space. Training is optimized using a supervised contrastive loss (Khosla et al., 2020), based on normalized temperature-scaled cosine similarity. A novel vowel excluded from the training set is utilized to combine with the target fricatives as test stimuli. After training, the encoder generates the four-dimensional feature embeddings for the test data. Discrimination performance is quantified using the silhouette score, which measures the degree of class separation in the embedding space. A higher silhouette score reflects greater perceptual distance. Each condition is repeated ten times with independently initialized models to reduce the influence of randomness. A linear regression model is then used to examine the effect of conditions on silhouette scores, followed by post-hoc pairwise comparisons.

Silhouette scores for each condition are displayed in Figure 1. The results demonstrate that, relative to phonemes, both contextual and semantic cues significantly reduce discrimination performance in the absence of a related pair (e.g., at steps 1 to 8, all p values $< .01$). Phonemes overall exhibit the best discrimination, while allophones in free variation show the weakest, with allophones in complementary distribution falling in between (e.g., at steps 5 to 8, all p values $< .05$). Notably, the influence of complementary contexts diminishes

as acoustic distance increases (e.g., at step 9, p = .184), indicating that acoustic cues can mitigate the perceptual attenuation caused by contextual predictability. In contrast, the impact of semantic cues on allophones in free variation remains robust and is harder to modulate with acoustic distance (e.g., p < .001 even at step 10). This pattern suggests that semantic cues exert a stronger and more persistent influence than contextual cues.

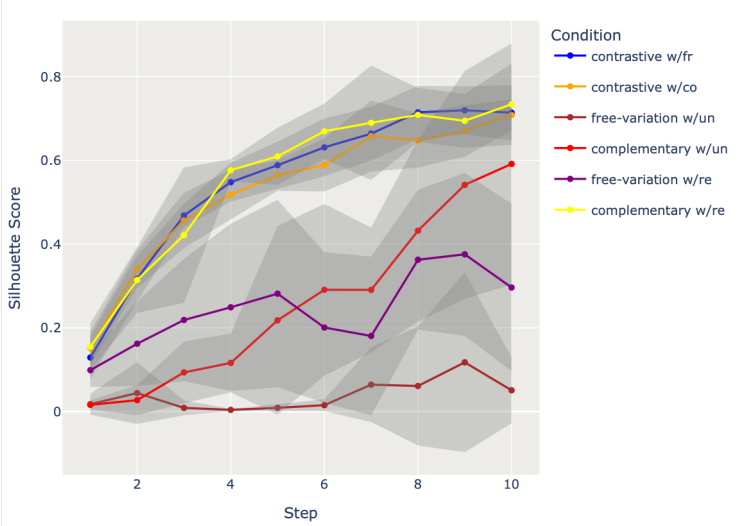


Figure 1 (Left) Silhouette scores across conditions. The labels 'contrastive w/fr' and 'contrastive w/co' refer to contrastive-phoneme conditions with an unrelated additional pair in free variation and in complementary distribution, respectively. The labels 'w/un' and 'w/re' refer to the non-contrastive conditions with an unrelated additional contrastive pair and a related additional contrastive pair, respectively. The y-axis depicts the silhouette score, while the x-axis indicates the acoustic distance of the two target sounds, referred to as 'step'.

In the non-contrastive conditions, adding related pairs with the same CoG distance significantly improves discrimination (all p values < .015 except for the free-variation condition at step 7 and the complementary condition at steps 9 and 10). Allophones in complementary distribution can approach contrastive performance through generalization (all p values > .05), whereas allophones in free variation improve yet remain below contrastive levels (all p values < .001 at steps 2 to 10), underscoring the pull of semantic information. Nevertheless, the improvement observed for allophones in free variation indicates that, even when the model is explicitly trained to categorize sounds together via cosine similarity in terms of semantic meaning, acoustic discrimination remains possible via Euclidean distance in the embedding space.

Taken together, the results indicate that allophonic discrimination is weakened by both contextual and semantic factors, with semantic cues exerting a more robust effect. Acoustic distance appears to modulate the influence of contextual cues more strongly, such that the reduced discrimination associated with complementary distribution is gradually diminished as the acoustic separation between sounds increases. Furthermore, related contrasts that share phonetic properties facilitate discrimination, suggesting a role for generalization across the sound system. These findings highlight speech perception as a cue-integration process in which acoustic, contextual, and semantic factors jointly shape the perceptual distance between sounds.

References

Barrios, S. L., Rodriguez, J. M., & Barriuso, T. A. (2023). The Acquisition of L2 Allophonic Variants: The Role of Phonological Distribution and Lexical Cues.

Boomershine, A., Currie Hall, K., Hume, E., & Johnson, K. (2008). The Impact of Allophony Versus Contrast on Speech Perception.

Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., & Krishnan, D. (2020). Supervised Contrastive Learning.

Noguchi, M., & Kam, C. L. H. (2018). The Emergence of the Allophonic Perception of Unfamiliar Speech Sounds: The Effects of Contextual Distribution and Phonetic Naturalness.

Peperkamp, S., Le Calvez, R., Nadal, J.-P., & Dupoux, E. (2006). The Acquisition of Allophonic Rules: Statistical Learning with Linguistic Constraints.

Peperkamp, S., Pettinato, M., & Dupoux, E. (2003). Allophonic Variation and the Acquisition of Phoneme Categories.

Seidl, A., & Cristia, A. (2012). Infants' Learning of Phonological Status.

The Malleability of Phonotactics in Processing

Alessandra Pintado-Urbanc[a] and Jason Shaw[a]
[a]*Department of Linguistics, Yale University, New Haven, Connecticut*

Overview: There is evidence that categorical phonotactic constraints present a barrier for processing, causing production difficulty and perceptual illusion[1,2,3]. Gradient phonotactics are also known to influence processing, i.e., in shadowing tasks, where high phonotactic probability nonwords are repeated faster and more accurately than low phonotactic probability nonwords[4]. However, phonotactic probabilities play an opposing role in learning. In word learning tasks, high probability sequences are more difficult to learn than low probability sequences[5]. This suggests that the barrier that low probability sequences present to processing is *short-lived*. Our work seeks to account for the malleability of gradient phonotactics on processing. We pursue the hypothesis that surprisal from low probability phonotactics, which aids in word learning, also plays a faciliatory role in processing over multiple repetitions. Specifically, we propose that the neural resources that ultimately facilitate learning of low probability sequences are first engaged to facilitate processing. We test this hypothesis in an adapted nonword shadowing experiment: items vary in phonotactic probability (as in previous work[4]) but biphones are repeated across multiple items. We predict that, with multiple repetitions, low probability items will gain a processing advantage over high probability items. This prediction derives from a neural process model, which implements our proposal in the framework of Dynamic Field Theory[6] (DFT). After describing model simulations, we test the predictions in two case studies, one using gradient phonotactics in English and the other using categorical phonotactics in Spanish with Spanish-English bilinguals.

DFT Simulations: We represent phonotactics as the coupling strength between neural nodes (representing phonemes). Higher levels of linguistic organization are also represented as nodes, which are excitatorily coupled to constituent phonemes. For example, a word node excites the phonemes that compose it. This configuration, representing wordhood with an additional node and phonotactic probability as coupling strength, derives two main findings of Vitevitch & Luce (1998): (1) real words have a faster speech onset time than nonwords and (2) high probability nonwords have faster speech onset times than low probability nonwords (Fig. 1). The same model derives new predictions for our case studies: adding the higher-level node to low probability sequences facilitates speech onset (3), beyond high probability sequences.

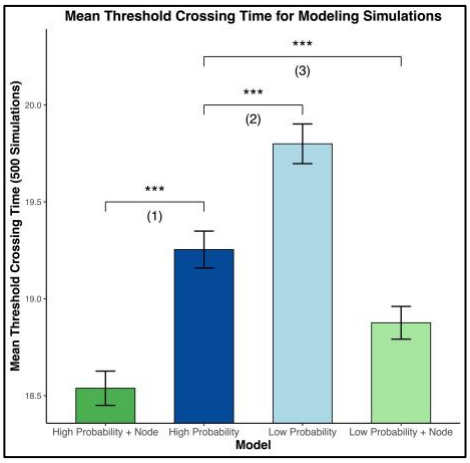


Figure 1: DFT Modeling Simulations

Experimental Methods: We conducted an adapted shadowing task with 28 participants (24F; 2M; 2NB; mean age=21), consisting of 12 English monolingual and 16 Spanish-English simultaneous bilingual speakers. Participants heard an English nonword and repeated it back as quickly and accurately as possible. Two sets of materials tested gradient phonotactics. Both cases

repeated target biphones 10 times throughout the experiment, affording the opportunity to recruit the neural resources (additional node) hypothesized to facilitate processing for low probability sequences. Speech onset times were automatically measured for each trial based on a threshold of acoustic intensity with a subset (11%) manually checked in Praat. Linear mixed-effects models were fit to speech onset time with random intercepts for subject and item, and fixed effects for language group (monolingual or bilingual), control variables (trial order, stimulus duration), and condition of interest: phonotactic probability (Case I) and Spanish well-formedness (Case II).

Case I – Gradient Phonotactics in English: Materials consisted of 226 bisyllabic nonwords, which varied in the pointwise mutual information (PMI) of the initial bigram: 113 high probability (PMI: -0.71 – 3.12, M = 1.20) and 113 low probability (PMI: -4.43 – 1.32, M = -1.43). The second syllable was sampled from a set of 10 high probability CVC sequences (Example nonwords – High Probability: 'lipis', 'pitin', 'tusik'; Low Probability: 'fipis', 'mutin', 'susik'). All nonwords were produced once, but the set of initial biphones (e.g., 'li', 'pi', 'tu', etc.) were repeated 10 times across nonwords that differed by virtue of the second syllable.

Case II – "Diluted" Categorical Phonotactics: Materials consisted of 60 nonwords, beginning in sC clusters which are illicit in Spanish (e.g., 'spupis', 'skitin', 'slusik'). We assume that simultaneous bilingual speakers have shared representations for overlapping phonemes across their languages such that phonotactic probabilities between these segments are influenced by *both* languages[7]. Thus, we expect phonotactic violations that are categorical for Spanish monolinguals to be gradient, i.e., "diluted", for our bilinguals, offering another way to test gradient phonotactics.

Results: For Case I, there was a significant effect of phonotactic probability: native English speakers produced low probability nonwords faster than high probability nonwords (β=-0.021, $p<0.001$). There was no interaction with language group. For Case II, there was a significant interaction between condition and language (β=0.014, $p<0.05$). Post-hoc comparisons (Tukey-adjusted) confirmed that speech onset times were faster for Spanish Illicit nonwords than for high probability nonwords but only for bilinguals (β=0.026, $p<0.05$) and not the monolingual speakers (β=0.013, $p=0.77$). Thus, across both cases, low probability nonwords, whether due to gradient phonotactics in English or "diluted" categorical phonotactics showed a processing advantage.

Conclusions: Both case studies suggest that speakers rapidly overcome low phonotactic probability sequences. These sequences have been shown to hinder processing, but, with repeated exposure they become even easier to process than high probability items. These results support our hypothesis that the brain responds to high surprisal by committing neural resources that facilitate processing, ultimately facilitating learning. This neural process may function to balance phonotactic entropy, aiding in the acquisition of low phonotactic probability lexical items.

References: [1] Dupoux, E., Kakehi, K., Hirose, Y., Pallier, C., & Mehler, J. (1999). Epenthetic vowels in Japanese: A perceptual illusion? Journal of Experimental Psychology: Human Perception and Performance. **[2]** Carlson, M. T., Goldrick, M., Blasingame, M., & Fink, A. (2016). Navigating conflicting phonotactic constraints in bilingual speech perception. **[3]** De Leeuw, E., Stockall, L., Lazaridou-Chatzigoga, D., & Gorba Masip, C. (2021). Illusory vowels in Spanish–English sequential bilinguals: Evidence that accurate L2 perception is neither necessary nor sufficient for accurate L2 production. Second Language Research. **[4]** Vitevitch, M. S., & Luce, P. A. (1998). When words compete: Levels of processing in perception of spoken words. **[5]** Storkel, H. L., Armbrüster, J., & Hogan, T. P. (2006). Differentiating Phonotactic Probability and Neighborhood Density in Adult Word Learning. **[6]** Schöner, G., & Spencer, J. (2016). Dynamic thinking: A primer on dynamic field theory. **[7]** Freeman, M. R., Blumenfeld, H. K., & Marian, V. (2016). Phonotactic Constraints are Activated across Languages in Bilinguals.

Children exhibit domain-narrowing in a variable phonological rule

Jessica Shepherd & Betsy Sneller

Introduction: Language change – including phonological change – on the community level occurs via intraspeaker variability, where individual speakers show variable application of the new process [1, 2]. Stratal frameworks of phonology and phonological change [3, 4] provide an additional theoretical expectation, which is that a phonological rule will enter a language variety first as a post-lexical (phrase-level) process. Over time (presumably across generations), this new rule will undergo domain narrowing and begin to apply earlier in the derivation: as a word-level process and ultimately as a stem-level process. In this paper, we provide evidence that children take a process available as a variable phrase level rule in their input data (but not variable in their word-level input) and produce output that shows variability applying at word level.

Background: Post-tonic /t/ (where /t/ occurs between a stressed and unstressed syllable) is the context that produces flapping in Americal Englishes, as in *butter* [bʌɾɚ]. However, flapping is not licit in post-tonic /t/ that precedes an alveolar nasal (*button*). Instead, the standard American English pronunciation of these forms is as [bʌʔn̩] or [bʌʔən]. In other words, the glottal replacement bleeds flapping in these contexts. However, as a phrase-level phenomenon, flapping is variably licit even for speakers who do not allow flapping in stem-level *button*. Contexts such as *ge**t** in the car* or *is the table**t** on* produce a post-tonic /tən/, which can be realized as [ɾ] or [ʔ] [6]. Additionally, the status of glottal replacement bleeding [ɾ] even in stem-level contexts like *button* seems to be weakening. Recent work using experiments [7] and large-scale naturalistic corpora [8] has found some younger speakers variably producing both [ʔ] and as [ɾ] in stem-level *button* words (both variants can occur either with a following [n̩] or a following [ən]).

We hypothesize that this cross-generational change in post-tonic /tən/ occurs via variability which begins first as a phrase-level process, which can then undergo the life cycle of phonological change and narrow in domain. In other words, we hypothesize that cross-generationally, first the glottal replacement rule becomes variable as a phrase-level process, enabling variability between *ge*[ʔ] *in the car* and *ge*[ɾ] *in the car*. Variability in this context on the phrase level in the input in turn enables the next cohort of speakers to allow variability in the glottal replacement rule in word level contexts. This weakening of the previously inviolable glottal rule will appear in the data as flapping variably being produced at both phrase and word level contexts. We hypothesize further that variable glottal rule at the word level is what has enabled the next cohort of speakers to weaken the glottal rule at stem level, resulting in variable [bʌʔən] / [bʌɾən]. pronunciation.

Results and Discussion: Here, we use data from the Providence corpus [9] to analyze how children's production of post-tonic /tən/ compares with their caregivers' inputs. We analyze four children from recordings at a range of ages: Lily (1;3-4), Ethan (1-2;11), Naima (1-3;10), William (1;6-3;4). Given the robust variability across varieties of American English in phrase-level post-tonic /tən/ (*ge*[ʔ] *in the car* and *ge*[ɾ] *in the car*), we assume variability in the input from the parents at phrase level.

We analyze word-level contexts, comprised primarily of words like *gettin'*. We note that these contexts must first be derived from the morphophonological variability between *-ing* [ɪŋ] and *-in* [ɪn]. We analyzed 3489 tokens of word-level post-tonic /tən/ (primarily consisting of *-in'* forms): 297 William + Mother; 1014 Lily + Mother; 362 Ethan + Mother; 1624 Naima + Mother. Speakers' realization of /t/ was auditorily coded as either deleted, flapped [ɾ], glottal [ʔ], or full release [tʰ]. Figure 1 shows the results for all children and their accompanying caregiver. Notably, most caregivers produce either categorical [ʔ] or sometimes a full burst release [tʰ] (William's parent). Every child, however, produces variation between glottal stops and flaps.

We interpret this data as evidence supporting a stratal life cycle of phonological processes. In this case, the new phonological process is the weakening of the glottal replacement rule, which opens the door for variability between the more broadly-applied flap and the existing glottal replacement rule. Children reanalyze their parents phrase-level variability as a process that can be variable at the word level as well, demonstrating domain narrowing.

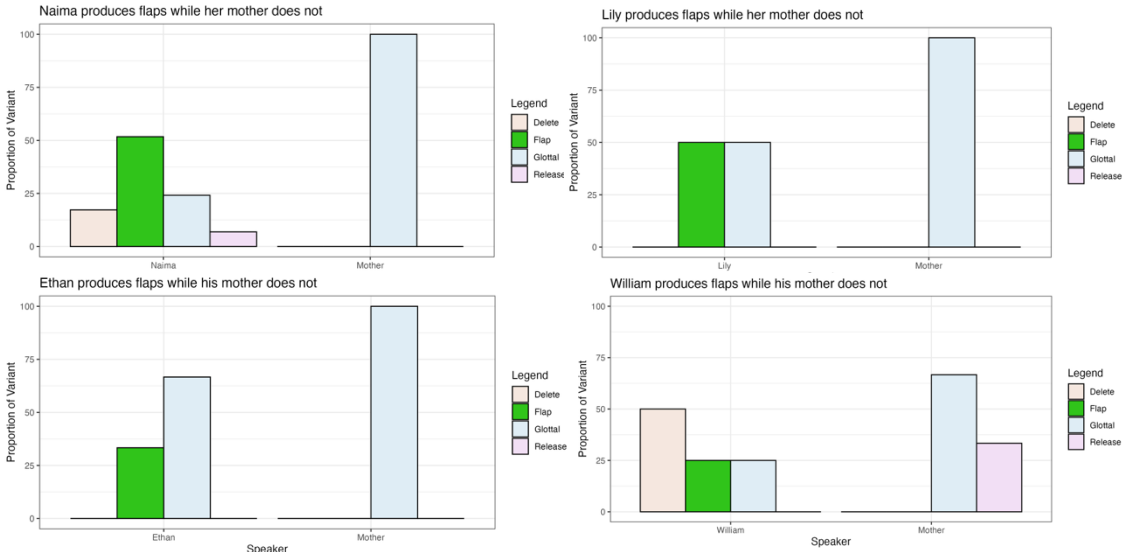


Figure 1: Child (left) production of /t/ in word-level forms (*gettin'; gotten*) compared to the input from their caregiver (right).

[1] **Weinreich, U., Labov, W., & Herzog, M**. **(1968)**. Empirical Foundations for a Theory of Language Change. In W.P. Lehmann & Y. Malkiel (Eds.) *Directions for Historical Linguistics: A symposium*. University of Texas Press. 95-195. [2] **Shepherd, J. (2025, November 7).** *Paths of sound change in the [mirən]: /rən/ in two varieties of Michigan English*. Paper presented at NWAV 53. Ann Arbor, Michigan. [3] **Bermúdez-Otero, R. & Trousdale, G. (2012)**. Cycles and continua: On unidirectionality and gradualness in language change. In T. Nevalainen & E. Closs Traugott (Eds.), *The Oxford handbook on the history of English*. 691-720. [4] **Bermúdez-Otero, R. (2015).** Amphichronic explanation and the life cycle of phonological processes. In P. Honeybone & J. C. Salmons (Eds.). *the Oxford handbook of historical phonology*. 374-399. [5] **Kiparsky, P. (2015).** Stratal OT: A synopsis and FAQs. In Y. E. Hsiao & L.-H. Wee (Eds.). *Capturing Phonological Shades*. 2-44. [6]. **Eddington, D. & Taylor, M. (2009).** T-Glottalization in American English. *American Speech*. 84(3). 298-314. [7]. **Repetti-Ludlow, C. (2026)**. The Realization of /t/ and /en/ in Words Like *button:* A Change in Progress on Long Island. *American Speech.* 101(3). 3-19. [8] **Author citation**. [9]. **Demuth, K., Culbertson, J., Alter, J. (2006)**. Word-minimality, Epenthesis and Coda Licensing in the Early Acquisition of English. *Language & Speech* 49(2). 137-173.

Integrating nominals into a metrical analysis of Kuria tone

Kai Schenck

University of California, Berkeley

Kuria, an interlacustrine Bantu language spoken in Kenya and Tanzania, is known for its complex melodic tone system that assigns inflectional H to succeeding moras of the verbal macrostem, up to the fourth mora (Mwita 2008, Marlo et al. 2015). The five patterns in affirmative tensed verbs are shown in (1). The macrostem is in brackets and the mora H is assigned to is underlined; the H in these tenses regularly spreads to the penult:

(1) Kuria affirmative verbal tone patterns (Mwita 2008).

1μ	Untimed Past Anterior	n-to-o-[tú̲rúúŋ̲ána]	'indeed we have already welcomed'
2μ	Hodiernal past progressive	n-to-oɣa-[turú̲úŋ̲áíni]	'we have been welcoming'
3μ	Remote future	n-to-ɾe-[turuú̲ŋ̲ána]	'we will welcome then'
4μ	Inceptive 1st/2nd person	to-ɾa-[turuuŋ̲á̲na]	'we are about to welcome'
1μ4μ	Inceptive 3rd person	a-ɾa-[tú̲ruuŋ̲á̲na]	'he is about to welcome'

Phonology has long been claimed to not be able to count beyond two (McCarthy & Prince 1986, among others; see also Paster 2019). Although ternary patterns are known in metrical and tonal phonology and can be accounted for with binary-branching structures (e.g. Martínez-Paricio & Kager 2015), the 1st/2nd person inceptive in Kuria is one of few examples of a quaternary pattern (Paster 2019). Accounting for it and the rest of Kuria's verbal tone system is a significant challenge for phonological theory, and as such previous accounts of Kuria tone typically focus almost exclusively on the elaborate verbal tone system (Mwita 2008, Marlo et al. 2015, Sande et al. 2020, Rolle & Lionnet 2020, Trommer 2022). However, the nominal tone system of Kuria has largely not been integrated into previous analyses, despite data being available and potentially providing crucial information on how to best model Kuria tone assignment. Examining the nominal tone data in Cammenga (2004) and Mwita (2012) reveals three key facts:

1. In isolation, Kuria contrasts underlyingly ∅-toned and H-toned nominal stems, unpredictable based on any phonological or morphological variable. One H tone is introduced automatically for nouns, resulting in ∅-toned stems having single H tone and H-toned stems having two independent H tones. For both types, the rightmost H spreads to the penult.
2. H-toned Kuria nominal stems match the μ1μ4 pattern of certain verb tenses, indicating that this pattern is the default when two H tones are present in the prosodic word, and should be accounted for as part of Kuria's default phonology. The examples in (2) show an ∅-toned noun, H-toned noun, and a verb in the 1μ4μ hodiernal future uncertain tense:

(2)

∅-toned noun		H-toned noun		1μ4μ verb	
e-ke-[hó̲óɾe]	'skull, oath'	o-mo-[hó̲óɾó̲]	'game'	ŋ-kaa-[βí̲imá̲]	'_ measure'
ee-n-[tɔ̲́βɛ́kɛ]	'porridge'	a-ma-[tá̲ɣítɔ̲̌]	'mud, clay'	ŋ-gaa-[té̲ɾɾékă̲]	'_ brew'
a-βa-[hó̲ɾééɾu]	'gentle people'	ee-ŋ-[gósiiɾá̲]	'hat'	ŋ-kaa-[hágaacá̲]	'_ build'
u-βu-[tí̲rímíku]	'calmness'	i-ɾi-[tɔ̲́ɾátoɾá̲]	'solon plant'	ŋ-gaa-[té̲rémeká̲]	'_ be calm'

3. In addition, alternations in nominal stems reveal that at least two L-toned moras must precede the first mora of the H-tone assignment domain, clearest in the class 9/10 alternations in (3):

(3)

Singular	Plural	Gloss	
e-∅-[haɾá̲ma]	i-ci-[há̲ɾáma]	'spot, mark'	(C. 110)
e-∅-[hamú̲ɾi]	i-ci-[há̲múɾi]	'command'	(C. 110)
e-∅-[hecé̲éɾi]	i-ci-[hé̲cééɾi]	'monitor lizard'	(C. 110)

However, when more than two moras precede the stem, the H surfaces on the first stem mora and not the third mora of the entire complex, evident in the alternations in (4):

(4)

Singular	Plural	Gloss	
o-ɾo-[βáɣɔ]	i-ciim-[báɣɔ]	‘hedge’	(M12. 160)
i-ɾii-[ɣéna]	a-ma-[ɣéna]	‘stone’	(M12. 160)

The principles governing nominal tone assignment in Kuria are: 1) at least two moras must precede the tonal assignment domain, 2) the leftmost H must be assigned to the first mora of the tonal assignment domain, and 3) two moras must intervene between the leftmost and rightmost H. This indicates that prosodic structure interacts with tonal assignment in nominals, and that an analysis incorporating abstract metrical structure, previously disregarded for Kuria (Hyman 1989, Marlo et al. 2015, Paster 2019), provides a parsimonious explanation of these facts, and accounts for the aspects of nominal tone assignment in a way that previous constraint-based analyses of solely verbal tone assignment (Sande et al. 2020, Rolle & Lionnet 2020, Trommer 2022) do not. I present an analysis of Kuria nominal tone assignment in Cophonologies by Phase (Sande et al. 2020):

1. Kuria n parses material in its spell-out domain into a prosodic word ω. The default 1μ4μ pattern reveals that Kuria nominal stems are parsed into maximally ternary trochaic layered feet (Martínez-Paricio & Kager 2015) without regard for syllable integrity.
2. Kuria D parses the existing prosodic word as well as the augment and class prefix. The alternations seen in nouns are due to a constraint compelling a foot to coincide with the left boundary of a phonological phrase ϕ, which may overwrite the existing prosodic structure.

The metrical Cophonologies by Phase analysis to Kuria verbal tenses can derive each of its tonal patterns by assuming that tonal autosegments may be introduced by heads at different phasal domains and that tonal autosegments license maximal feet. I also derive the 4μ tenses, assuming a phonological L is introduced in the 1st/2nd person at T. Although other analyses have accounted for Kuria verbal tone without mora counting (Rolle & Lionnet 2020, Trommer 2022), this analysis provides a more cohesive account of nominal *and* verbal tone. The CbP derivation for the verb stem [hootootɛra] ‘reassure’ in the major melodic patterns is below, with each head shown:

Pattern	Mood	Focus	C
1μ	[$_\phi$ [$_\omega$ (hoo)tootɛra	[$_\phi$ [$_\omega$ ((hóó)to)otɛra	[$_\phi$ [$_\omega$ ((hóó)tó)ótéɾà
3μ	[$_\phi$ [$_\omega$ (hoo)((tóó)te)ɾa	[$_\phi$ [$_\omega$ (hoo)((tóó)te)ɾa	[$_\phi$ [$_\omega$ (hòò)((tóó)té)ɾà
1μ4μ	[$_\phi$ [$_\omega$ (hoo)((tóó)te)ɾa	[$_\phi$ [$_\omega$ ((hóó)tó)((óte)ɾa)	[$_\phi$ [$_\omega$ ((hóó)tó)((óté)ɾà)
4μ	[$_\phi$ [$_\omega$ (hoo)((tóó)te)ɾa	[$_\phi$ [$_\omega$ ((hòò)tò)((óte)ɾa)	[$_\phi$ [$_\omega$ ((hòò)tò)((óté)ɾà)

References: **Cammenga**, Jelle. 2004. *Igikuria Phonology and Morphology: A Bantu Language of South-West Kenya and North-West Tanzania*. Cologne: Rüdiger Köppe **Hyman**, Larry M. 1989. Accent in Bantu: An Appraisal. *SLS* 19(2). 116–134 **Marlo**, Michael R., Leonard Chacha **Mwita** & Mary **Paster**. 2015. Problems in Kuria H tone assignment. *NLLT* 33(1). 251–265 **Martínez-Paricio**, Violeta & René **Kager**. 2015. The binary-to-ternary rhythmic continuum in stress typology: layered feet and non-intervention constraints. *Phonology* 32(3). 459–504 **McCarthy**, John J. & Alan **Prince**. 1986. Prosodic morphology. Ms. **Mwita**, Leonard Chacha. 2008. *Verbal Tone in Kuria*. UCLA **Mwita**, Leonard Chacha. 2012. Noun Tonology in Kuria. *IJHSS* 8(2). 159–168 **Paster**, Mary. 2019. Phonology counts. *Radical* 1. 1–61 **Rolle**, Nicholas & Florian **Lionnet**. 2020. Phantom structure: A representational account of floating tone association. In *Proceedings of AMP 2019*. Washington, DC: LSA **Sande**, Hannah, Peter **Jenks** & Sharon **Inkelas**. 2020. Cophonologies by Ph(r)ase. *NLLT* 38(4). 1211–1261 **Trommer**, Jochen. 2022. The stratal structure of Kuria morphological tone. *Phonology* 39(4). 679–723

Effects of Lexical Frequency and Movement Direction on Metathesis in American Sign Language

Yadi Peng

Background: Metathesis plays a key role in the phonological variation of words. The phenomenon is well-researched in spoken languages but rarely discussed in the context of signed language phonology. In North American English, consonant [ɹ] and vowel [ə] can be switched in words beginning with 'pre' or 'pro' (e.g. [pɹəpoʊz] ~ [pəɹpoʊz] 'propose'). Similarly, in American Sign Language (ASL), the initial and final locations of some signs (e.g., DEAF) can be switched. Previous research has explored predictors of metathesis; lexical frequency and movement direction were proposed but remain untested (Liddell & Johnson, 1989; Lucas, 2014). Building on Lucas (2014) and Bybee (2007), I hypothesize 1) higher-frequency words tend to undergo metathesis more often and 2) compared to signs with vertical movements, those with horizontal movements tend to undergo metathesis more often in ASL. This study shows that lexical frequency does not predict metathesis; horizontal signs tend to be metastasized more often, especially when produced on the chin.

Methods: Using corpus data from ASL Citizen (Desai et al., 2023) and ASL-LEX 2.0 (Sehyr et al., 2021), I built two mixed logistic regression models using 719 isolated tokens of 24 metathesizable signs from 44 signers. In the first model, the fixed effect was lexical frequency coded as a categorical variable (high, medium, or low); in the second model, the fixed effect was movement direction coded as a categorical variable (horizontal or vertical). In both models, the outcome was "metathesis" or "no metathesis", a binomial variable. Sign and subject were the random effects.

Results: Five signs did not exhibit metathesis: HOME, DORMITORY, PARENTS, TEENAGER, STAFF. The differences between including and excluding them in the models are substantial. When included, results from the frequency model suggest a weak negative association (β = −0.49, p = .091, 95% CI [−1.065, 0.079]). When excluded, the effect of word frequency is gone (β = −0.31, p = .184, 95%CI [−0.765, 0.147]). For the movement-direction models, including the five signs, direction significantly predicted metathesis occurring more often (β = −2.327, p = .006**, 95%CI [−4.040, −0.614]). Excluding them makes direction not matter (β = −0.43, p = .634, 95%CI [−2.209, 1.335]).

Meanwhile, sign, as a random effect, contributed substantially to model variance. Figure 1 shows that ARIZONA, TWINS1, TWINS2, RESTAURANT, and CAFETERIA are the most metathesis-prone signs, all produced around the chin.

Conclusion: Lexical frequency only has a weak association with the probability of metathesis in ASL, which challenges long-held assumptions of lexical frequency's effects on sound change in spoken languages. Movement direction can predict

metathesis rate. More specifically, signs at the chin are the most metathesizable, suggesting that, even though signers perceive “landmarks” on the face as distinctive locations, they do not distinguish between the left and right sides of the landmarks.

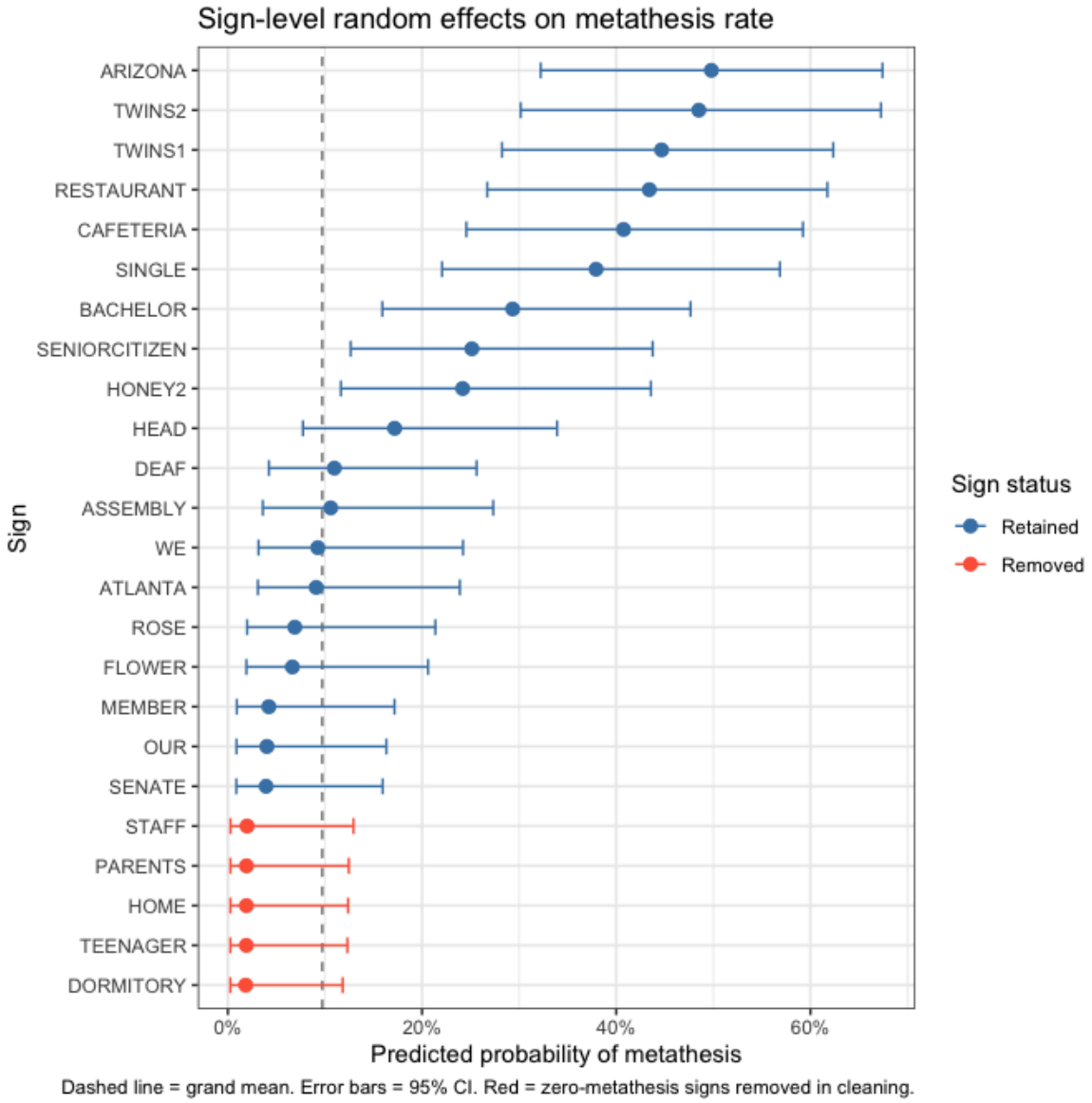


Figure 1. Sign-level random effects

References: Bybee, J. (2007, January). Frequency of use and the organization of language.Oxford University Press. **Desai**, A., Berger, L., Minakov, F., Milano, N., Singh, C., Pumphrey, K., Ladner, R., Daume III, H., Lu, A. X., Caselli, N., & Bragg, D. (2023). Aslcitizen: A community-sourced dataset for advancing isolated sign language recognition. In A. Oh, **Lucas**, C. (2014). Sociolinguistics in Deaf Communities. Washington: Gallaudet University Press. **Sehyr**, Z. S., Caselli, N., Cohen-Goldberg, A. M., & Emmorey, K. (2021). The ASL-Lex 2.0 project: A database of lexical and phonological properties for 2,723 signs in American Sign Language. The Journal of

Deaf Studies and Deaf Education, 26 (2), 263–277. Liddell, S. K., & Johnson, R. E. (1989). American Sign Language: The Phonological Base [Publisher: Gallaudet University Press]. Sign Language Studies, (64), 195–278.

Alternation-sensitive Phonemic Learning and (non-)phonologization in Japanese loanwords

Drake Howard, howar379@msu.edu

Phonologization, non-phonologization, and merger (Kiparsky, 2015) are central problems in phonological theory, and Japanese loanword adaptation strategies from the past 150-odd years have changed in ways that allow us to study all three. I argue that Alternation-sensitive Phonemic Learning (henceforth 'APL'; Richter, 2021) makes predictions that offer an explanatory account for the diachronic changes observed in Japanese. I then show preliminary evidence in favor of this claim from the CEJC-Child corpus (Koiso et al., 2022).

Historically, /ts/ and /tɕ, ɕ, ʑ/ were conditional allophones of /t̪, s, d̪, z/; in the present day, (near-)minimal pairs suggest that they are best analyzed as separate phonemes. However, a difference has emerged between /t̪, d̪/ and /s, z/, which both historically neutralized with respect to their historical allophones in their previous conditioning environments. While /s, z/ still merge with /ɕ, ʑ/ before /i/, /t̪, d̪/ can now surface before /i, u/ in recent loanwords. I assume the following processes (1) to capture the data.

(1) a. Affrication:
/t̪, d̪/ → [ts, z] /_$V_{[+\text{high}]}$

b. Palatalization:
/ts, s, z/ → [tɕ, ɕ, ʑ] /_i

Under my analysis, /si, zi/ are only subject to (1b), but /t̪i, d̪i/ are subject to a feeding relationship: (1a) yields intermediate /tsi, zi/, which become subject to (1b). These processes are still observable in the morphological paradigms of some native words, e.g. in (2).

(2) a. /hanás-/ 'talk'
i. /hanás-imás-u/ (-POL-NPST)
[hanaɕimásɯ̥]
ii. /hanás-u/ (-NPST)
[hanásɯ̥]
iii. /hanás-anái/ (-NEG)
[hanasanái]

b. /hanát̪-/ 'let x loose'
i. /hanát̪-imás-u/ (-POL-NPST)
[hanatɕimásɯ̥]
ii. /hanát̪-u/ (-NPST)
[hanátsɯ̥]
iii. /hanát̪-anái/ (-NEG)
[hanat̪anái]

In many loanwords, default epenthetic /u/ was changed to /o/ so as to distinguish /t̪, d̪/ from /ts, z/ when not followed by a vowel. (1a, b) still applied, though, as (3) shows.

(3) a. 'ticket' → [tɕi̥két̪ːo]
(∼archaic [t̪eketsɯ̥];
later, uncommon [t̪i̥két̪ːo])

b. 'tourist' → [tsúːɾisɯ̥t̪o]

c. 'trip' → [t̪oríp:ɯ̥]

d. 'credit card' → [kuɾeʑit̪ːoká:d̪o]

e. 'zipper' → [ʑíp:a:]

f. 'Dulcin' [dʊlˈtsiːn] → [zúɾutɕiɴ]

Given the dates of borrowing, I claim this means that many Japanese speakers have analyzed /t̪, ts/ as distinct since the 1880s if not much earlier (Arakawa, 1968); (1a) just kept the two in complementary distribution. But while (1b) still prevents *[si, zi], (1a) has stopped applying in more recent loanwords like those shown in (4), allowing [t̪i, d̪i] and even [t̪u, d̪u].

(4) a. ‘city’ → [ɕíʈi]
b. ‘tutti’ → [ʈúʈːi] (∼[tsúʈːi], [ʧúʈːi])
c. ‘CD’ → [ɕiːɖíː]
d. ‘duet’ → [ʤjuéʈːo]

If modern acquirers of Japanese only posited (1b) as a productive rule, not (1a), that would allow /ʈi, ɖi/ to surface as [ʈi, ɖi] without allowing *[si, zi] to surface, accounting for the data in (4). But then how do speakers seem to apply both (1a) and (1b) to native words in (2), even if they only apply the latter to loanwords?

Under APL (Richter, 2021), acquirers start with the null hypothesis (H_0) that each surface phone they’re hearing has a one-to-one mapping to its underlying representation; that is to say, [X] → /X/ and [X] ← /X/, i.e. [X] ↔ /X/. Surface alternations like in (2) are treated as exceptions to this H_0, and acquirers must decide if the amount of exceptions (e) present in their linguistic input is worth the cognitive load of rejecting H_0 and reanalyzing the data to look for a better generalization. This decision is modeled using the Tolerance Principle (Yang, 2016): acquirers reject H_0 if and only if $e > \theta_N := \frac{N}{\ln N}$.

I contend that the existence of a process where [X] ← /Y/ does not entail that [X] ↔ /X/ is never true, even if [Y] is in complementary distribution with [X]. When there are categorical contextual allophonic alternations in the input of the acquirer above θ_N, APL biases learners to treat them as contextual mergers because they can be exceptions to H_0, not proof against it. And when they’re below θ_N, APL predicts they’ll be treated as exceptional allomorphy.

I therefore propose that the reason [ʈi, ɖi] can surface but *[si, zi] cannot is that the evidence for (1a) in the language is now below θ_N, but evidence for (1b) is still above θ_N. Only ten common verbs remain in modern Japanese whose stems end in /ʈ/, but far more end in /s/, so it’s plausible for this to be true of the Japanese modern-day acquirers are exposed to.

As a proof of concept, I used data from CEJC-Child to test whether its recordings of parents’ everyday speech to their children contained a high enough proportion of [ɕi, ʧi, tsu] tokens that alternate with other segments for acquirers to be confident that there’s an exception to H_0. I found that of the 843 vocabulary items parents pronounced with [ɕi], the number of words whose underlying form does not contain /ɕi/ is above the Tolerance Principle’s threshold ($e = 137 > 125 = \theta_{843}$), so acquirers of modern Japanese still have enough evidence to deduce that [ɕi] doesn’t necessarily correspond to /ɕi/ in all cases and posit that /si/ → [ɕi]. But in the case of [ʧi], the number of exceptions to H_0 is well below θ_N ($e = 15 < 65 = \theta_{390}$), suggesting that acquirers of modern Japanese do not have sufficient evidence to reassess their assumption that [ʧi] ↔ /ʧi/, and that they therefore store the cases of alternation that go against this rule as allomorphs. As for [tsu], the number that could disprove H_0 is similarly negligible ($e = 16 < 84 = \theta_{522}$), suggesting the same holds for the assumption [tsu] ↔ /tsu/.

This presentation thus applies Richter (2021) to synchronic data from Japanese containing latent diachronic patterns. I find that APL makes correct predictions about which segments will phonemicize in a specific phonological context given the amount of alternation-based evidence there is in the input. This shows APL has explanatory power regarding the acquisition of apparently conflicting patterns that come from diachronic change; the bias acquirers have in favor of a simpler input-output interface under this framework leads to the fossilization of historically active processes and the formation of novel contrasts that loanwords can reveal.

Arakawa, S. (1968). Kadokawa gairaigo-jiten (kadokawa loanword dictionary). Kadokawa Shoten.

Kiparsky, P. (2015). Phonologization (P. Honeybone & J. Salmons, Eds.). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199232819.013.017

Koiso, H., Amatani, H., Den, Y., & al. (2022). Design and evaluation of the corpus of everyday japanese conversation. Proceedings of the Thirteenth Language Resources and Evaluation Conference, 5587–5594. https://aclanthology.org/2022.lrec-1.599/

Richter, C. (2021). Alternation-sensitive phoneme learning: Implications for children's development and language change [Doctoral dissertation, University of Pennsylvania]. https://repository.upenn.edu/handle/20.500.14332/31462

Yang, C. (2016). The price of linguistic productivity: How children learn to break the rules of language. The MIT Press. https://doi.org/10.7551/mitpress/9780262035323.001.0001

Edge alignment in stress priming

Intro. Stress representations are hidden structure. While we can tell from its surface form that *banána* has stress on the middle syllable, this stress pattern could be represented underlyingly as left-aligned ("peninitial syllable", or "left-aligned Iamb"), or right-aligned: ("penultimate syllable", or "right-aligned Trochee"), or without reference to particular edges as σσ́σ. Representations based on edge alignment seem to be important in adequately describing the productivity and typology of stress patterns, but relatively little is known about the cognitive reality of these types of structures. The existence of any psychologically real abstract representation of stress has actually been called into question by priming results in lexical decision (e.g. Protopapaps et al., 2016). Evidence in favor of an abstract representation of stress (e.g. Moore-Cantwell et al., 2024; Shaw, 2012), does not rule out a simle "metrical frame" approach (Levelt, 1993), in which stress pattern and number of syllables are integrated together (as σσ́σ above). In this study, we find evidence for edge-aligned representations in English.

We use a priming paradigm following Moore-Cantwell et al. (2024, 2026), who argue that priming of abstract prosodic structure is strongest when participants must make a choice of stress pattern (rather than processing a known stress pattern). We allowed participants to make choices of stress in two ways: **(1)** Participants chose stress in nonwords, and **(2)** they interpreted visually-presented homographs (*record*) as initial or final stress. Crucially, we manipulated whether prime and target matched in syllable count. For example, does a prime like *banána* cause target *record* to be interpreted more as *récord* (right aligned), or as *recórd* (left aligned)? In both words and nonwords, we find responses consistent with left-aligned stress representation.

(1)

3 σ Prime	**2 σ Target** *project*	
	próject	projéct
sílica	242	135
vanílla	254	214

(2)

	3 σ Target ræ fæ zʌ	
	ræfəzə	rəfæzə
sílica	181	200
vanílla	195	196

(3)

	4 σ Target væ ræ sæ zi	
	værəsæzi	vəræsəzi
sílica	210	134
vanílla	215	120

(4)

NW 3 σ	**3 σ Target** ræ fæ zʌ	
	ræfəzə	rəfæzə
túməsə	95	79
təmúsə	61	165

(5)

NW 2 σ	**4 σ Target** væ ræ sæ zi	
	værəsæzi	vəræsəzi
fúʒuk	131	38
fuʒúk	157	75

Methods. 66 participants were recorded as they spoke visually-presented English words and auditorily-presented English-like nonwords. Nonwords were presented as syllables separated by pauses. These were pronounced as individual prosodic words, and were resynthesized to avoid any acoustic differences interpretable as stress (ex. [ræ] [fæ] [zʌ]). Real word three-syllable primes were matched with targets of two syllables (real words like *project*, **(1)**), three syllables (nonwords, **(2)**), and four syllables (nonwords, **(3)**). Because we also found an unanticipated strong effect of word status (priming only occurs word-to-word and nonword-to-nonword), we also analyzed two filler conditions with nonword primes and nonword targets. These either matched in syllable length (3 syllables, **(4)**) or mismatched (2 syllable primes, 4 syllable targets, **(5)**). Additional fillers were included so that neither syllable length nor word status could be predicted from prime to target across the study. No segments of a prime's stressed syllable overlapped with segments in the target. Stress was transcribed automatically, using the montreal forced aligner

(McAuliffe et al., 2017) to "choose" between transcriptions. This methodology was validated against human transcriptions of around 1/3 of the data, with a mismatch rate of 16%.

Results. Prime-target pairs were coded for whether or not they matched in stress based on the left edge ([σσ́ or [σ́), and whether or not they matched in stress based on the right edge (σ́] , σσ́] or σσσ́]). For example, *vanílla* followed by *projéct*, or [fuʒúk] followed by [vərǽsəzi], would be matches on the left but not on the right. *Vanílla ~ próject* would match on the right only. When syllable length matched, as in **(2)** and **(4)**, both right and left edges matched. Two mixed-effects logistic regression models, with random intercepts for subjects, were run with left-edge match and right-edge match as dependent variables (0 for mismatch, 1 for match). Predictors were length match, and word/nonword match (MATCH = both words, or both nonwords).

(6)

	Left-edge match		**Right-edge match**	
	β	p	β	p
Intercept	-0.19	**>0.001**	-0.42	**>0.001**
length MATCH	0.13	0.132	0.28	**0.002**
w/nw MATCH	0.31	**>0.001**	-0.31	**>0.001**
len. MATCH :w/nw MATCH	0.37	**0.011**	1.08	**>0.001**

Both models have negative intercepts, indicating that when neither w/nw nor syllable length matches, the target's stress is more likely to *mismatch* the prime's stress than to match it. In the left-edge match model, there is a significant main effect of w/nw match, with a positive coefficient, indicating that prime-target pairs which are both words or are both nonwords are more likely to match in stress, regardless of whether they match in length. The interaction term is significant as well, indicating that pairs that match in both length and w/nw status are even more likely to match in stress. In the right-edge match model, w/nw match actually has a negative coefficient, while both length match and the interaction term have positive coefficients. Prime-target pairs are more likely to match in right-edge stress only when their length in syllables also matches - exactly where they would also match on the left edge.

Conclusion. We find clear evidence for abstract representation of stress, in the absence of segmental overlap, and even when syllable length does not match. This is, to our knowledge, the first direct evidence for the psychological reality of edge-alignment in stress representations. Interestingly, while English stress is typically thought to be a right-aligned system (e.g. Hayes, 1982), it appears in our experiment that stress is represented as left-aligned: *banána* has the stress pattern [σσ́, not σσ́]. We find also that priming only occurs when prime and target are both words, or both nonwords. This could be an effect of the presentation (visual presentation vs. auditory), or of a difference in how stress is represented in novel words vs. in known words.

Guion, S. G., et al. (2003). Factors affecting stress placement for English nonwords include syllabic structure, lexical class, and stress patterns of phonologically similar words. *Language and Speech*, *46*(4), 403–427. **Hayes, B. (1982).** Extrametricality and English stress. *Linguistic Inquiry*, *13*(2), 227–276. **Levelt, W. J. (1993).** *Speaking: From intention to articulation* (Vol. 1). MIT press. **McAuliffe, M., et al. (2017).** *Montreal Forced Aligner*. **Moore-Cantwell, C., et al. (2024)**. Production priming of stress in nonwords. *Laboratory Phonology*, *15*(1), 1–31. **Protopapaps, A., et al. (2016).** Priming stress patterns in word recognition. *Journal of Experimental Psychology: Human Perception and Performance*, *42*(11), 1739. **Schiller, N. O., et al. (2004).** Stress priming in picture naming: An SOA study. *Brain and Language*, *90*(1–3), 231–240. **Shaw, J. (2012).** *metrical rhythm in speech planning: Priming or predictability*.

Bayesian hierarchical grammars: a case study in the English comparative/superlative alternation

Canaan Breiss, University of Chicago - cbreiss@uchicago.edu

Summary: We propose that Bayesian hierarchical grammars that jointly model item-specific indexed constraints and grammar-wide phonological constraints are well-suited to modeling lexical idiosyncrasy, and further have conceptual and empirical benefits for phonological learning and generalization that extend beyond this domain. They resolve the challenge of learning restrictive grammars in the absence of negative evidence (the "subset problem"), recast the "exception vs. rule" paradigm in terms of a quantitative gradient between generalizable and non-generalizable behavior, and exhibit the "blessing of abstraction" common to Bayesian cognitive models that allow for learning of abstract principles to outstrip item-specific information, despite only observing item-level data. Below, we present the model, validate it on previously-presented data on the English comparative and superlative alternation (*more, most* vs. *-er, -est*), and demonstrate that the model's predictions are borne out in a new experiment with existing and novel adjectives.

Problem setting: It has long been observed that learning data is inherently ambiguous – each datapoint provides evidence for multiple possible generalizations (also known as the "subset problem", Chomsky & Halle 1968; Dell 1981; Berwick 1985). Probabilistic phonological patterns make this problem more dire: how does a learner decide which items constitute the "default" or "generalizable" grammar, and which constitute the "exception"? Responses to this challenge of ambiguity have come in the form of indexed constraints (Pater 2000; Moore-Cantwell & Pater 2016), which are posited to account for item-specific behavior, while the general grammar captures tendencies which are generalizable. This approach, where item-specific and general constraints are equally valid candidate explanations for the behavior of one data point, have been shown to yield pathological results in the batch-learning regime (Zymet 2018), where in the limit, indexed constraints bear all the explanatory weight, and the constraints of the general grammar drop to zero, predicting random generalization for new items (thus a failed *wug* test).

Our model: Building on the approach of Zymet (2018) we propose a novel *hierarchical* architecture for the phonological grammar that brings it in line with well-established models of non-phonological linguistic cognition (see Tenenbaum et al. 2025 for an overview). The model resolves the tension of having to attribute explanatory power to *either* the general phonological grammar *or* the indexed constraints by allowing the learner to "hedge their bets", assigning credit to general grammar-wide constraints *and* item-specific constraints jointly, in proportion to their utility. Specifically, we model the item-specific constraints as drawn from a population whose mean is centered on the behavior predicted by the grammar-wide constraints – that is, the general grammar acts as a Bayesian prior or "fallback" for items whose indexed constraints are weak (or, in novel words, don't exist). This means that as learning progresses and more data are observed, items which are high-frequency and/or strongly divergent from the mean behavior in the data will come to have strong weights for their indexed constraints, while items which are either less-divergent or less frequent will have their behavior explained by the general grammar, having their item-specific constraints "shrunk" towards zero. Because these two levels of hierarchy are yoked, fitting the model to all items at once allows for statistically-optimal allocation of credit to item-specific vs. broadly-applicable constraints, thus encoding the behavior of individual items as a blend of "rule-following" (phonological grammar) and "exceptional" behavior.

A second consequence of this nesting of item-specific constraints "inside" grammar-wide constraints means that one level acts as regularization for the other: the Bayesian approach allows us to place priors on *all* constraints that favor weights nearer zero, and so *any* constraint (or family thereof) is not used unless warranted by the data: item-specific behavior (that is, not generalizing beyond the training data) is a good candidate for the behavior of all items, until sufficient evidence accrues to render it better explained as a general phonological pattern (predicting generalization at a wug test).

Finally, a consequence of the hierarchical nature of the grammar is that evidence from many diverse sources – low-frequency or possibly-exceptional items, few in types or tokens – is automatically brought to bear in aggregate on the general grammar level, allowing for faster learning of general phonological constraints than the behavior of any individual item. This is a phenomenon termed the "blessing of abstraction" in the cognitive science literature, which is argued to underpin a wide range of phenomena where rapid humans exhibit rapid learning and generalization of abstract knowledge without clear mastery of the specifics (Gershman 2017).

Data: We fit our model to corpus and experimental data from the English comparative and superlative alternations (presented in Breiss 2024), which has been shown to exhibit extensive lexical idiosyncrasy: for any given adjective (209 types), the degree to which it prefers analytic vs. morphological comparatives and superlatives is partially explainable by general phonological principles (ex., monosyllabic morphological, final stress analytic, etc.) and also item-specific variation which is not phonologically-patterned. We also present novel data from a replication and extension of Breiss (2024)'s experiment as a model validation (52 participants), gathering preference judgments for both the previously-studied real adjectives (n=209), as well as nonce adjectives (n=80).

Results: First, we fit our grammar to the data from Breiss (2024), demonstrating that the grammar learns that there is a wide degree of lexical idiosyncrasy (population SD = 2.3), as well as strong grammar-wide phonological predictors of the alternation, replicating Breiss (2024). Second, we fit the grammar on sequentially-increasing amounts of the training data (sampled according to token frequency), and observe the rate at which the certainty of each parameter increases as more data are observed. We find that the model exhibits faster, more sure learning of abstractions (phonological constraints, the correlation of lexical idiosyncrasy between comparative and superlative) than item-level information – the "blessing of abstraction", and also that high-frequency items are learned faster than low-frequency items (the "token advantage", Jarosz et al. 2025). Third, the weight correlation between both phonological constraints and indexed constraints in Breiss (2024)'s real word experimental data and our replication real word data are extremely high (phon. constraints r = 0.94, indexed constraints r = 0.95(comp.) r = 0.96 (sup.)), indicating the model architecture reliably attributes credit to grammar-wide vs. item-specific sources. Last, we evaluate the predictions of the fitted model to the novel-adjective wug-test data and show that as expected, nonce items exhibit extremely little idiosyncrasy (i.e., they have no strong indexed constraint, pop. SD = 0.1), but the extent of their inferred idiosyncrasy is well-correlated across comparative and superlative (r = 0.7), mirroring the paradigm-level generalization learned by the model.

Discussion: Hierarchical grammars have several benefits over other approaches to jointly modeling lexical idiosyncrasy and the general grammar: they do so using models that are common currency in other domains of cognition, they address the challenge of subset learning and of parsing rule-following from exceptional items, and they allow accelerated learning of abstraction from diverse and noisy data.

Learning Opaque and Transparent Vowel Harmony in Early Development

Chunzi Li[1], Zhenting Liu[1], and Regine Lai[1]

[1]*Department of Linguistics and Modern Languages, The Chinese University of Hong Kong*

Introduction. This study examines how 14-month-old toddlers learn two vowel harmony patterns that differ in the role of a neutral vowel. In opaque vowel harmony, the neutral vowel blocks feature spreading, whereas in transparent vowel harmony, it does not participate in the harmony process, allowing harmony to skip the neutral vowel, so that nonadjacent vowels agree despite an intervening neutral vowel [1]. An artificial-grammar learning study with adults suggests that transparent patterns are more difficult to learn than opaque patterns [2], an asymmetry often attributed to differences in dependency locality and consistent with typological observations that vowel harmony favors local agreement [1, 3] While adult learners are sensitive to this difference, it remains unclear whether toddlers are similarly sensitive to locality in vowel harmony learning. To address this question, we examine 14-month-old Cantonese-learning toddlers' learning of opaque and transparent vowel harmony. Cantonese does not exhibit vowel harmony, allowing us to test learners' ability to acquire novel phonological dependencies. Prior developmental work shows that sensitivity to vowel harmony emerges early: Turkish-learning infants as young as 6 months prefer harmonic patterns [4–5], and by 10–13 months, French-learning infants detect non-local vowel dependencies [6]. These findings suggest that such sensitivities develop within the first year of life, motivating the present focus on 14-month-old learners.

Method. A total of seventy-two 14-month-old Cantonese-learning toddlers were recruited; 24 were excluded due to fussiness or technical issues, resulting in a final sample of 48 toddlers. Participants were tested in a central-fixation paradigm using Habit2 and randomly assigned to either the Opaque or Transparent condition (24 per condition). Stimuli were trisyllabic nonwords (CVCV stem + CV suffix) instantiating a round vowel harmony pattern (Table 1-2). Vowels were drawn from /i, ɛ, a, ɔ, u/, in which non-low vowels contrast in [±round] and the low vowel /a/ serves as a neutral vowel, as it lacks a [+round] counterpart. In the Opaque condition, the suffix agreed with the neutral vowel in roundness, whereas in the Transparent condition, it agreed with the initial vowel across the neutral vowel. All items were phonotactically legal in Cantonese and unfamiliar to toddlers. During the experiment, toddlers completed a 2-minute familiarization phase followed by 12 novel test trials (6 grammatical, 6 ungrammatical relative to the trained pattern). Looking time was measured for each trial.

Table 1. Example trisyllabic training stimuli (CVCV stem + CV suffix) used in the Opaque and Transparent vowel harmony conditions. Vowels were drawn from /i, ɛ, a, ɔ, u/ ([−round]: /i, ɛ/; [+round]: /ɔ, u/) with /a/ [+low, -round] serving as a neutral vowel (N-). The suffix alternated as *-kɛ* ~ *-kɔ*.

Stem type	Example stem	Opaque condition	Transparent condition
Harmonic (- round)	*kisɛ* (− −)	*kisɛ-kɛ* (− − −)	*kisɛ-kɛ* (− − −)
Harmonic (+ round)	*zɔpu* (+ +)	*zɔpu-kɔ* (+ + +)	*zɔpu-kɔ* (+ + +)
Harmonic (neutral)	*gɛka* (- N-)	*gɛka-kɛ* (− N- −)	*gɛka-kɛ* (− N- −)
Disharmonic (neutral)	*zɔda* (+N-)	*zɔda-kɛ* (+ N- −)	*zɔda-kɔ* (+ N- +)

Note. Training stimuli included 16 items distributed across the four stem types (4 in each type). Consonant frames and vowel sequences were matched across conditions, and all syllables carried Cantonese high-level tone.

Table 2. Example test stimuli in the Opaque and Transparent conditions (✓ grammatical, ✗ ungrammatical).

Stem type	Example item	Opaque condition	Transparent condition
Harmonic (- round)	*sɛki-kɛ* (− − −)	✓	✓
	sɛki-kɔ (− − +)	✗	✗
Harmonic (+ round)	*puzɔ-kɔ* (+ + +)	✓	✓
	puzɔ-kɛ (+ + −)	✗	✗
Harmonic (neutral)	*gɛda-kɛ* (− N- −)	✓	✓
	gɛda-kɔ (− N- +)	✗	✗
Disharmonic (neutral)	*pɔda-kɛ* (+ N- −)	✓	✗

pɔda-kɔ (+ N- +)	✗	✓

Note. Test stimuli included 12 items (6 grammatical, 6 ungrammatical) in each condition. Identical items were used across conditions. Consonant frames, vowel sequences, and tonal patterns (all high-level tone) were controlled across conditions.

Result. Looking times were analysed using linear mixed-effects models with grammaticality as a fixed effect and random intercepts for participants. In the Opaque condition, toddlers showed a significant effect of grammaticality, looking longer at ungrammatical than grammatical items, $F(1, 10) = 6.00$, $p = .034$. In the Transparent condition, toddlers also showed a significant effect of grammaticality, looking longer at grammatical than ungrammatical items, $F(1, 9.34) = 6.88$, $p = .027$ (Figure 1). These results indicate that 14-month-old toddlers discriminated grammatical from ungrammatical vowel harmony patterns in both conditions, although the direction of preference differed across conditions.

Discussion. The results show that 14-month-old Cantonese-learning toddlers distinguished both opaque and transparent vowel harmony patterns, but with different preference directions across conditions. Such differences in preference direction have been discussed in the literature [7] and may reflect differences in processing demands or learning dynamics. One possible explanation relates to dependency locality: opaque harmony involves a local dependency, whereas transparent harmony requires tracking a non-local dependency across a neutral vowel, hence different learning mechanisms might have been adopted. This difference may lead to variation in processing demands, which could in turn influence the direction of infants' looking preferences. While the present findings do not directly demonstrate differences in learnability, the observed asymmetry is broadly compatible with adult findings suggesting that transparent harmony may be more difficult to learn, pointing to early sensitivity to differences between local and non-local vowel dependencies.

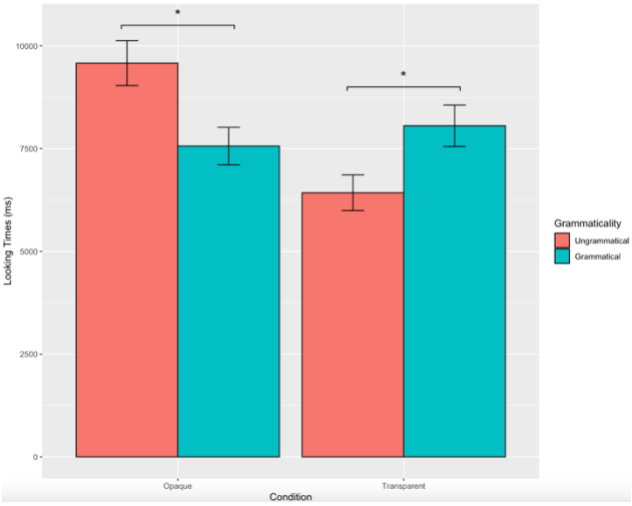


Figure 1. Mean looking times (in milliseconds) to grammatical and ungrammatical test items in the Opaque and Transparent vowel harmony conditions for 14-month-old Cantonese-learning toddlers. Error bars indicate ±1 standard error.

References

[1] Gafos, A., & Dye, A. (2011). Vowel harmony: Transparent and opaque vowels. The Blackwell companion to phonology, 4, 2164-2189.

[2] Finley, S. (2015). Learning nonadjacent dependencies in phonology: Transparent vowels in vowel harmony. Language, 91(1), 48-72.

[3] Kiparsky, P., & Pajusalu, K. (2003). Towards a typology of disharmony.

[4] Altan, A., Kaya, U., & Hohenberger, A. (2016). Sensitivity of Turkish Infants to Vowel Harmony in Stem-suffix Sequences: Preference Shift from Familiarity to Novelty. Proceedings of the 40th Boston University Conference on Language Development.

[5] Hohenberger, A., Kaya, U., & Altan, A. (2017). Discrimination of Vowel-Harmonic vs Vowel-Disharmonic Words by Monolingual Turkish Infants in the First Year of Life. Proceedings of the 41st Annual Boston University Conference on Language Development, 309–322.

[6] Gonzalez-Gomez, N., & Nazzi, T. (2016). Delayed acquisition of non-adjacent vocalic distributional regularities. Journal of Child Language, 43(1), 186-206.

[7] DePaolis, R. A., Keren-Portnoy, T., & Vihman, M. (2016). Making sense of infant familiarity and novelty responses to words at lexical onset. Frontiers in psychology, 7, 715.

Underlying representations affect allophone judgements in Canadian Raising

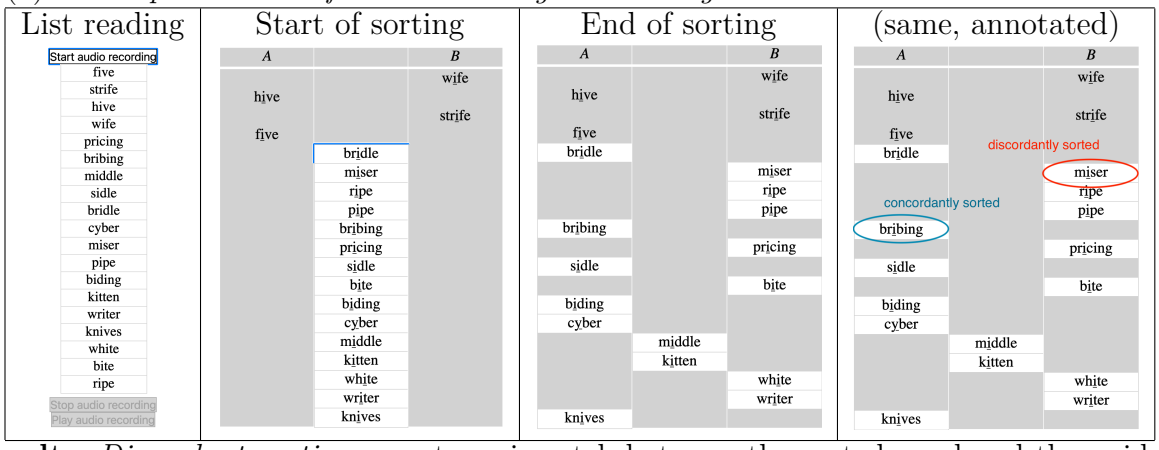
Elliott Moreton, Jeff Lamontagne, and Monica Nesbitt

One view of allophony in generative phonology posits an underlying segment /X/ that is identical to the elsewhere allophone [X] but is changed to [Y] by the grammar in a specific environment (e.g., [1]). An alternative is that sounds pronounced as [Y] are stored as /Y/ (Lexicon Optimization, [2]). Analogous views occur in exemplar-based frameworks ([3; 4; 5]). If speakers can use these representations in judging which allophone occurs in a familiar word, the Underlying Theory predicts a bias to judge [Y] as [X] more than [X] as [Y], while the Surface Theory does not. Using Canadian Raising, which here can be adequately described as /**aɪ, aʊ**/ ⟶ [**ʌɪ, ʌʊ**] before a voiceless coda (including ambisyllabic, [6]), we found that speakers sorted raised-vowel words into the class of unraised-vowel words more often than the reverse, supporting the Underlying Theory.

Materials consisted of 116 English words arranged in 6 lists. List 1, had /**i**/ and /**æ**/ words (e.g., *meet, flat*) for practice with the sorting task. List 2 had 16 /**aɪ**/ monomorphemic /**aɪ**/ monosyllables (e.g., *rice, rise*) for practice with sorting /**aɪ**/. Lists 3–5 had a total of 28 more /**aɪ**/ monosyllables, plus 25 trochees. Half of the words had an underlying voiceless obstruent /p t f s/ after the /**aɪ**/, and half had an underlying voiced one /b d v z/ ("VLSS" vs. "ELSW"). List 6 had 18 monomorphemic /**eɪ**/ monosyllables (e.g., *fate, fade*), not discussed here. (Results for flapped /t/ and /p f s/ are combined in this abstract because they were similar. Known lexical exceptions like *tiger* were excluded from this analysis.) Each list included two "catch" words with an unrelated vowel.

Participants were 100 self-identified L1 English speakers in Canada, recruited on-line. Data was excluded from analysis for poor sound quality, clearly non-Canadian English, failure on the practice, or sorting more than two catch words across Lists 2–5, leaving 70. **Procedure:** Speakers read each list aloud, then they sorted the list words into columns headed by different guide words, if they judged the guide words to have different vowel sounds (Figure 1). The guide words were monomorphemic monosyllables that did not rhyme with any sortable list words. No feedback was given. Speakers judged and sorted written words, not audio recordings. Recordings were segmented using the Montreal Forced Aligner [7], and durations and formant tracks were extracted with Praat [8], both human-supervised.

(1) *Participant's view of the list-reading and sorting tasks.*



Results: *Discordant sorting* meant a mismatch between the sorted word and the guide words as to whether the structural description for Raising was met (e.g., ELSW *miser* sorted under the VLSS guides *wife* and *strife* in Figure 1). The *relative risk* — the ratio of the discordance rate in the VLSS to that in the ELSW condition — is predicted by the Underlying Theory

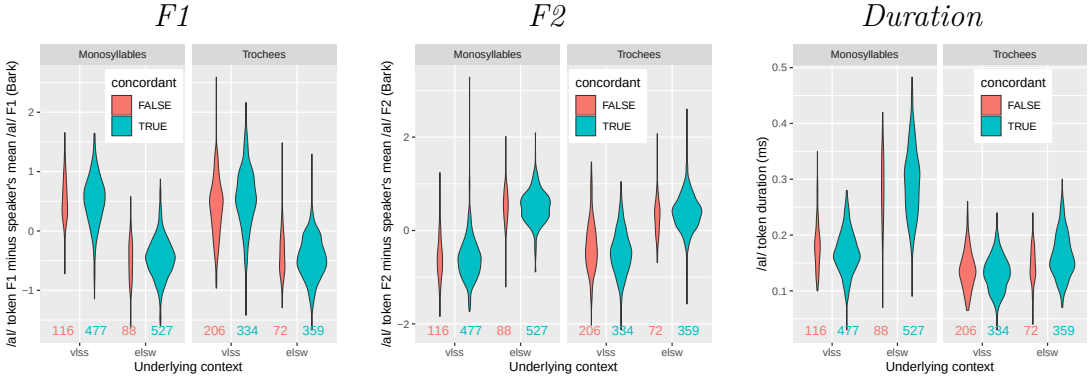
to be > 1. That was in fact the case (Table 2). Bootstrap resampling of (but not within) speakers was used to take into account non-normality and within-participant dependencies (10,000 replicates).

(2) *Discordant sorting was more common in the* VLSS *than the* ELSW *words.*

	Sorted as:			Relative risk (VLSS/ELSW)	
Context	VLSS	ELSW	Discordance	mean [95% CI]	times > 1
Monosyllables					
VLSS/p,t,f,s/	507	119	0.196 [0.144, 0.251]	1.37 [0.99, 1.92]	9706/10000
ELSW/b,d,v,z/	93	606	0.143 [0.101, 0.188]		
Trochees					
VLSS/p,t,f,s/	364	209	0.381 [0.322, 0.444]	2.28 [1.69, 3.32]	10000/10000
ELSW/b,d,v,z/	72	390	0.167 [0.113, 0.227]		

Since Raising is thoroughly phonologized and can have lexical exceptions [9], we must ask: Were participants *sorting* words discordantly because they were *pronouncing* them discordantly? No: Figure (3) shows that /aɪ/ in discordantly-sorted words was pronounced like /aɪ/ in concordantly-sorted words with the same underlying context (VLSS or ELSW).

(3) *Within a word type, con- and discordantly-sorted* /aɪ/ *aren't pronounced differently.*



Discussion: Introspective judgements classify the contextual allophone as the elsewhere allophone oftener than the other way around, especially in trochees where shortening abbreviates phonetic evidence (Figure 3). Together with eye-tracking evidence that VLSS [aɪ]-words are recognized slower than ELSW ones [10], these results favor the Underlying Theory(that words with either allophone are represented with the elsewhere allophone, at some level in speakers' actual linguistic knowledge), over the Surface Theory.

[1] M. Kenstowicz and C. Kisseberth, *Generative phonology: description and theory*, Academic Press, San Diego, 1979.

[2] A. Prince and P. Smolensky, *Optimality Theory: constraint interaction in generative grammar*, Blackwell Publishers, Malden, Massachusetts, 2004.

[3] M. Ernestus and R. H. Baayen, The handbook of phonological theory , 374 (2011).

[4] J. B. Pierrehumbert, Annual Review of Linguistics **2**, 33 (2016).

[5] R. Kirchner, R. K. Moore, and T.-Y. Chen, Journal of Phonetics **38**, 540 (2010).

[6] C. Paradis, Canadian Journal of Linguistics **25**, 35 (1980).

[7] M. McAuliffe, M. Socolof, S. Mihuc, M. Wagner, and M. Sonderegger, Montreal Forced Aligner: Trainable Text-Speech Alignment Using Kaldi, in *Proc. Interspeech 2017*, pages 498–502, 2017.

[8] P. Boersma and D. Weenink, PRAAT Version 6.3.08, Software, www.praat.org, 2021.

[9] K. C. Hall, Defining phonological rules over lexical neighbourhoods: Evidence from Canadian Raising, in *Proceedings of the 24th West Coast Conference on Formal Linguistics*, edited by J. Alderete, C. hye Han, and A. Kochetov, pages 191–199, Somerville, Massachusetts, 2005, Cascadilla Press.

[10] A. Farris-Trimble and A. Tessier, Language **95**, e136 (2019).

Domain-Specific Phonology of Shekgalagari Diminutives

Rebecca Everson
University of California, Berkeley

Introduction: Cross-linguistic evidence of domain-specific phonology suggests that a language's phonology has access to morphosyntactic information, and that morphology can directly or indirectly condition phonology. Bantu languages provide a testing ground for the interaction between morphology and phonology, as they are well-known for having complex, agglutinative morphology and tone systems. In this study, I investigate nominal tone spreading as a morphologically conditioned phonological process in Shekgalagari, a Bantu (Sotho-Tswana) language of Botswana, by analyzing tonal alternations of the diminutive suffix. I show that nominal tone spreading is different from verbal spreading, and that strata, rather than a single, global phonology, are necessary to account for the Shekgalagari facts. This work also fills a descriptive gap of Shekgalagari tone, constituting the first analysis of nominal word tone.

Background on Shekgalagari Tone: Work on Shekgalagari tone has been limited to Crane's (2009) description of the verbal tone system and Hyman and Monaka's (2010) work on phrasal tone. The tone system has previously been described as privative. In verbs, lexical H-spreading is bounded, with H-toned object markers or H-toned roots spreading one TBU to the right without sensitivity to morphological domain (Crane 2009: 228). Grammatical H's spread unboundedly. Throughout the language, utterance-final /H.H/ sequences surface as [HL.H], accompanied by penultimate lengthening (Hyman and Monaka 2010:7).

Tonal Alternations in Shekgalagari Diminutives: Shekgalagari is a typologically rare Bantu language in that nominal morphology includes derivational suffixes rather than class-marking prefixes for diminutive noun formation (see Gibson et al. 2017 for a typological overview of Bantu diminutives). I show that the Shekgalagari diminutive suffix has three possible surface-level tonal patterns, [L.L] (1), [H.L] (2), and [HL.L] (3), and that its surface tone is predictable based on the tonal properties of the noun (called the "base" here) to which the suffix attaches. The predictable segmental allomorphy of the diminutive suffix is independent of the tonal alternations. All data come from an unpublished Filemaker Pro dictionary of Shekgalagari words created by Larry Hyman and Kemmonye Monaka, an L1 speaker of the Ngologa variety of Shekgalagari. I consider 90 nouns and their diminutive forms.

(1) [lɪ-hoqo] 'word' [lɪ-hoqw-ana] 'short word'
(2) [ma-bú] 'sand' [ma-bú-ɲána] 'small amount of sand'
(3) [phalá] 'impala' [phal-âna] 'young of an impala'

Analysis: I propose that the diminutive suffix has underlying /L.L/ tone, evidenced by (1) where the suffix follows a L-toned base, and that H-spreading is a domain-specific process that occurs across the suffix boundary or within the diminutive suffix, but not in noun bases due to base tone faithfulness. Examples (2) and (3) provide evidence of H-spreading on the diminutive in two scenarios: in (2), the base-final H spreads one tone-bearing unit (TBU) rightwards onto the first vowel of the diminutive suffix, and in (3), the base-final H re-links to the first vowel of the suffix

after its base mora is deleted, then the re-linked H spreads to the final vowel, creating a /H.H/ sequence that is realized as [HL.L]. H-spreading does not occur within noun bases, as exemplified in (4). If H-spreading were active in noun bases, the H on the first TBU of the noun base in (4) would spread to the second TBU.

(4) [lɪ-bɛ́lɛ] 'breast' [lɪ-bɛ́-gʲana] 'small breast'

I propose that there are two morphological domains that condition phonology: the base and the nominal word. I model the tonal processes using Stratal OT (Bermudez-Otero 1999, 2012; Kiparsky 2000, 2008, 2015) in which one phonology applies to a nominal base, and a different phonology applies after affixation. The derivation in (5) demonstrates that a stratal account is necessary to explain how nominal base tone faithfulness prevents H-spreading.

(5) [bɛ́lɛ]	*Step 1: Tone class /H.L/ assignment to base morae*
[lɪ- + bɛ́lɛ + -ana]	*Step 2: Affixes added*
[lɪ-bɛ́gʲ-ana]	*Step 3: Segmental changes due to phonotactic rules*
[lɪ-bɛ́gʲ-ana]	*Step 4: De-linked base-final L relinks to diminutive, no H-spreading due to base tone faithfulness*

The derivation for the diminutive form of [mabú] shows that H-spreading occurs when nominal base tone faithfulness is maintained, which is represented in (6).

(6) [bú]	*Step 1: Tone class /H/ assignment to base morae*
[ma- + bú + -ana]	*Step 2: Affixes added*
[ma-bú-ɲana]	*Step 3: Segmental changes due to phonotactic rules*
[ma-bú-ɲána]	*Step 4: Base-final H spreads once to first mora of diminutive suffix*

Implications: The diminutive tonal alternations demonstrate that nominal words are also affected by a process previously described as a property of verbal phonology, bounded H-spreading. Contrary to verbal words, H-spreading in nominal words is domain-specific, demonstrating the interleaving between phonology and morphology in Shekgalagari. Findings on category-specific phonology (Smith 2011) suggest that nouns have more phonological privilege than verbs. Hyman (2022) examines category-specific (noun vs. verb) phonology in Bantu at the root, word, and phrasal levels. He finds that verbal roots and phrases show less "prosodic variation" (2022: 141) than nouns. However, he claims that verbal words show more contrasts than nominal words, which he partially attributes to the relatively minimal nominal morphology in Bantu, though he did not consider Bantu languages with productive nominal suffixes. This study warrants further investigation into category-specific processes in Bantu languages with more nominal morphology, as languages like Shekgalagari show that nouns have more prosodic contrasts than previously thought.

Selected References: Crane, Thera (2009). Tense, aspect, and mood in Shekgalagari. *UC Berkeley PhonLab Annual Report.* ● Hyman, Larry M (2022). Prosodic asymmetries in nominal vs. verbal phrases in Bantu. *Prosody and Prosodic Interfaces.* ● Hyman, Larry M., and Kemmonye C. Monaka (2010). Tonal and non-tonal intonation in Shekgalagari. *Prosodic Categories: Production, Perception and Comprehension.* ● Smith, Jennifer L. (2011). Category‐specific effects. *The Blackwell Companion to Phonology*.

Phonetic substance in the neural network implementation of phonological learning: the case of vowel harmony and vowel disharmony

Shuang Zheng[1], Youngah Do[1]

[1]The University of Hong Kong (Hong Kong SAR)

A longstanding question in phonology concerns how the phonetic basis of sound patterns drives typology (Blevins, 2004; Hayes et al., 2004; Wilson, 2006; Ohala, 1993). While artificial language learning studies reported learning asymmetries favoring phonetically natural patterns (Moreton & Pater, 2012; Zheng & Do, 2025), the learnability differences between patterns in the lab experiments were highly intertwined with perceptual factors (Greenwood, 2016). A recent modeling study demonstrated a learning advantage for vowel harmony over disharmony using text-based inputs and outputs, indicating the role of phonetic substance in phonological learning independently of perceptual factors (Zheng & Do, 2026). The authors proposed that the models developed representational organizations mirroring the similarity relations underlying vowel-to-vowel coarticulation, thereby interpreting the observed learning bias as a structurally encoded substantive bias. The current study introduces the transmission channel into such a learning scenario to investigate how phonetic substance in continuous speech signals shapes the learned representations, as well as how perceptual salience interacts with and contributes to the cognitive bias in phonological learning.

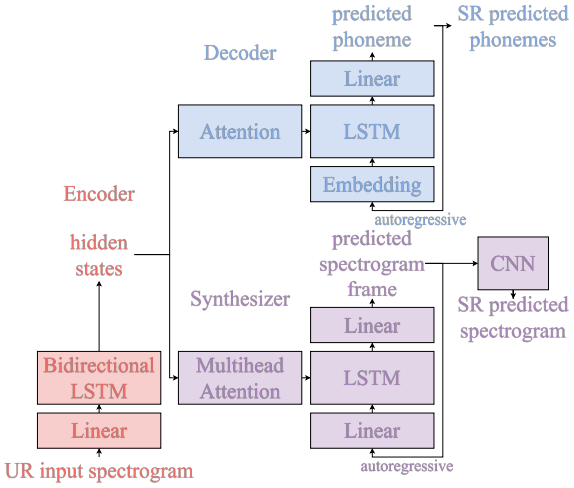
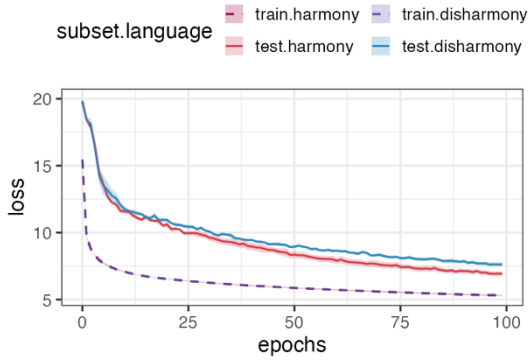
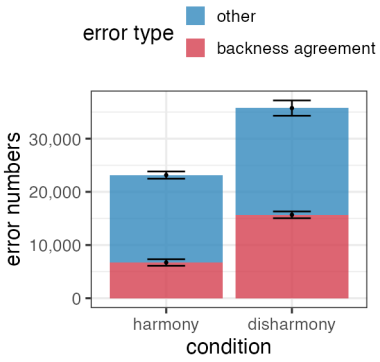


Figure 1. Model architecture.

We adapted the Google speech-to-speech translation model (Jia et al., 2019; Figure 1) to learn the underlying-surface mapping of backness harmony (e.g., /i.mo/ → [i.me]) and backness disharmony (e.g., /i.me/ → [i.mo]). The encoder processed the underlying mel-spectrogram into hidden states, simulating the perception of the acoustic signals and the abstraction of the underlying representation. The hidden states simultaneously conditioned the phonological decoder and the phonetic synthesizer through attention-based alignments (Bahdanau et al., 2014; Vaswani et al., 2017). The decoder predicted the surface phonemes, while the synthesizer generated the surface mel-spectrogram. The model was trained for 100 epochs using a combination of cross-entropy loss on phoneme prediction and L1 loss on spectrogram generation, and evaluated on a novel test set throughout. The analysis included 5 independent runs per condition.



(left) Figure 2. L1 loss of mel-spectrogram generation as a function of training epochs.



(right) Figure 3. Number of vowel prediction errors across training and evaluation.

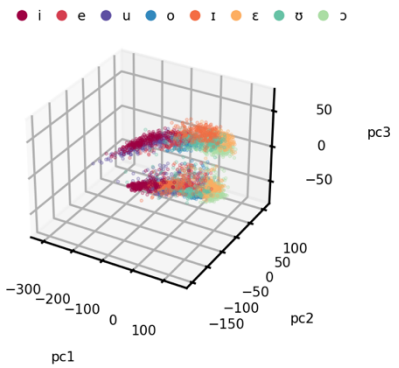
Post hoc results of a generalized additive model on the mel-spectrogram generation loss (Figure 2) indicated significantly lower loss in harmony evaluation compared to disharmony evaluation (β = -0.57, t = -5.84, $p < 0.001$). This suggests that the learned acoustic cues in vowel harmony

can be better generalized to unseen data compared to disharmony. We then ran linear models on the number of vowel prediction errors across training and testing (Figure 3). The harmony condition produced significantly fewer vowel errors compared to the disharmony condition (β = -12595, t = -7.23, $p < 0.001$), within which there were even fewer backness (dis)agreement errors (β = -8962, t = -10.15, $p < 0.001$). The phonetic and phonological learning results converge in showing better acquisition of vowel harmony compared to disharmony.

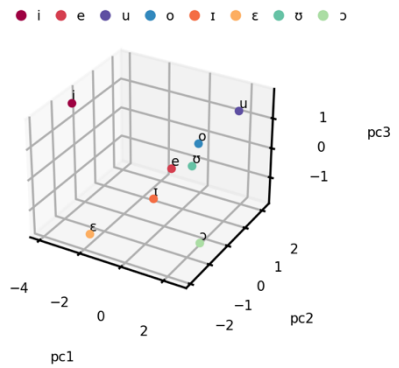
Distance	Condition	Spectrogram	Mean	SD
Euclidean	Harmony	Target	160.59	45.92
		Predicted	129.03	49.42
	Disharmony	Target	181.66	43.61
		Predicted	135.91	47.98
Cosine	Harmony	Target	0.82	0.10
		Predicted	0.88	0.08
	Disharmony	Target	0.77	0.09
		Predicted	0.88	0.07

Table 1. Average distance between vowel pairs in the target and predicted spectrograms.

The preference for harmony likely stems from the input spectrograms, where vowel-to-vowel coarticulation appears as similar formant values and is therefore readily identifiable. Inspection of the acoustic data further revealed that, across training conditions, the model increased the similarity between the first and second vowels in the predicted spectrograms relative to the target spectrograms (e.g., [i_e] in [i.me]; Table 1). This tendency effectively amplifies the vowel-to-vowel coarticulation present in the input, thereby facilitating the perception and production of vowel harmony. By the end of training, the embedding spaces of both the predicted spectrograms and the phonemes exhibited structured organization resembling natural classes, with vowel backness generally coinciding with one principal component in three-dimensional Principal Component Analysis (PCA; Figures 4 and 5). As the model extracts relevant phonological features from the acoustic signal, the enhanced coarticulation may be reflected in the phonological representations, lending further support to the learning of vowel harmony.



(left) Figure 4. 3D PCA results of the embedding space of all predicted spectrograms from a disharmony run.



(right) Figure 5. 3D PCA results of the embedding space of all phonemes from a disharmony run.

Taken together, our findings substantiate the learning advantage of vowel harmony over disharmony, consistent with previous behavioral and text-based modeling studies (Martin & Peperkamp, 2020; Martin & White, 2021; Zheng & Do, 2026), while corroborating Zheng and Do's (2026) characterization of the learning asymmetry as a structurally encoded substantive bias. Our model simulation illustrates how a bias toward vowel-to-vowel coarticulation in the acoustic input may be encoded in the phonological system and, consequently, gives rise to the higher learnability of vowel harmony. In other words, phonetic optimization naturally propagates across production and phonological learning through the shared representations. Although the repeated consideration of phonetic substance in the transmission channel and the phonological grammar appears theoretically redundant (Hale & Reiss, 2000; Blevins, 2004), it may constitute a psychologically realistic property of a learning system.

Selected References

Hayes, B., Kirchner, R., & Steriade, D. (2004). *Phonetically based phonology*. Cambridge University Press.

Zheng, S., & Do, Y. (2026). *Phonetic substance is encoded in the phonological featural system: neural network modeling of vowel harmony and vowel disharmony* [Conference presentation abstract]. 23rd Old World Conference in Phonology, Cambridge, United Kingdom.

Sound-symbolic forms predict diachronic shifts in the gender association of personal names

Overview: Gradient phonological differences between English male (M) and female (F) names have been widely documented (e.g. Cutler et al., 1990; Slater & Feinman, 1985). Crucially, these patterned phonological differences are a productive component of the grammar: Cassidy et al. (1999) demonstrate that stress placement, name length, and word-final segment impact the gender that both children and adults assign to nonce names. I developed a model that captures gradient phonological differences between M and F names using gender-category-indexed constraints in a Maximum Entropy (MaxEnt) Harmonic Grammar. **I found that names whose gender association has dramatically shifted in the past 70 years tend to be more harmonic when evaluated by the constraint set indexed to the gender category toward which they shifted.** The significance of this result is confirmed with a 100,000-trial Monte Carlo simulation.

Methods: Past work documents the following gradient differences between English M and F names: (1) All names prefer primary stress to fall on the first syllable; M names are less likely to have non-initial stress. (2) F names are less likely to have a final obstruent (Slater and Feinman, 1985). (3) F names have a higher ratio of open syllables. (4) Most names are polysyllabic; F names are less likely to be monosyllabic. (5) Stressed syllables in M names are less likely to have an [i] nucleus. (6) Stressed syllables in F names are less likely to have an [ɑ] or [ɔ] nucleus. (7) M names are less likely to end with [ə] or [i] (Cutler et al., 1990). I formulated the following constraints corresponding to (1)–(7). I don't necessarily take all seven constraints to represent linguistic universals; at minimum, I take them to reflect patterns that English speakers could plausibly acquire as part of a statistically learned personal name grammar.

Constraints based on (1)–(7):	*(Learned weights discussed subsequently)*	
INITSTR:	Penalize names where primary stress does not fall on the first syllable.	
		***w*♂= 2.8** ***w*♀=1.0**
***OBS#:**	Penalize every name with a final [-sonorant] segment.	
		***w*♂=0.4** ***w*♀=2.0**
wdBIN:	Penalize every name that is monosyllabic.	
		***w*♂=1.2** ***w*♀=2.2**
***HF:**	Penalize names with the high front vowel [i] in their primary stressed syllable.	
		***w*♂=2.7** ***w*♀=2.1**
***LB:**	Penalize names with the low back vowels [ɑ] or [ɔ] in their primary stressed syllable.	
		***w*♂=1.2** ***w*♀=1.9**
***[ə,i]#:**	Penalize names that end with [ə] or [i].	
		***w*♂=1.1** ***w*♀=0.0**
O:T	Penalize names with a ratio *open syllables:total syllables* < 0.5.	
		***w*♂=1.6** ***w*♀=2.6**

I extracted the top 300 M and F names between 1955–2024 from public Social Security Administration (SSA) data. Constraint weights were then estimated with the MaxEnt Grammar Tool (Hayes, 2009) using a single input and a categorical candidate space permitting full competition among the dimensions in (1)–(7); each competitor category is defined by a distinct violation profile, and every violation profile has a corresponding category. Names were assigned to exactly one category. For example:

"Amelia" **/əˈmi.li.ə/** ∈ { *polysyllabic & non-initial stress & stressed syllable contains [i] & name ends with [i/ə] & majority open syllables* }

I performed the category sorting and weight calculation separately for M and F names to derive the M- and F-indexed constraint weights. Weights were calculated from SSA token counts (i.e. count of people), rather than type counts (i.e. count of names), reflecting the assumption that the modeled grammar is shaped by associations across individuals, rather than name lists.

Results: Predicted and observed category counts were aligned (R^2 fit to $x = y$: M=0.88, F=0.81), indicating that the limited model generally captures the distributional patterns in (1)–(7). Next, to test whether these differences in M and F names contribute to diachronic shifts in gender association, I extracted names that shifted M→F between 1955–2024 (>70% M and given to >100 children one year, >70% F and given to >100 children in a later year). This yielded 27 names, with a mean harmony improvement of 0.535 when evaluated by the F-indexed ("destination" gender) rather than the M-indexed constraints. A 100,000-trial Monte Carlo simulation found that random samples of 27 M names reached this level of harmony improvement under F-indexed constraints with a probability of 0.00001—an exceptionally unlikely result. The corresponding 12-name F→M comparison set showed a mean harmony improvement under M-indexed ("destination") constraints of 0.96, with probability 0.0002 under the analogous simulation procedure:

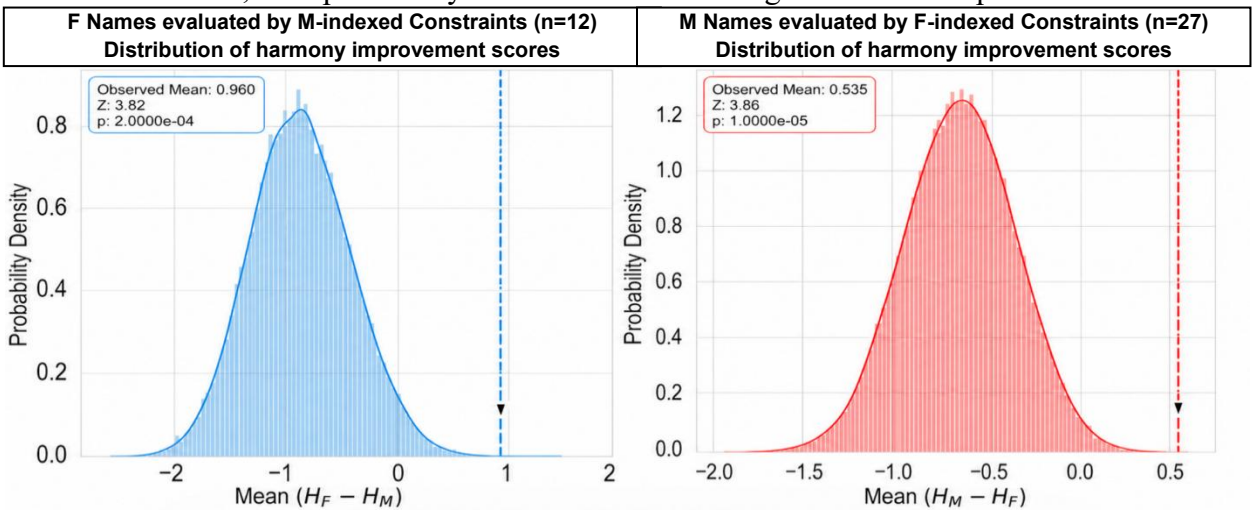


Figure 1. *Mean harmony improvement in random name samples evaluated by opposite-gender-indexed constraints. Blue and pink dotted lines: observed mean harmony improvement under "destination" gender-indexed constraints for names that shifted F→M and M→F, respectively.*

Beyond the many social factors that shape naming trends regarding gender association, these results indicate that phonology plays a robust and measurable role. The gradient differences between M and F names have many of the characteristics of sound-symbolism; these results support and build upon Shih (2020) who argues that sound-symbolic patterns should not be treated as exceptions to "core" phonology but can be accounted for using the same formal models applied to core phonological patterns. In fact, she illustrates this point with M and F names using two hand-weighted constraints. I extend Shih's approach by formulating a larger constraint set and estimating weights from real name data. The resulting grammar not only captures broad distributional patterns across English M and F names, but it also provides predictive and explanatory power concerning historical shifts in the gender association of names.

References: Cassidy, K. W., Kelly, M. H., & Sharoni, L. J. (1999). Inferring gender from name phonology. *Journal of Experimental Psychology: General, 128* (3), 362–381. • Cutler, A., McQueen, J., & Robinson, K. (1990). Elizabeth and John: Sound patterns of men's and women's names. *Journal of Linguistics, 26* (2), 471–482. • Shih, S. (2020). Gradient categories in lexically-conditioned phonology: An example from sound symbolism. *Proceedings of the Annual Meetings on Phonology.* • Hayes, B. (2009). Manual for Maxent Grammar Tool. UCLA. • Slater, A.S., Feinman, S. (1985). Gender and the phonology of North American first names. *Sex Roles* 13, 429–440.

Phonological category adaptation to non-native speech in preschoolers with symptoms of ADHD

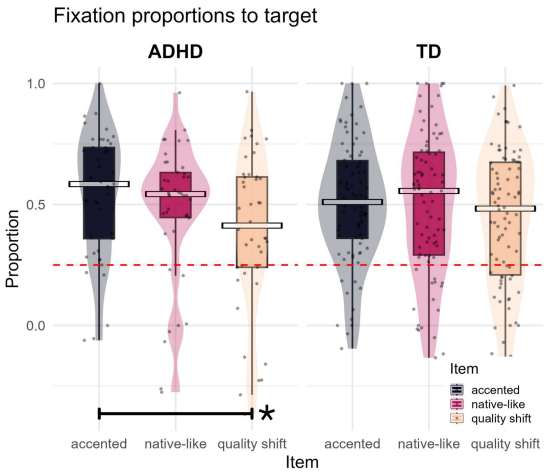
Monika Kučerová[1,2], Kateřina Chládková[1,2]
[1] Institute of Czech Language and Theory of Communication, Charles University
[2] Institute of Psychology, Czech Academy of Sciences

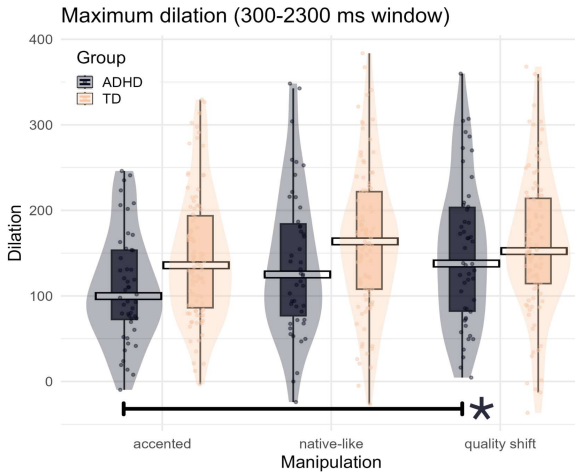
Adults recognize linguistic units in an unfamiliar accent quickly and reliably (Cristia et al., 2012), however, children tolerate phonemic deviations to a lesser degree (Cooper et al., 2023). Accent adaptation has been studied in typically developing preschoolers, who recognize accented word forms, albeit more slowly than familiar pronunciations (Creel, 2012). Despite not being officially recognized as a symptom (American Psychiatric Association, 2021), language problems in various domains have been reported in individuals with ADHD (Korrel et al., 2017). Current research focuses on phonological skills as a predictor of reading ability. However, on-line processing of accented speech likely relies on different cognitive processes than metalinguistic phonological ability. Further, individuals with ADHD may experience higher cognitive load during speech processing (Derawi et al., 2023). We test whether children with symptoms of ADHD differ from typically developing (TD) children in phoneme adaptation strategy after exposure to a non-native accent. We examine word recognition using fixations and cognitive load using pupil size in the visual world paradigm (VWP).

Participants are Czech monolingual children: 16 with symptoms of ADHD (mean age=71 months, SD=11 months), and 30 TD (mean age=69 months, SD=9 months). Parents complete the ADHD-IV scale in Czech; children with symptoms of inattention and/or hyperactivity-impulsivity are assigned to the ADHD group (cf. Theiner et al. 2013). Vowel length is phonemic in Czech, but not in Ukrainian. Ukrainian-accented Czech features stress shifts that result in vowel lengthening. After exposure to a short video narrated by an L1 Ukrainian talker, we test word recognition using the VWP with one novel and three real objects. Words are presented in the Ukrainian-accented talker's voice in carrier phrases as (1) produced naturally with a lengthened vowel ([bɛruːʃka] for /bɛruʃka/, *ladybug*), (2) manipulated with a shortened vowel to sound native-like ([bɛruʃka]), or (3) produced with a shifted vowel quality to sound unfamiliar ([bɛraʃka]). Eye gaze data are collected with Eyelink Portable Duo.

We calculate the proportion of time fixating the target image relative to time fixating any image, in a pre-naming window (0-2000 ms before target onset) and a post-naming window (300-2300 ms after). Target fixation proportion is equal to post-naming proportion minus pre-naming proportion. Fixation proportion is predicted using a linear mixed-effects model with predictors Group (ADHD, TD) and Manipulation (1-3 above); by-participant varying intercepts are fitted. Pupil dilation in the post-naming window is calculated as pupil size during the post-naming window minus pupil size in a pre-naming interval. Maximum pupil dilation in the post-naming window was estimated with the same predictors and random effect as in the fixation model. Preliminary results show that TD and ADHD children fixate target images above chance for all manipulations. ADHD children fixate the target less after hearing a word with a vowel quality shift than after hearing an accented word with a long vowel (see Figure 1). ADHD children also show larger maximum pupil dilation on the trials with a vowel quality shift compared to trials with a lengthened vowel (see Figure 2).

Both groups recognized Ukrainian-accented words, and also unfamiliar words with vowel quality changes. However, ADHD children were more likely to recognize the target word in its accented form than in a form with a shifted vowel. Smaller relative fixation proportions for quality-shifted words suggest that while the ADHD group may employ a broad relaxation of their lexical matching criteria, recent input facilitates more efficient recognition of accented word forms. In other words, their lexical matching process is sensitive to familiarity rather than being an across-the-board relaxation of phonological categories. Further, our pupil size results indicate that it may be more cognitively demanding for ADHD children to process unfamiliar word forms, in comparison to familiar accented forms. Our findings underscore the impact of cognitive processing differences underlying ADHD on phonological processing and word recognition.





Left: Figure 1. Fixation proportions to target. The x-axis shows Manipulation (native-like pronunciation, lengthened pronunciation, quality shift). Red dashed line indicates the chance level. Right: Figure 2. maximum pupil dilation (in arbitrary units) in the post-target onset window. Boxplots show medians and extend from the 25th to the 75th percentile.

References

American Psychiatric Association. (2021). Diagnostic and Statistical Manual of Mental Disorders-5 (DSM-5). In Encyclopedia of Gerontology and Population Aging (pp. 1414–1425).

Cooper, A., Paquette-Smith, M., Bordignon, C., & Johnson, E. K. (2023). The Influence of Accent Distance on Perceptual Adaptation in Toddlers and Adults. *Language Learning and Development*, *19*(1), 74–94.

Creel, S. C. (2012). Phonological similarity and mutual exclusivity: On-line recognition of atypical pronunciations in 3–5-year-olds. *Developmental Science*, *15*(5), 697–713.

Cristia, A., Seidl, A., Vaughn, C., Schmale, R., Bradlow, A., & Floccia, C. (2012). Linguistic Processing of Accented Speech Across the Lifespan. *Frontiers in Psychology*, *3*.

Derawi, H., Reinisch, E., & Gabay, Y. (2023). Internal Cognitive Load Differentially Influences Acoustic and Lexical Context Effects in Speech Perception: Evidence From a Population With Attention-Deficit/Hyperactivity Disorder. *Journal of Speech, Language, and Hearing Research*, *66*(10), 3721–3734.

Korrel, H., Mueller, K. L., Silk, T., Anderson, V., & Sciberras, E. (2017). Research Review: Language problems in children with Attention-Deficit Hyperactivity Disorder – a systematic meta-analytic review. *Journal of Child Psychology and Psychiatry*, *58*(6), 640–654.

Phonological opacity: challenging the pathological account

Peter Nyhuis [✉], Sacha Beniamine & Erich Round
Surrey Morphology Group, University of Surrey

In phonology, *opaque process interactions* have been taken to represent a high level of complexity, since they involve generalisations which are difficult to motivate directly from surface wordforms, and are theorised to be prone to reanalysis in transparent terms (Kiparsky 1973). Similarly in morphology, *inflection class systems* have been claimed to show an unnecessary degree of complexity, as in Aronoff's (2016) claim that morphology is 'a disease, a pathology of language'. Recent research in 'abstractive-discriminative-implicational' morphology (Blevins 2020) has challenged the pathological account of inflection systems, demonstrating that their organisation tends to make paradigm cells predictable from one another, even in systems with a high degree of fusion and syncretism (Ackerman & Malouf 2013; Round et al. 2025). In a similar vein, and from a similar perspective, we wish to challenge the pathological account of opacity.

We argue that all phonological alternations, including opaque ones, can be understood as a type of paradigm inference problem akin to that posed by inflection classes, which we illustrate with a putative overapplication interaction (Baković 2007) in the phonology of Wubuy, a Gunwinyguan language of northern Australia (Table 1). We demonstrate this in an implemented model that treats phonological and morphological generalisations alike as a set of bidirectional alternations emerging from pairwise comparisons between paradigm cells. 'Phonological' classes of alternations differ from 'morphological' ones only in applying to a set of stems that share some phonological property, and both types present difficulty for the language user only as a function of pattern competition (measured in terms of entropy) — not as a function of opacity.

In traditional analyses of Wubuy (Baker 2009; Heath 1984), underlying /k/ is deleted preceding any other consonant: this is 'Velar Stop Deletion' (see Table 2). Meanwhile, underlying continuants are realised as stops when they follow a non-continuant such as a stop or a nasal: this is 'Hardening'. Hardening feeds Velar Stop Deletion, such that underlying /a-ɻaŋa**k-r**uc/ 'Obl-tree-Loc' is realised as [aɻaŋa**t**uc] on the surface. This is classically opaque: Velar stop deletion has removed the trigger for Hardening, making it appear unmotivated on the surface. A recent reanalysis of these interactions couched in Relational Morphology (Nyhuis 2023) argues that the generalisation here is fundamentally paradigmatic. The locative suffix allomorph in nouns like [aɻaŋatuc] can be predicted not by the phonological form of the base, but rather by the form it takes when it is word-final — that is, the nominative form, which ends in /k/.

While we agree with the general thrust of that analysis, we find the formalisation too permissive to implement computationally. Instead, we infer alternation patterns automatically using `Qumin` (Beniamine 2018; Beniamine & Bouton 2025). Qumin evaluates each pair of paradigm cells and produces patterns which specify both alternating material and a context of application obtained by minimal generalisation (Albright & Hayes 2002). It then uses these patterns to calculate how difficult it is on average to guess the forms of a cell knowing those of another cell (implicative entropy, Bonami & Beniamine 2016), and to analyze inflection classes. These are represented as a semi-lattice of shared patterns across lexemes (Beniamine 2021). For nouns previously analysed as containing a final underlying /k/, we find an alternation pattern where _k in the nominative alternates with _tuc in the locative. In these terms, the final /k/ in [ɻaŋak] is diagnostic of nominative case, and uniquely identifies the class to which the form belongs (making all other cells predictable). Likewise, all paradigm cells are predictable on the basis of the locative, even though this cell constitutes the evidence for opacity (Figure 1), since a continuant followed by [...tuc] is unique to the class. And yet the instrumental, which only shows 'transparent' application of Velar Stop Deletion (Table 2), does introduce entropy, as the suffix fails to distinguish the classes. Thus while Qumin has hitherto been exploited to address more properly morphological questions, here we demonstrate that it can offer novel and illuminating perspectives on questions at the heart of phonology also.

	marija ‘food’	l̪aŋac ‘bait’	ɻaŋak ’tree’
Ntop.Nom	marija	l̪aŋac	ɻaŋak
Top.Nom	anamarija	anal̪aŋac	anaɻaŋak
Obl.Nom	amarija	al̪aŋac	aɻaŋak
Obl.Dat	amarijawuj	al̪aŋackuj	aɻaŋakuj
Obl.Abl	amarijawala	al̪aŋackala	aɻaŋakala
Obl.Loc	amarijaruc	al̪aŋactuc	aɻaŋatuc
Obl.Instr	marijamiri	l̪aŋacmiri	ɻaŋamiri

Table 1: Partial paradigm for some representative Wubuy nouns.

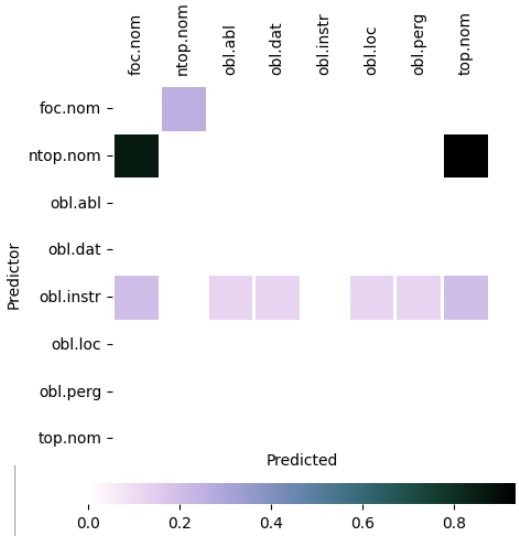


Figure 1: Entropy for pairwise alternations in Wubuy Nouns. Less predictable cells are darker.

		locative			instrumental	
Input:	/a-marija-ruc/	/a-l̪aŋac-ruc/	/a-ɻaŋak-ruc/	/marija-miri/	/l̪aŋac-miri/	/ɻaŋak-miri/
Hardening	—	a-l̪aŋac-tuc	a-ɻaŋak-tuc	—	—	—
Velar deletion	—	—	a-ɻaŋa-tuc	—	—	ɻaŋa-miri
Output:	amarijaruc	al̪aŋactuc	aɻaŋatuc	marijamiri	l̪aŋacmiri	ɻaŋamiri

Table 2: Serial derivations for locative and instrumental nouns.

References

Ackerman, Farrell & Robert Malouf (2013). ‘Morphological Organization: The Low Conditional Entropy Conjecture’. *Language* 89.3, pp. 429–464.

Albright, Adam & Bruce Hayes (2002). ‘Modeling English Past Tense Intuitions with Minimal Generalization’. *Proceedings of the ACL-02 Workshop on Morphological and Phonological Learning* -. The ACL-02 Workshop. Vol. 6. Not Known: Association for Computational Linguistics, pp. 58–69.

Aronoff, Mark (2016). ‘Unnatural Kinds’. *The Morphome Debate*. Ed. by Ana R. Luís & Ricardo Bermúdez-Otero. First edition. Oxford: Oxford University Press, pp. 11–32.

Baker, Brett J. (2009). ‘Two Opaque Processes in Wubuy: Issues in Learnability and the Lexicon’. OzPhon. University of Melbourne.

Baković, Eric (2007). ‘A Revised Typology of Opaque Generalisations’. *Phonology* 24.2, pp. 217–259.

Beniamine, Sacha (2018). *Classifications flexionnelles. Étude quantitative des structures de paradigmes*. PhD thesis. Université Sorbonne Paris Cité - Université Paris Diderot (Paris 7).

— (2021). ‘One Lexeme, Many Classes: Inflection Class Systems as Lattices’. *One-to-Many-Relations in Morphology, Syntax, and Semantics*. Ed. by Berthold Crysmann & Manfred Sailer. Berlin: Language Science Press, pp. 23–51.

Beniamine, Sacha & Jules Bouton (2025). *Qumin*. Version 3.0.10.

Blevins, Jim (2020). ‘Two Frameworks of Morphological Analysis’. *Linguistic Analysis*.

Bonami, Olivier & Sacha Beniamine (2016). ‘Joint Predictiveness in Inflectional Paradigms’. *Word Structure* 9.2, pp. 156–182.

Heath, Jeffrey (1984). *Functional Grammar of Nunggubuyu*. Canberra: Australian Institute of Aboriginal Studies.

Kiparsky, Paul (1973). ‘Abstractness, Opacity and Global Rules’. *Three Dimensions of Linguistic Theory*. Ed. by Osamu Fujimura. Tokyo: Taikusha, pp. 1–136.

Nyhuis, Peter Christopher (2023). *Lexical Representation in Wubuy*. PhD thesis. Melbourne, Australia: University of Melbourne.

Round, Erich, Louise Esher & Sacha Beniamine (2025). ‘The Natural Stability of Autonomous Morphology: How an Attraction–Repulsion Dynamic Emerges from Paradigm Cell Filling’. *Morphology* 35.1, pp. 1–49.

Does"stress deafness" have a basis in the learnability of layered hidden structure?

Ryan Sandell, Ludwig-Maximilians-Universität München

A substantial body of psycholinguistic and neurolinguistic work has accumulated evidence for a phenomenon termed "stress deafness": speakers of L1s with fixed-stress word-prosodic systems exhibit worse performance on stress identification tasks (Dupoux et al. 2001, Peperkamp and Dupoux 2002, Peperkamp et al. 2010, Domahs 2012, i.a.). These studies distinguish three broad classes of performance in stress identification: Finnish-like ("stress deaf"), Polish-like ("stress impaired"), and Spanish-like (lexical stress). To account for these differences, Peperkamp (2004) advanced the intuitively plausible hypothesis that "stress deafness" is directly connected to the ease of learnability and/or speed with which a stress system can be identified early in phonological acquisition: the extent to which stress is (redundantly) encoded in the lexicon depends upon whether a language's stress pattern is SURFACE OBSERVABLE, i.e., whether it "can be inferred from bare utterances", without knowledge of (prosodic) word boundaries (Peperkamp 2004:103–7). According to Peperkamp, the surface-observability of an algorithmic stress pattern is primarily disrupted by the presence of monosyllabic content words that can result in inconsistencies in the stress patterns of at the edges of bare utterances. Yet, despite research into the learnability of stress grammars from unlabeled surface representations (Tesar and Smolensky 2000, Jarosz 2013, i.a.), the surface-observability hypothesis, and how it might interact with known differences in the learnability known for stress windows (Staubs 2014, Stanton 2016) remains uninvestigated.

The present paper tests the SURFACE-OBSERVABILITY HYPOTHESIS as an account of "stress deafness" via computational simulation of the learnability of stress grammars in the pre-lexical stage (i.e., "phonotactic learning") employing Robust Interpretive Parsing (RIP; Tesar and Smolensky 2000:Ch. 4). Novel is the inclusion of two levels of hidden structure to be learned simultaneously from overt forms: both the structure of feet and of prosodic word boundaries are to be imputed from overt forms in which syllabic strings may contain multiple syllables bearing primary stress. Example parses for an overt token [σ1 σ1 s s σ1 s] are shown in the table in (1), where curly brackets delimit prosodic words.

(1) Example parses of [σ1 σ1 σ σ σ1 σ] mapped faithfully to UR //σ1 σ1 σ σ σ1 σ//

Overt Form		**Parse**	Comment
[σ1 σ1 σ σ σ1 σ]	⇒	{(σ1) (σ1 σ) σ (σ1 σ)}	violates CULMINATIVITY
[σ1 σ1 σ σ σ1 σ]	⇒	{(σ1)} {(σ1)} {σ σ} {(σ1 σ)}	violates OBLIGATORINESS
[σ1 σ1 σ σ σ1 σ]	⇒	{(σ1)} {(σ1 σ) σ} {(σ1 σ)}	exactly 1 primary stress per ω

In order to assess the differences that different patterns of overt data (viz., different languages) might have on the makeup of lexical representations, the constraint IDENT(Stress) is included alongside standard prosodic markedness constraints (see Kager 1999:Ch. 4). Conceptually, the strength of the weight learned for IDENT(stress) may act as a proxy for the extent to which lexical representations maintain encoding of information on stress. The relative strength of IDENT(stress) may similarly indicate how readily new data encountered later (e.g., in L2 acquisition) that conflicts with the stress grammar of the L1 may be learned, i.e., how relatively "malleable" the stress grammar constructed for the L1 is.

Two specific questions were assessed through computational simulation: 1) does larger window for primary stress results in a higher weight for IDENT(stress) when learning from a fixed N of tokens?; 2) does a higher weight for IDENT(stress) result when the learning data contain monosyllabic prosodic words, i.e., strings containing elements intended to be parsed as $[(\acute{\sigma})]_{\omega}$? Accordingly, twelve distinct languages with quantity-insensitive metrical stress varying along three parameters were tested: edge orientation, default position of primary stress (1/2/3), and whether stressed monosyllabic prosodic words occur. Table (2) lists eight of twelve tested combinations of these parameters and natural languages that (approximately) instantiate them (cf. Gordon 2002); see further below on the rightmost column.

(2)

Code	Edge	Edge Dist.	Min. Wd. Req.?	Language	μ(μ(MaxCandProb))
L1N	Left	1	No	Hungarian	89.8%
L1Y	Left	1	Yes	Lardil	78.7%
R1N	Right	1	No	Azerbaijani	88%
R1Y	Right	1	Yes	Diegueño	78.6%
L2N	Left	2	No	Tocharian B	57.5%
L2Y	Left	2	Yes	Mohawk	67.7%
R2N	Right	2	No	Polish	55.2%
R2Y	Right	2	Yes	Tiriyó	66.7%

All simulations (conducted in R v. 4.5.3) employed Maximum Entropy Grammar trained through Stochastic Gradient Ascent (Jäger 2007; compare Staubs 2014). The basis for all simulations is a set of 502 tableaux, reflecting all URs consisting of all possible strings composed of primary stresses (σ1) and unstressed syllables (s) 1–8 syllables in length with at least one primary stress (σ1); unfaithful mappings and all alternatives in parses resulted in a candidate set of 7,364,641 OVERT ⇒ PARSE pairs. Sets of overt forms for the training data representing the twelve languages have between 33 (L1Y and R1Y) and 285 (L3N and R3N) distinct patterns; all overt forms set as equiprobable. 20 grammars per language were trained on 50,000 tokens each. Full documentation and code for the preparation and execution of these simulations is available at `https://tinyurl.com/mpeu49vv`. Overall, L1 and R1 languages were learned correctly, regardless of whether monosyllabic prosodic words are include or not: sum squared error (SSE) between the grammar's outputs and target overt representations was < 1. This is also true of the languages L2Y and R2Y, which are surface observable by Peperkamp's definition. The non-surface-observable languages L2N, R2N, and L3/R3 failed to achieve the convergence criterion of SSE < 1 within 50,000 tokens. Larger edge distance correlates with greater uncertainly in the ideal parse: μ(μ(MaxCandProb)) in (2) gives the average across all 20 simulations of the average probability of the single highest probability candidate for each tableaux. Whether or not monosyllabic prosodic words are present in a language robustly correlates with both the weight of IDENT(Stress) (r = -0.97, t = -14.13, df = 10, $p < 0.01$) and number of grammar updates (r = -0.97, t = -14.13, df = 10, $p < 0.01$); .These results broadly support the SURFACE-OBSERVABILITY HYPOTHESIS and suggest that RIP is sufficient for simultaneously learning two layers of hidden structure (foot structure and prosodic word boundaries) from stress alone, but only when all strings in the learning data exhibit surface-observable stress.

References

Domahs, Ulrike. 2012. Stress "deafness" in a Language with Fixed Word Stress: An ERP Study on Polish. *Frontiers in Psychology* 3.1–15.

Dupoux, Emmanuel, Sharon Peperkamp, and Nuría Sebastián-Gallés. 2001. A Robust Method to Study Stress "Deafness". *Journal of the Acoustical Society of America* 110.1606–18.

Gordon, Matthew. 2002. A Factorial Typology of Quantity-Insensitive Stress. *Natural Language and Linguistic Theory* 20.491–552.

Jäger, Gerhard. 2007. Maximum Entropy Models and Stochastic Optimality Theory. In Jane Grimshaw, J. Maling, Christopher D. Manning Manning, J. Simpson and A. Zaenen (eds.), *Architectures, Rules, and Preferences: A Festschrift for Joan Bresnan*, 467–79. Stanford: CSLI Publications.

Jarosz, Gaja. 2013. Learning with Hidden Structure in Optimality Theory and Harmonic Grammar: Beyond Robust Interpretive Parsing. *Phonology* 30.27–71.

Kager, René. 1999. *Optimality Theory*. Cambridge: Cambridge University Press.

Peperkamp, Sharon. 2004. Lexical Exceptions in Stress Systems: Arguments from Early Language Acquisition and Adult Speech Perception. *Language* 80.98–126.

Peperkamp, Sharon, and Emmanuel Dupoux. 2002. A Typological Study of Stress "Deafness". In Carlos Gussenhoven and Natasha Warner (eds.), *Laboratory Phonology 7*, 203–40. Berlin: Mouton de Gruyter.

Peperkamp, Sharon, Inga Vendelin, and Emmanuel Dupoux. 2010. Perception of Predictable Stress: A Cross-linguistic Investigation. *Journal of Phonetics* 38.422–30.

Stanton, Juliet. 2016. Learnability Shapes Typolgy: The Case of the Midpoint Pathology. *Language* 92.753–91.

Staubs, Robert. 2014. Computational Modeling of Learning Biases in Stress Typology. Ph.D. diss., University Of Massachusetts, Amherst.

Tesar, Bruce, and Paul Smolensky. 2000. *Learnability in Optimality Theory*. Cambridge, MA: MIT Press.

Interacting and variable processes in French liaison: GSRs at the word and phrase levels

Jeffrey Lamontagne[1,2,3] & Anne-Michelle Tessier[4]

[1]Concordia University, [2]Indiana University, [3]McGill University, [4]University of British Columbia

Introduction. French liaison is a notoriously complex phenomenon: in restricted syntactic or prosodic contexts, a liaison consonant (LC) may surface when two words are adjacent (e.g. [gʁɑ̃tami] *grand ami*) but not when in isolation (e.g. [gʁɑ̃] *grand* 'tall/great', [ami] *ami* 'friend') [e.g., 1,5]. Debates have especially focused on the affiliation of the LC (word 1, word 2, or both) and its underlying representation (e.g. floating, deleted). To further complicate analysis, while LCs are most straightforwardly interpreted as being part of word 1 because word 1 generally determines which consonant is realised, the LC does not exhibit the coda-like behaviour of stable word-final consonants, which is the case we seek to model. In Quebec French, stable final consonants categorically trigger the laxing of preceding short high vowels (e.g. [p͡tsɪt] *petite* 'small.FEM'; cf. [p͡tsi] *petit* 'small.MASC'), but this process does not occur before LCs [3]. Additionally, /t/ undergoes affrication before high vowels at a higher rate when a LC compared to when a resyllabified stable final /t/ -- but not categorically like initial or medial /t/ do [1,6].

Under Gradient Symbolic Representations (GSR) [5,7], segments can have different activations (i.e., different degrees of presence in cognitive representations), and many constraint violations incur penalties proportional to the segment's activation. Variability in liaison rates is captured by specifying partially activated LCs on both word 1 and word 2 and, if either is not specified, LCs do not surface [5,7]. We extend this analysis to demonstrate that GSR can also capture the different behaviours of liaison and stable consonants, the floating-segment behaviour of the former emerging from partially activated consonants' dispreference for syllabification.

Corpus study. Spontaneous speech from 85 native speakers of Quebec French (877k total words [2]) was coded using automated classification for /t d/ affrication (categorical within words; variable across them) and liaison (variable). After preliminary confirmation, high-vowel laxing was not made variable in final syllables of liaison words. Results support Côté's [1] findings from read speech that *enchaînement* consonants undergo affrication less often than do *liaison* consonants, and both less than CV sequences within a word (rates in tableaux below).

Formal analysis. Combining GSR with Harmonic Grammars and computing constraint weights using Maximum Entropy models (as in [5,7]), we find that two strata are needed. In the first stratum, stable final consonants trigger high-vowel laxing (Tableau 1), but LCs do not because the partially activated segment – which is not deleted – is not syllabified (Tableau 2). This effectively floating LC then gets parsed in the onset of the following word and coalesces with word 2's LC in stratum 2 (Tableau 3), affricating at a higher rate than do stable final consonants (Tableau 4) because of affrication of word 2's initial LC in stratum 1.

…/i_1t_1/	IDENT(tns) 45.96	*[tns]C# 94.66	DEP 100.00	MAX 100.00	*UNSYLL 100.00	*PARTMEL 8.08	Pred.
[tsi.]				1			0%
[tsit.]		1					0%
[tsɪt.]	1						100%

Tableau 1: Stratum 1 computation of *petite* (stable consonant with categorical laxing).

Discussion. This analysis expands GSR by introducing strata with evidence that partially activated segments are present at two levels of computation, with strata having been analysed for phenomena outside of GSR and here further motivated by the overlap between liaison contexts and phonological spell-out domains (cf. [4]). It is critical for the partially activated LC to be unsyllabified at the lexical level (therefore never triggering laxing) and yet be present at the phrasal level (where it reaches sufficient activation for the LC to surface). This analysis thereby resolves the issue of apparent opacity that is a challenge for rule-based approaches and Strict CV, for example. The analysis further elucidates the status of partially activated segments: they not only exhibit variable realisation, but a dispreference for association with melodic structure (our novel *PARTMEL constraint), yielding floating-consonant behaviour. The analysis also supports intuitions that stable consonants – unlike LCs – should not be interpreted as covertly undergoing coalescence, but instead exhibiting deletion of word 2's partially activated LC.

…/$i_1t_{0.61}$/	IDENT(tns) 45.96	*[tns]C# 94.66	DEP 100.00	MAX 100.00	*UNSYLL 100.00	*PARTMEL 8.08	Pred.
[tsi.]				0.61			0%
[tsit.]		1	0.39				0%
pe.[$tsit_{0.61}$.]		1				1	0%
pe.[tsɪt.]	1		0.39				0%
pe.[$tsɪt_{0.61}$.]	1					1	0%
pe.[tsi<t_1>]		1	0.39		1		0%
pe.[tsi<$t_{0.61}$>]		1			0.61	1	100%
pe.[tsɪ<t_1>]	1		0.39		1		0%
pe.[tsɪ<$t_{0.61}$>]	1				0.61	1	0%

Tableau 2: Stratum 1 computation of *petit* (variable liaison; consonant must be in Stratum 2, but no laxing).

…i<$t_{0.61}$>+$ts_{0.39}$i…	IDENT(str) 2.30	*ti 1.00	DEP 86.11	MAX 42.15	UNIF 100	Pred.	Actual
[i.i]				1		10%	7%
[i.ti] (coal.)	0.39	1			0.39	34%	32%
[i.tØi] (del. C2)		1	0.39	0.39		0%	
[i.tsi] (coal.)	0.61				0.39	56%	61%
[i.tsØi] (del. C2)			0.39	0.39		0%	

Tableau 3: Stratum 2 computation of a case like *petit innocent* (LC, more affrication).

…ɪt. + $ts_{0.39}$i…	IDENT(str) 2.30	*ti 1.00	DEP 86.11	MAX 42.15	UNIF 100	Pred.	Actual
[ɪi]						0%	0%
[ɪti] (coal.)	0.39	1			0.39	0%	63%
[ɪtØi] (del. C2)		1		0.39		60%	
[ɪtsi] (coal.)	1				0.39	0%	37%
[ɪtsØi] (del. C2)	1			0.39		40%	

Tableau 4: Stratum 2 computation of a case like *petite innocente* (stable consonant, less affrication).

*UNSYLL: Assign violation equal to unsyllabified segment's activation.

*PARTMEL: Assign violation per segment not associated with melodic structure.

[1] Côté, M.-H. (2014). Liaison et assibilation en français laurentien. *La liaison: Approches contemporaines*, 9–32. Berne: Peter Lang. **[2] Durand, J., Laks, B. & Lyche, C. (2002).** La phonologie du français contemporain: usages, variétés et structure. *Romanistische Korpuslinguistik- Korpora und gesprochene Sprache*, 93-106. Tübingen: Gunter Narr Verlag. **[3] Lamontagne, J. (2019)**. Acoustic evidence of phonemicization: Lax high vowels in Quebec French. *LSA Annual Meeting*, New York City. **[4] Sande, H., Jenks, P. & Inkelas, S. (2020).** Cophonologies by Ph(r)ase. *Nat Lang Lg Theory* 38: 1211–1261. **[5] Smolensky, P., & M. Goldrick. (2016).** Gradient symbolic representations in grammar: The case of French liaison. Ms, ROA 1286. **[6] Storme, B. (2024).** Paradigm uniformity effects on French liaison. *Nat Lang Lg Theory* 42:1307-1352. **[7] Tessier, A.-M., & K. Jesney. (2021).** Learning French liaison with gradient symbolic representations: Errors, predictions, consequences. *Proc. of the 2020 AMP*, 4940.

Evidence for the Maximum Onset Principle in morpheme segmentation

Dhanya Charan[1] and Jennifer Kuo[2]
[1]University of California, San Diego (UCSD), [2]Cornell University

Background: In early acquisition, morpheme segmentation is known to be sensitive to both distributional cues [1, 2] and syllable structure preferences. Specifically, children exhibit a preference for onsets over codas [3], formalized as the Maximal Onset Principle (MOP). While adults are known to use phonotactics in morpheme segmentation [4, 5], it is unclear if they still prefer onset maximization; for example, is "progress" syllabified as [pra.**gr**ɛs] (onset-maximizing) or [pra**g.r**ɛs] (non-maximizing)? In this study, we investigate whether adults utilize the MOP in morpheme segmentation using an artificial language learning paradigm, in which adults are taught languages where phonotactics and MOP variably aid in segmentation. We find that speakers do utilize the MOP, but only if doing so results in a phonotactically licit output.

Experiment: The study was run online through Prolific (n=60 self-reported native English speakers, data collection in progress). In the training phase, participants learned a new suffix via auditory and orthographic presentation of CVCVC(C) stem-plural pairs (e.g. pəpag ~ pəpagram; N=30). To manipulate the effect of phonotactics/MOP on morpheme segmentation, we varied two between-speaker factors: the suffix ([ram] or [kpam]) and type of stem-final consonant participants saw in training ({b,d,g} or {s,n,l}). The factors were crossed, resulting in four conditions (see Table 1).

Table 1. *Experimental stimuli design*

Suffix	*Stem-final C*	*Example word* (underline=stem, red=bad syllable)
ram	{b,d,g}	[**pə.pa.g**ram] or [**pə.pag**.ram]
ram	{s,n,l}	*[**pə.pa.l**ram] or [**pə.pal**.ram]
kpam	{b,d,g}	*[**pə.pag**k.pam] or *[**pə.pag**.kpam]
kpam	{s,n,l}	*[**pə.pal**.kpam] or [**pə.palk**.pam]

We hypothesize that speakers use both phonotactics and the MOP in segmentation. Based on this hypothesis, Table 2 shows the conditions arranged in order of predicted learning accuracy: A (phonotactics+MOP are helpful) > B (MOP is helpful) > C (MOP is harmful) > D (phonotactics+MOP are harmful). Conditions are also color-coded by whether phonotactics/MOP support the correct segmentation (green), are neutral w.r.t. segmentations (yellow), or harmful (red). In Cond. A, there is only one phonotactically licit syllabification of the suffixed form ([pə.pal.ram]). Assuming syllabification guides morpheme segmentation, participants should be biased towards correctly segmenting the suffix as [-ram]. In contrast, in Cond. D, phonotactics biases speakers towards an incorrect segmentation of the suffix as [-pam] based on the syllabification [pə.palk.pam]. In Conds. B & C, phonotactics do not bias learners towards any segmentation, but the MOP should bias speakers towards the onset-maximizing syllabification (e.g. in Cond. C, [pə.pa.gram] > [pə.pag.ram]). In the testing phase, participants were asked to provide the suffixed forms of 30 new stems (10 x {b,d,g}, {s,n,l}, and vowel-final stems). Two-thirds of items end in a segment that participants were never exposed to in training, allowing us to test how the plural suffix is generalized across new phonological contexts.

Results: Results, summarized in Fig. 1, suggest effects of both MOP and phonotactics in the predicted direction; participants performed the best in Condition A (P=98%), where both phonotactics and MOP agreed with the correct segmentation. Participants also performed better in Conditions A/B (MOP is "helpful") than in Conditions C/D (MOP is "harmful"). However, participants' error patterns provide only partial support for an MOP effect. In Cond. C, 75% of

incorrect responses had a suffix of the form [{b,d,g}ram] (cf. correct = [ram]); this is exactly the segmentation supported by the MOP. However, in Cond. B, the most common error was learning the suffix as /pam/ (cf. correct /kpam/), contrary to predictions of the MOP. Results are supported by a mixed effects logistic regression model, which found an effect of TrainingFinalC ({b,d,g} vs. {s,n,l}) and a significant interaction of TrainingFinalC and suffix (ram vs. kpam).

Table 2. *Experimental design and conditions, ordered by predicted learning accuracy*

Condition	A	B	C	D
Correct suffix	ram	kpam	ram	kpam
Stem-final C	{s,n,l}	{b,d,g}	{b,d,g}	{s,n,l}
Syllabifications	*[**pə.pa.l**ram] or [**pə.pal**.ram]	*[**pə.pag**k.pam] or *[**pə.pag**.kpam]	[**pə.pa.g**ram] or [**pə.pag**.ram]	*[**pə.pal**.kpam] or [**pə.palk**.pam]
Phonotactic bias	ram	none	none	pam
MOP bias	ram	kpam	gram	pam

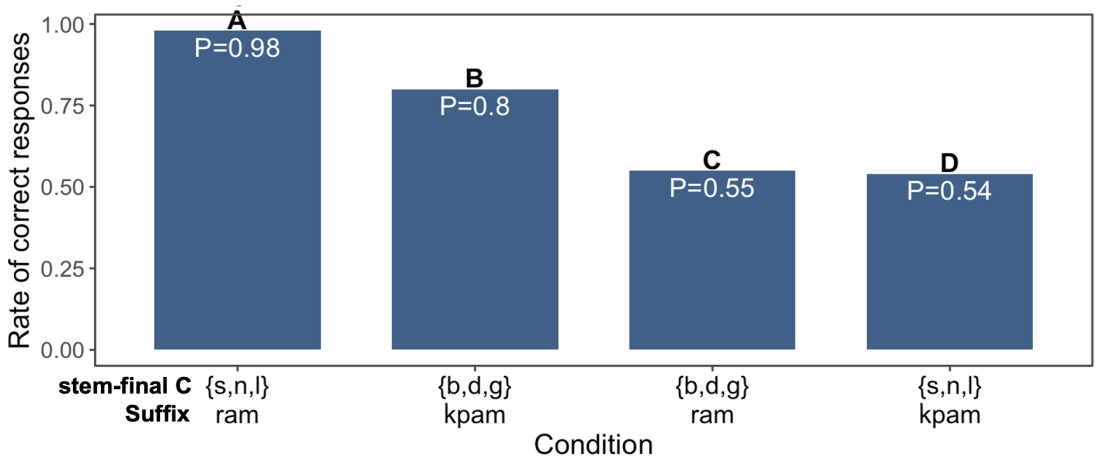


Figure 1. *Proportion of correct responses by experimental condition*

Discussion: In Cond. A, where any segmentation results in a phonotactically licit suffix, adults appear to utilize the MOP. However, when there is no phonotactically licit segmentation of the suffixed form (Cond. C), participant errors suggest a bias towards learning the suffix as the non-onset-maximizing [pam]. This could be because while the MOP is active in segmentation, speakers prioritize learning a phonotactically well-formed morpheme. Note that in the results, onset-maximization always causes the suffix to form a separate syllable from the stem. As such, these results could be explained by a preference for syllabifying affixes separately from stems. This is especially likely for speakers of English, where affixes tend to form a separate syllable from the stem. To test for this possibility, we are running a follow-up study in French, which has more vowel-initial suffixes that are syllabified with the final consonant of the stem.

Selected references: [1] Jusczyk, P. & Aslin, R. (1995). Infants′ detection of the sound patterns of words in fluent speech. *Cognitive psychology*. [2] Saffran et al. (1996). Statistical learning by 8-month-old infants. *Science*. [3] Seidl, A., & Johnson, E. (2008). Boundary alignment enables 11-month-olds to segment vowel initial words from speech. *JCL*. [4] McQueen, J. (1998). Segmentation of continuous speech using phonotactics. *JML*. [5] Dahan, D., & Brent, M. (1999). On the discovery of novel wordlike units from utterances: an artificial-language study with implications for native-language acquisition. *Journal of Experimental Psychology*.

Secwepemctsín non-local retraction harmony: Cyclically derived postvelar contexts

Julia Schillo
University of British Columbia

Overview: A formal analysis is presented for Secwepemctsín (Shuswap; Salishan) retraction harmony in an infixing reduplication context—a morphophonological context which leads to non-local harmonized vowels. Secwepemctsín retraction harmony is generally local, however, it can skip over an infixed reduplicated consonant depending on the type of harmony trigger involved. This complex interaction is accounted for in a stratal Optimality Theory (OT) analysis. All data was collected through original fieldwork.

Data: The Secwepemctsín postvelar inventory comprises uvular and pharyngeal consonants (Bessell 1992; Kuipers 1974), as well as a retracted counterpart to each unretracted vowel /i, i̱, e, e̱, u, u̱/. Retracted vowels occur underlyingly, but may also be derived through consonant–vowel retraction harmony. Uvular consonants trigger regressive retraction harmony on directly preceding vowels (1a), while pharyngeals trigger bidirectional harmony on directly adjacent vowels (1b–c) (Kuipers 1974). Outside of reduplication contexts, retraction harmony only applies locally, to directly adjacent vowels.

(1)	Input	Output	Gloss	
a.	/xł**u**qʷmn̓/	[xł**ú̱**qʷmən̓]	‘cup’	(CC)
b.	/m**e**ʕxn/	[m**é̱**ʕxən]	‘moon’	(GD)
c.	/ʕʷ**u**yt/	[ʕʷ**ú̱**yt]	‘he is tired’	(GD)

Infixing diminutive reduplication—optionally used in first person constructions to convey politeness—copies the consonant preceding a stressed vowel to the right side of the vowel: $C_1\acute{V}<C_1>C_2$ (Nakamura 2000) (additional schwas appear in some reduplicants due to a regular epenthesis process; Kuipers 1974). When a uvular or pharyngeal is the C_2, the intervening reduplicated C_1 means that the postvelar C_2 is no longer adjacent to the vowel. When the C_2 is a uvular, the vowel in the reduplicant surfaces as unretracted (2). However, when the C_2 is a pharyngeal, the vowel is systematically retracted in reduplicants (3a–b). In the Northern dialect, the high front vowel /i/ exceptionally surfaces as unretracted when a pharyngeal consonant has been moved away by reduplication, but /i/ surfaces as retracted in the base when the pharyngeal is adjacent to the vowel (3c).

(2)	Base	Gloss	Reduplicant	Gloss	
	[p**í̱**x̌məs]	‘he is hunting’	[p**íp**x̌əmwən]	‘I am hunting’	(GD)

(3)	Base	Gloss	Reduplicant	Gloss	
a.	[məʕm**é̱**ʕt]	‘gray’	[məʕm**é̱**məʕts]	‘a little bit gray’	(GD)
b.	[c̓l**í̱**ʕʷən̓s]	‘he scratches (it)’	[c̓l**í̱**ləʕʷən̓s]	‘I scratch (it)’	(GD)
c.	[s**í̱**ʕʷən̓s]	‘he loosens (it)’	[s**í**sʕʷən̓]	‘I loosen (it)’	(JW)

Analysis: A stratal OT (Bermúdez-Otero 2018; Kiparsky 2015; Mellesmoen 2025) analysis is used to account for non-local harmonized vowels, because these only occur in a specific morphophonological context, i.e., with the addition of a diminutive reduplication morpheme. Two strata are needed to account for the data: An earlier stratum (stratum 1) at which pharyngeal triggered retraction harmony applies, and a later stratum (stratum 2) at which both uvular triggered harmony and diminutive reduplication apply. Because pharyngeal triggered harmony occurs at an earlier stratum than reduplication, this allows harmonized retracted vowels to serve

as the input of the later stratum, accounting for non-local harmonized vowels, as in (3a–b). This analysis adopts a feature geometry based on the Laryngeal Articulator Model (Esling et al. 2019; Moisik et al. 2012), whereby the [RETRACTED] feature is dually linked to both the [PHARYNGEAL] and [DORSAL] place nodes, allowing ways to keep uvulars and pharyngeals distinct while also unifying them as a retracted natural class.

Pharyngeal triggered harmony: The input vowel at stratum 1 is an unretracted vowel. When there is a pharyngeal trigger, the winning candidate will have a harmonized retracted vowel due to the crucial ranking AGREE(PHARYNGEAL) >> DEP[RETRACTED]. The retracted vowel winner of stratum 1 then serves as the input of stratum 2. The winning candidate of stratum 2, with or without reduplication, is a faithful retracted vowel (Table 1a). This accounts for the bases and reduplicants of (3a–b).

Northern dialect: In the Northern dialect, a *i̲ constraint is ranked above AGREE(PHARYNGEAL) in stratum 1 to prevent /i/ from harmonizing. At stratum 2, *i̲ is demoted to a lower ranking than AGREE(PHARYNGEAL), causing /iʕ/ sequences to retract when there is no reduplication, and a faithful unretracted [i] when there is a reduplicated consonant intervening between the [i] and pharyngeal consonant (Table 1b). This accounts for the base and reduplicant in (3c).

Uvular triggered harmony: When there is a uvular trigger, a faithful unretracted vowel is the winner at stratum 1 due to the crucial ranking DEP[RETRACTED] >> *VC̲ (a markedness constraint against unretracted vowel–retracted consonant sequences). At stratum 2, the crucial ranking is reversed, with *VC̲ >> DEP[RETRACTED]. When there is no reduplication, the winning candidate has a harmonized retracted vowel to avoid a *VC̲ violation. When there is reduplication, and the uvular is no longer adjacent to the vowel, the winning candidate has a faithful unretracted vowel, which does not violate *VC̲ (Table 1c). This accounts for the base and reduplicant in (2).

Table 1: Inputs and outputs at each stratum, accounting for vowel retraction in reduplicants and bases.

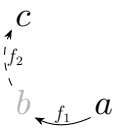
		a) Pharyn. harmony		*b) Northern dialect*		*c) Uvular harmony*	
Stratum 1	*Input:*	/CV́ʕ/		/Cíʕ/		/CV́q/	
	Output:	[CV̲́ʕ]		[Cíʕ]		[CV́q]	
Stratum 2	*Input:*	/CV̲́ʕ/	/RED+CV̲́ʕ/	/Cíʕ/	/RED+Cíʕ/	/CV́q/	/RED+CV́q/
	Output:	[CV̲́ʕ]	[CV̲́<C>ʕ]	[Ci̲ʕ]	[Cí<C>ʕ]	[CV̲́q]	[CV́<C>q]

References * Bermúdez-Otero, Ricardo. (2018). Stratal Phonology. In S.J. Hannahs, & Anna R. K. Bosch (Eds.), *The Routledge handbook of phonological theory* (pp. 100–134). Routledge. * Bessell, Nicola Jane. (1992). *Towards a phonetic and phonological typology of post-velar articulation* [Doctoral dissertation]. Department of Linguistics, University of British Columbia. * Esling, John H., Scott R. Moisik, Allison Brenner, & Lise Crevier-Buchman. (2019). *Voice quality: The Laryngeal Articulator Model*. Cambridge University Press. * Kiparsky, Paul. (2015). Stratal OT: A synopsis and FAQs. In Yuchau E. Hsiao, & Lian-Hee Wee (Eds.), *Capturing phonological shades.* Cambridge Scholars Publishing. * Kuipers, Aert H. (1974). *The Shuswap language: Grammar, texts, dictionary*. Mouton. * Mellesmoen, Gloria Madeleine. (2025). *The grammar of Salish reduplication* [Doctoral dissertation]. Department of Linguistics, University of British Columbia. * Moisik, Scott, Ewa Czaykowska-Higgins, & John H. Esling. (2012). The epilaryngeal articulator: A new conceptual tool for understanding lingual-laryngeal contrasts. *McGill Working Papers in Linguistics 22* (1), 1–15. * Nakamura, Yumiko. (2000). Secwepemctsin (Shuswap) reduplication. In, *Proceedings of the 35th International Conference on Salish and Neighbouring Languages* (pp. 171–184). UBCWPL.

Symmetrical and asymmetrical re-pairing in vowel harmony

Eric Baković, UC San Diego (AMP 2026, UMD)

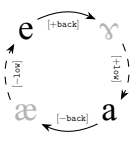
Overview. *Re-pairing* refers to situations where an alternation primarily affecting one feature f_1 causes a change in an additional feature f_2 (Baković 2000). This additional feature change is typically understood to be made necessary by the absence in the segmental inventory of a direct f_1 counterpart (or 'pair') b for the segment a undergoing the alternation; instead of leading to blocking of the motivated change in f_1, there is a further change in f_2 to 're-pair' a with another segment c. Re-pairing is common in vowel harmony patterns, but a typology of attested/possible types of re-pairing has not been described and formalized. This work aims to address this gap in the literature.



Symmetrical re-pairing. One attested type of re-pairing is referred to here as *symmetrical*, and is exemplified by the behavior of [−high] vowels in Turkish [±back] harmony (Clements & Sezer 1982). The data in (1) illustrate. The [+high] vowel of the 1P.SG possessive suffix in the examples in (1a) alternates between [−back] [i] and [+back] [ɯ], a perfect harmonic pair that requires no re-pairing. The [−high] vowel of the plural suffix in (1b), on the other hand, alternates between [−back, −low] [e] and [+back, +low] [a]. These vowels consistently form a [±back] harmonic pair despite the fact that they also differ in terms of [±low].

(1) a. **ip-im** 'my rope' **el-im** 'my hand' **kɯz-ɯm** 'my daughter' **ʧan-ɯm** 'my bell'
b. **ip-ler** 'ropes' **el-ler** 'hands' **kɯz-lar** 'daughters' **ʧan-lar** 'bells'

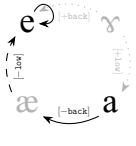
The symmetry of this re-pairing is understood to be due to the absence in the Turkish vowel inventory of *both* the [+low, −back] vowel [æ] *and* the [−low, +back] vowel [ɤ]. When a hypothetical input vowel /a/ is harmonically prompted to change from [+back] to [−back], there is no direct harmonic pair in the vowel inventory for it to become, and likewise for hypothetical /e/ to change from [−back] to [+back]. An additional change in [±low] to both hypothetical inputs leads these vowels to be re-paired with each other, as illustrated to the left. Indeed, the underlying [−high] vowel in Turkish suffixes like the plural could hypothetically be *any* of these four vowels, with harmony and/or re-pairing processes ensuring that the only vowels to surface are [a] or [e].



Asymmetrical neutral re-pairing. Another attested type of re-pairing is here called *asymmetrical neutral*, and is exemplified by the behavior of [−high] vowels in Hungarian [±back] harmony (Siptár & Törkenczy 2000). The data in (2) illustrate. The vowels of the 1P.PL.POSS and allative suffixes in (2a) alternate between [u] and [ü] and between [o] and [ö], respectively, both perfect harmonic pairs. Much as in Turkish, [+low] /a, aː/ raise to [−low] [e, eː] when harmonically prompted to become [−back], as shown by the alternations of the inessive and superessive + 3P.S.POSS suffixes in (2b). However, /eː/ is independently neutral to vowel harmony, surfacing unchanged even in [+back] contexts (2c).

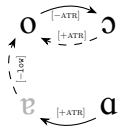
(2) a. **haːz-unk** 'our house' **föld-ünk** 'our land' **haːz-hoz** 'to the house' **föld-höz** 'to the land'
b. **haːz-ban** 'in the house' **föld-ben** 'in the land' **haːz-aːn** 'on his/her house' **föld-eːn** 'on his/her land'
c. **haːz-eːrt** 'the house's' **föld-eːrt** 'the land's' **haːz-eː** 'for the house' **föld-eː** 'for the land'

Putting aside vowel length, the same two vowel qualities that are absent from the Turkish vowel inventory are absent from the Hungarian vowel inventory. However, only underlying /a/ undergoes harmony and re-pairing to become surface [e]; /e/ instead resists harmony, as illustrated to the right. (Whether hypothetical /ɤ/ lowers to [a] or fronts to [e] cannot be decided, hence the dotted arrows from this vowel.)



Asymmetrical alternating re-pairing. A third attested type of re-pairing is referred to here as *asymmetrical alternating*, and is exemplified by the behavior of [−high] vowels

in Maasai (rightward) [±ATR] harmony (Tucker & Mpaayei 1955). The data in (3) illustrate. The vowels of the dative and applicative suffixes in (3a) alternate between [ɔ] and [o] and between [ɪ] and [i], respectively, both perfect harmonic pairs. The [+low] vowel /ɑ/ of the plural suffix in (3b), on the other hand, becomes [−low] [o] when harmonically prompted to become [+ATR]. The vowel [o] is thus both harmonically paired as expected with [ɔ] (viz. the dative suffix) and re-paired with [ɑ] (viz. the plural suffix), due to the absence of [ɑ]'s direct [+ATR] counterpart, [ɐ], from the inventory (right).



(3) a. ɑ-lɪpɔŋ-ɔk-ɪn 'INF-female-DAT-APPL' ɑ-mudoŋ-ok-in 'INF-relative-DAT-APPL'
b. ɪn-lɪpɔŋ-ɑ 'FEM.PL-female-PL' in-mudoŋ-o 'FEM.PL-relative-PL'

Formalization. We formalize this typology in terms of *Boolean Monadic Recursive Schemes* (BMRS; Chandlee & Jardine, 2021), a computational formalism that determines the output value of each feature f for a segment x by an IF . . . THEN . . . ELSE . . . function $\phi_f(x)$, returning either $\top$ (true, $x = [+f]$) or $\perp$ (false, $x = [-f]$). The three re-pairing patterns require two functions, one determining the output value of the harmonic feature and the other that of the re-pairing feature. The number and placement of *redundancy conditions* in these functions is what distinguishes each type of re-pairing from the others (cf. Nelson & Baković 2025).

Symmetrical re-pairing in Turkish. The output value of [±back] is determined by (4a): root vowels are faithful to their input value (= back(x)) and vowels otherwise agree with the output value of their predecessor (= $\phi_{\texttt{back}}(p(x))$). The output value of [±low] is determined by (4b): [+high] and [+round] vowels are redundantly [−low] (= the first $\perp$); remaining [+back] vowels are otherwise redundantly [+low] (= $\top$), accounting for /e/→|ɤ|→[a]; and remaining vowels are [−low] (= the last $\perp$), symmetrically accounting for /a/→|æ|→[e].

(4) a. $\phi_{\texttt{back}}(x) \coloneqq$ IF root(x) THEN back(x) ELSE $\phi_{\texttt{back}}(p(x))$
b. $\phi_{\texttt{low}}(x) \coloneqq$ IF $\left\{\begin{smallmatrix}\texttt{+high}\\ \texttt{+round}\end{smallmatrix}\right\}(x)$ THEN $\perp$ ELSE IF $\phi_{\texttt{back}}(x)$ THEN $\top$ ELSE $\perp$

Asymmetrical neutral re-pairing in Hungarian. The output value of [±back] is determined by (5), differing from Turkish (4a) by first determining that [−low, −round] vowels are redundantly [−back] (= $\perp$) before propogating [±back] from roots to suffixes, thus ensuring the neutrality of [e] (and, as it happens, that |ɤ|→[e]). The output value of [±low] in Hungarian is determined by the same function as in Turkish, (4b), accounting for /a/→|æ|→[e].

(5) $\phi_{\texttt{back}}(x) \coloneqq$ IF $\left[\begin{smallmatrix}\texttt{−low}\\ \texttt{−round}\end{smallmatrix}\right](x)$ THEN $\perp$ ELSE IF root(x) THEN back(x) ELSE $\phi_{\texttt{back}}(p(x))$

Asymmetrical alternating re-pairing in Maasai. The output value of [±ATR] is in determined like [±back] is in Turkish, replacing 'back' with 'ATR' in (4a). The output value of [±low] is then determined by (6): [+ATR] vowels are all redundantly [−low] (= $\perp$), accounting for /ɑ/→|ɐ|→[o]; and remaining vowels are faithful to their underlying [±low] value, allowing [ɑ] and [ɔ] to contrast in [−ATR] contexts even as they neutralize to [o] in [+ATR] contexts. (Note that [-low, +back] vowels must also be redundantly specified as [+round].)

(6) $\phi_{\texttt{low}}(x) \coloneqq$ IF $\phi_{\text{ATR}}(x)$ THEN $\perp$ ELSE low(x)

Conclusion. This is a first attempt to describe this typology and to understand it formally. The presentation will clarify and expand upon much that was cut in this abstract for space.

References cited. Baković. 2000. *Harmony, Dominance and Control*: Rutgers dissertation. • Chandlee & Jardine. 2021. Computational universals in linguistic theory. *Lg.* 97. • Clements & Sezer. 1982. Vowel and consonant disharmony in Turkish. In van der Hulst & Smith (eds.), *The Structure of Phonological Representations, Pt. 2*, Foris. • Nelson & Baković. 2025. Feature spreading, redundancy, and blocking behavior. lingbuzz/009295. • Siptár & Törkenczy. 2000. *The Phonology of Hungarian.* OUP. • Tucker & Mpaayei. 1955. *A Maasai Grammar, with Vocabulary.* Longmans, Green & Co.

The representation of vocoid-fricative vowel allophony in Nantong Chinese

Larry Lyu, University of California, Santa Cruz

Overview. The phonological status of fricative vowels (FVs), tone-bearing, nuclear segments produced with concomitant fricative-like noise, is a topic of considerable controversy (see van Hugte et al. 2025). In this study, we consider the FVs of Nantong Chinese, which (i) occur after both fricative and non-fricative onsets, and (ii) exhibit phonotactically conditioned allophony with high vowels (vocoids). We propose that Nantong FVs are underlyingly complex (specified for both [dorsal] and non-dorsal fricative place features), FVs and vocoid allophones each surface as a subset of the features in the UR.

Background. Existing phonological treatments of FVs largely fall within two camps. One camp argues that they are allophones of vocoid phonemes based on the restricted distribution of FVs in languages such as Mandarin, where FVs surface exclusively after sibilant onsets (Lee-Kim 2014). Opponents to this account propose that FVs are phonemes independent from vocoids, citing the substantive phonetic differences between surface FVs and their vocoid counterparts (e.g. [z̩] vs. [i]). Moreover, the allophony analysis appears unsuitable for FVs which surface following both fricative and non-fricative onsets and contrast with vocoids, in languages like Jixi Hui Chinese (Shao 2020) and Mambila (Connell 2007).

Nantong Chinese is a variety of Tong-Tai Chinese with three FVs [ʒ̩ ʒ̩ʷ ɣ̩], which are reflexes of historical high vowels [i y u]; we focus on the coronal FV [ʒ̩] in this abstract. As seen in (1a~b, g~h), Nantong [ʒ̩] surfaces outside post-fricative contexts. Frication in the FV must thus be inherent, since it cannot be attributed to coarticulation or assimilation to an outside segment. However, the vocoid [i] surfaces in place of [ʒ̩] following an alveolar onset, as seen in (1c-f). Compare with their cognates in the closely related Rudong Chinese, where [ʒ̩] surfaces after all onsets in (1).

(1) *Nantong and Rudong cognates containing [ʒ̩]*

	Rudong	Nantong	Gloss		Rudong	Nantong	Gloss
a.	pʒ̩˧	pʒ̩˥	‘compare’	e.	nʒ̩˩	ni˩	‘mud’
b.	mʒ̩˧	mʒ̩˥	‘rice’	f.	lʒ̩˩	li˩	‘to leave’
c.	tʒ̩˧	ti˥	‘bottom’	g.	ʒ̩˩	ʒ̩˩	‘ten million’
d.	tʰʒ̩˨˧	tʰi˨˦	‘ground’	h.	ʃʒ̩˥	ʃʒ̩˩	‘to play’

The complementary distribution between [ʒ̩] and [i] (such that [pʒ̩] [ti] but *[pi] *[tʒ̩]) an indicates allophonic relationship. Moreover, in the same phonotactic environments where Nantong [ʒ̩] alternates with [i], in an apparent conspiratorial fashion, the related Fengli variety retains the FV but shows affrication (2c) and deletion (2d-e) of the alveolar onset. This suggests the combination of alveolar onsets with [ʒ̩] results in ill-formed syllables, which may be resolved cross-linguistically via different processes. According to this view, Nantong [ʒ̩] surfaces as [i] in, for example, (2c) to avoid an otherwise marked syllable *[tʒ̩]. The markedness is also predicated on onset place, since [pʒ̩] with a [labial] onset surfaces without variation across Tong-Tai in (2).

(2)

	Rudong	Nantong	Fengli	*Gloss*
a.	ʒ̩˨˦	ʒ̩˨˦	ʒ̩˨˦	‘clothes’
b.	pʒ̩˧	pʒ̩˥	pʒ̩˧	‘compare’
c.	tʒ̩˧	t**i**˥	t**ʃ**ʒ̩˨˦	‘bottom’
d.	nʒ̩˩	n**i**˩	Øʒ̩˩	‘mud’
e.	lʒ̩˩	l**i**˩	Øʒ̩˩	‘to leave’

This account is further consistent with the idea that FVs and vocoids are allophonous in Nantong. However, FV-vocoid allophony raises considerable challenges. In the absence of an external source of frication, we must account for the phonetically disparate FV and vocoidal allophones within the same phonological representation.

Proposal. We propose that Nantong FVs are underlyingly specified for multiple place features. [ʑ̩] is thus underlyingly /ʒ^{i}/, specified for both [coronal] and [dorsal], as schematized in Figure 1. Following Padgett (2004), I assume stricture and anteriority features are closely bundled in phonological processes. In Fig. 1, association lines between features indicate close bundling as feature classes rather than dependency. Thus [coronal] is associated with [–anterior] and [+strident], and [dorsal] bundles with the vowel feature [–back]. Following a common heuristic for primary articulation (van de Weijer 2011; Smith 2018, p.23), the place feature associated with the greatest stricture (i.e. [coronal]) is specified as primary.

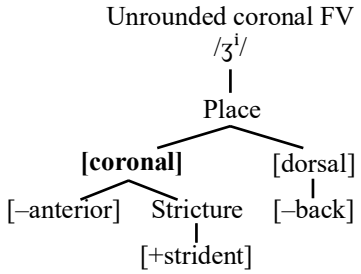


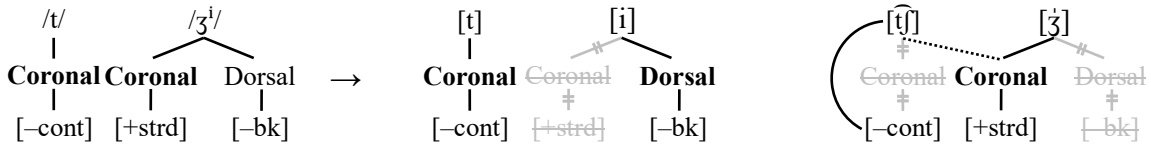
Figure 1

In this way, [ʑ̩] surfaces when [dorsal] and its associated feature are deleted, realizing a segment resembling a syllabic postalveolar fricative. Whereas [i] surfaces when [coronal] and its associated [–anterior] and [+strident] are deleted, recovering a front vocoid.

Phonotactically-conditioned 'de-fricativization' We argue that the mapping from /ʒ^{i}/→[ʑ̩] is driven by two constraints: *COMPLEXSEGMENT, which requires surface segments to have no more than one place feature, and IDENT(PRIMARY), which requires the SR and UR to have the same primary place feature. Given input /ʒ^{i}/, these constraints demand the deletion of [dorsal] and the associated [–back], giving rise to the FV [ʑ̩].

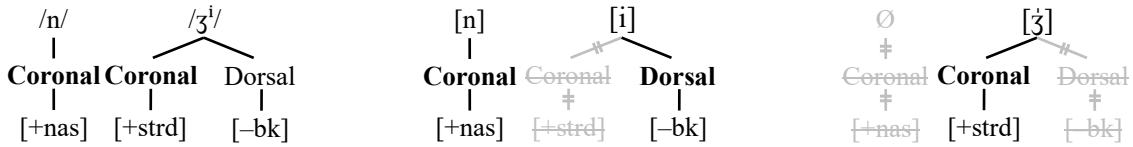
The mapping /ʒ^{i}/→[i] in Nantong is analyzed as a repair for marked syllables with an alveolar (i.e. [coronal, +anterior]) onset followed by [ʑ̩]. We propose such repairs attested across Tong-Tai varieties are driven by an OCP constraint, which prohibits adjacent identical place features. As illustrated in (3), given the input /tʒ^{i}/, a language must either delete [coronal] of the FV and its associated features, yielding Nantong [ti], or spread the [coronal] of the FV to the onset, yielding Fengli [tʃʑ̩].

(3) *Input:* *Output a: Nantong* *Output b: Fengli*



When the onset is nasal, however, spreading à la (3b) results in a nasalized affricate onset, which we propose is eliminated by a phonetically grounded constraint against nasalized fricatives. Thus onset deletion is adopted instead, as attested in (2).

(4) *Input:* *Output a: Nantong* *Output b: Fengli*



Crucially, the present analysis predicts that the FV in a heterorganic syllable like /pʒ^{i}/ should surface as [ʑ̩], since it does not incur an OCP violation. This prediction is in line with the attested data across Tong-Tai Chinese.

Bibliography

Connell, Bruce. 2007. Mambila fricative vowels and Bantu spirantisation. Persée - Portail des revues scientifiques en SHS. https://doi.org/10.3406/aflin.2007.969.

Hugte, Thom van, Yiya Chen & Li Guo. 2025. The contradictory nature of fricative vowels in Chinese and beyond. *Rarities in phonetics and phonology: Structural, typological, evolutionary, and social dimensions*. Language Science Press 395–436.

Padgett, Jaye. 1994. Stricture and Nasal Place Assimilation. *Natural Language & Linguistic Theory*. Springer 12(3). 465–513.

Shao, Bowei. 2020. *The apical vowel in Jixi-Hui Chinese : phonology and phonetics*. Université de la Sorbonne nouvelle - Paris III phdthesis. https://theses.hal.science/tel-03425234. (13 October, 2025).

Smith, Caitlin Michele. 2018. *Harmony in Gestural Phonology*.

Van De Weijer, Jeroen. 2011. Secondary and Double Articulation. In Marc Oostendorp, Colin J. Ewen, Elizabeth Hume & Keren Rice (eds.), *The Blackwell Companion to Phonology*, 1–17. 1st edn. Wiley. https://doi.org/10.1002/9781444335262.wbctp0029.

From Denasalization to Backcopying in Nyandarua Gĩkũyũ:
the Emergence of Base-Reduplicant Identity from Inventory Change

Chelsea (Yuen Chun) Tang | Massachusetts Institute of Technology

Overview: This paper reports novel fieldwork data on the interaction between nasal-prefix consonant mutation, Meinhof's Law, and reduplication in Nyandarua Gĩkũyũ (Bantu, Kenya). The findings make two theoretical contributions. First, it confirms that both variable and obligatory backcopying are present in the language, strengthening the case for Base-Reduplicant Correspondence Theory (BRCT; McCarthy & Prince 1995) while posing challenges to alternative analyses. Second, comparison with Standard Gĩkũyũ suggests that the emergence of backcopying in Nyandarua follows from a change in the stop contrast system.

Theoretical Background: Backcopying is an overapplication effect in which the reduplicant undergoes a process and the base "copies back" its effect, even when the triggering environment is absent in the base (McCarthy & Prince 1995; henceforth M&P). Such pattern has been central to debates in theories of reduplication because they bear on whether a more general model of phonology under- or over-generates (M&P 1995; Inkelas & Zoll 2005; Kiparsky 2010; and McCarthy, Kimper, & Mullin 2012, among others). Therefore, confirming the empirical status of these patterns remains theoretically important. This paper presents a new case of backcopying which contributes to this theoretical debate.

BR correspondence, as a subtype of Output-Output correspondence, often interacts with Input-Output correspondence. McCarthy's (1998) paradigm occultation analysis proposes that the presence or absence of Output-Output identity effects can depend on the language-particular ranking of OO-faithfulness relative to independently motivated markedness constraints and IO-faithfulness. The present analysis extends this logic to BR-correspondence: backcopying in Nyandarua surfaces due to a reranking of markedness and IO-faithfulness constraints relative to BR-faithfulness constraint.

Data: Nyandarua Gĩkũyũ is reported to have lost the prenasalization over the last few decades and has only voiced stops in its inventory, while the earlier, Standard Gĩkũyũ has prenasalized stops and lacks voiced stops. Under nasal prefixation, Standard Gĩkũyũ forms prenasalized stops: /N+ɾɛɣa/ → [ⁿdɛɣa] 'refuse me', /N+tuuɾa/ → [ⁿduuɾa] 'hurt me'. Nyandarua Gĩkũyũ instead forms voiced stops: [dɛɣa], [duuɾa]. Both varieties exhibit Meinhof's Law, where a prenasalized/voiced stop must become a plain nasal when followed by a prenasalized/voiced stop or a nasal: /N+ɾɛma/ → [nɛma] 'defeat me'.

In reduplication, only Nyandarua shows consonant backcopying. Voiced stop backcopying is variable (1). When the RED undergoes Meinhof's law and results in a plain nasal, backcopying of the nasal is obligatory (2).

(1)	Stem	N+Stem	N+RED+Stem[1]	Notes
(a)	[tɛi͡ðia] 'help'	[dɛi͡ðia] 'help me'	[**d**ɛi͡ða=**d**ɛi͡ðia] or [**d**ɛi͡ða=**t**ɛi͡ðia] 'help me a little'	*[**t**ɛi͡ða=**d**ɛi͡ðia]
(b)	[ʃɔɾa] 'draw'	[dʒɔɾa] 'draw me'	[**dʒ**ɔɾa=**dʒ**ɔɾa] or [**dʒ**ɔɾa=**ʃ**ɔɾa] 'draw me a little'	*[**ʃ**ɔɾa=**dʒ**ɔɾa]
(c)	[ɾiɾikana] 'remember'	[diɾikana] 'remember me'	[**d**iɾa=**d**iɾikana] or [**d**iɾa=**ɾ**iɾikana] 'remember me a little'	*[**ɾ**iɾa=**d**iɾikana]

[1] The verbal morpheme order of Gĩkũyũ is …-Object-Reduplicant-Stem-…. In the examples, the underspecified 'N' represents the 1SG object pronoun. The underlined copy is the Reduplicant and the non-underlined copy is the Base.

(2)	Stem	N+Stem	N+RED+Stem	Notes
(a)	[ɾɛma] ‘defeat’	[nɛma] ‘defeat me’	[**n**ɛma=**n**ɛma] ‘defeat me a little’	*[**n**ɛma=**ɾ**ɛma] *[ɾɛma=nɛma]
(b)	[ɾomia] ‘feed’	[nomia] ‘feed me’	[**n**oma=**n**omia] ‘feed me a little’	*[**n**oma=**ɾ**omia] *[ɾoma=nomia]
(c)	[ɾadoɾa] ‘remind sb. of a debt’	[nadoɾa] ‘remind me of a debt’	[**n**ada=**n**adoɾa] ‘remind me to pay you back a little’	*[**n**ada=**ɾ**adoɾa] *[ɾada=nadoɾa]

Analysis: The difference between the two varieties follows from reranking among *NC, IO-Id[nas], and BR-Id[nas]. In Standard Gĩkũyũ, prenasalized stops are contrastive which requires a ranking where IO-Id[nas] outranks *NC, BR-Id[nas]. This ranking blocks any consonant backcopying because all nasal-prefixed consonants are [+nas] when nasal prefixation on consonants forms either a prenasalized stop or plain nasal.

In Nyandarua, *NC is highly ranked to avoid NC sequences across the language, and IO-Id[nas] needs to rank low to allow for *NC repairs: *NC, BR-Id[nas] >> IO-Id[nas]. With the dominance of BR-Id[nas], nasal backcopying becomes obligatory and surfaces in the language. Voiced-stop backcopying is not obligatory but variable because voiced stops do not bear a [+nas] feature (i.e., no violation of BR-Id[nas]). The variability follows from lower-ranked general IO-Id and BR-Id, variably ranked as in Anttila (1995).

Alternative Analyses: Serial Templatic Satisfaction (McCarthy, Kimper, & Mullin 2012) faces a fundamental problem with the Nyandarua data: no ordering of constraint satisfaction can simultaneously derive the optionality in some cases and obligatoriness in others. A second alternative, framed within Morphological Doubling Theory (Inkelas & Zoll 2005), posits that the *N* prefix is inserted twice—once on the Base and once on the RED—before subsequent processes apply. This is implausible because other object pronouns surface only once in reduplicated forms and never intervene between Base and RED. For example, the stem [tɛ͡iðia] ‘help’ in (1a) can take different object pronouns and reduplicate: **d**ɛiða=**d**ɛiðia [1SG], where the object prefix becomes a part of the RED-initial onset and backcopied; versus **to**-tɛiða=tɛiðiɛ [1PL] and **ɣo**-tɛiða=tɛiðia [2SG], where the prefixes do not interact with RED at all. To defend this alternative analysis, one must first explain why only the 1SG object marker would double and not others. By contrast, BRCT (M&P 1995) captures both variable and obligatory backcopying: BR-faithfulness motivates the Base to match the RED, while IO-Id[nas] and variable ranking determines when that identity is enforced. These data show that neither process ordering nor morphological doubling alone can account for backcopying and BR-correspondence is empirically necessary for a theory of reduplication. Furthermore, Gĩkũyũ connects reduplicative identity to sound change: changes in the consonant inventory can affect constraint ranking in a way that would determine whether BR-Identity may surface.

Selected references: Anttila, Arto. 1995. *Deriving variation from grammar: A study of Finnish genitives*. Ms., Stanford University. • Inkelas, Sharon & Cheryl Zoll. 2005. *Reduplication: Doubling in Morphology*. Cambridge: Cambridge University Press. • McCarthy, John J. 2002. Morpheme structure constraints and paradigm occultation. In: Carpenter, Angela, Andries Coetzee & Paul de Lacy (eds.), *Papers in Optimality Theory II*. Amherst, MA: GLSA. 123–150. • McCarthy, John J., Wendell Kimper & Kevin Mullin. 2012. Reduplication in Harmonic Serialism. *Morphology* 22. 173–232. • McCarthy, John J. & Alan Prince. 1995. Faithfulness and reduplicative identity. ROA-60.

Modeling Stem-Internal ATR Harmony in Ìyíṇnọ́ with Tier-Based Strictly Local Grammar

Olalekan Malik Adebayo & Aniello De Santo
University of Utah

Introduction

Despite growing interest in the phonology of Niger-Congo languages, Ìyíṇnọ́ (Niger-Congo; Nigeria) remains largely understudied, and its ATR harmony system has received no formal computational attention. Subregular phonology classifies phonological patterns in terms of grammatical complexity, with the goal of identifying fundamental representational primitives that go beyond typological description alone (Heinz 2018; Aksënova et al. 2024). Within this scope of research, a range of long-distance phonological dependencies have been characterized as Tier-Based Strictly Local (TSL, henceforth; Heinz et al 2011), including ATR harmony in Niger-Congo languages such as Lokaa (Aksënova et al. 2024) and Dagaare (Angsongna 2025). This paper extends this scope of research to Ìyíṇnọ́, where ATR harmony presents a typologically distinct challenge: the transparent segment is the low vowel /a/, which is neither a consonant nor a high vowel, yet fails to participate in harmony. I show that this system is analyzable as TSL-2 within the subregular hierarchy.

ATR Harmony in Ìyíṇnọ́

Data come from original fieldwork conducted with three speakers of Ìyíṇnọ́ in Ondo State, Nigeria (2017–2018), elicited via the Ibadan 400-wordlist through recorded dialogue sessions. Ìyíṇnọ́ lacks underlying consonant clusters, though surface clusters may arise via glide formation (e.g., [ùsjā̌k]). Stems may be monosyllabic or multisyllabic, meaning harmonizing vowels may be separated by an arbitrary number of intervening consonants. ATR harmony in Ìyíṇnọ́ is therefore a long-distance pattern. The language vowel inventory is divided into two harmonic sets: [+ATR] = {i, e, u, o} and [−ATR] = {ɪ, ɛ, ʊ, ɔ}. Within stems, vowels obligatorily agree in [ATR] features, as illustrated in Table 1.

Form	**Gloss**	**Pattern**	
[ìfò]	'mortal'	[+ATR]	✓
[otutug]	'dust'	[+ATR]	✓
[èkiù]	'mate'	[+ATR]	✓
[kɔ̀ríh]	'break'	[−ATR]	✓
[èmɔ̌]	'hunger'	[−ATR]	✓
[ɔ̀sɛ̀kàdʒ]	'king'	[−ATR]	✓
[àhwè]	'money'	/a/ with [+ATR]	✓
[àʃʊ]	‘cloth’	/a/ with [−ATR]	✓
[ògwàdò]	'maize'	[+ATR]… /a/…[+ATR]	✓
[ɔ̀kàdʒì]	'stranger'	[−ATR]…/a/… [−ATR]	✓
*[ekiʊ]	—	mixed ATR	✗
*[ɔ̀sekàdʒ]	—	mixed ATR	✗

Table 1: Stem-internal ATR harmony in Ìyíṇnọ́

Crucially, as seen in Table 1, the low vowel /a/ behaves both neutrally and transparently. In [àhwè] and [àʃʊ], /a/ is neutral, as it co-occurs with vowels from both harmonic sets. In [ògwàdò] and [ɔ̀kàdʒì], /a/ is transparent, as it intervenes between ATR-participating vowels that agree in [ATR] without triggering or blocking harmony. This transparency further extends the distance between harmonizing vowels, since /a/ may intervene between them in addition to consonants. The ill-formed forms *[ekiʊ] and *[ɔ̀sekàdʒ] show that mixed-ATR vowel sequences are systematically excluded.

Subregular Analysis

TSL grammars (Heinz et al. 2011) capture non-local dependencies by projecting relevant segments onto a tier, over which a strictly local constraint applies. For Ìyíɲnọ́, I propose the natural class of non-low vowels as the harmony tier, *T = V[-low]* = {i, e, u, o, ɪ, ɛ, ʊ, ɔ}, excluding low /a/ and consonants, thereby encoding the transparency of /a/. Let H_{ATR} be the set of all mixed-ATR bigrams of the form [α ATR][−αATR]. The grammar bans substrings in H_{ATR} i.e., any pair of tier-adjacent vowels that disagree in [ATR] (e.g., *iɪ, *ɪi, *eɛ, *ɛe, *uʊ, ... *ɔo). Because the tier ignores all irrelevant segments, relevant vowels are always tier-adjacent regardless of intervening consonants or /a/, and the harmony constraint is strictly local over the tier. Since the banned substrings are tier-adjacent bigrams (k=2), ATR harmony in Ìyíɲnọ́ is a TSL-2 grammar. Although /a/-transparency can be represented in autosegmental or constraint-based approaches to vowel harmony, the TSL analysis here characterizes the formal complexity of the surface pattern as strictly 2-local. Figure 1 illustrates the analysis.



Figure 1: ATR Harmony in Ìyíɲnọ́

Given Ìyíɲnọ́'s surface (C)(C)V(C) syllable structure, an SL grammar (Chandlee, 2014) might be expected to model the pattern using bounded windows. However, the transparent behavior of /a/ means that harmonizing vowels may be separated by an unbounded number of segments. No fixed k-SL grammar can capture such dependencies, since SL grammars operate over contiguous substrings of fixed length. The limitation is not one of window size but of representational domain: SL cannot abstract away from irrelevant intervening material. The TSL analysis resolves this by projecting only ATR-active vowels to the tier, where the dependency is strictly local.

Conclusion

This study offers the first computational analysis of Ìyíɲnọ́ phonology, showing that stem-internal ATR harmony with a transparent low vowel is TSL-2. Importantly, TSL-2 grammars are efficiently learnable from positive data alone (Jardine & McMullin 2017; Lambert et al. 2021), suggesting that the phonotactic knowledge modeled here is acquisitionally plausible. Future work will extend the analysis to cross-morphemic vowel harmony and tonal patterns in the language.

References

Aksënova, A., Rawski, J., Graf, T. & Heinz, J. 2024. The computational power of harmonic forms. In van der Hulst & Ritter (eds.), Oxford Handbook of Vowel Harmony, 437–451. Oxford: OUP.

Angsongna, A. 2025. Phonotactic constraints and learnability: Analyzing Dagaare vowel harmony with tier-based strictly local grammar. Open Linguistics 11(1).

Chandlee, J. 2014. Strictly local phonological processes. PhD dissertation, University of Delaware.

Heinz, J. 2018. The computational nature of phonological generalizations. In Hyman & Plank (eds.), Phonological Typology, 126–195. Berlin: De Gruyter.

Heinz, J., Rawal, C. & Tanner, H. G. 2011. Tier-based strictly local constraints for phonology. Proceedings of ACL 2011, 58–64.

Jardine, A. & McMullin, K. 2017. Efficient learning of tier-based strictly k-local languages. In LATA 2017, 64–76. Springer.

Lambert, D., Rawski, J. & Heinz, J. 2021. Typology emerges from simplicity in representations and learning. Journal of Language Modelling 9, 151–194.